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A New Approach to Mixed Methods for Kirchhoff-Love Plates and Shells

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Abstract

In this thesis, we introduce a new approach to mixed methods for fourth-order problems with focus on Kirchhoff plates and Kirchhoff-Love shells. The main achievement of this work is the derivation of new mixed variational formulations that are only based on standard H^1 spaces. This offers the possibility to apply well-known techniques for second-order problems both regarding the discretization and the solution strategy.

In the first part, we consider the Kirchhoff plate bending problem with mixed boundary conditions involving clamped, simply supported, and free boundary parts. For this problem a new mixed variational formulation is derived, which satisfies Brezzi's conditions and is equivalent to the original problem, without additional convexity assumptions on the domain. These important properties come at the cost of involving an appropriate nonstandard Sobolev space for the auxiliary variable, the bending moment tensor, which is related to the Hessian of the vertical displacement. For the vertical displacement the standard Sobolev space H^1 (with appropriate boundary conditions) is used. Based on a regular decomposition of this nonstandard space, the fourth-order problem can be equivalently written as a system of three (consecutively to solve) second-order elliptic problems in standard Sobolev spaces.
This decomposition result on the continuous level leads in the discrete setting to new discretization methods, which are flexible in the sense, that any existing and well-working discretization method and solution strategy for standard second-order problems can be used as modular building blocks of the new method.

In the second part, we consider the Kirchhoff-Love shell problem, which is a more general fourth-order problem including beside the fourth-order derivative term (present in the Kirchhoff-plate bending problem) additional lower-order derivative terms. By extending the technique introduced for plates a new mixed formulation solely based on standard H^1 spaces is obtained. However, due to the additional lower-order derivative terms, the system can no longer be solved consecutively.
Nevertheless, this allows for flexibility in the construction of discretization spaces, e.g., standard C^0-coupling of multi-patch isogeometric spaces is sufficient. In terms of solution strategies, efficient methods for standard second-order problems like multigrid can be used as building blocks of preconditioners for iterative solvers.

The performance of the resulting discretization methods for plates and shells is demonstrated by numerical experiments. In case of plates, under the assumption of a polygonal domain, a rigorous numerical analysis is performed. All considered methods are implemented in the C++ library G+Smo.

Zusammenfassung

Diese Dissertation widmet sich der Entwicklung eines neuen Zugangs zu gemischten Methoden für Probleme vierter Ordnung, wobei der Fokus auf Kirchhoff-Platten und Kirchhoff-Love-Schalen liegt. Das Hauptergebnis dieser Arbeit ist die Herleitung einer neuen gemischten variationellen Formulierung, welche nur auf üblichen Sobolev-Räumen erster Ordnung ("standard" H^1-Räumen) basiert. Dies ermöglicht die Anwendung bekannter Methoden für Probleme zweiter Ordnung sowohl für die Diskretisierung als auch für die Lösungsstrategie.

Im ersten Teil betrachten wir das Kirchhoff-Plattenproblem für die Durchbiegung mit gemischten Randbedingungen, welche eingespannte, gelenkig gelagerte und freie Randteile beinhalten. Für dieses Problem leiten wir eine neue gemischte variationelle Formulierung her, die Brezzis Bedingungen erfüllt und äquivalent zum ursprünglichen Problem ist, ohne zusätzliche Konvexitätsvoraussetzung an das Gebiet. Diese wichtigen Eigenschaften fordern ihren Preis, nämlich die Verwendung eines entsprechenden "nonstandard" Sobolev-Raums für die Hilfsvariable, den Tensor der Biegemomente, welcher mit der Hesse-Matrix der vertikalen Verschiebung zusammenhängt. Für die vertikale Verschiebung wird der "standard" H^1-Raum (mit entsprechenden Randbedingungen) verwendet. Basierend auf einer regulären Zerlegung dieses "nonstandard" Sobolev-Raums kann das Problem vierter Ordnung äquivalent als System von drei (hintereinander zu lösenden) elliptischen Problemen zweiter Ordnung in "standard" H^1-Räumen formuliert werden.

Dieses Zerlegungsresultat am kontinuierlichen Level führt im Diskreten zu neuen Diskretisierungsmethoden, die flexibel sind im Sinn, dass jede existierende und gut funktionierende Diskretisierungsmethode und Lösungsstrategie für Probleme zweiter Ordnung als Bestandteile für die Konstruktion einer neuen Methode verwendet werden können.

Im zweiten Teil betrachten wir das Kirchhoff-Love-Schalenproblem, dies ist ein allgemeineres Problem vierter Ordnung, das neben dem Term vierter Ordnung (vorhanden im Kirchhoff-Plattenproblem für die Durchbiegung) zusätzliche Ableitungsterme niedrigerer Ordnung enthält. Durch eine Erweiterung der für Platten eingeführten Technik erhalten wir eine neue gemischte Formulierung basierend auf "standard" H^1-Räumen. Auf Grund der zusätzlichen Ableitungsterme niedrigerer Ordnung kann das System nicht mehr hintereinander gelöst werden.

Dennoch erlaubt dies Flexibilität in der Konstruktion der Diskretisierungsräume, z.B. C^0-Kopplung von "multi-patch" isogeometrischen Räumen ist ausreichend. Im Bezug auf Lösungsstrategien können effiziente Methoden für Probleme zweiter Ordnung wie Mehrgittermethoden zur Konstruktion von Vorkonditionierern für iterative Löser verwendet werden.

Die Leistungsfähigkeit der resultierenden Diskretisierungsmethoden für Platten und Schalen wird durch numerische Experimente demonstriert. Im Fall von Platten wird unter der Voraussetzung eines polygonalen Gebiets eine rigorose numerische Analyse durchgeführt. Alle betrachteten Methoden sind in der C++ Bibliothek G+Smo implementiert.

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Contents

Chapter 1

Introduction

Partial differential equations (PDEs) are used to mathematically describe complex processes in natural sciences and engineering, to name just some areas. An essential attribute of a PDE is its order, which is the order of the highest order derivative that appears in the PDE. The collection of terms containing derivatives of this highest order is called the principal part of the PDE.
A general second-order linear elliptic PDE (in divergence form) has the following principal part
$$\operatorname{div}(C(x)\nabla u(x)),$$
where div denotes the standard divergence of a vector-valued function, ∇ the gradient and $C(x)$ a coefficient matrix. As second-order model problem consider

$$\begin{aligned} -\operatorname{div}(C\nabla u) &= f \quad \text{in } \Omega, \\ u &= 0 \quad \text{on } \Gamma, \end{aligned} \tag{1.1}$$

with the unknown function u, the source term f and the computational domain $\Omega \subset \mathbb{R}^d$ with boundary Γ. If C is the identity, problem (1.1) reduces to the Poisson problem with Dirichlet boundary conditions.
In this thesis, we turn our attention to fourth-order linear elliptic PDEs with the principal part given by
$$\operatorname{div}\operatorname{Div}(\mathcal{C}(x)\nabla^2 u(x)),$$
where Div denotes the row-wise divergence of a matrix-valued function, ∇^2 the Hessian and $\mathcal{C}(x)$ a fourth-order coefficient tensor. As fourth-order model problem consider

$$\begin{aligned} \operatorname{div}\operatorname{Div}(\mathcal{C}\nabla^2 u) = f \qquad & \text{in } \Omega, \\ u = 0, \quad \partial_n u = 0 \qquad & \text{on } \Gamma, \end{aligned} \tag{1.2}$$

where ∂_n denotes the derivative in normal direction. Note that in contrast to the second-order problem (1.1) for a fourth-order problem we have to prescribe two conditions at the same part of the boundary. Similar as above, if $\mathcal{C}$ is the identity, problem (1.2) reduces to the well-known first biharmonic boundary value problem.

In this thesis we restrict ourselves to space dimension $d = 2$. This is in accordance with the applications we have in mind in the field of plate and shell analysis. A shell is a thin-walled structure, where one dimension is considerably smaller that the other two. A plate is the special case of a shell with no curvature.
The governing equation of the Kirchhoff plate bending problem is of the form (1.2), where the unknown u describes the vertical displacement, the source term f the vertical loading and $\mathcal{C}$ the material tensor. The boundary conditions in (1.2) correspond to the situation of a purely clamped plate, meaning the displacement is fully restricted along the boundary. In the context of plates two other types of boundary conditions are of importance, called free and simply supported. Free means we have no restriction of the displacement, and in case of simply supported we only impose partial restriction of the displacement. In the general situation, which we consider in this thesis, we have mixed boundary conditions involving clamped, simply supported, and free boundary parts. Kirchhoff-Love shell problems lead to more general fourth-order problems, which include beside the principle part in (1.2) additional terms involving lower-order derivatives.

State of the art

In literature different formulations of the model problem (1.2) have been considered.

Primal formulation

In the standard (primal) variational formulation of (1.2) the displacement is sought in the Sobolev space H^2 of twice weakly differentiable functions with appropriate boundary conditions. For the obtained symmetric and coercive problem existence and uniqueness of a solution can be guaranteed by standard application of the theorem of Lax-Milgram.
The analysis of the continuous formulation is straight-forward, but this formulation poses some difficulties for the further considerations: First of all, for an H^2-conforming discretization C^1-continuity between elements is needed. Second, the order four of the PDE has additionally an impact on the properties of the obtained system matrix after discretization. More precisely, for discretizations with piecewise polynomials the condition number of the resulting system matrix is of order h^{-4}, where h is the mesh size. For direct solvers this leads to an increased loss of significant digits in the solution. Furthermore, for the construction of efficient iterative solvers well-known standard techniques from second-order problems cannot be directly applied.
For both of the mentioned difficulties, many approaches to cope with them are discussed in literature.

Discretization method

Finite element discretizations range from conforming (like the Argyris element) and classical non-conforming finite element methods (like the Morley element) for fourth-order problems, see, e.g., [27, 59] to discontinuous Galerkin (dG) methods for fourth-order problems as introduced in [34, 21].
Another aspect is the demand to perform analysis directly on the geometry representation provided by the Computer Aided Design (CAD) model, and, thereby, avoiding unnecessary and costly geometry approximations. Obviously, reproducing the exact geometry is of importance for shells and also for plates with a curved boundary, as underlined very clearly in [4]. This is enabled by isogeometric analysis proposed by Hughes and co-workers in [48, 29], where B-Splines or Non Uniform Rational B-Splines (NURBS) are used for the geometry description as well as for the representation of the unknown fields.
Another feature of isogeometric discretizations is that they easily allow for inter-element continuity beyond the classical C^0-continuity within so-called patches. This permits the straight-forward construction of conforming isogeometric Kirchhoff-Love type shell elements, which are hard to obtain by means of standard Lagrange basis functions. The contribution [52] was one of the first to exploit this.
For these approaches the application on a single patch is straight-forward. However, for practical computations involving complex geometries usually geometry representations by means of several subdomains (patches) are necessary. Such geometry representations are called multi-patch domains. Then the continuity between patches is an issue, for which several techniques have been developed. In [51] the so-called bending strip method is introduced, which was the starting point for the improved technique in [38]. Alternatively, dG techniques can be used patch-wise, see, e.g., [41]. Another approach are analysis suitable C^1 multi-patch isogeometric spaces, see, e.g., [50].

Solution strategy

Iterative solution techniques proposed for the linear systems, which show mesh-independent or nearly mesh-independent convergence rates are typically based on multilevel additive or multiplicative Schwarz methods (including multigrid methods), see, e.g., [65, 18, 19, 20, 22, 72, 73, 71], and the references therein. For sparse direct methods we refer the interested reader to [30].

Mixed formulation

A possible remedy to working with a fourth-order problem are mixed variational formulations. A common choice is to introduce beside the original unknown u as additional unknown $\boldsymbol{M} = -\mathcal{C}\nabla^2 u$, which is in the context of plates (and for shells with additional lower-order derivative terms) called the bending moment tensor. Note, since it has a physical meaning the bending moment tensor is in applications often a quantity of interest on its own. This allows to rewrite the fourth-order equation in (1.2) as a system of two second-order equations

$$\begin{aligned} \boldsymbol{M} &= -\mathcal{C}\nabla^2 u, \\ -\operatorname{div}\operatorname{Div}(\boldsymbol{M}) &= f. \end{aligned}$$

It turns out to be surprisingly hard to come up with a pair of function spaces $\boldsymbol{V}, Q$ for the bending moment tensor $\boldsymbol{M}$ and the displacement u, with Q being a subset of the Sobolev space H^1 equipped with the standard norm, that satisfies the well-posedness conditions of Brezzi's theory.
So far in literature, mixed methods for (1.2) in u and $\boldsymbol{M}$ have been formulated as linear operator equations in function spaces for which the associated linear operator is not an isomorphism, see, e.g., [15, 35, 3, 14]. In order to obtain existence of a solution, these papers rely on regularity properties of the primal problem, which are (except for [14]) ensured by the assumption that Ω is convex. This lack of easy-to-access knowledge on the mapping properties of the involved operators on the continuous level makes it hard to design efficient preconditioners on the discrete level.

Discretization method

In the papers mentioned above for the discretization a method referred to as the Hellan-Herrmann-Johnson (HHJ) method is considered, which goes back to [44, 45, 49, 23]. Despite the missing validity of Brezzi's conditions on the continuous level, a complete error analysis providing optimal error estimates is available, see, e.g., [35, 3, 14]. In particular, the approach in [3, 14] is of special interest because by using a suitable mesh-dependent variant of the H^2-norm for u it was shown that on the discrete level Brezzi's conditions are satisfied. Keep in mind, all of the mentioned papers (except for [14]) require Ω to be convex and only consider the simple situation of a purely clamped boundary. The HHJ method is strongly related to the non-conforming Morley finite element, cf. [2], and to the mixed dG method introduced in [47].
There exist alternative mixed methods with other additional unknowns. In [8, 10] a method was proposed that uses only standard C^0 finite element spaces for second-order problems for a formulation in the kinematic variables u and $\boldsymbol{\theta} = \nabla u$. By introducing beside u and $\boldsymbol{\theta} = \nabla u$ also $\boldsymbol{M} = \nabla^2 u$ and $\operatorname{Div} \boldsymbol{M}$ as unknowns a mixed method for the biharmonic problem is obtained in [5].

Solution strategy

In most papers mentioned above solution strategies for the linear system resulting from the considered mixed formulation are not addressed. We refer to [42] for a two-level additive Schwarz preconditioning scheme for solving the discrete system obtained by the HHJ mixed method.

On this work

In this thesis we investigate a new approach to mixed methods for fourth-order problems, in particular we apply it to Kirchhoff plates and Kirchhoff-Love shells. More precisely, we focus on mixed methods for (1.2) with the original unknown u and $\boldsymbol{M} = -\mathcal{C}\nabla^2 u$ as additional unknown. A built-in feature of such a method is the direct computation (without post-processing) of the bending moment tensor $\boldsymbol{M}$. The goal is to derive a mixed formulation that is only based on standard H^1 spaces. This offers the possibility to apply standard techniques for second-order problems both regarding the discretization and the solution strategy.

Main achievements

The new results of this thesis, which have not been previously known, can be grouped into the following three topics: New mixed variational formulation, decomposition result of the Kirchhoff plate bending problem and extensibility of the approach to more general fourth-order problems, in more detail:

New mixed variational formulation

As previously discussed, only very few papers consider other boundary conditions than the case of a purely clamped plate. Based on the classical concept of densely-defined (possibly unbounded) operators we present a procedure to obtain a new mixed variational formulation for Kirchhoff plate bending problems for all possible combinations of mixed boundary conditions involving clamped, simply supported, and free boundary parts. If for the original variable u a subspace of the standard Sobolev space H^1 (with appropriate boundary conditions) is specified, the classical framework of densely defined operators provides us with an appropriate nonstandard Sobolev space $\boldsymbol{V}$ for the auxiliary variable $\boldsymbol{M}$. With this choice for $\boldsymbol{V}$ our new mixed formulation satisfies Brezzi's conditions. Therefore, the associated linear operator is an isomorphism. Moreover, the equivalence of the new mixed formulation to the original problem follows rather straightforwardly and without convexity assumptions on Ω, while most of the papers in literature require Ω to be convex.

Decomposition result of the Kirchhoff plate bending problem

Based on a regular decomposition of the nonstandard Sobolev space $\boldsymbol{V}$, the Kirchhoff plate bending problem can be equivalently rewritten as a sequence of three (consecutively to solve) second-order problems in standard Sobolev spaces. Two of these problems are Poisson problems, the remaining one is a planar linear elasticity problem. Hence, the fourth-order Kirchhoff plate bending problem can be decomposed into three second-order elliptic problems.
This decomposition result on the continuous level leads to new discretization methods, which are flexible in the sense, that any existing and well-working discretization

method and solution strategy for second-order problems can be used as modular building blocks of the new method. One option would be to choose standard C^0 finite elements for each of the three second-order elliptic problems (for two scalar fields and one vector field) resulting in approximate solutions to u and $\boldsymbol{M}$.
In [8, 10] a different method was proposed that also uses only standard C^0 finite element spaces for second-order problems for a formulation in the kinematic variables u and $\boldsymbol{\theta} = \nabla u$. For reaching approximate solutions of comparable accuracy the method in [8, 10] requires the approximation of one scalar field less than the approach presented here. However, the linear system resulting from the method in [8, 10] is a coupled system of all degrees of freedoms of one scalar and one vector field, while the method presented here requires to solve linear systems for the degrees of freedom separately for each of the two scalar fields and one vector field. This reduces the computational costs for direct solvers. For the use of iterative solvers efficient methods for standard second-order problems like multigrid methods can be directly used for each of the three linear systems. Preconditioning is not addressed in [8, 10]. So we feel that our method is competitive with respect to the overall computational efficiency.

Extensibility of the approach to more general fourth-order problems

We extend our approach to the Kirchhoff-Love shell problem, which is a more general fourth-order problem involving beside the principle part of the form (1.2) (present in the Kirchhoff plate bending problem) additional lower-order derivative terms. By adapting the technique introduced for plates we derive again a new mixed formulation only based on standard H^1 spaces. Due to the additional lower-order derivative terms, the equations no longer decouple, hence, no decomposition result is obtained. Nevertheless, this leads to flexibility in the construction of discretization spaces, e.g., standard C^0-coupling of multi-patch isogeometric spaces is sufficient. In terms of solution strategies, efficient methods for standard second-order problems like multigrid can be used as building blocks of preconditioners for iterative solvers.

Outline

The thesis is organized as follows: In Chapter 2 we consider Kirchhoff plate problems.

- In Section 2.1 the Kirchhoff plate model and the basic notations used in this chapter are introduced. Starting from the three-dimensional elasticity problem using the strain representation (resulting from the Kirchhoff kinematical assumptions) and the constitutive equation (for the plane stress state) the variational formulation of Kirchhoff plates is derived by a dimension reduction. This problem reduces to two independent problems, namely, the membrane problem and the bending problem. The membrane problem for the in-plane part of the displacement is a standard second-order problem, which can be solved independently using standard techniques. The bending problem for the transverse (vertical) part of the displacement is of the form of the fourth-order model problem (1.2) with the more general boundary conditions mentioned above.

- In Section 2.2 the mathematically precise (primal) variational formulation of the Kirchhoff plate bending problem is stated. In Section 2.3 first a brief introduction to densely-defined (possibly unbounded) operators is given. Based on this abstract concept we derive our new mixed formulation, which satisfies Brezzi's conditions and is equivalent to the original problem. These important properties come at the cost of an appropriate nonstandard Sobolev space $\boldsymbol{V}$ for $\boldsymbol{M}$. For completeness, a characterization of this space in case of smooth functions is provided.

- Section 2.4 contains a regular decomposition of this nonstandard space $\boldsymbol{V}$, which makes it computationally accessible. The regular decomposition involves a coupling condition for the two components in the decomposition. In case of a polygonal domain the coupling condition can be interpreted as essential boundary conditions which prepare for the incorporation by a standard Nitsche method (for second-order problems), see Section 2.5. We obtain an equivalent reformulation as a system of three (consecutively to solve) second-order elliptic problems. The decomposed formulation leads in a natural way to the construction of a new discretization method for which a-priori error estimates are derived.

  The extension of this approach to curvilinear polygonal domains poses severe difficulties. Therefore, we propose in Section 2.6 an alternative approach based on Lagrange multipliers, where we add the coupling condition as additional constraint. We obtain a similar system that can be solved consecutively with the difference that the second problem is now a saddle point problem.

- In Section 2.7 we outline how to extend our techniques to inhomogeneous boundary conditions.

- In Section 2.8 we illustrate the theoretical results for polygonal domains by numerical experiments. Furthermore, we show first promising numerical tests for the extensions to curvilinear polygonal domains and inhomogeneous boundary conditions.

Chapter 3 treats the extension to Kirchhoff-Love shell problems, which are more general fourth-order problems.

- Section 3.1 is organized like Section 2.1. We additionally provide a brief introduction to surface differential geometry as it is of essential importance for the shell model. In contrast to plates, the Kirchhoff-Love shell problem does not separate into a membrane and bending problem, since membrane and bending strain depend on both the tangential and transverse part of the displacement. Therefore, all three components of the displacement have to be considered at once, leading to a vector-valued space of displacements.

- In Section 3.2 the mathematically precise (primal) variational formulation of the Kirchhoff-Love shell problem is stated. As indicated above, this problem consists of a membrane and a bending part. The only fourth-order derivative term appears in the bending part and is of the same structure as (1.2), where u is the transverse component of the displacement. An extension of the approach proposed in Chapter 2 is possible, as outlined in Section 3.3. In Section 3.4 a combination of the proposed mixed formulation of the bending part with a popular mixed formulation of the membrane part to avoid membrane locking is considered. For both mixed formulations well-posedness and equivalence to the original problem is shown.

- In Section 3.5 we reformulate both mixed formulations as formulations in standard H^1 spaces using the regular decomposition of Section 2.4. Due to the additional lower-order derivative terms, in contrast plates the equations cannot be solved consecutively. The new mixed formulation leads in a natural way to the construction of a new discretization method, which is introduced in Section 3.6.

- The chapter closes with numerical experiments in Section 3.7.

In Chapter 4 we draw a brief conclusion and discuss further work including possible extensions of the presented approach.

Parts of this thesis have already been published in peer-reviewed journals or conference proceedings:

- Parts of Chapter 2 are presented in

  [54] W. Krendl, K. Rafetseder, and W. Zulehner. A decomposition result for biharmonic problems and the Hellan-Herrmann-Johnson method. *ETNA, Electron. Trans. Numer. Anal.*, 45:257–282, 2016.

  [66] K. Rafetseder and W. Zulehner. A decomposition result for Kirchhoff plate bending problems and a new discretization approach. *SIAM J. Numer. Anal.*, 2018. (to appear).

  [68] K. Rafetseder and W. Zulehner. On a new mixed formulation of Kirchhoff plates on curvilinear polygonal domains. In *Numerical mathematics and advanced applications – ENUMATH 2017.* Springer, 2018. (to appear).

- Parts of Chapter 3 are addressed in

  [67] K. Rafetseder and W. Zulehner. A new mixed isogeometric approach to Kirchhoff-Love shells. 2018. (submitted).

Chapter 2

Kirchhoff plates

In this chapter we consider the Kirchhoff plate problem, which turns out to decouple into two independent problems. The membrane problem for the in-plane displacement is a second-order problem, which can be solved independently using standard techniques. The bending problem for the transverse (vertical) displacements is of the form of the fourth-order model problem (1.2), with general boundary conditions involving clamped, simply supported, and free boundary parts.
This chapter is organized as follows. In Section 2.1 the Kirchhoff plate model is introduced. The (primal) variational formulation of the Kirchhoff plate bending problem is stated in Section 2.2. Section 2.3 contains a new mixed formulation, for which well-posedness and equivalence to the original problem is shown. A regular decomposition of the nonstandard Sobolev space for $\boldsymbol{M}$ is derived in Section 2.4 and the resulting formulation is presented. The regular decomposition involves a coupling condition for the two components in the decomposition. In case of a polygonal domain the coupling condition can be interpreted as essential boundary conditions which prepare for the incorporation by a standard Nitsche method (for second-order problems). We obtain an equivalent reformulation as a system of three (consecutively to solve) second-order elliptic problems, as outlines in Section 2.5. This leads in a natural way to the construction of a new discretization method, for which a-priori error estimates are derived. The extension of this approach to curvilinear polygonal domains poses severe difficulties. Therefore, we propose in Section 2.6 an alternative approach based on Lagrange multipliers. In Section 2.7 we outline how to extend our techniques to inhomogeneous boundary conditions. The chapter closes with numerical experiments in Section 2.8.

2.1 Kirchhoff plate model

The description of the Kirchhoff plate model is based on the presentations in [69, 16]. A plate is a planar thin-walled structure. One dimension, in our case x_3, is significantly smaller than the other two. The 3D plate in the undeformed configuration is defined by

$$\Omega^{3\mathrm{D}} = \left\{ (x_1, x_2, x_3) \in \mathbb{R}^3 \colon (x_1, x_2) \in \Omega,\ x_3 \in \left(-\frac{t}{2}, \frac{t}{2} \right) \right\},$$

where $\Omega \subset \mathbb{R}^2$ defines the midsurface of the plate and t denotes the thickness, which we assume for simplicity to be constant. In the following considerations the plate is subject to loads that cause bending and stretching deformations. Before we introduce the Kirchhoff plate model, we fix some notations.

2.1.1 Notations

In this chapter, scalars are denoted by simple letters and vectors and matrices by boldface letters. Throughout this chapter, we use the standard Cartesian coordinate system and denote the Cartesian components of a vector $\boldsymbol{a}$ and a matrix $\boldsymbol{A}$ using subscripts, i.e., $\boldsymbol{a} = (a_i)$ and $\boldsymbol{A} = (A_{ij})$. Latin indices take values $\{1, 2, 3\}$ and Greek indices $\{1, 2\}$. Furthermore, Einstein's summation convention is applied to indices appearing twice within a product. When ever we use the term (second-order) tensor we mean its representation as matrix with respect to the Cartesian coordinate system. In this sense, we do not distinguish between second-order tensors and matrices in this chapter.

We use the dot for the inner product between vectors $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{R}^2$ and $\boldsymbol{c}, \boldsymbol{d} \in \mathbb{R}^3$

$$\boldsymbol{a} \cdot \boldsymbol{b} = a_\alpha b_\alpha, \quad \boldsymbol{c} \cdot \boldsymbol{d} = a_i b_i,$$

and the colon for the inner product between matrices $\boldsymbol{A}, \boldsymbol{B} \in \mathbb{R}^{2\times 2}$ and $\boldsymbol{C}, \boldsymbol{D} \in \mathbb{R}^{3\times 3}$

$$\boldsymbol{A} : \boldsymbol{B} = A_{\alpha\beta} B_{\alpha\beta}, \quad \boldsymbol{C} : \boldsymbol{D} = C_{ij} D_{ij},$$

where usually the dimension is clear from the context and otherwise stated explicitly.

Let the midsurface Ω be a domain with a Lipschitz boundary Γ. In what follows let $\boldsymbol{n} = (n_\alpha)$ and $\boldsymbol{t} = (t_\alpha)$, with $t_1 = -n_2$ and $t_2 = -n_1$, represent the unit outer normal vector and the unit counterclockwise tangent vector to Γ, respectively. Furthermore, $\partial_{\boldsymbol{n}}$ denotes the normal derivative and $\partial_{\boldsymbol{t}}$ the tangential derivative along Γ.

For a vector $\boldsymbol{a} = (a_\alpha)$ we introduce the following notations

$$a_n = \boldsymbol{a} \cdot \boldsymbol{n}, \qquad a_t = \boldsymbol{a} \cdot \boldsymbol{t}$$

for the normal component and the tangential component, and for a matrix $\boldsymbol{A} = (A_{\alpha\beta})$

$$A_{nn} = \boldsymbol{A}\boldsymbol{n} \cdot \boldsymbol{n}, \qquad A_{nt} = \boldsymbol{A}\boldsymbol{n} \cdot \boldsymbol{t}$$

for the normal-normal component and the normal-tangential component.

Let v represent a scalar field $\Omega \to \mathbb{R}$, $\boldsymbol{\psi}$ a vector field $\Omega \to \mathbb{R}^2$, and $\boldsymbol{L}$ a matrix field $\Omega \to \mathbb{R}^{2\times 2}$, all sufficiently smooth. Here and in the following we use the notation

$\partial_\alpha = \partial_{x_\alpha}$. Then we have the following classical definitions of the differential operators:

$$\begin{aligned}
\nabla v &= \begin{pmatrix} \partial_1 v \\ \partial_2 v \end{pmatrix}, & \operatorname{curl} v &= \begin{pmatrix} \partial_2 v \\ -\partial_1 v \end{pmatrix}, \\
\nabla \boldsymbol{\psi} &= \begin{pmatrix} \partial_1 \psi_1 & \partial_2 \psi_1 \\ \partial_1 \psi_2 & \partial_2 \psi_2 \end{pmatrix}, & \operatorname{Curl} \boldsymbol{\psi} &= \begin{pmatrix} \partial_2 \psi_1 & -\partial_1 \psi_1 \\ \partial_2 \psi_2 & -\partial_1 \psi_2 \end{pmatrix}, \\
\operatorname{div} \boldsymbol{\psi} &= \partial_1 \psi_1 + \partial_2 \psi_2, & \operatorname{rot} \boldsymbol{\psi} &= \partial_1 \psi_2 - \partial_2 \psi_1, \\
\operatorname{Div} \boldsymbol{L} &= \begin{pmatrix} \partial_1 L_{11} + \partial_2 L_{12} \\ \partial_1 L_{21} + \partial_2 L_{22} \end{pmatrix}, & \operatorname{Rot} \boldsymbol{L} &= \begin{pmatrix} \partial_1 L_{12} - \partial_2 L_{11} \\ \partial_1 L_{22} - \partial_2 L_{21} \end{pmatrix}.
\end{aligned}$$

Moreover, the symmetric gradient and the symmetric Curl are introduced by

$$\varepsilon(\boldsymbol{\psi}) = \frac{1}{2}(\nabla \boldsymbol{\psi} + (\nabla \boldsymbol{\psi})^T), \quad \operatorname{symCurl} \boldsymbol{\psi} = \frac{1}{2}(\operatorname{Curl} \boldsymbol{\psi} + (\operatorname{Curl} \boldsymbol{\psi})^T).$$

2.1.2 Plate kinematics and constitutive equation

The invention of the classical plate theory dates back to 1850 and is accredited to Gustav Kirchhoff (1824-1887). In his model, the Kirchhoff plate model, the displacement field is based on the Kirchhoff kinematical assumptions, which consist of the following three parts:

1. Straight lines orthogonal to the midsurface (i.e. transverse normals) in the undeformed configuration remain straight during the deformation.
2. Transverse normals remain unstreched during the deformation.
3. Transverse normals rotate such that they remain orthogonal to the midsurface after the deformation.

For thick plates this theory is too restrictive, since also transverse shear deformations have to be taken into account. Transverse shear deformations can be understood as the sliding over each other of the surfaces parallel to the midsurface. These effects were first incorporated in their plate models by Eric Reissner (1913-1996) in 1945 and by Raymond Mindlin (1906-1987) in 1951. The idea was to consider the rotations as additional unknown beside the displacement of the midsurface. This means the third of the Kirchhoff assumptions was dropped, leading to the Reissner-Mindlin kinematical assumptions.
The Reissner-Mindlin kinematical assumptions imply the following form of the 3D displacement $\boldsymbol{U} = (U_i)$:

$$\begin{aligned}
U_\alpha(x_1, x_2, x_3) &= u_\alpha(x_1, x_2) - x_3\, \theta_\alpha(x_1, x_2), \\
U_3(x_1, x_2, x_3) &= u_3(x_1, x_2),
\end{aligned} \tag{2.1}$$

where $\boldsymbol{u} = (u_i)$ denotes the displacement of the midsurface of the plate, and $\boldsymbol{\theta} = (\theta_\alpha)$ the rotation of a transverse normal. In the following we refer to $\underline{\boldsymbol{u}} = (u_\alpha)$ and u_3 as in-plane and transverse (vertical) part of the displacement, respectively.

Throughout the thesis we consider small displacements, i.e., linear analysis. Then the 3D strain tensor $\boldsymbol{E}(\boldsymbol{U})$ is given by its components $E_{ij} = \frac{1}{2}(\partial_j U_i + \partial_i U_j)$ with $\partial_i = \partial_{x_i}$. For the specific displacement in (2.1) we obtain

$$\begin{aligned} E_{\alpha\beta}(\boldsymbol{U}) &= \frac{1}{2}(\partial_\beta u_\alpha + \partial_\alpha u_\beta) - x_3 \frac{1}{2}(\partial_\beta \theta_\alpha + \partial_\alpha \theta_\beta) = \varepsilon_{\alpha\beta}(\underline{\boldsymbol{u}}) + x_3 \kappa_{\alpha\beta}(\boldsymbol{\theta}), \\ E_{\alpha 3}(\boldsymbol{U}) &= \frac{1}{2}(\partial_\alpha u_3 - \theta_\alpha) = \gamma_\alpha(u_3, \boldsymbol{\theta}), \end{aligned} \tag{2.2}$$

where

$$\varepsilon_{\alpha\beta}(\underline{\boldsymbol{u}}) = \frac{1}{2}(\partial_\beta u_\alpha + \partial_\alpha u_\beta), \qquad \kappa_{\alpha\beta}(u_3) = -\frac{1}{2}(\partial_\beta \theta_\alpha + \partial_\alpha \theta_\beta). \tag{2.3}$$

The tensors $\boldsymbol{\varepsilon}(\underline{\boldsymbol{u}})$, $\boldsymbol{\kappa}(\boldsymbol{\theta})$ and $\boldsymbol{\gamma}(u_3, \boldsymbol{\theta})$ with components introduced above are called membrane strain, bending strain, and shear strain tensor, respectively. The third part of the Kirchhoff kinematical assumptions implies

$$\theta_\alpha = \partial_\alpha u_3.$$

Substituting this expression for the rotation in (2.2) leads to

$$\begin{aligned} E_{\alpha\beta}(\boldsymbol{U}) &= \frac{1}{2}(\partial_\beta u_\alpha + \partial_\alpha u_\beta) - x_3 \partial_{\alpha\beta} u_3 = \varepsilon_{\alpha\beta}(\underline{\boldsymbol{u}}) + x_3 \kappa_{\alpha\beta}(u_3), \\ E_{\alpha 3}(\boldsymbol{U}) &= 0 \end{aligned} \tag{2.4}$$

with the bending strain tensor $\boldsymbol{\kappa}(u_3)$ given by

$$\kappa_{\alpha\beta}(u_3) = -\partial_{\alpha\beta} u_3. \tag{2.5}$$

For an isotropic linear elastic material Hooke's law provides for the components of the 3D stress tensor $\boldsymbol{\sigma}$

$$\sigma_{ij} = 2\mu E_{ij} + \lambda(E_{11} + E_{22} + E_{33})\delta_{ij},$$

with the Lamé constants λ and μ given by

$$\lambda = \frac{\nu E}{(1+\nu)(1-2\nu)} \quad \text{and} \quad \mu = \frac{E}{2(1+\nu)}$$

if we denote Young's modulus by E and Poisson's ratio by ν, as usual.
By applying the plane stress assumption $\sigma_{33} = 0$ we obtain

$$E_{33} = -\frac{\nu}{1-\nu}(E_{11} + E_{22}),$$

which leads to the 2D constitutive equation for the plane stress state

$$\sigma_{\alpha\beta} = \frac{E}{1+\nu}\left(E_{\alpha\beta} + \frac{\nu}{1-\nu}(E_{11} + E_{22})\delta_{\alpha\beta}\right). \tag{2.6}$$

In short we write for $\boldsymbol{\sigma}_{2D} = (\sigma_{\alpha\beta})$ and $\boldsymbol{E}_{2D} = (E_{\alpha\beta})$

$$\boldsymbol{\sigma}_{2D} = \mathcal{C}\, \boldsymbol{E}_{2D}$$

with the application of the fourth-order material tensor given by

$$\mathcal{C}\boldsymbol{A} = \frac{E}{1+\nu}\left(\boldsymbol{A} + \frac{\nu}{1-\nu}\operatorname{tr}(\boldsymbol{A})\boldsymbol{I}\right) \quad \text{for all } \boldsymbol{A} \in \mathbb{R}^{2\times 2}, \tag{2.7}$$

where $\boldsymbol{I}$ is the identity matrix and tr is the trace operator for matrices. The theory presented in the following is independent of the particular structure of the material tensor. We only assume that the material tensor is symmetric and positive definite on symmetric matrices, i.e.,

$$\begin{aligned} &\mathcal{C}\boldsymbol{A} : \boldsymbol{B} = \boldsymbol{A} : \mathcal{C}\boldsymbol{B} && \text{for all } \boldsymbol{A}, \boldsymbol{B} \in \mathbb{R}^{2\times 2}_{sym} \\ &\mathcal{C}\boldsymbol{A} : \boldsymbol{A} > 0 && \text{for all } 0 \neq \boldsymbol{A} \in \mathbb{R}^{2\times 2}_{sym}, \end{aligned}$$

where $\mathbb{R}^{2\times 2}_{sym}$ denotes the space of symmetric matrices in $\mathbb{R}^{2\times 2}$. Furthermore, $\lambda_{min}(\mathcal{C})$ and $\lambda_{max}(\mathcal{C})$ denote the minimal and maximal eigenvalue of the material tensor, respectively, then

$$\lambda_{min}(\mathcal{C})\, \boldsymbol{A} : \boldsymbol{A} \;\leq\; \mathcal{C}\boldsymbol{A} : \boldsymbol{A} \;\leq\; \lambda_{max}(\mathcal{C})\, \boldsymbol{A} : \boldsymbol{A} \quad \text{for all } \boldsymbol{A} \in \mathbb{R}^{2\times 2}_{sym}.$$

2.1.3 Derivation of the plate model

In this section we follow the lines of [43] to derive the dimension reduced counterparts of the elasticity operator and the forces involved in a standard 3D linear elasticity problem. In the following we recall the results and refer for details to [43]. Throughout this section all functions are assumed to be sufficiently smooth.
We consider the standard linear elasticity problem in the domain $\Omega^{3D} \subset \mathbb{R}^3$ with boundary Σ. The undeformed midsurface $\Omega \subset \mathbb{R}^2$ is assumed to have a Lipschitz boundary Γ. Let the boundary Γ be written in the form

$$\Gamma = \mathcal{V}_\Gamma \cup \mathcal{E}_\Gamma \quad \text{with } \mathcal{E}_\Gamma = \bigcup_{k=1}^{K} E_k,$$

where E_k, $k = 1, 2, \ldots, K$, are the edges of Γ, considered as open possibly curvilinear segments and $\mathcal{V}_\Gamma$ denotes the set of corner points in Γ. The edges are numbered consecutively in counterclockwise direction. We denote the vertex at the start point of $\overline{E}_k$ by x_k, where $\overline{E}_k$ denotes the closure of E_k. Since we consider a closed boundary curve, the index $k = 0$ is in the following always identified with $k = K$.
The 3D plate is considered to be clamped on a part $\Sigma_c = \Gamma_c \times (-t/2, t/2)$, simply supported on a part $\Sigma_s = \Gamma_s \times (-t/2, t/2)$, and free on a part $\Sigma_f = \Gamma_f \times (-t/2, t/2)$, with $\Gamma = \Gamma_c \cup \Gamma_s \cup \Gamma_f$ and pairwise disjoint subsets Γ_c, Γ_s and Γ_f. We assume that each edge $E \in \mathcal{E}_\Gamma$ is contained in exactly one of the sets Γ_c, Γ_s, Γ_f, and the edges are maximal in the sense that two edges with the same type of boundary condition do not meet at an angle of π. We suppose that no kinematical boundary conditions for the displacement are prescribed on the top and bottom boundaries of Ω^{3D}. Tractions

$\boldsymbol{g} = (g_i)$ are prescribed on the whole boundary Σ, and body forces $\boldsymbol{f} = (f_i)$ are given in Ω^{3D}.
Thus, the total energy of the elasticity problem reads as

$$J(\boldsymbol{U}) = \frac{1}{2}\int_{\Omega^{3D}} \boldsymbol{\sigma}(\boldsymbol{U}) : \boldsymbol{E}(\boldsymbol{U})\; dV - \int_{\Omega^{3D}} \boldsymbol{f}\cdot\boldsymbol{U}\; dV - \int_{\Sigma} \boldsymbol{g}\cdot\boldsymbol{U}\; dA,$$

where the first part is the strain energy corresponding to a displacement $\boldsymbol{U}$, the second part is the potential energy related to the body forces $\boldsymbol{f}$, and the third part is the energy resulting from the prescribed boundary tractions $\boldsymbol{g}$.
The displacement $\boldsymbol{U} = (\underline{\boldsymbol{U}}, U_3)$ is then obtained by minimizing the energy $J(\boldsymbol{U})$ with respect to the space of kinematically admissible displacements $\boldsymbol{U} = (\underline{\boldsymbol{U}}, U)$ satisfying the kinematical boundary conditions

$$\begin{aligned} &\underline{\boldsymbol{U}} = \underline{\hat{\boldsymbol{U}}} \quad U_3 = \hat{U}_3 \quad \text{on } \Sigma_c,\\ &\underline{U}_t = \underline{\hat{U}}_t,\; U_3 = \hat{U}_3 \quad \text{on } \Sigma_s. \end{aligned} \tag{2.8}$$

This is equivalents to solving the variational formulation: find $\boldsymbol{U} = (\underline{\boldsymbol{U}}, U_3)$ satisfying the kinematical boundary conditions (2.8) such that

$$\int_{\Omega^{3D}} \boldsymbol{\sigma}(\boldsymbol{U}) : \boldsymbol{E}(\boldsymbol{V})\; dV = \int_{\Omega^{3D}} \boldsymbol{f}\cdot\boldsymbol{V}\; dV + \int_{\Sigma} \boldsymbol{g}\cdot\boldsymbol{V}\; dA \tag{2.9}$$

for all $\boldsymbol{V}$ satisfying the homogeneous counter part of the kinematical boundary conditions.

Remark 2.1. *From the physical meaning the terms clamped and simply supported, typically refer to homogeneous kinematical boundary conditions. However, for later use, we also allow for inhomogeneous boundary data.*
In literature the type of boundary condition we impose on Σ_s is referred to as hard simple support. In the case of soft simple support only the condition

$$U_3 = \hat{U}_3$$

is imposed.

Using the strain representation (2.4), which is a consequence of the kinematical assumptions, and the constitutive equation for the plane stress state (2.6) the variational formulation (2.9) becomes by performing explicit integration with respect to x_3:
Find $\boldsymbol{u} = (\underline{\boldsymbol{u}}, u_3)$ satisfying the essential boundary conditions

$$\begin{aligned} &\underline{\boldsymbol{u}} = \underline{\hat{\boldsymbol{u}}}, \quad u_3 = \hat{u}_3,\; \partial_{\boldsymbol{n}} u_3 = \hat{\theta}_n \quad \text{on } \Gamma_c,\\ &\underline{u}_t = \underline{\hat{u}}_t, \quad u_3 = \hat{u}_3 \qquad\qquad\quad \text{on } \Gamma_s \end{aligned} \tag{2.10}$$

such that

$$\begin{aligned} &\int_{\Omega} \left(t\,\mathcal{C}\boldsymbol{\varepsilon}(\underline{\boldsymbol{u}}) : \boldsymbol{\varepsilon}(\underline{\boldsymbol{v}}) + \frac{t^3}{12}\mathcal{C}\boldsymbol{\kappa}(u_3) : \boldsymbol{\kappa}(v_3) \right)\, dx \\ &\quad = \int_{\Omega} \left(\underline{\hat{\boldsymbol{f}}}\cdot\underline{\boldsymbol{v}} - \hat{\boldsymbol{c}}\cdot\nabla v_3 + \hat{f}_3\, v_3 \right)\, dx \\ &\quad + \int_{\Gamma_s} \left(\hat{N}_n \underline{v}_n - \hat{M}_n\, \partial_{\boldsymbol{n}} v_3 \right)\, ds + \int_{\Gamma_f} \left(\hat{\boldsymbol{N}}\cdot\underline{\boldsymbol{v}} - \hat{\boldsymbol{M}}\cdot\nabla v_3 + \hat{Q}\, v_3 \right)\, ds \end{aligned} \tag{2.11}$$

for all $\boldsymbol{v} = (\underline{\boldsymbol{v}}, v_3)$ satisfying the homogeneous counterpart of the essential boundary conditions. Here the applied distributed forces $\hat{\boldsymbol{f}} = (\underline{\hat{\boldsymbol{f}}}, \hat{f}_3)$ and couples $\hat{\boldsymbol{c}} = (\hat{c}_\alpha)$ are given by

$$\hat{f}_\alpha = \int_{-\frac{t}{2}}^{\frac{t}{2}} f_\alpha \, dx_3 + \langle g_\alpha \rangle, \quad \hat{f}_3 = \int_{-\frac{t}{2}}^{\frac{t}{2}} f_3 \, dx_3 + \langle g_3 \rangle, \quad \hat{c}_\alpha = \int_{-\frac{t}{2}}^{\frac{t}{2}} f_\alpha \, x_3 \, dx_3 + \langle g_\alpha \, x_3 \rangle,$$

with the operator $\langle . \rangle$ defined by

$$\langle a(x_1, x_2, x_3) \rangle = a(x_1, x_2, -t/2) + a(x_1, x_2, t/2),$$

and the boundary forces $\hat{\boldsymbol{N}} = (\hat{N}_\alpha)$, $\hat{Q}$ and moments $\hat{\boldsymbol{M}} = (\hat{M}_\alpha)$ read

$$\hat{N}_\alpha = \int_{-\frac{t}{2}}^{\frac{t}{2}} g_\alpha \, dx_3, \quad \hat{Q} = \int_{-\frac{t}{2}}^{\frac{t}{2}} g_3 \, dx_3, \quad \hat{M}_\alpha = \int_{-\frac{t}{2}}^{\frac{t}{2}} g_\alpha \, x_3 \, dx_3.$$

Remark 2.2. *There exists numerous literature devoted to the justification of plate models and the estimation of the model error. We mention here, in a totally non-exhaustive way, the works [26, 17].*

The variational problem (2.11) decouples into two independent problems:

Membrane problem

Find $\underline{\boldsymbol{u}} = (u_\alpha)$ satisfying the essential boundary conditions

$$\begin{aligned} \underline{\boldsymbol{u}} &= \underline{\hat{\boldsymbol{u}}} \quad \text{on } \Gamma_c, \\ \underline{u}_t &= \underline{\hat{u}}_t \quad \text{on } \Gamma_s \end{aligned}$$

such that

$$\int_\Omega t \, \mathcal{C}\boldsymbol{\varepsilon}(\underline{\boldsymbol{u}}) : \boldsymbol{\varepsilon}(\underline{\boldsymbol{v}}) \, dx = \int_\Omega \underline{\hat{\boldsymbol{f}}} \cdot \underline{\boldsymbol{v}} \, dx + \int_{\Gamma_s} \hat{N}_n \underline{v}_n \, ds + \int_{\Gamma_f} \hat{\boldsymbol{N}} \cdot \underline{\boldsymbol{v}} \, ds \tag{2.12}$$

for all $\underline{\boldsymbol{v}} = (v_\alpha)$ satisfying the homogeneous counterpart of the essential boundary conditions.

Bending problem

Find u_3 satisfying the essential boundary conditions

$$\begin{aligned} u_3 &= \hat{u}_3, \; \partial_{\boldsymbol{n}} u_3 = \hat{\theta}_n \quad && \text{on } \Gamma_c, \\ u_3 &= \hat{u}_3 && \text{on } \Gamma_s \end{aligned}$$

such that

$$\begin{aligned} &\int_\Omega \frac{t^3}{12} \mathcal{C}\boldsymbol{\kappa}(u_3) : \boldsymbol{\kappa}(v_3) \, dx \\ &\quad = \int_\Omega \left(\hat{f}_3 \, v_3 - \hat{\boldsymbol{c}} \cdot \nabla v_3 \right) \, dx - \int_{\Gamma_s} \hat{M}_n \, \partial_{\boldsymbol{n}} v_3 \, ds + \int_{\Gamma_f} \left(\hat{Q} \, v_3 - \hat{\boldsymbol{M}} \cdot \nabla v_3 \right) \, ds, \end{aligned} \tag{2.13}$$

for all v_3 satisfying the homogeneous counterpart of the essential boundary conditions. Note, the membrane problem only involves first-order derivatives. The bending problem involves second-order derivatives, since the bending strain is defined by $\boldsymbol{\kappa}(u_3) = -\nabla^2 u_3$, see (2.5). The membrane problem can be solved (independently) using standard techniques for second-order problems. Therefore, we restrict our considerations in the following to the plate bending problem for the transverse displacement u_3, as this is the only unknown we skip the subscript and just write u for the rest of this chapter.
For completeness we derive the strong formulation of the plate bending problem (2.13). For this we define the bending moment tensor $\boldsymbol{M}$, which is related to the bending strain through the constitutive equation

$$\boldsymbol{M} = \frac{t^3}{12}\,\mathcal{C}\boldsymbol{\kappa}(u) = -\frac{t^3}{12}\,\mathcal{C}\nabla^2 u. \tag{2.14}$$

Two times integration by parts (see Appendix A.3) of the left-hand side in (2.13) provides

$$\begin{aligned} -\int_\Omega \boldsymbol{M} : \nabla^2 v\; dx &= \int_\Omega \operatorname{Div}\boldsymbol{M}\cdot\nabla v\; dx - \int_\Gamma (\boldsymbol{M}\boldsymbol{n})\cdot\nabla v\; ds \\ &= -\int_\Omega (\operatorname{div}\operatorname{Div}\boldsymbol{M})\, v\; dx + \int_\Gamma (\operatorname{Div}\boldsymbol{M}\cdot\boldsymbol{n})\, v\; ds - \int_\Gamma (\boldsymbol{M}\boldsymbol{n})\cdot\nabla v\; ds. \end{aligned}$$

Using the representation $\nabla v = (\partial_{\boldsymbol{n}} v)\,\boldsymbol{n} + (\partial_{\boldsymbol{t}} v)\,\boldsymbol{t}$ we obtain

$$\int_\Gamma (\boldsymbol{M}\boldsymbol{n})\cdot\nabla v\; ds = \int_\Gamma (M_{nn}\,\partial_{\boldsymbol{n}} v + M_{nt}\,\partial_{\boldsymbol{t}} v)\; ds.$$

In the next step we perform integration by parts along the boundary in tangent direction. For this as well as for later use, we first define the following notations:

Definition 2.3. *With the notations introduced at the beginning of Section 2.1.3 we define the restriction of a function f to the boundary of an edge E_{k-1} by*

$$f\big|_{\partial E_{k-1}} = f(x_k) - f(x_{k-1}) \quad \text{for } k = 1, 2, \ldots, K \tag{2.15}$$

and the jump at the corner point x_k by

$$[\![f]\!]_{x_k} = f(x_k^-) - f(x_k^+) \quad \text{for } k = 1, 2, \ldots, K, \tag{2.16}$$

with the one-sided limits given by

$$f(x_k^-) = \lim_{\varepsilon\to 0} f(x_k - \varepsilon\,\boldsymbol{t}_{k-1}) \quad \text{and} \quad f(x_k^+) = \lim_{\varepsilon\to 0} f(x_k + \varepsilon\,\boldsymbol{t}_k),$$

where $\boldsymbol{t}_{k-1}$ and $\boldsymbol{t}_k$ are the tangent vectors on the edges E_{k-1} and E_k, respectively, in case of a polygonal boundary and an appropriate adaption for curved boundaries.

Then we receive

$$\begin{aligned}\int_\Gamma (\boldsymbol{M}\boldsymbol{n}) \cdot \nabla v \; ds &= \int_\Gamma (M_{nn}\, \partial_{\boldsymbol{n}} v - \partial_{\boldsymbol{t}} M_{nt}\, v) \; ds + \sum_{E \in \mathcal{E}_\Gamma} (M_{nt}\, v(x))\big|_{\partial E} \\ &= \int_\Gamma (M_{nn}\, \partial_{\boldsymbol{n}} v - \partial_{\boldsymbol{t}} M_{nt}\, v) \; ds + \sum_{x \in \mathcal{V}_\Gamma} [\![M_{nt}]\!]_x \, v(x).\end{aligned}$$

We end up with

$$\begin{aligned}-\int_\Omega \boldsymbol{M} : \nabla^2 v \; dx = &-\int_\Omega (\operatorname{div} \operatorname{Div} \boldsymbol{M})\, v \; dx \\ &+ \int_{\Gamma_f} (\partial_{\boldsymbol{t}} M_{nt} + \operatorname{Div} \boldsymbol{M} \cdot \boldsymbol{n})\, v \; ds - \int_{\Gamma_s \cup \Gamma_f} M_{nn}\, \partial_{\boldsymbol{n}} v \; ds - \sum_{x \in \mathcal{V}_{\Gamma,f}} [\![M_{nt}]\!]_x \, v(x),\end{aligned}$$

where we incorporate the boundary conditions of v. Here $\mathcal{V}_{\Gamma,f}$ denotes the set of corner points whose two adjacent edges belong to Γ_f.
By integration by parts of the term involving $\hat{\boldsymbol{c}}$ of right-hand side in (2.13) and again using the representation $\nabla v = (\partial_{\boldsymbol{n}} v)\, \boldsymbol{n} + (\partial_{\boldsymbol{t}} v)\, \boldsymbol{t}$ and integration by parts along the boundary we obtain

$$\begin{aligned}&\int_\Omega \Big(\hat{f}_3\, v - \hat{\boldsymbol{c}} \cdot \nabla v \Big) \; dx - \int_{\Gamma_s} \hat{M}_n\, \partial_{\boldsymbol{n}} v \; ds + \int_{\Gamma_f} \Big(\hat{Q}\, v - \hat{\boldsymbol{M}} \cdot \nabla v \Big) \; ds \\ &= \int_\Omega (\hat{f}_3 + \operatorname{div} \hat{\boldsymbol{c}})\, v \; dx - \int_{\Gamma_s \cup \Gamma f} \hat{M}_n\, \partial_{\boldsymbol{n}} v \; ds + \int_{\Gamma_f} \Big((\hat{Q} - \hat{c}_n)\, v - \hat{M}_t\, \partial_{\boldsymbol{t}} v \Big) \; ds \\ &= \int_\Omega (\hat{f}_3 + \operatorname{div} \hat{\boldsymbol{c}})\, v \; dx - \int_{\Gamma_s \cup \Gamma f} \hat{M}_n\, \partial_{\boldsymbol{n}} v \; ds + \int_{\Gamma_f} (\hat{Q} + \partial_{\boldsymbol{t}} \hat{M}_t - \hat{c}_n)\, v \; ds - \sum_{x \in \mathcal{V}_{\Gamma,f}} [\![\hat{M}_t]\!]_x \, v(x).\end{aligned}$$

Summing up, we obtain

$$\begin{aligned}&\int_\Omega \Big(-(\operatorname{div} \operatorname{Div} \boldsymbol{M}) - (\hat{f}_3 + \operatorname{div} \hat{\boldsymbol{c}}) \Big) \; v \; dx - \int_{\Gamma_s \cup \Gamma_f} (M_{nn} - \hat{M}_n)\, \partial_{\boldsymbol{n}} v \; ds \\ &+ \int_{\Gamma_f} \Big((\partial_{\boldsymbol{t}} M_{nt} + \operatorname{Div} \boldsymbol{M} \cdot \boldsymbol{n}) - \hat{V}_n \Big) \; v \; ds - \sum_{x \in \mathcal{V}_{\Gamma,f}} ([\![M_{nt}]\!]_x - [\![\hat{M}_t]\!]_x)\, v(x) = 0\end{aligned}$$

with $\hat{V}_n = \partial_{\boldsymbol{t}} \hat{M}_t + \hat{Q} - \hat{c}_n$.
The procedure to deduce the strong formulation is to first consider test functions v that vanish on the boundary Γ, which leads to

$$-\operatorname{div} \operatorname{Div} \boldsymbol{M} = \hat{f}_3 + \operatorname{div} \hat{\boldsymbol{c}} \quad \text{in } \Omega.$$

In order to receive the natural boundary conditions we consider in a first step test functions v with $v = 0$ and arbitrary normal derivative $\partial_{\boldsymbol{n}} v$, which leads to the following boundary condition for the normal-normal component of $\boldsymbol{M}$:

$$M_{nn} = \hat{M}_n \quad \text{on } \Gamma_s \cup \Gamma_f.$$

Next we consider test functions v with $v(x) = 0$ for all $x \in \mathcal{V}_{\Gamma,f}$, which leads to the boundary condition for the Kirchhoff shear force

$$\partial_{\boldsymbol{t}} M_{nt} + \operatorname{Div} \boldsymbol{M} \cdot \boldsymbol{n} = \hat{V}_n \quad \text{on } \Gamma_f.$$

Finally, using that $v(x)$ can be chosen arbitrary at the corners $x \in \mathcal{V}_{\Gamma,f}$ we deduce the corner conditions

$$[\![M_{nt}]\!]_x = [\![\hat{M}_t]\!]_x \quad \text{for all } x \in \mathcal{V}_{\Gamma,f}.$$

In summary, the strong form of the plate bending problem (2.13) reads as

$$-\operatorname{div} \operatorname{Div} \boldsymbol{M} = \hat{f}_3 + \operatorname{div} \hat{\boldsymbol{c}} \quad \text{in } \Omega, \quad \text{with } \boldsymbol{M} = -\frac{t^3}{12}\, \mathcal{C}\nabla^2 u \tag{2.17}$$

with the boundary conditions

$$\begin{aligned} u &= \hat{u}_3, & \partial_{\boldsymbol{n}} u &= \hat{\theta}_n & &\text{on } \Gamma_c, \\ u &= \hat{u}_3, & M_{nn} &= \hat{M}_n & &\text{on } \Gamma_s, \\ M_{nn} &= \hat{M}_n, & \partial_{\boldsymbol{t}} M_{nt} + \operatorname{Div} \boldsymbol{M} \cdot \boldsymbol{n} &= \hat{V}_n & &\text{on } \Gamma_f, \end{aligned} \tag{2.18}$$

with $\hat{V}_n = \partial_{\boldsymbol{t}} \hat{M}_t + \hat{Q} - \hat{c}_n$, and the corner conditions

$$[\![M_{nt}]\!]_x = [\![\hat{M}_t]\!]_x \quad \text{for all } x \in \mathcal{V}_{\Gamma,f}, \tag{2.19}$$

where $\mathcal{V}_{\Gamma,f}$ denotes the set of corner points whose two adjacent edges belong to Γ_f.

Remark 2.4. *There is a fourth type of boundary condition given by*

$$\partial_{\boldsymbol{n}} u = \hat{\theta}_n, \quad \partial_{\boldsymbol{t}} M_{nt} + \operatorname{Div} \boldsymbol{M} \cdot \boldsymbol{n} = \hat{V}_n$$

with corner conditions of the form (2.19)*, which appears, e.g., in boundary value problems of the Cahn-Hilliard equation. The theory presented in the following can easily be extended to mixed boundary conditions including also this fourth type.*

2.2 Variational formulation (plate bending problem)

So far we have assumed all functions to be sufficiently smooth. In this section we state mathematically precisely what is meant by the term smoothness. To do this, we equip the variational formulation of the plate bending problem with appropriate function spaces.
Then the (primal) variational formulation of the plate bending problem derived in (2.13) becomes: find $u \in W_g$ such that

$$\int_\Omega \hat{\mathcal{C}} \nabla^2 u : \nabla^2 v \, dx = \langle \hat{F}, v \rangle \quad \text{for all } v \in W_0 \tag{2.20}$$

with the right-hand side

$$\langle \hat{F}, v \rangle = \langle F, v \rangle - \int_{\Gamma_s \cup \Gamma f} \hat{M}_n \, \partial_{\boldsymbol{n}} v \; ds + \int_{\Gamma_f} \hat{V}_n \, v \; ds - \sum_{x \in \mathcal{V}_{\Gamma,f}} \hat{R}_x \, v(x),$$

where

$$\begin{aligned} \langle F, v \rangle &= \int_\Omega f \, v \; dx \quad \text{with } f = \hat{f}_3 + \operatorname{div} \hat{\boldsymbol{c}}, \\ V_n &= \partial_{\boldsymbol{t}} \hat{M}_t + \hat{Q} - \hat{c}_n, \\ R_x &= [\![\hat{M}_t]\!]_x. \end{aligned}$$

The application of the modified material tensor $\hat{\mathcal{C}} = \frac{t^3}{12} \mathcal{C}$ is given by

$$\hat{\mathcal{C}} \boldsymbol{A} = D \left(\boldsymbol{A} + \frac{\nu}{1-\nu} \operatorname{tr}(\boldsymbol{A}) \boldsymbol{I} \right) \quad \text{for all } \boldsymbol{A} \in \mathbb{R}^{2\times 2} \tag{2.21}$$

with

$$D = \frac{t^3}{12} \frac{E}{1+\nu}.$$

Here the function spaces are given by

$$W_0 = \{ v \in H^2(\Omega) : \; v = 0, \; \partial_{\boldsymbol{n}} v = 0 \text{ on } \Gamma_c, \quad v = 0 \text{ on } \Gamma_s \}, \tag{2.22}$$

$$W_g = \{ v \in H^2(\Omega) : \; v = \hat{u}_3, \; \partial_{\boldsymbol{n}} v = \hat{\theta}_n \text{ on } \Gamma_c, \quad v = \hat{u}_3 \text{ on } \Gamma_s \} \tag{2.23}$$

with associated norm $\|v\|_W = \|v\|_2$. For the further considerations we make the following assumptions on the boundary data: We consider $\hat{M}_n \in L^2(\Gamma_s \cup \Gamma_f)$ and $\hat{V}_n \in L^2(\Gamma_f)$, well aware that by using appropriate duality products this requirements can be weakened. Furthermore, we assume that there exists an extension $\overline{u} \in H^2(\Omega)$ of the boundary data $\hat{u}_3$ and $\hat{\theta}_n$ such that $\overline{u} = \hat{u}_3$ on $\Gamma_c \cup \Gamma_s$ and $\partial_{\boldsymbol{n}} \overline{u} = \hat{\theta}_n$ on Γ_c, i.e,

$$W_g = \overline{u} + W_0. \tag{2.24}$$

Following, e.g., [1, 58], here and throughout the thesis $L^2(\Omega)$ and $H^m(\Omega)$ denote the standard Lebesgue and Sobolev spaces of functions on Ω with corresponding norms $\|.\|_0$ and $\|.\|_m$ for positive integers m. For functions on Γ we use $L^2(\Gamma)$ and $H^{\frac{1}{2}}(\Gamma)$ to denote the Lebesgue space and the trace space of $H^1(\Omega)$ with corresponding norms $\|.\|_{0,\Gamma}$ and $\|.\|_{\frac{1}{2},\Gamma}$. Moreover, $H^1_{0,\Gamma'}(\Omega)$ denotes the set of functions in $H^1(\Omega)$ which vanish on a part Γ' of Γ. The L^2-inner product on Ω and Γ' are always denoted by $(.,.)$ and $(.,.)_{\Gamma'}$, respectively, no matter whether it is used for scalar, vector-valued, or matrix-valued functions.
For scalar functions v, vector-valued functions $\boldsymbol{\psi}$, and matrix-valued functions $\boldsymbol{L}$ the first order differential expressions

$$\nabla v, \nabla \boldsymbol{\psi}, \operatorname{curl} v, \operatorname{Curl} \boldsymbol{\psi}, \operatorname{div} \boldsymbol{\psi}, \operatorname{Div} \boldsymbol{L}, \operatorname{rot} \boldsymbol{\psi}, \operatorname{Rot} \boldsymbol{L}$$

are defined in the weak sense on the corresponding domains of definition

$$H^1(\Omega), (H^1(\Omega))^2, H(\mathrm{curl}, \Omega), H(\mathrm{Curl}, \Omega), \ldots .$$

In case that all components are in $H^1(\Omega)$ they take on their classical form given at the end of Section 2.1.1.
In order to avoid technicalities, we assume throughout this thesis that the essential boundary conditions in W_0 are such that no rigid body motion is possible, i.e., the only element $v \in W_0$ that satisfies $\nabla^2 v = 0$ is $v = 0$. Then existence and uniqueness of a solution u to (2.20) are guaranteed by the theorem of Lax-Milgram (see, e.g., [55, 60]) for general right-hand sides $\langle \hat{F}, v\rangle$, where $\hat{F} \in W_0^*$. Here we use H^* to denote the dual of a Hilbert space H and $\langle ., .\rangle$ for the duality product on $H^* \times H$. Moreover, the solution u depends continuously on $\hat{F}$

$$\|u\|_W \leq c_{st}\, \|\hat{F}\|_{W_0^*}, \tag{2.25}$$

with $c_{st} = c'/\lambda_{min}(\hat{\mathcal{C}})$, where c' depends only on the constant c_F of Friedrichs' inequality (see Appendix A.4). In case that the essential boundary conditions in W_0 are such that rigid body motions are possible, we need additional compatibility conditions of the right-hand side $\hat{F}$ and scaling conditions of the solution to guarantee again existence and uniqueness of a solution.
For simpler presentation purposes, we make at first the following assumptions:

Assumption 1. We assume that the boundary conditions in the primal formulation (2.20) are homogeneous. Hence, the right-hand sides $\hat{F}$ and F, and the spaces W_g and W_0 coincide. Therefore, we skip in the following the subscript and just write

$$W = W_0 = W_g.$$

The legitimate question of treatment of inhomogeneous boundary conditions is treated in Section 2.7.

Assumption 2. We assume that Γ_c contains at least one non-trivial edge $E \in \mathcal{E}_\Gamma$. All results of this thesis can easily be extended to the case $\Gamma_c = \emptyset$, as outlined in Remark 2.25.

2.3 Mixed variational formulation

We introduce as a new unknown the bending moment tensor $\boldsymbol{M}$, as defined in (2.14), which is given in terms of the modified material tensor $\hat{\mathcal{C}}$, see (2.21), by

$$\boldsymbol{M} = -\hat{\mathcal{C}}\nabla^2 u.$$

This leads to a first mixed formulation: find $\boldsymbol{M} \in \boldsymbol{L}^2(\Omega)_{\mathrm{sym}}$ and $u \in W$ such that

$$\begin{aligned} \int_\Omega \hat{\mathcal{C}}^{-1}\boldsymbol{M} : \boldsymbol{L}\; dx + \int_\Omega \nabla^2 u : \boldsymbol{L}\; dx &= 0 && \text{for all } \boldsymbol{L} \in \boldsymbol{L}^2(\Omega)_{\mathrm{sym}}, \\ \int_\Omega \boldsymbol{M} : \nabla^2 v\; dx &= -\langle F, v\rangle && \text{for all } v \in W, \end{aligned} \tag{2.26}$$

where the first line results from the relation $\hat{\mathcal{C}}^{-1}\boldsymbol{M} + \nabla^2 u = 0$ and the second line originates from the primal variational formulation (2.20). Here $\boldsymbol{L}^2(\Omega)_{\text{sym}}$ denotes the space of symmetric matrix-valued functions given by

$$\boldsymbol{L}^2(\Omega)_{\text{sym}} = \{\boldsymbol{L} = (L_{\alpha\beta}) : L_{\alpha\beta} = L_{\beta\alpha} \in L^2(\Omega)\}$$

equipped with the standard L^2-norm $\|\boldsymbol{L}\|_0$ for a matrix-valued function $\boldsymbol{L}$. This first mixed formulation is trivially equivalent to the original problem, but brings no advantage. The goal of this section is to derive a reformulation of (2.26) that allows us to use a different space for the displacement u (and test functions v) with less smoothness. For this the starting point is the bilinear form

$$b(\boldsymbol{L}, v) = \int_\Omega \nabla^2 v : \boldsymbol{L}\ dx,$$

which appears in the first line of (2.26) with $v = u$ and in the second line with $\boldsymbol{L} = \boldsymbol{M}$. The operator B associated to this bilinear form is the second-order differential operator Hessian ∇^2 considered as an operator from W to the dual of $\boldsymbol{L}^2(\Omega)_{\text{sym}}$ given by

$$\langle Bv, \boldsymbol{L}\rangle = b(\boldsymbol{L}, v) = \int_\Omega \nabla^2 v : \boldsymbol{L}\ dx.$$

In order to better motivate the next step, let us first consider the standard Poisson problem.

2.3.1 Mixed formulation of the Poisson problem

For the Poisson equation with pure Dirichlet boundary conditions the (primal) variational formulation reads: find $u \in H_0^1(\Omega)$ such that

$$\int_\Omega \nabla u \cdot \nabla v = \langle F, v\rangle \quad \text{for all } v \in H_0^1(\Omega).$$

Introducing as new unknown $\boldsymbol{p} = -\nabla u$ leads analogously as above to a first mixed formulation: find $\boldsymbol{p} \in (L^2(\Omega))^2$ and $u \in H_0^1(\Omega)$ such that

$$\begin{aligned}
\int_\Omega \boldsymbol{p}\cdot\boldsymbol{q}\ dx \quad + \int_\Omega \nabla u \cdot \boldsymbol{q}\ dx &= 0 && \text{for all } \boldsymbol{q} \in (L^2(\Omega))^2,\\
\int_\Omega \boldsymbol{p}\cdot\nabla v\ dx \qquad\qquad\qquad &= -\langle F, v\rangle && \text{for all } v \in H_0^1(\Omega).
\end{aligned}$$

Then the corresponding operator B is the first-order differential operator gradient ∇ considered as an operator from $H_0^1(\Omega)$ to the dual of $(L^2(\Omega))^2$ given by

$$\langle Bv, \boldsymbol{q}\rangle = b(\boldsymbol{q}, v) = \int_\Omega \nabla v \cdot \boldsymbol{q}\ dx.$$

The well-known mixed formulation of the Poisson problem with $u \in L^2(\Omega)$ and $\boldsymbol{p} \in H(\mathrm{div},\Omega) = \{\boldsymbol{q} \in (L^2(\Omega))^2 : \mathrm{div}\, \boldsymbol{q} \in L^2(\Omega)\}$, see, e.g., [15], is obtained by means of integration by parts of the form

$$\langle Bv, \boldsymbol{q}\rangle = \int_\Omega \nabla v \cdot \boldsymbol{q}\; dx = -\int_\Omega v \,\mathrm{div}\, \boldsymbol{q}\; dx = \langle B^* \boldsymbol{q}, v\rangle.$$

In an abstract way, integration by parts can be viewed as the change from the operator B to its adjoint operator B^*. The appropriate framework to understand the adjoint operator B^* is the classical concept of densely defined (possibly unbounded) linear operators. Therefore, B and B^* are defined by

$$\begin{aligned} B = \nabla :& \quad D(B) = H_0^1(\Omega) \quad\;\; \subset L^2(\Omega) \quad\; \to ((L^2(\Omega))^2)^*,\\ B^* = -\,\mathrm{div} :& \, D(B^*) = H(\mathrm{div},\Omega) \subset (L^2(\Omega))^2 \to (L^2(\Omega))^*, \end{aligned}$$

where we distinguish between their domain of definition and a larger base space, here $L^2(\Omega)$ and $(L^2(\Omega))^2$. Note that the domain of definition is a dense subset of the base space. For the Poisson problem with $B = \nabla$ this abstract point of view is not necessary, since the choice for the space of the auxiliary variable $\boldsymbol{p}$, namely $H(\mathrm{div},\Omega)$, is quite clear. For the well-understood space $H(\mathrm{div},\Omega)$ integration by parts formulas are available, which allow for straight-forward derivation of mixed formulations also for problems with more general boundary conditions, e.g., mixed Dirichlet and Neumann boundary conditions. Note that integration by parts formulas are obtained from the corresponding integration by parts formulas for smooth functions using trace theorems and density arguments. Both are readily available for the space $H(\mathrm{div},\Omega)$.

In our situation with $B = \nabla^2$ a derivation based on integration by parts is not straight-forward. There are approaches based on appropriate integration by parts formulas. However, this is not as clear as for the Poisson problem, since trace theorems and density results are a quite subtle issue. We refer in this context to the works [63, 64] and to Remark 2.16.
From our point of view the more appropriate technique to find the adjoint operator B^* and the space for the auxiliary variable $\boldsymbol{M}$ (given by the domain of definition of B^*) is by means of the classical concept of densely defined (possibly unbounded) linear operators B. A brief introduction to densely defined (possibly unbounded) operators is given in the next section. Later we apply this framework to the particular operator $B = \nabla^2$ and define the operator $\mathrm{div}\,\mathrm{Div}$ as its adjoint B^*.

2.3.2 Densely defined (possibly unbounded) linear operators

We consider an operator

$$B\colon D(B) \subset X \to Y^*,$$

where X and Y are Hilbert spaces and $D(B)$ is the domain of definition of B. We assume that $D(B)$ is dense in X. The adjoint

$$B^*\colon D(B^*) \subset Y \to X^*,$$

with domain of definition $D(B^*)$, is then defined as follows: $y \in D(B^*)$ if and only if $y \in Y$ and there is a linear functional $G \in X^*$ such that

$$\langle Bx, y\rangle = \langle G, x\rangle \quad \text{for all } x \in D(B). \tag{2.27}$$

For $y \in D(B^*)$ we define $B^*y = G$. Note, the relation (2.27) used to define $D(B^*)$ can be expressed in a more explicit form, namely,

$$\begin{aligned} D(B^*) = \{y \in Y\colon\ &\text{the functional} \\ &G\colon x \mapsto \langle Bx, y\rangle,\ x \in D(B),\ \text{is bounded w.r.t. the } X\text{-norm}\}. \end{aligned} \tag{2.28}$$

Note that $\langle B^*y, x\rangle$ is well-defined for $x \in X$ and $y \in D(B^*)$ and we have in particular the following well-known relation for an operator B and its adjoint B^*:

$$\langle B^*y, x\rangle = \langle Bx, y\rangle \quad \text{for all } x \in D(B),\ y \in D(B^*). \tag{2.29}$$

The domain $D(B^*)$ is a Hilbert space w.r.t. the graph norm $\|y\|_{D(B^*)} = (\|y\|_Y^2 + \|B^*y\|_{X^*}^2)^{\frac{1}{2}}$. For further information on the concept of densely defined (possibly unbounded) operators, see, e.g., [12].

2.3.3 Derivation of the new mixed formulation

As already indicated above, it is quite natural to choose $D(B) = W$, with W given by (2.22), and to define $B = \nabla^2$ as an operator mapping to $Y^* = (\boldsymbol{L}^2(\Omega)_{\text{sym}})^*$ given by

$$\langle \nabla^2 v, \boldsymbol{L}\rangle = \int_\Omega \nabla^2 v : \boldsymbol{L}\ dx \quad \text{for } v \in D(B) = W,\ \boldsymbol{L} \in Y = \boldsymbol{L}^2(\Omega)_{\text{sym}}. \tag{2.30}$$

Keep in mind, in the following we always fix $D(B) = W$ and obtain depending on the particular choice of X different domains of definition $D(B^*)$ for the adjoint B^*. There are several options how to choose X. However, according to Section 2.3.2, there is one restriction to meet: $D(B)$ is a dense subset of X. Now we discuss three possible choices for X:

- A first and trivial option would be $X = W$. Then it is easy to see that $D(B^*) = \boldsymbol{L}^2(\Omega)_{\text{sym}}$. Note that for this choice we end up with the first mixed formulation (2.26) stated above, which brings no advantage compared to the primal formulation (2.20). We still work with a second-order Sobolev space for u.

- A second option would be $X = L^2(\Omega)$. Then it turns out that $D(B^*) \subset \boldsymbol{H}(\operatorname{div}\operatorname{Div}, \Omega)_{\text{sym}} = \{\boldsymbol{L} \in \boldsymbol{L}^2(\Omega)_{\text{sym}} : \operatorname{div}\operatorname{Div}\boldsymbol{L} \in L^2(\Omega)\}$, where here div Div is defined in the distributional sense. This time the disadvantage for the mixed method is to work with a second-order Sobolev space for $\boldsymbol{M}$.

- The idea for the new mixed formulation is to distribute the smoothness requirements evenly among u and $\boldsymbol{M}$ by choosing the space X as an intermediate space between W and $L^2(\Omega)$. In particular, we propose to set X equal to Q, given by

$$Q = H^1_{0,\Gamma_c\cup\Gamma_s}(\Omega) = \{v \in H^1(\Omega) : v = 0 \text{ on } \Gamma_c \cup \Gamma_s\}, \tag{2.31}$$

equipped with the norm $\|v\|_Q = \|v\|_1$.

Remark 2.5. *The space Q is the interpolation space between W and $L^2(\Omega)$. Note that we only use interpolation as motivation, but do not rely in the following on results from interpolation theory.*

This choice for X meets the required density condition:

Lemma 2.6. *The subspace W is dense in Q.*

Proof. We follow the lines of the proof of [39, Theorem 1.6.1]. In order to verify the density, we have to check that the trace space $\gamma(W)$ is dense in the trace space $\gamma(Q)$, where γ is the standard H^1-trace representing the value on the boundary. By considering the situation locally near each corner in $\mathcal{V}_\Gamma$, the required density follows from the density of $C_0^\infty(\mathbb{R}^+)$ in $H^{\frac{1}{2}}(\mathbb{R}^+)$ and $\tilde{H}^{\frac{1}{2}}(\mathbb{R}^+)$; see [40]. □

Now we leave the abstract framework and use, from now on, the notations

$$\operatorname{div}\operatorname{Div} \quad \text{and} \quad \boldsymbol{H}(\operatorname{div}\operatorname{Div},\Omega;Q^*)_{\text{sym}} \tag{2.32}$$

instead of B^* and $D(B^*)$, respectively, where $X = Q$ and unchanged $D(B) = W$ and $Y = \boldsymbol{L}^2(\Omega)_{\text{sym}}$. In consistence with the abstract framework elements of the Hilbert space $\boldsymbol{H}(\operatorname{div}\operatorname{Div},\Omega;Q^*)_{\text{sym}}$ are characterized in the following way (cf.(2.27)): $\boldsymbol{L} \in \boldsymbol{H}(\operatorname{div}\operatorname{Div},\Omega;Q^*)_{\text{sym}}$ if and only if $\boldsymbol{L} \in \boldsymbol{L}^2(\Omega)_{\text{sym}}$ and there is a linear functional $G \in Q^*$ such that

$$\langle \nabla^2 v, \boldsymbol{L}\rangle = \int_\Omega \nabla^2 v : \boldsymbol{L}\, dx = \langle G, v\rangle \quad \text{for all } v \in W. \tag{2.33}$$

For $\boldsymbol{L} \in \boldsymbol{H}(\operatorname{div}\operatorname{Div},\Omega;Q^*)_{\text{sym}}$ we set $\operatorname{div}\operatorname{Div}\boldsymbol{L} = G$. More explicitly the space $\boldsymbol{H}(\operatorname{div}\operatorname{Div},\Omega;Q^*)_{\text{sym}}$ is given by (cf. (2.28))

$$\begin{aligned}\boldsymbol{H}(\operatorname{div}\operatorname{Div},\Omega;Q^*)_{\text{sym}} = \{&\boldsymbol{L} \in \boldsymbol{L}^2(\Omega)_{\text{sym}} : \text{ the functional}\\ &G\colon v \mapsto \langle \nabla^2 v, \boldsymbol{L}\rangle = \int_\Omega \nabla^2 v : \boldsymbol{L}\, dx,\ v \in W, \text{ is bounded w.r.t. the } Q\text{-norm}\},\end{aligned} \tag{2.34}$$

equipped with the norm

$$\|\boldsymbol{L}\|_{\operatorname{div}\operatorname{Div};Q^*} = (\|\boldsymbol{L}\|_0^2 + \|\operatorname{div}\operatorname{Div}\boldsymbol{L}\|_{Q^*}^2)^{1/2}. \tag{2.35}$$

Furthermore, we have the following well-known relation for an operator and its adjoint (cf. (2.29)):

$$\langle \operatorname{div}\operatorname{Div}\boldsymbol{L}, v\rangle = \langle \nabla^2 v, \boldsymbol{L}\rangle \quad \text{for all } v \in W,\ \boldsymbol{L} \in \boldsymbol{H}(\operatorname{div}\operatorname{Div},\Omega;Q^*)_{\text{sym}}. \tag{2.36}$$

This motivates the new mixed formulation as follows: For $F \in Q^*$, find $\boldsymbol{M} \in \boldsymbol{V}$ and $u \in Q$ such that

$$\begin{aligned} &\int_\Omega \hat{\mathcal{C}}^{-1}\boldsymbol{M} : \boldsymbol{L}\,dx + \langle \operatorname{div}\operatorname{Div}\boldsymbol{L}, u\rangle = 0 && \text{for all } \boldsymbol{L} \in \boldsymbol{V}, \\ &\langle \operatorname{div}\operatorname{Div}\boldsymbol{M}, v\rangle \qquad\qquad = -\langle F, v\rangle && \text{for all } v \in Q, \end{aligned} \tag{2.37}$$

with the function spaces

$$\boldsymbol{V} = \boldsymbol{H}(\operatorname{div}\operatorname{Div}, \Omega; Q^*)_{\mathrm{sym}}, \quad Q = H^1_{0,\Gamma_c\cup\Gamma_s}(\Omega), \tag{2.38}$$

equipped with the norms

$$\|\boldsymbol{L}\|_{\boldsymbol{V}} = \|\boldsymbol{L}\|_{\operatorname{div}\operatorname{Div};Q^*} \quad \text{and} \quad \|v\|_Q = \|v\|_1. \tag{2.39}$$

Note that we require additional regularity of F, namely $F \in Q^*$. In contrast, for the primal problem (2.20) we need only $F \in W^*$.

Remark 2.7. *The operator* $\operatorname{div}\operatorname{Div}$ *as defined in* (2.32) *and the composition of the first order operators* div *and* Div *introduced in Section 2.2 differ in two ways. First of all, their domains of definition are different. While* $\operatorname{div}(\operatorname{Div}\boldsymbol{L})$ *is only well-defined for functions* $\boldsymbol{L} \in \boldsymbol{L}^2(\Omega)$*, where* $\operatorname{div}\boldsymbol{L}$ *and* $\operatorname{div}\operatorname{Div}\boldsymbol{L}$ *are* L^2*-functions as well, the domain of definition of* $\operatorname{div}\operatorname{Div}$ *in* (2.34) *contains functions* $\boldsymbol{L} \in \boldsymbol{L}^2(\Omega)_{\mathrm{sym}}$ *with the less restrictive requirement* $\operatorname{div}\operatorname{Div}\boldsymbol{L} \in Q^*$*. The domain of definition of* $\operatorname{div}\operatorname{Div}$ *in* (2.34) *includes the boundary conditions* $L_{nn} = 0$ *on* $\Gamma_s \cup \Gamma_f$ *and* $[\![L_{nt}]\!]_x = 0$ *for all* $x \in \mathcal{V}_{\Gamma,f}$ *in a weak sense, as we show later in Theorem 2.15.*
But even on the intersection of the domains of definition $\operatorname{div}(\operatorname{Div}\boldsymbol{L})$ *coincides with* $\operatorname{div}\operatorname{Div}\boldsymbol{L}$*, only if* $\boldsymbol{L}$ *satisfies the boundary condition* $\partial_{\boldsymbol{t}} L_{nt} + \operatorname{Div}\boldsymbol{L}\cdot\boldsymbol{n} = 0$ *on* Γ_f*.*

Remark 2.8. *In [62, 63, 64] a similar nonstandard Sobolev space is introduced. Note, our way of definition is different and well-suited for the further considerations.*

2.3.4 Mathematical analysis

In this section we show well-posedness of our new mixed formulation. Problem (2.37) has the typical structure of a saddle point problem

$$\begin{aligned} a(\boldsymbol{M}, \boldsymbol{L}) + b(\boldsymbol{L}, u) &= \langle F_{\boldsymbol{L}}, \boldsymbol{L}\rangle && \text{for all } \boldsymbol{L} \in \boldsymbol{V}, \\ b(\boldsymbol{M}, v) &= \langle F_v, v\rangle && \text{for all } v \in Q, \end{aligned}$$

whose associated linear operator $\mathcal{A}\colon \boldsymbol{V} \times Q \longrightarrow (\boldsymbol{V} \times Q)^*$ is given by

$$\langle \mathcal{A}(\boldsymbol{M}, u), (\boldsymbol{L}, v)\rangle = a(\boldsymbol{M}, \boldsymbol{L}) + b(\boldsymbol{L}, u) + b(\boldsymbol{M}, v).$$

If the bilinear form a is symmetric, i.e., $a(\boldsymbol{M}, \boldsymbol{L}) = a(\boldsymbol{L}, \boldsymbol{M})$, and non-negative, i.e., $a(\boldsymbol{L}, \boldsymbol{L}) \geq 0$, which is fulfilled for (2.37), it is well-known that $\mathcal{A}$ is an isomorphism from $\boldsymbol{V} \times Q$ onto $(\boldsymbol{V} \times Q)^*$, if and only if the following conditions are satisfied; see, e.g., [15]:

1. a is bounded: There is a constant $\|a\| > 0$ such that
$$|a(\boldsymbol{M}, \boldsymbol{L})| \leq \|a\| \, \|\boldsymbol{M}\|_{\boldsymbol{V}} \, \|\boldsymbol{L}\|_{\boldsymbol{V}} \quad \text{for all } \boldsymbol{M}, \ \boldsymbol{L} \in \boldsymbol{V}.$$

2. b is bounded: There is a constant $\|b\| > 0$ such that
$$|b(\boldsymbol{L}, v)| \leq \|b\| \, \|\boldsymbol{L}\|_{\boldsymbol{V}} \|v\|_Q \quad \text{for all } \boldsymbol{L} \in \boldsymbol{V}, \ v \in Q.$$

3. a is coercive on the kernel of b: There is a constant $\alpha > 0$ such that
$$a(\boldsymbol{L}, \boldsymbol{L}) \geq \alpha \, \|\boldsymbol{L}\|_{\boldsymbol{V}}^2 \quad \text{for all } \boldsymbol{L} \in \mathrm{Ker} B$$
with $\mathrm{Ker}B = \{\boldsymbol{L} \in \boldsymbol{V} \colon b(\boldsymbol{L}, v) = 0 \quad \text{for all } v \in Q\}$.

4. b satisfies the inf-sup condition: There is a constant $\beta > 0$ such that
$$\inf_{0 \neq v \in Q} \sup_{0 \neq \boldsymbol{L} \in \boldsymbol{V}} \frac{b(\boldsymbol{L}, v)}{\|\boldsymbol{L}\|_{\boldsymbol{V}} \, \|v\|_Q} \geq \beta.$$

We refer to these conditions as Brezzi's conditions with constants $\|a\|$, $\|b\|$, α, and β. (We tacitly assume that $\|a\|$ and $\|b\|$ are the smallest constants for estimating the bilinear forms a and b. Then $\|a\|$ and $\|b\|$ match the standard notation for the norms of the bilinear forms a and b.)
In order to verify Brezzi's conditions for (2.37), we need the following result on the relation between the primal problem (2.20) and the new mixed problem (2.37).

Theorem 2.9. *Let $u \in W$ be the solution of the primal problem (2.20) for $F \in Q^*$. Then we have $\boldsymbol{M} = -\hat{\mathcal{C}} \nabla^2 u \in \boldsymbol{V}$ and $(\boldsymbol{M}, u)$ solves the mixed problem (2.37).*

Proof. Since $u \in W$ solves (2.20), it follows that $\boldsymbol{M} \in \boldsymbol{L}^2(\Omega)_{\mathrm{sym}}$ and
$$\langle \nabla^2 v, \boldsymbol{M} \rangle = \int_\Omega \boldsymbol{M} : \nabla^2 v \; dx = -\int_\Omega \hat{\mathcal{C}} \nabla^2 u : \nabla^2 v \; dx = \langle -F, v \rangle \quad \text{for all } v \in W.$$
From the definition of $\boldsymbol{V}$ in (2.33) we obtain that $\boldsymbol{M} \in \boldsymbol{V}$ with $\operatorname{div} \operatorname{Div} \boldsymbol{M} = -F \in Q^*$, which shows that the second equation of (2.37) is satisfied. Since $u \in W$, we receive from (2.36) that
$$\langle \operatorname{div} \operatorname{Div} \boldsymbol{L}, u \rangle = \langle \nabla^2 u, \boldsymbol{L} \rangle = \int_\Omega \boldsymbol{L} : \nabla^2 u \; dx = -\int_\Omega \boldsymbol{L} : \hat{\mathcal{C}}^{-1} \boldsymbol{M} \; dx$$
for all $\boldsymbol{L} \in \boldsymbol{V}$, which proves the first equation of (2.37). □

Theorem 2.10. *The bilinear forms*
$$a(\boldsymbol{M}, \boldsymbol{L}) = \int_\Omega \hat{\mathcal{C}}^{-1} \boldsymbol{M} : \boldsymbol{L} \; dx \quad \textit{and} \quad b(\boldsymbol{L}, v) = \langle \operatorname{div} \operatorname{Div} \boldsymbol{L}, v \rangle$$

satisfy Brezzi's conditions on $\boldsymbol{V} = \boldsymbol{H}(\operatorname{div}\operatorname{Div},\Omega;Q^*)_{\mathrm{sym}}$ *and* Q*, equipped with the norms* $\|\boldsymbol{L}\|_{\boldsymbol{V}} = \|\boldsymbol{L}\|_{\operatorname{div}\operatorname{Div};Q^*}$ *and* $\|v\|_Q = \|v\|_1$*, respectively, with the constants*

$$\|a\| = 1/\lambda_{min}(\hat{\mathcal{C}}),\;\; \|b\| = 1,\;\; \alpha = 1/\lambda_{max}(\hat{\mathcal{C}}), \textit{and } \beta = (1+c)^{-1/2},$$

where $c = c'\,\lambda_{max}(\hat{\mathcal{C}})/\lambda_{min}(\hat{\mathcal{C}})$ *and* c' *as in the stability estimate of the primal problem* (2.25)*.*

Proof. 1. Let $\boldsymbol{M},\boldsymbol{L} \in \boldsymbol{H}(\operatorname{div}\operatorname{Div},\Omega;Q^*)_{\mathrm{sym}}$. Then

$$\begin{aligned}|(\hat{\mathcal{C}}^{-1}\boldsymbol{M},\boldsymbol{L})| &\le (1/\lambda_{min}(\hat{\mathcal{C}}))\|\boldsymbol{M}\|_0\|\boldsymbol{L}\|_0\\ &\le (1/\lambda_{min}(\hat{\mathcal{C}}))\|\boldsymbol{M}\|_{\operatorname{div}\operatorname{Div};Q^*}\|\boldsymbol{L}\|_{\operatorname{div}\operatorname{Div};Q^*}.\end{aligned}$$

2. Let $\boldsymbol{L} \in \boldsymbol{H}(\operatorname{div}\operatorname{Div},\Omega;Q^*)_{\mathrm{sym}}$ and $v \in Q$. Then

$$|\langle \operatorname{div}\operatorname{Div}\boldsymbol{L}, v\rangle| \le \|\operatorname{div}\operatorname{Div}\boldsymbol{L}\|_{Q^*}\|v\|_Q \le \|\boldsymbol{L}\|_{\operatorname{div}\operatorname{Div};Q^*}\|v\|_Q.$$

3. On $\mathrm{Ker}B = \{\boldsymbol{L} \in \boldsymbol{V} : \langle \operatorname{div}\operatorname{Div}\boldsymbol{L}, v\rangle = 0 \quad \text{for all } v \in Q\}$ we have

$$(\hat{\mathcal{C}}^{-1}\boldsymbol{L},\boldsymbol{L}) \ge (1/\lambda_{max}(\hat{\mathcal{C}}))\|\boldsymbol{L}\|_0^2 = (1/\lambda_{max}(\hat{\mathcal{C}}))\|\boldsymbol{L}\|^2_{\operatorname{div}\operatorname{Div};Q^*}.$$

4. It remains to show the inf-sup condition

$$\sup_{0\neq \boldsymbol{L}\in\boldsymbol{V}} \frac{\langle \operatorname{div}\operatorname{Div}\boldsymbol{L}, v\rangle}{\|\boldsymbol{L}\|_{\operatorname{div}\operatorname{Div};Q^*}} \ge \beta \quad \text{for all } v \in Q.$$

For a fixed but arbitrary $v \in Q$, let u^v be the solution of the primal problem (2.20) with the right-hand side $\langle F^v, .\rangle = -(v, .)_Q \in Q^*$. From Theorem 2.9 it follows that $\boldsymbol{M}^v = -\hat{\mathcal{C}}\nabla^2 u^v \in \boldsymbol{V}$, and $(\boldsymbol{M}^v, u^v)$ is solution of the corresponding mixed problem (2.37). From the second line of the mixed formulation (2.37) we obtain

$$\langle \operatorname{div}\operatorname{Div}\boldsymbol{M}^v, v\rangle = (v,v)_Q = \|v\|_Q^2$$

and

$$\|\operatorname{div}\operatorname{Div}\boldsymbol{M}^v\|_{Q^*} = \sup_{q\in Q}\frac{\langle \operatorname{div}\operatorname{Div}\boldsymbol{M}^v, q\rangle}{\|q\|_Q} = \sup_{q\in Q}\frac{(v,q)_Q}{\|q\|_Q} = \|v\|_Q.$$

Using that u^v is the solution of the primal problem (2.20) and the stability estimate (2.25) we receive

$$\begin{aligned}\|\boldsymbol{M}^v\|_0^2 = \|\hat{\mathcal{C}}\nabla^2 u^v\|_0^2 &\le \lambda_{max}(\hat{\mathcal{C}})(\hat{\mathcal{C}}\nabla^2 u^v, \nabla^2 u^v) = \lambda_{max}(\hat{\mathcal{C}})\langle F^v, u^v\rangle\\ &\le \lambda_{max}(\hat{\mathcal{C}})\|F^v\|_{W^*}\|u^v\|_W \le c\;\|F^v\|^2_{W^*} \le c\;\|F^v\|^2_{Q^*} = c\;\|v\|_Q^2\end{aligned}$$

with $c = c'\,\lambda_{max}(\hat{\mathcal{C}})/\lambda_{min}(\hat{\mathcal{C}})$. Hence, we obtain from the last two inequalities

$$\|\boldsymbol{M}^v\|^2_{\operatorname{div}\operatorname{Div};Q^*} = \|\boldsymbol{M}^v\|_0^2 + \|\operatorname{div}\operatorname{Div}\boldsymbol{M}^v\|^2_{Q^*} \le (1+c)\,\|v\|_Q^2.$$

Therefore,

$$\sup_{0 \neq \boldsymbol{L} \in \boldsymbol{V}} \frac{\langle \operatorname{div} \operatorname{Div} \boldsymbol{L}, v \rangle}{\|\boldsymbol{L}\|_{\operatorname{div}\operatorname{Div};Q^*}} \geq \frac{\langle \operatorname{div} \operatorname{Div} \boldsymbol{M}^v, v \rangle}{\|\boldsymbol{M}^v\|_{\operatorname{div}\operatorname{Div};Q^*}} \geq (1+c)^{-1/2} \, \|v\|_Q,$$

which completes the proof.

□

Remark 2.11. *The choice of $\boldsymbol{M}^v$ for proving the inf-sup condition is different to the choice $\boldsymbol{M}^v = v\,\boldsymbol{I}$ as it is used, e.g., in [23, 54], which would not work here, since $v\,\boldsymbol{I} \notin \boldsymbol{V}$ for problems with a free boundary part.*

Corollary 2.12. *For $F \in Q^*$, the primal problem* (2.20) *and the mixed problem* (2.37) *are equivalent in the following sense: If $u \in W$ solves* (2.20)*, then $\boldsymbol{M} = -\hat{\mathcal{C}}\nabla^2 u \ \in \boldsymbol{V}$ and $(\boldsymbol{M}, u)$ solves* (2.37)*. Vice versa, if $(\boldsymbol{M}, u) \in \boldsymbol{V} \times Q$ solves* (2.37)*, then $u \in W$ and solves* (2.20)*.*

Proof. The first part has already been shown in Theorem 2.9. Since, both problems are uniquely solvable the reverse direction is true as well. □

Remark 2.13. *It is important to note that in Corollary 2.12 we do not need any assumptions on the domain Ω, like convexity, to obtain the equivalence of our mixed formulation to the original problem. Most other mixed formulations in literature require Ω to be convex, see [15, 35, 3].*

Remark 2.14. *From a more abstract point of view, the just obtained results, well-posedness in Theorem 2.10 and equivalence to the original problem in Corollary 2.12 mainly rely on the following two basic assumptions:*

- *The density of the chosen space Q in W, the space of the primal problem.*
- *The well-posedness of the primal problem in W.*

This shows that these results can straight-forward be extended to different choices of Q or even other primal problems, e.g., other high-order problems.

2.3.5 The space $\boldsymbol{H}(\operatorname{div}\operatorname{Div}, \Omega; Q^*)_{\text{sym}}$ revisited

In the case of a purely clamped plate, where $W = H_0^2(\Omega)$, we have $Q = H_0^1(\Omega)$. So, in the mixed setting, $u = 0$ on Γ is treated as an essential boundary condition, while $\partial_{\boldsymbol{n}} u = 0$ on Γ becomes a natural boundary condition incorporated in the variational formulation. According to (2.18), no boundary conditions are prescribed for $\boldsymbol{M}$, which makes the definition of an appropriate space $\boldsymbol{V}$ for $\boldsymbol{M}$ much easier. The space $\boldsymbol{V}$ can be introduced directly as $\{\boldsymbol{L} \in \boldsymbol{L}^2(\Omega)_{\text{sym}} \colon \operatorname{div}\operatorname{Div}\boldsymbol{L} \in H^{-1}(\Omega)\}$ with $H^{-1}(\Omega) = (H_0^1(\Omega))^*$, where $\operatorname{div}\operatorname{Div}\boldsymbol{L}$ is to be interpreted in the distributional sense, see [54, 53]. It can easily be verified that this space coincides with the space $\boldsymbol{H}(\operatorname{div}\operatorname{Div}, \Omega; Q^*)_{\text{sym}}$ as defined in (2.34) for $W = H_0^2(\Omega)$ and $Q = H_0^1(\Omega)$.

The situation is more involved for mixed boundary conditions, where W is given by (2.22). Here we have $Q = H^1_{0,\Gamma_c \cup \Gamma_s}(\Omega)$. So $u = 0$ on $\Gamma_c \cup \Gamma_s$ is treated as an essential boundary condition, while $\partial_{\boldsymbol{n}} u = 0$ on Γ_c and the conditions for the Kirchhoff shear force $\partial_{\boldsymbol{t}} M_{nt} + \operatorname{Div} \boldsymbol{M} \cdot \boldsymbol{n} = 0$ on Γ_f become natural boundary conditions incorporated in the variational formulation. The remaining boundary conditions (cf. (2.18)) for the normal-normal component of $\boldsymbol{M}$, $M_{nn} = 0$ on $\Gamma_s \cup \Gamma_f$, and the corner conditions (2.19) are treated as essential boundary conditions. They are not explicitly visible but are hidden in the definition of the space $\boldsymbol{H}(\operatorname{div}\operatorname{Div}, \Omega; Q^*)_{\text{sym}}$.
However, for sufficiently smooth functions $\boldsymbol{L}$, these essential boundary conditions can be explicitly extracted using the definition of $\boldsymbol{H}(\operatorname{div}\operatorname{Div}, \Omega; Q^*)_{\text{sym}}$, as we see in the next theorem.

Theorem 2.15. *Let $\boldsymbol{L} \in \boldsymbol{L}^2(\Omega)_{\text{sym}} \cap \boldsymbol{C}^1(\overline{\Omega})$. Then $\boldsymbol{L} \in \boldsymbol{H}(\operatorname{div}\operatorname{Div}, \Omega; Q^*)_{\text{sym}}$ if and only if*

$$L_{nn} = 0 \quad \text{on } \Gamma_s \cup \Gamma_f \quad \text{and} \quad [\![L_{nt}]\!]_x = 0 \quad \text{for all } x \in \mathcal{V}_{\Gamma,f}. \tag{2.40}$$

Furthermore, for $v \in Q$ we obtain the following representation of the functional $\operatorname{div}\operatorname{Div} \boldsymbol{L}$*:*

$$\langle \operatorname{div}\operatorname{Div} \boldsymbol{L}, v \rangle = -\int_\Omega \operatorname{Div} \boldsymbol{L} \cdot \nabla v \, dx - \int_{\Gamma_f} (\partial_{\boldsymbol{t}} L_{nt})\, v \, ds. \tag{2.41}$$

Proof. Recall the representation of $\boldsymbol{H}(\operatorname{div}\operatorname{Div}, \Omega; Q^*)_{\text{sym}}$ in (2.34). For $v \in W$, we obtain by integration by parts (see Appendix A.3), using the representation $\nabla v = (\partial_{\boldsymbol{n}} v)\, \boldsymbol{n} + (\partial_{\boldsymbol{t}} v)\, \boldsymbol{t}$ and finally integration by parts along the boundary of the term involving L_{nt}

$$\begin{aligned}
\langle G, v \rangle &= \int_\Omega \boldsymbol{L} : \nabla^2 v \, dx \\
&= -\int_\Omega \operatorname{Div} \boldsymbol{L} \cdot \nabla v \, dx + \int_\Gamma L_{nn}\, \partial_{\boldsymbol{n}} v + L_{nt}\, \partial_{\boldsymbol{t}} v \, ds \\
&= -\int_\Omega \operatorname{Div} \boldsymbol{L} \cdot \nabla v \, dx + \int_{\Gamma_s \cup \Gamma_f} L_{nn}\, \partial_{\boldsymbol{n}} v \, ds - \int_{\Gamma_f} (\partial_{\boldsymbol{t}} L_{nt})\, v \, ds + \sum_{x \in \mathcal{V}_{\Gamma,f}} [\![L_{nt}]\!]_x v(x).
\end{aligned} \tag{2.42}$$

Assume now that $\boldsymbol{L}$ satisfies the boundary conditions (2.40). Then we have

$$\langle G, v \rangle = -\int_\Omega \operatorname{Div} \boldsymbol{L} \cdot \nabla v \, dx - \int_{\Gamma_f} (\partial_{\boldsymbol{t}} L_{nt})\, v \, ds,$$

which is obviously bounded w.r.t. the H^1-norm. Hence $\boldsymbol{L} \in \boldsymbol{H}(\operatorname{div}\operatorname{Div}, \Omega; Q^*)_{\text{sym}}$ with the representation (2.41).
On the other hand, if $\boldsymbol{L} \in \boldsymbol{H}(\operatorname{div}\operatorname{Div}, \Omega; Q^*)_{\text{sym}}$, then the functional G given by (2.42) is bounded w.r.t. the H^1-norm. For $v \in H^2_{N,\mathcal{V}}(\Omega) \cap W$, with

$$H^2_{N,\mathcal{V}}(\Omega) = \{ v \in H^2(\Omega) : \partial_{\boldsymbol{n}} v = 0 \text{ on } \Gamma, \quad v(x) = 0 \text{ for all } x \in \mathcal{V}_\Gamma \}$$

we obtain from (2.42)

$$\langle G, v\rangle = -\int_\Omega \operatorname{Div} \boldsymbol{L} \cdot \nabla v \; dx - \int_{\Gamma_f} (\partial_{\boldsymbol{t}} L_{nt})\, v \; ds. \tag{2.43}$$

Note that all expressions in (2.43) are continuous in v w.r.t. the H^1-norm. Analogously to the proof of Lemma 2.6, one can show that $H^2_{N,\mathcal{V}}(\Omega) \cap W$ is dense in W w.r.t. the H^1-norm. Then it follows that (2.43) is valid for all $v \in W$. This implies together with (2.42) that

$$\int_{\Gamma_s \cup \Gamma_f} L_{nn} \partial_{\boldsymbol{n}} v \; ds + \sum_{x \in \mathcal{V}_{\Gamma,f}} [\![L_{nt}]\!]_x v(x) = 0 \quad \text{for all } v \in W.$$

From this the boundary conditions in (2.40) follow by standard arguments. Moreover, W is a dense subset of Q, and, therefore, (2.43) holds also for $v \in Q$, from which the representation (2.41) follows. □

Remark 2.16. *For smooth functions $\boldsymbol{L}$ Theorem 2.15 provides us with a precise characterization of $\boldsymbol{H}(\operatorname{div}\operatorname{Div}, \Omega; Q^*)_{\mathrm{sym}}$ in terms of boundary conditions* (2.40) *and a representation of the operator* div Div (2.41).
Therefore, the question arises, if there is a super-space such that $\boldsymbol{H}(\operatorname{div}\operatorname{Div}, \Omega; Q^*)_{\mathrm{sym}}$ can be characterized as subspace of this larger space where particular traces vanish. Such a way of definition has been used in [62, 63, 64] to define a similar nonstandard Sobolev space.

2.4 Regular decomposition

The rather simple proof of the well-posedness of the new mixed formulation (2.37) comes at the cost of the nonstandard Sobolev space $\boldsymbol{H}(\operatorname{div}\operatorname{Div}, \Omega; Q^*)_{\mathrm{sym}}$. The next theorem provides a regular decomposition of the space $\boldsymbol{H}(\operatorname{div}\operatorname{Div}, \Omega; Q^*)_{\mathrm{sym}}$, which makes this space computationally accessible. In order to derive our decomposition, we need a characterization of the kernel of the distributional div Div, given in the next lemma.

Lemma 2.17. *Let Ω be simply connected. For $\boldsymbol{L} \in \boldsymbol{L}^2(\Omega)_{\mathrm{sym}}$, we have $\operatorname{div}\operatorname{Div} \boldsymbol{L} = 0$ in the distributional sense if and only if $\boldsymbol{L} = \operatorname{symCurl} \boldsymbol{\psi}$ for some function $\boldsymbol{\psi} \in (H^1(\Omega))^2$. The function $\boldsymbol{\psi}$ is unique up to an element from $RT_0 = \{ax + b \colon a \in \mathbb{R},\ b \in \mathbb{R}^2\}$.*

Proof. From [9, Lemma 1] it follows that $\boldsymbol{L}$ with $\operatorname{div}\operatorname{Div} \boldsymbol{L} = 0$ can be written in the following way:

$$\boldsymbol{L} = \begin{pmatrix} 0 & -\rho \\ \rho & 0 \end{pmatrix} + \operatorname{Curl} \boldsymbol{\psi}$$

for some $\rho \in L^2_0(\Omega)$ and $\boldsymbol{\psi} \in \left(H^1(\Omega)\right)^2$. The proof in [9] relies on the characterization of the kernel of the divergence operator in $\left(L^2(\Omega)\right)^2$, which is well-known (see, e.g., [37, Theorem 3.1]) and in $(H^{-1}(\Omega))^2$, for which we refer to Lemma 2.18 below. To ensure symmetry of $\boldsymbol{L}$, it follows (see [46]) that

$$-\rho - \partial_1 \boldsymbol{\psi}_1 = \rho + \partial_2 \boldsymbol{\psi}_2,$$

hence

$$\rho = -\frac{1}{2} \operatorname{div} \boldsymbol{\phi}.$$

Summing up, we obtain

$$\begin{aligned}
\boldsymbol{L} &= \begin{pmatrix} 0 & \frac{1}{2}(\partial_1\psi_1 + \partial_2\psi_2) \\ -\frac{1}{2}(\partial_1\psi_1 + \partial_2\psi_2) & 0 \end{pmatrix} + \begin{pmatrix} \partial_2\psi_1 & -\partial_1\psi_1 \\ \partial_2\psi_2 & -\partial_1\psi_2 \end{pmatrix} \\
&= \begin{pmatrix} \partial_2\psi_1 & \frac{1}{2}(-\partial_1\psi_1 + \partial_2\psi_2) \\ \frac{1}{2}(-\partial_1\psi_1 + \partial_2\psi_2) & -\partial_1\psi_2 \end{pmatrix} = \operatorname{symCurl} \boldsymbol{\psi}.
\end{aligned}$$

□

The next lemma provides the outsourced result for divergence-free distributions in $(H^{-1}(\Omega))^2$ used in the proof of the previous lemma.

Lemma 2.18. *Let Ω be a simply connected, open and bounded set in $\mathbb{R}^2$ with Lipschitz boundary Γ. For each $f \in (H^{-1}(\Omega))^2$ with $\operatorname{div} f = 0$ there exists a function $\rho \in L^2_0(\Omega)$ such that*

$$f = \operatorname{curl} \rho.$$

Proof. Let $\langle f, v\rangle = \langle f_1, v_1\rangle + \langle f_2, v_2\rangle \in (H^{-1}(\Omega))^2$ with $\operatorname{div} f = 0$, i.e.

$$\langle f_1, \partial_1\varphi\rangle + \langle f_2, \partial_2\varphi\rangle = 0 \quad \text{for all } \varphi \in C_0^\infty(\Omega).$$

Then, by continuity, it follows that

$$\langle f_1, \partial_1\varphi\rangle + \langle f_2, \partial_2\varphi\rangle = 0 \quad \text{for all } \varphi \in H_0^2(\Omega).$$

Now, let $v \in (H_0^1(\Omega))^2$ with $\operatorname{div} v = 0$. Then from [37, Corollary 3.2] it follows that there is a function $\varphi \in H_0^2(\Omega)$ with

$$v = \operatorname{curl} \varphi.$$

Therefore, for $\langle g, v\rangle \equiv \langle f_2, v_1\rangle - \langle f_1, v_2\rangle$, we have

$$\langle g, v\rangle = \langle f_2, v_1\rangle - \langle f_1, v_2\rangle = \langle f_2, \partial_2\varphi\rangle + \langle f_1, \partial_1\varphi\rangle = 0.$$

Then [37, Lemma 2.1] implies that there is a function $\rho \in L_0^2(\Omega)$ with

$$g = \nabla\rho, \quad \text{i.e.,} \quad f = \text{curl } \rho.$$

□

In preparation for the next theorem, observe that $\operatorname{Div}\operatorname{Curl}\boldsymbol{\psi} = 0$, since for $\boldsymbol{\psi} \in (H^1(\Omega))^2$ according to the definition of the weak derivative and integration by parts (see Appendix A) we have

$$\int_\Omega \operatorname{Div}\operatorname{Curl}\boldsymbol{\psi}\cdot\boldsymbol{\xi}\;dx = -\int_\Omega \operatorname{Curl}\boldsymbol{\psi} : \nabla\boldsymbol{\xi}\;dx = -\int_\Omega \boldsymbol{\psi}\cdot\operatorname{Rot}\nabla\boldsymbol{\xi}\;dx = 0$$

for all $\boldsymbol{\xi} \in (C_0^\infty(\Omega))^2$. Therefore, $\operatorname{Curl}\boldsymbol{\psi} \in \boldsymbol{H}(\operatorname{Div},\Omega) = \{\boldsymbol{L} \in \boldsymbol{L}^2(\Omega) : \operatorname{Div}\boldsymbol{L} \in (L^2(\Omega))^2\}$. Hence, the normal component of $\operatorname{Curl}\boldsymbol{\psi}$ is well-defined on the boundary

$$(\operatorname{Curl}\boldsymbol{\psi})\boldsymbol{n} \in (H^{-\frac{1}{2}}(\Gamma))^2 \quad \text{with} \quad H^{-\frac{1}{2}}(\Gamma) = (H^{\frac{1}{2}}(\Gamma))^*,$$

and we have by integration by parts (see Appendix A.3)

$$\int_\Omega \operatorname{Curl}\boldsymbol{\psi} : \nabla\boldsymbol{\xi} = -\int_\Omega \underbrace{\operatorname{Div}\operatorname{Curl}\boldsymbol{\psi}}_{=0}\cdot\boldsymbol{\xi}\;dx + \langle(\operatorname{Curl}\boldsymbol{\psi})\boldsymbol{n},\boldsymbol{\xi}\rangle_\Gamma, \tag{2.44}$$

for all $\boldsymbol{\psi} \in (H^1(\Omega))^2$ and $\boldsymbol{\xi} \in (H^1(\Omega))^2$. For smooth functions $(\operatorname{Curl}\boldsymbol{\psi})\boldsymbol{n}$ coincides with the tangential derivative of $\boldsymbol{\psi}$, so we use the notation

$$\partial_{\boldsymbol{t}}\boldsymbol{\psi} = (\operatorname{Curl}\boldsymbol{\psi})\boldsymbol{n}.$$

Theorem 2.19. *Let Ω be simply connected. For each $\boldsymbol{L} \in \boldsymbol{H}(\operatorname{div}\operatorname{Div},\Omega;Q^*)_{\text{sym}}$ there exists a decomposition*

$$\boldsymbol{L} = q\boldsymbol{I} + \operatorname{symCurl}\boldsymbol{\psi} \tag{2.45}$$

with $q \in Q = H^1_{0,\Gamma_c\cup\Gamma_s}(\Omega)$ and $\boldsymbol{\psi} \in (H^1(\Omega))^2$ satisfying the coupling condition

$$\langle\partial_{\boldsymbol{t}}\boldsymbol{\psi},\nabla v\rangle_\Gamma = -\int_\Gamma q\;\partial_{\boldsymbol{n}}v\;ds \quad \text{for all } v \in W. \tag{2.46}$$

The function $q \in Q$ is the unique solution of the Poisson problem

$$\int_\Omega \nabla q\cdot\nabla v\;dx \quad = -\langle\operatorname{div}\operatorname{Div}\boldsymbol{L},v\rangle \quad \text{for all } v \in Q, \tag{2.47}$$

and $\boldsymbol{\psi} \in (H^1(\Omega))^2$ is unique up to an element from RT_0. Vice versa, for each $\boldsymbol{L}$ given by (2.45) *with $q \in Q$ and $\boldsymbol{\psi} \in (H^1(\Omega))^2$ satisfying* (2.46)*, it follows that $\boldsymbol{L} \in \boldsymbol{H}(\operatorname{div}\operatorname{Div},\Omega;Q^*)_{\text{sym}}$ with the same representation of* $\operatorname{div}\operatorname{Div}\boldsymbol{L}$ *as in* (2.47)*, i.e.,*

$$\langle\operatorname{div}\operatorname{Div}\boldsymbol{L},v\rangle = -\int_\Omega \nabla q\cdot\nabla v\;dx \quad \text{for all } v \in Q.$$

Moreover,

$$\underline{c}\;(\|q\|_1^2 + \|\operatorname{symCurl}\boldsymbol{\psi}\|_0^2) \le \|\boldsymbol{L}\|^2_{\operatorname{div}\operatorname{Div};Q^*} \le \overline{c}\;(\|q\|_1^2 + \|\operatorname{symCurl}\boldsymbol{\psi}\|_0^2), \tag{2.48}$$

with positive constants $\underline{c}$ and $\overline{c}$, which depend only on the constant c_F of Friedrichs' inequality.

Proof. Let $q \in Q$ be the unique solution of the variational problem

$$\int_\Omega \nabla q \cdot \nabla v = -\langle \operatorname{div} \operatorname{Div} \boldsymbol{L}, v\rangle \quad \text{for all } v \in Q. \tag{2.49}$$

For $v \in C_0^\infty(\Omega)$ we receive for $\operatorname{div}\operatorname{Div} q\boldsymbol{I}$ in the distributional sense from integration by parts

$$\langle \operatorname{div}\operatorname{Div} q\boldsymbol{I}, v\rangle = \int_\Omega q\boldsymbol{I} : \nabla^2 v \; dx = -\int_\Omega \operatorname{Div}(q\boldsymbol{I}) \cdot \nabla v \; dx = -\int_\Omega \nabla q \cdot \nabla v \; dx.$$

This implies $\operatorname{div}\operatorname{Div}(\boldsymbol{L} - q\boldsymbol{I}) = 0$ in the distributional sense. According to Lemma 2.17, there exists a function $\boldsymbol{\psi} \in (H^1(\Omega))^2$ such that

$$\boldsymbol{L} - q\boldsymbol{I} = \operatorname{symCurl} \boldsymbol{\psi}.$$

For $v \in W$ the relation (2.36), exploiting the symmetry of $\nabla^2 v$, and integration by parts using (2.44) provides

$$\begin{aligned}
\langle \operatorname{div}\operatorname{Div} \boldsymbol{L}, v\rangle = \langle \nabla^2 v, \boldsymbol{L}\rangle &= \int_\Omega (q\boldsymbol{I} + \operatorname{symCurl}\boldsymbol{\psi}) : \nabla^2 v \; dx = \int_\Omega (q\boldsymbol{I} + \operatorname{Curl}\boldsymbol{\psi}) : \nabla^2 v \; dx \\
&= -\int_\Omega \left(\operatorname{Div}(q\boldsymbol{I}) + \underbrace{\operatorname{Div}\operatorname{Curl}\boldsymbol{\psi}}_{=0} \right) \cdot \nabla v \; dx + \int_\Gamma (q\boldsymbol{I})\boldsymbol{n} \cdot \nabla v \; ds + \langle \partial_{\boldsymbol{t}} \boldsymbol{\psi}, \nabla v\rangle_\Gamma \\
&= -\int_\Omega \nabla q \cdot \nabla v \; dx + \int_\Gamma q \; \partial_{\boldsymbol{n}} v \; ds + \langle \partial_{\boldsymbol{t}} \boldsymbol{\psi}, \nabla v\rangle_\Gamma.
\end{aligned}$$

With (2.49) it follows that

$$\langle \partial_{\boldsymbol{t}} \boldsymbol{\psi}, \nabla v\rangle_\Gamma = -\int_\Gamma q \; \partial_{\boldsymbol{n}} v \; ds \quad \text{for all } v \in W.$$

For the reverse direction, assume that (2.45) and (2.46) hold. By using the integration by parts formula from above we obtain

$$\langle \nabla^2 v, \boldsymbol{L}\rangle = -\int_\Omega \nabla q \cdot \nabla v \; dx,$$

where (2.46) makes the boundary contributions vanish. This immediately implies that $G\colon v \mapsto \langle \nabla^2 v, \boldsymbol{L}\rangle$ is bounded w.r.t. the Q-norm, i.e. $\boldsymbol{L} \in \boldsymbol{H}(\operatorname{div}\operatorname{Div}, \Omega; Q^*)_{\text{sym}}$ (cf. (2.34)) , with $\operatorname{div}\operatorname{Div} \boldsymbol{L} = G \in Q^*$, i.e.,

$$\langle \operatorname{div}\operatorname{Div} \boldsymbol{L}, v\rangle = \langle G, v\rangle = -\int_\Omega \nabla q \cdot \nabla v \; dx \quad \text{for all } v \in Q.$$

In order to show (2.48), note that (2.47) implies

$$\|\operatorname{div}\operatorname{Div} \boldsymbol{L}\|_{Q^*} = \sup_{v \in Q} \frac{\int_\Omega \nabla q \cdot \nabla v \; dx}{\|v\|_1}.$$

Hence,

$$(1+c_F^2)^{-1/2}\ |q|_1 \le \|\operatorname{div}\operatorname{Div}\boldsymbol{L}\|_{Q^*} \le |q|_1,$$

using Friedrichs' inequality $\|q\|_0 \le c_F\,|q|_1$, where $|q|_1$ denotes the H^1-semi-norm (see Appendix A.4).
With these inequalities we obtain for the estimate from above

$$\begin{aligned}\|\boldsymbol{L}\|^2_{\operatorname{div}\operatorname{Div};Q^*} &= \|\boldsymbol{L}\|_0^2 + \|\operatorname{div}\operatorname{Div}\boldsymbol{L}\|^2_{Q^*} = \|q\,\boldsymbol{I} + \operatorname{symCurl}\boldsymbol{\psi}\|_0^2 + \|\operatorname{div}\operatorname{Div}\boldsymbol{L}\|^2_{Q^*}\\ &\le 2\,\|q\boldsymbol{I}\|_0^2 + 2\,\|\operatorname{symCurl}\boldsymbol{\psi}\|_0^2 + |q|_1^2 \le 4\,\|q\|_1^2 + 2\,\|\operatorname{symCurl}\boldsymbol{\psi}\|_0^2,\end{aligned}$$

and for the estimate from below using again Friedrichs' inequality

$$\begin{aligned}\|q\|_1^2 + \|\operatorname{symCurl}\boldsymbol{\psi}\|_0^2 &= \|q\|_1^2 + \|\boldsymbol{L} - q\boldsymbol{I}\|_0^2\\ &\le \|q\|_1^2 + 2\,\|\boldsymbol{L}\|_0^2 + 2\,\|q\,\boldsymbol{I}\|_0^2 \le 2\,\|\boldsymbol{L}\|_0^2 + (1+5c_F^2)|q|_1^2\\ &\le 2\,\|\boldsymbol{L}\|_0^2 + (1+5c_F^2)(1+c_F^2)\,\|\operatorname{div}\operatorname{Div}\boldsymbol{L}\|^2_{Q^*}.\end{aligned}$$

Therefore, (2.48) holds with $1/\underline{c} = \max(2,(1+5c_F^2)(1+c_F^2))$ and $\overline{c} = 4$. □

Remark 2.20. *By applying Korn's inequality (see Appendix A.4) to* $\boldsymbol{\psi}^\perp = (-\boldsymbol{\psi}_2, \boldsymbol{\psi}_1)^T$ *we obtain*

$$\|\operatorname{symCurl}\boldsymbol{\psi}\|_0 = \|\varepsilon(\boldsymbol{\psi}^\perp)\|_0 \ge c_K\,|\boldsymbol{\psi}^\perp|_1 = c_K\,|\boldsymbol{\psi}|_1 \quad \textit{for all } \boldsymbol{\psi} \in (H^1(\Omega))^2/RT_0,$$

where H/RT_0 *denotes the* L^2*-orthogonal complement of* RT_0 *in* H *for spaces* $H \subset (H^1(\Omega))^2$*. Then it follows that*

$$c_K(1+c_F^2)^{-1/2}\,\|\boldsymbol{\psi}\|_1 \le \|\operatorname{symCurl}\boldsymbol{\psi}\|_0 \le \|\boldsymbol{\psi}\|_1.$$

So, provided the unique element $\boldsymbol{\psi} \in (H^1(\Omega))^2/RT_0$ *is chosen for the decomposition, stability follows from* (2.48) *in standard* H^1*-norms.*

Remark 2.21. *The proof of Theorem 2.19 relies on two main components. The first step is the construction of* q*, which can be viewed as homogenization, since it provides*

$$\boldsymbol{L} = q\boldsymbol{I} + \boldsymbol{L}_0,$$

where $\operatorname{div}\operatorname{Div}\boldsymbol{L}_0 = 0$*. The second essential ingredient is a characterization of the elements in the kernel of* $\operatorname{div}\operatorname{Div}$*. As outlined above, there is a potential function* $\boldsymbol{\psi} \in (H^1(\Omega))^2$ *such that*

$$\boldsymbol{L}_0 = \operatorname{symCurl}\boldsymbol{\psi}.$$

- *In literature, another form of homogenization has been considered. The classical Helmholtz decomposition* $\boldsymbol{L} = \nabla^2 q + \operatorname{symCurl}\boldsymbol{\psi}$*, see, e.g., [9, 46], has the same second component. However, the first component is different and requires the solution of a fourth-order problem, which brings no benefit. In contrast, the decomposition introduced here only requires the solution of a second-order Poisson problem for the first component.*

- *Note the analogy to the well-known characterization of the stress tensor* $\boldsymbol{\sigma}$ *with* $\operatorname{Div}\boldsymbol{\sigma} = 0$ *by means of the Airy stress function* φ *in* $2D$

  $$\boldsymbol{\sigma} = \operatorname{Curl}\operatorname{curl}\varphi,$$

  and a similar result in $3D$ *with the Beltrami stress functions.*

2.4.1 The resulting mixed formulation

According to Theorem 2.19 elements of the space $\boldsymbol{V} = \boldsymbol{H}(\operatorname{div}\operatorname{Div},\Omega;Q^*)_{\mathrm{sym}}$ can be characterized in the following way: $\boldsymbol{L} \in \boldsymbol{V} = \boldsymbol{H}(\operatorname{div}\operatorname{Div},\Omega;Q^*)_{\mathrm{sym}}$ if and only if it can be represented by (2.45) with a pair of functions $(q,\boldsymbol{\psi}) \in \mathcal{V}$, where

$$\mathcal{V} = \{(q,\boldsymbol{\psi}) \in Q \times (H^1(\Omega))^2 \colon q,\boldsymbol{\psi} \text{ satisfy the coupling condition}$$
$$\langle \partial_{\boldsymbol{t}}\boldsymbol{\psi}, \nabla v\rangle_\Gamma = -\int_\Gamma q\, \partial_{\boldsymbol{n}} v\; ds \quad \text{for all } v \in W \quad \text{(cf. (2.46))}\}.$$

Observe that the coupling condition involves only traces of q and $\boldsymbol{\psi}$ on Γ. Moreover, for $\boldsymbol{L} \in \boldsymbol{V} = \boldsymbol{H}(\operatorname{div}\operatorname{Div},\Omega;Q^*)_{\mathrm{sym}}$ we have the following description of $\operatorname{div}\operatorname{Div}\boldsymbol{L}$

$$\langle \operatorname{div}\operatorname{Div}\boldsymbol{L}, v\rangle = -\int_\Omega \nabla q \cdot \nabla v\; dx \quad \text{for all } v \in Q.$$

Using the following representations of the solution $\boldsymbol{M}$ and test functions $\boldsymbol{L}$

$$\boldsymbol{M} = p\boldsymbol{I} + \operatorname{symCurl}\boldsymbol{\phi}, \quad \boldsymbol{L} = q\boldsymbol{I} + \operatorname{symCurl}\boldsymbol{\psi},$$

with $(p,\boldsymbol{\phi}) \in \mathcal{V}$ and $(q,\boldsymbol{\psi}) \in \mathcal{V}$, leads to an equivalent formulation of (2.37): For $F \in Q^*$, find $(p,\boldsymbol{\phi}) \in \mathcal{V}$ and $u \in Q$ such that

$$\begin{aligned} \int_\Omega \hat{\mathcal{C}}^{-1}(p\boldsymbol{I} + \operatorname{symCurl}\boldsymbol{\phi}) : (q\boldsymbol{I} + \operatorname{symCurl}\boldsymbol{\psi})\; dx - \int_\Omega \nabla u \cdot \nabla q\; dx &= 0, \\ -\int_\Omega \nabla p \cdot \nabla v\; dx \qquad\qquad &= -\langle F, v\rangle, \end{aligned} \tag{2.50}$$

for all $(q,\boldsymbol{\psi}) \in \mathcal{V}$ and $v \in Q = H^1_{0,\Gamma_c\cup\Gamma_s}(\Omega)$.

Remark 2.22. *If the whole boundary is clamped, then for $v \in W$ we have $\nabla v = 0$ on $\Gamma_c = \Gamma$, hence, the space $\mathcal{V}$ reduces to $Q \times (H^1(\Omega))^2$. If we only have clamped and simply supported boundary parts, q does not appear in the coupling condition* (2.46) *since $q = 0$ on $\Gamma_c \cup \Gamma_s = \Gamma$.*

In the sequel we use the following notations:

- For simplicity of notation, we drop in the following the hat in the notation of the material tensor and write $\mathcal{C}$, where it is still defined as in (2.21).
- In order to make equations more compact, we use in the following the short notation $(.,.)$ for the L^2-inner product on Ω instead of an integral and put the material tensor as subscript, i.e.,

$$(\boldsymbol{M},\boldsymbol{L})_{\mathcal{C}^{-1}} = (\mathcal{C}^{-1}\boldsymbol{M},\boldsymbol{L}) = \int_\Omega \mathcal{C}^{-1}\boldsymbol{M} : \boldsymbol{L}\; dx,$$
$$(\nabla u, \nabla q) = \int_\Omega \nabla u \cdot \nabla q\; dx.$$

2.4.2 Coupling condition

If $\partial_{\boldsymbol{t}}\boldsymbol{\psi} \in (L^2(\Gamma))^2$, the coupling condition (2.46) given by

$$\langle \partial_{\boldsymbol{t}}\boldsymbol{\psi}, \nabla v\rangle_\Gamma = -\int_\Gamma q\; \partial_{\boldsymbol{n}} v \; ds \quad \text{for all } v \in W$$

can be rewritten as follows

$$\int_{\Gamma_s \cup \Gamma_f} (\partial_{\boldsymbol{t}}\boldsymbol{\psi}\cdot\boldsymbol{n} + q)\; \partial_{\boldsymbol{n}} v \; ds + \int_{\Gamma_f} \partial_{\boldsymbol{t}}\boldsymbol{\psi}\cdot\boldsymbol{t}\; \partial_{\boldsymbol{t}} v \; ds = 0 \quad \text{for all } v \in W, \tag{2.51}$$

where we used the representation $\nabla v = (\partial_{\boldsymbol{n}} v)\,\boldsymbol{n} + (\partial_{\boldsymbol{t}} v)\,\boldsymbol{t}$ and incorporated the boundary conditions for $v \in W$. Note, this is the weak form of the boundary conditions

$$\begin{aligned} &\partial_{\boldsymbol{t}}\boldsymbol{\psi}\cdot\boldsymbol{n} = 0 && \text{on } \Gamma_s,\\ &\partial_{\boldsymbol{t}}\boldsymbol{\psi}\cdot\boldsymbol{n} = -q, \quad \partial_{\boldsymbol{t}}(\partial_{\boldsymbol{t}}\boldsymbol{\psi}\cdot\boldsymbol{t}) = 0 && \text{on } \Gamma_f, \end{aligned} \tag{2.52}$$

where we used that $q = 0$ on Γ_s and the corner conditions

$$[\![\partial_{\boldsymbol{t}}\boldsymbol{\psi}\cdot\boldsymbol{t}]\!]_x = 0 \quad \text{for all } x \in \mathcal{V}_{\Gamma,f}. \tag{2.53}$$

2.5 Coupling condition as standard boundary conditions

In this section we make as a first step the following assumption on the domain Ω:

Assumption 3. We consider a domain Ω with a polygonal boundary, i.e., the normal and tangent vectors $\boldsymbol{n}$, $\boldsymbol{t}$ are piecewise constant.

In this case the boundary conditions (2.52) can be rewritten as

$$\partial_{\boldsymbol{t}}(\boldsymbol{\psi}\cdot\boldsymbol{n}) = 0 \qquad \text{on } \Gamma_s, \tag{2.54}$$

$$\partial_{\boldsymbol{t}}(\boldsymbol{\psi}\cdot\boldsymbol{n}) = -q, \quad \partial_{\boldsymbol{t}}^2(\boldsymbol{\psi}\cdot\boldsymbol{t}) = 0 \qquad \text{on } \Gamma_f. \tag{2.55}$$

These conditions already reveal that the coupling condition can be written as a Dirichlet boundary condition for the normal component $\boldsymbol{\psi}\cdot\boldsymbol{n}$ on Γ_s and for $\boldsymbol{\psi}$ on Γ_f, since on Γ_f we obtain conditions for the normal and tangential component $\boldsymbol{\psi}\cdot\boldsymbol{n}$ and $\boldsymbol{\psi}\cdot\boldsymbol{t}$. In the remainder of this section we outline how to do this mathematically rigorously without assuming additional smoothness of $\boldsymbol{\psi}$.

2.5.1 The coupling condition revisited

For this purpose we alternatively characterize $\mathcal{V}$ as follows

$$\mathcal{V} = \{(q, \boldsymbol{\psi}) \in Q \times (H^1(\Omega))^2 : \boldsymbol{\psi} \in \Psi_q\},$$

where Ψ_q is given by the following definition:

Definition 2.23. *For fixed $q \in Q$ we define*

$$\Psi_q = \{\boldsymbol{\psi} \in (H^1(\Omega))^2 : \langle \partial_{\boldsymbol{t}} \boldsymbol{\psi}, \nabla v \rangle_\Gamma = -\int_\Gamma q \, \partial_{\boldsymbol{n}} v \, ds \quad \text{for all } v \in W\}.$$

So Ψ_q consists of those functions $\boldsymbol{\psi} \in (H^1(\Omega))^2$ that fulfill the coupling condition (2.46) for given q, which becomes in this context a boundary condition for $\boldsymbol{\psi}$. In particular, functions from Ψ_0, the space associated to $q = 0$, satisfy the corresponding homogeneous boundary conditions.
It is essential for deriving a decoupled formulation that $\mathcal{V}$ is a direct sum of two subspaces, which we show next. For this we construct, for each $q \in Q$, a specific function $\boldsymbol{\psi}[q] \in \Psi_q$ as follows: Let E be a fixed edge on Γ_c with boundary $\partial E = \{x_A, x_B\}$, where x_A, x_B are ordered in counterclockwise direction. We set

$$\boldsymbol{\psi}_\Gamma[q](x) = -\int_0^\sigma q \, \boldsymbol{n} \, ds \quad \text{with} \quad x = \gamma(\sigma) \quad \text{on } \Gamma \setminus E. \tag{2.56}$$

Here $\sigma \mapsto \gamma(\sigma)$ denotes the arc length parametrization of Γ in counterclockwise direction with $x_B = \gamma(0)$ and $x_A = \gamma(\sigma_E)$ for some $\sigma_E > 0$, see Figure 2.1 for an illustration. Next we extend $\boldsymbol{\psi}_\Gamma[q]$ on the whole boundary Γ by connecting its values

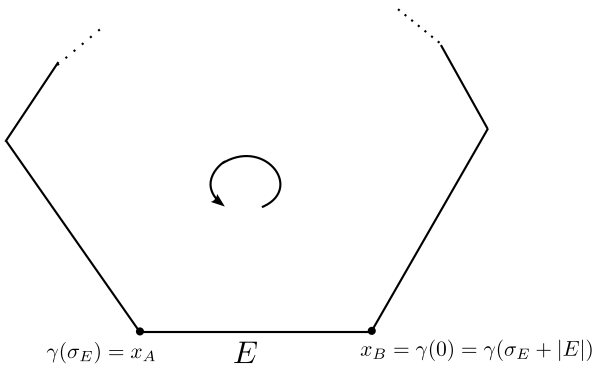


Figure 2.1: Illustration parametrization of boundary Γ.

at ∂E linearly on E. By this $\boldsymbol{\psi}_\Gamma[q]$ becomes a continuous function on Γ with a weak tangential derivative in $(L^2(\Gamma))^2$ satisfying

$$\partial_{\boldsymbol{t}} \boldsymbol{\psi}_\Gamma[q] = -q \, \boldsymbol{n} \quad \text{on } \Gamma \setminus E.$$

Finally, let $\boldsymbol{\psi}[q] \in (H^1(\Omega))^2$ be the harmonic extension of $\boldsymbol{\psi}_\Gamma[q]$, which is well-defined, since we obviously have $\boldsymbol{\psi}_\Gamma[q] \in (H^{\frac{1}{2}}(\Gamma))^2$ (see Appendix A.2).

Lemma 2.24. *For each $q \in Q$, let $\boldsymbol{\psi}[q] \in (H^1(\Omega))^2$ be given as described above. Then $\boldsymbol{\psi}[q] \in \Psi_q$ and we have*

$$\mathcal{V} = \mathcal{V}_0 \oplus \mathcal{V}_1 \quad \text{with} \quad \mathcal{V}_0 = \{0\} \times \Psi_0, \quad \mathcal{V}_1 = \{(q, \boldsymbol{\psi}[q]) : q \in Q\}. \tag{2.57}$$

Moreover, there is a constant $c > 0$ such that

$$\|\boldsymbol{\psi}[q]\|_1 \le c \|q\|_1 \quad \text{for all } q \in Q. \tag{2.58}$$

Proof. For all $v \in W$, we have

$$\langle \partial_{\boldsymbol{t}} \boldsymbol{\psi}[q], \nabla v \rangle_\Gamma = -\int_{\Gamma \setminus E} q \boldsymbol{n} \cdot \nabla v \; ds = -\int_\Gamma q \, \partial_{\boldsymbol{n}} v \; ds,$$

since $\nabla v = 0$ on E, which shows that $\boldsymbol{\psi}[q] \in \Psi_q$. The decomposition (2.57) can easily be derived from the representation

$$(q, \boldsymbol{\psi}) = (q, \boldsymbol{\psi}[q]) + (0, \boldsymbol{\psi}_0) \quad \text{with} \quad \boldsymbol{\psi}_0 = \boldsymbol{\psi} - \boldsymbol{\psi}[q] \in \Psi_0.$$

The sum is direct, since $\boldsymbol{\psi}[q] = 0$ for $q = 0$. Finally, for showing (2.58), observe that

$$\boldsymbol{\psi}_\Gamma[q](x) = \big(1 - (\sigma - \sigma_E)/|E|\big) \, \boldsymbol{\psi}_\Gamma[q](x_A) \quad \text{with} \quad x = \gamma(\sigma) \quad \text{on } E,$$

where in the counterclockwise parametrization of E we have $x_A = \gamma(\sigma_E)$ and $x_B = \gamma(\sigma_E + |E|) = \gamma(0)$ and $|E|$ denotes the length of E. From this and (2.56) it easily follows that

$$\|\boldsymbol{\psi}_\Gamma[q]\|_{1,\Gamma} \le c \, \|q\|_{0,\Gamma},$$

where $H^1(\Gamma)$ denotes the space of functions with weak tangential derivative with corresponding norm $\|.\|_{1,\Gamma}$, see, e.g., [58]. For the proof of the just stated inequality note that

$$\|\boldsymbol{\psi}_\Gamma[q]\|_{0,\Gamma}^2 = \int_{\Gamma \setminus E} \left| \int_0^\sigma q \boldsymbol{n} \; ds \right|^2 \; d\sigma + \int_E |\boldsymbol{\psi}_\Gamma[q](x)|^2 \; ds.$$

In both integrals the integrand can be bounded from above independently of σ:

$$\left| \int_0^\sigma q \boldsymbol{n} \; ds \right|^2 \le \left(\int_0^{\sigma_E} |q| \; ds \right)^2 \le |\Gamma| \; \|q\|_{0,\Gamma}^2$$

and

$$|\boldsymbol{\psi}_\Gamma[q](x)|^2 \le |\boldsymbol{\psi}_\Gamma[q](x_A)|^2 = |\boldsymbol{\psi}_\Gamma[q](\gamma(\sigma_E))|^2 \le \left(\int_0^{\sigma_E} |q| \; ds \right)^2 \le |\Gamma| \; \|q\|_{0,\Gamma}^2.$$

In summary we end up with

$$\|\boldsymbol{\psi}_\Gamma[q]\|_{0,\Gamma}^2 \le |\Gamma|^2 \|q\|_{0,\Gamma}^2.$$

Using similar arguments we obtain

$$\begin{aligned} \|\partial_{\boldsymbol{t}} \boldsymbol{\psi}_\Gamma[q]\|_{0,\Gamma}^2 &= \int_{\Gamma \setminus E} |q\boldsymbol{n}|^2 \; d\sigma + \int_E \left| \frac{1}{|E|} \int_0^{\sigma_E} q \boldsymbol{n} \; ds \right|^2 \; d\sigma \\ &\le c \; \|q\|_{0,\Gamma}^2. \end{aligned}$$

The rest follows from standard trace and inverse trace inequalities (see Appendix A.2). $\square$

Remark 2.25. *The above construction relies on the Assumption 2 that we have at least one clamped edge. However, this idea can be extended to configurations without a clamped edge using the freedom we have in the choice of $\boldsymbol{\psi}_\Gamma[q]$ on simply supported and free edges. Now we consider the case $\Gamma_c = \emptyset$ and make the following ansatz*

$$\boldsymbol{\psi}_\Gamma[q](x) = \int_0^\sigma (-q\,\boldsymbol{n} + f\,\boldsymbol{t})\ ds \quad \text{with} \quad x = \gamma(\sigma) \quad \text{on } \Gamma, \tag{2.59}$$

where $f \in L^2(\Gamma)$ with

$$f(x) = \begin{cases} \text{an arbitrary function } f_s(x) & \text{on } \Gamma_s, \\ \text{a constant } a_C & \text{on each connected component } C \subset \Gamma_f. \end{cases}$$

For all $v \in W$, we receive by incorporating the boundary conditions for v

$$\begin{aligned} \langle \partial_{\boldsymbol{t}} \boldsymbol{\psi}_\Gamma[q], \nabla v \rangle_\Gamma &= \int_\Gamma (-q\,\boldsymbol{n} + f\,\boldsymbol{t}) \cdot \nabla v\ ds = \int_\Gamma (-q\,\partial_{\boldsymbol{n}} v + f\,\partial_{\boldsymbol{t}} v)\ ds \\ &= \int_\Gamma -q\,\partial_{\boldsymbol{n}} v\ ds + \int_{\Gamma_s} f_s\ \underbrace{\partial_{\boldsymbol{t}} v}_{=0}\ ds + \underbrace{\int_{\Gamma_f} a_C\,\partial_{\boldsymbol{t}} v\ ds}_{=0}. \end{aligned}$$

The last term vanishes since we have

$$\int_{\Gamma_f} a_C\,\partial_{\boldsymbol{t}} v\ ds = \sum_{E \subset \Gamma_f} \int_E \underbrace{\partial_{\boldsymbol{t}} a_C}_{=0}\ v\ ds + \sum_{x \in \mathcal{V}_{\Gamma,f}} \underbrace{[\![a_C]\!]_x}_{=0}\ v(x) = 0,$$

where $\mathcal{V}_{\Gamma,f}$ denotes the set of corner points whose two adjacent edges belong to Γ_f and we used that for corner points on the interface of Γ_f with Γ_s we have $v(x) = 0$. Summing up, the proposed extended ansatz for $\boldsymbol{\psi}_\Gamma[q]$ satisfies the compatibility condition (2.46). In order to obtain an extension $\boldsymbol{\psi}[q] \in \Psi_q$, it remains to show continuity, which is according to the definitions satisfied at every point in Γ except for $x = \gamma(0)$. In order to ensure the continuity of $\boldsymbol{\psi}_\Gamma[q]$ in $x = \gamma(0)$ we have to construct data f such that

$$\int_\Gamma \boldsymbol{\psi}_\Gamma[q](x)\ ds = 0. \tag{2.60}$$

As stated in Section 2.2, we restrict ourselves in this thesis to the situation where the essential boundary conditions in W are such that no rigid-body motion is possible, hence in the absence of a clamped edge we have at least two simply supported edges. Therefore, we only have to consider the following two configurations:

- *We have two simply supported edges that are not parallel, which are denoted by E_1 and E_2 with normal and tangent vectors $\boldsymbol{n}_1$, $\boldsymbol{n}_2$ and $\boldsymbol{t}_1$, $\boldsymbol{t}_2$, respectively. Then the choice*

$$f(x) = \begin{cases} \text{the constant } a_1 = \frac{\boldsymbol{c}_q \cdot \boldsymbol{n}_2}{|E_1| \boldsymbol{t}_1 \cdot \boldsymbol{n}_2} & \text{on } E_1, \\ \text{the constant } a_2 = \frac{\boldsymbol{c}_q \cdot \boldsymbol{n}_1}{|E_2| \boldsymbol{t}_2 \cdot \boldsymbol{n}_1} & \text{on } E_2, \\ 0 & \text{on } \Gamma_s \setminus \{E_1 \cup E_2\}, \\ 0 & \text{on } \Gamma_f, \end{cases}$$

where

$$\boldsymbol{c}_q = \int_0^\sigma q\,\boldsymbol{n}\;ds$$

ensures (2.60).

- *We only have simply supported edges that are parallel, and the rest of the boundary is free. In order to make rigid-body motions vanish, at least two of the parallel simply supported edges necessarily do not lie on a common straight line. Let us consider those two simply supported edges, and denote one by* E_1 *with normal and tangent vector* $\boldsymbol{n}_1$ *and* $\boldsymbol{t}_1$*, respectively. Then we have necessarily in between these two edges a connected component* C *of* $\Gamma_f \cup \Gamma_s$ *such that* $\boldsymbol{c}_C \neq 0$ *and* $\boldsymbol{c}_C$ *is not parallel to the simply supported edges, where* $\boldsymbol{c}_C$ *is the vector connecting the two considered simply supported edges given by*

$$\boldsymbol{c}_C = \sum_{E_i \subset C} |E_i| \boldsymbol{t}_i.$$

Then the choice

$$f(x) = \begin{cases} \text{the constant } a_1 = \frac{\boldsymbol{c}_q \cdot \boldsymbol{n}_C}{|E_1| \boldsymbol{t}_1 \cdot \boldsymbol{n}_C} & \text{on } E_1, \\ \text{the constant } a_C = \frac{\boldsymbol{c}_q \cdot \boldsymbol{n}_1}{\boldsymbol{c}_C \cdot \boldsymbol{n}_1} & \text{on } C, \\ 0 & \text{on } \Gamma \setminus (E_1 \cup C), \end{cases}$$

with $\boldsymbol{n}_C$ *such that* $\boldsymbol{n}_C \cdot \boldsymbol{c}_C = 0$*, ensures* (2.60).

The construction can also be extended to situations where rigid body motions are possible, e.g., fully free boundary, by using the corresponding compatibility conditions of the right-hand side of the primal problem, cf. Section 2.2.

So far we have constructed a particular function $\boldsymbol{\psi}[q]$ satisfying the coupling condition (2.46), which can be used to perform homogenization. The missing second part is a precise characterization of the homogeneous space Ψ_0, which is provided by the next lemma.

Lemma 2.26. *The space* Ψ_0 *is equal to the set of all functions* $\boldsymbol{\psi} \in (H^1(\Omega))^2$ *that satisfy the following boundary conditions:*

$$\begin{aligned} \boldsymbol{\psi} \cdot \boldsymbol{n} &= c_E \quad \text{on each edge } E \subset \Gamma_s, \\ \boldsymbol{\psi} &= r_C \quad \text{on each connected component } C \text{ of } \Gamma_f, \end{aligned} \tag{2.61}$$

with some constants $c_E \in \mathbb{R}$ *for each edge* $E \subset \Gamma_s$ *and functions* $r_C \in RT_0$ *for each connected component* $C \subset \Gamma_f$ *satisfying the compatibility conditions* $c_E = r_C(x) \cdot \boldsymbol{n}_E$*, if* $E \subset \Gamma_s$ *is an adjacent edge to a component* C *of* Γ_f*, where* x *is the enclosed corner point and* $\boldsymbol{n}_E$ *is the normal on* E*. Moreover, for each set of constants* $c_E \in \mathbb{R}$ *and functions* $r_C \in RT_0$ *that satisfy the compatibility conditions, there is a function* $\boldsymbol{\psi} \in (H^1(\Omega))^2$*, for which the boundary conditions* (2.61) *hold.*

Proof. For $\boldsymbol{\psi} \in (H^1(\Omega))^2$ and $v \in C^\infty(\overline{\Omega}) \cap W$ we obtain by integration by parts (see Appendix A.3)

$$\begin{aligned}
\langle \partial_{\boldsymbol{t}} \boldsymbol{\psi}, \nabla v \rangle_\Gamma &= -\int_\Omega \underbrace{\operatorname{Div}\operatorname{Curl}\boldsymbol{\psi}}_{=0} \cdot \nabla v \; dx + \langle \partial_{\boldsymbol{t}} \boldsymbol{\psi}, \nabla v \rangle_\Gamma \\
&= \int_\Omega \operatorname{Curl}\boldsymbol{\psi} : \nabla^2 v \; dx \\
&= \int_\Omega \boldsymbol{\psi} \cdot \underbrace{\operatorname{Rot}\nabla^2 v}_{=0} \; dx - \int_\Gamma \boldsymbol{\psi} \cdot (\nabla^2 v)\, \boldsymbol{t} \; ds.
\end{aligned} \tag{2.62}$$

Using the representation $\nabla v = (\partial_{\boldsymbol{n}} v)\, \boldsymbol{n} + (\partial_{\boldsymbol{t}} v)\, \boldsymbol{t}$ and that the boundary of Ω is polygonal leads to

$$(\nabla^2 v)\, \boldsymbol{t} = \nabla(\nabla v)\, \boldsymbol{t} = \partial_{\boldsymbol{t}}(\nabla v) = \partial_{\boldsymbol{t}}(\partial_{\boldsymbol{n}} v)\boldsymbol{n} + \partial_{\boldsymbol{t}}(\partial_{\boldsymbol{t}} v)\boldsymbol{t}.$$

Hence, by incorporating the boundary conditions for v we have

$$\begin{aligned}
\langle \partial_{\boldsymbol{t}} \boldsymbol{\psi}, \nabla v \rangle_\Gamma &= -\sum_{E \subset \Gamma_s} \int_E (\boldsymbol{\psi} \cdot \boldsymbol{n})\; \partial_{\boldsymbol{t}}(\partial_{\boldsymbol{n}} v) \; ds - \sum_{E \subset \Gamma_f} \int_E \boldsymbol{\psi} \cdot \partial_{\boldsymbol{t}}(\nabla v) \; ds \\
&= -\sum_{E \subset \Gamma_s \cup \Gamma_f} \int_E (\boldsymbol{\psi} \cdot \boldsymbol{n})\; \partial_{\boldsymbol{t}}(\partial_{\boldsymbol{n}} v) \; ds - \sum_{E \subset \Gamma_f} \int_E \boldsymbol{\psi} \cdot \boldsymbol{t}\; \partial_{\boldsymbol{t}}^2 v \; ds.
\end{aligned} \tag{2.63}$$

Now let $\boldsymbol{\psi} \in \Psi_0$ and let E be an edge from $\Gamma_s \cup \Gamma_f$. For each function $\varphi \in C_0^\infty(E)$, one can easily construct a function $v \in C_0^\infty(\Omega \cup E)$ with $v = 0$ and $\partial_{\boldsymbol{n}} v = \varphi$ on E. Since $\boldsymbol{\psi} \in \Psi_0$, we have $\langle \partial_{\boldsymbol{t}} \boldsymbol{\psi}, \nabla v \rangle_\Gamma = 0$, which reduces using (2.63) to

$$\int_E (\boldsymbol{\psi} \cdot \boldsymbol{n})\; \partial_{\boldsymbol{t}} \varphi \; ds = 0 \quad \text{for all } \varphi \in C_0^\infty(E).$$

Hence, $\boldsymbol{\psi} \cdot \boldsymbol{n}$ is equal to a constant c_E on E. By a similar argument it follows that $\partial_{\boldsymbol{t}}(\boldsymbol{\psi} \cdot \boldsymbol{t})$ is equal to a constant a_E on each edge $E \in \Gamma_f$.
Using that $\boldsymbol{\psi} \cdot \boldsymbol{n}$ is edgewise constant on $\Gamma_s \cup \Gamma_f$ and $\partial_{\boldsymbol{t}}(\boldsymbol{\psi} \cdot \boldsymbol{t})$ is edgewise constant on Γ_f, we obtain from (2.63) by edgewise integration by parts:

$$\begin{aligned}
\langle \partial_{\boldsymbol{t}} \boldsymbol{\psi}, \nabla v \rangle_\Gamma &= -\sum_{E \subset \Gamma_s \cup \Gamma_f} \big((\boldsymbol{\psi} \cdot \boldsymbol{n})\; \partial_{\boldsymbol{n}} v\big)\big|_{\partial E} - \sum_{E \subset \Gamma_f} \big((\boldsymbol{\psi} \cdot \boldsymbol{t})\; \partial_{\boldsymbol{t}} v - \partial_{\boldsymbol{t}}(\boldsymbol{\psi} \cdot \boldsymbol{t})\, v\big)\big|_{\partial E} \\
&= -\sum_{E \subset \Gamma_s} \big((\boldsymbol{\psi} \cdot \boldsymbol{n})\; \partial_{\boldsymbol{n}} v\big)\big|_{\partial E} - \sum_{E \subset \Gamma_f} \big(\boldsymbol{\psi} \cdot \nabla v - \partial_{\boldsymbol{t}}(\boldsymbol{\psi} \cdot \boldsymbol{t})\, v\big)\big|_{\partial E}
\end{aligned} \tag{2.64}$$

with the notations of Definition 2.3. Now let E and E' be two adjacent edges from Γ_f with the common corner point x and let $v \in C_0^\infty(\Omega \cup E \cup E' \cup \{x\}) \cap W$. Then the condition $\langle \partial_{\boldsymbol{t}} \boldsymbol{\psi}, \nabla v \rangle_\Gamma = 0$ reduces using (2.64) to

$$0 = [\![\boldsymbol{\psi} \cdot \nabla v - \partial_{\boldsymbol{t}}(\boldsymbol{\psi} \cdot \boldsymbol{t})\, v]\!]_x = [\![\boldsymbol{\psi}]\!]_x \cdot \nabla v(x) - [\![\partial_{\boldsymbol{t}}(\boldsymbol{\psi} \cdot \boldsymbol{t})]\!]_x \, v(x).$$

Observe that $\boldsymbol{\psi} \in P_1$ on each edge of Γ_f, where P_1 is the set of polynomials of degree ≤ 1, and is the trace of an H^1-function. Therefore, $\boldsymbol{\psi}$ must be continuous on Γ_f, which implies $[\![\boldsymbol{\psi}]\!]_x = 0$. Since $v(x)$ can be chosen arbitrarily, it follows that $[\![\partial_{\boldsymbol{t}}(\boldsymbol{\psi} \cdot \boldsymbol{t})]\!]_x = 0$. So $\partial_{\boldsymbol{t}}(\boldsymbol{\psi} \cdot \boldsymbol{t})$ is not only constant on each edge E of Γ_f but it is equal to the same constant a_C on each edge of a connected component C of Γ_f. Since we additionally know from above that $\boldsymbol{\psi} \cdot \boldsymbol{n}$ is constant on such an edge, we have

$$\partial_{\boldsymbol{t}}\boldsymbol{\psi} \cdot \boldsymbol{n} = \partial_{\boldsymbol{t}}(\boldsymbol{\psi} \cdot \boldsymbol{n}) = 0 \quad \text{and} \quad \partial_{\boldsymbol{t}}\boldsymbol{\psi} \cdot \boldsymbol{t} = \partial_{\boldsymbol{t}}(\boldsymbol{\psi} \cdot \boldsymbol{t}) = a_C.$$

This easily implies that $\partial_{\boldsymbol{t}}\widetilde{\boldsymbol{\psi}} = 0$ on each edge of C with $\widetilde{\boldsymbol{\psi}}(x) = \boldsymbol{\psi}(x) - a_C x$, since

$$\partial_{\boldsymbol{t}}\tilde{\boldsymbol{\psi}} \cdot \boldsymbol{n} = a_C\, \boldsymbol{t} \cdot \boldsymbol{n} = 0 \quad \text{and} \quad \partial_{\boldsymbol{t}}\boldsymbol{\psi} \cdot \boldsymbol{t} = a_C - a_C\, \boldsymbol{t} \cdot \boldsymbol{t} = 0.$$

Since $\boldsymbol{\psi}$ is continuous on Γ, $\widetilde{\boldsymbol{\psi}}$ is continuous, too. Then it follows that $\widetilde{\boldsymbol{\psi}}$ is equal to a common constant $b_C \in \mathbb{R}^2$ on C. Hence, $\boldsymbol{\psi}(x) = a_C x + b_C$ on C.
Next let $E \subset \Gamma_s$ and $E' \subset \Gamma_f$ be two adjacent edges with the common corner point x (ordered counterclockwise, i.e., $E_{k-1} = E$ and $E_k = E'$) and let $v \in C_0^\infty(\Omega \cup E \cup E' \cup \{x\}) \cap W$. With $\nabla v = (\partial_{\boldsymbol{n}_E} v)\, \boldsymbol{n}_E + (\partial_{\boldsymbol{t}_E} v)\, \boldsymbol{t}_E$ and $v(x) = 0$ the condition $\langle \partial_{\boldsymbol{t}}\boldsymbol{\psi}, \nabla v\rangle_\Gamma = 0$ reduces using (2.64) to

$$\begin{aligned} 0 &= -\big((\boldsymbol{\psi} \cdot \boldsymbol{n})\ \partial_{\boldsymbol{n}} v\big)(x^-) + \big(\boldsymbol{\psi} \cdot \nabla v\big)(x^+) \\ &= -c_E\ \partial_{\boldsymbol{n}_E} v(x) + r_C(x) \cdot \boldsymbol{n}_E\ \partial_{\boldsymbol{n}_E} v(x) + r_C(x) \cdot \boldsymbol{t}_E\ \underbrace{\partial_{\boldsymbol{t}_E} v(x)}_{=0} \\ &= (-c_E + r_C(x) \cdot \boldsymbol{n}_E)\ \partial_{\boldsymbol{n}_E} v(x). \end{aligned} \tag{2.65}$$

Since $\partial_{\boldsymbol{n}_E} v(x)$ can be chosen arbitrarily, this implies the compatibility condition $c_E = r_C(x) \cdot \boldsymbol{n}_E$. For the reverse case $E_{k-1} = E'$ and $E_k = E$ the result follows analogously.
For proving the reverse direction, let $\boldsymbol{\psi} \in (H^1(\Omega))^2$ satisfy the boundary conditions (2.61). In order to verify that $\boldsymbol{\psi} \in \Psi_0$ we have to show that

$$\langle \partial_{\boldsymbol{t}}\boldsymbol{\psi}, \nabla v\rangle_\Gamma = 0$$

for all $v \in C^\infty(\overline{\Omega}) \cap W$. Using (2.64) we have

$$\langle \partial_{\boldsymbol{t}}\boldsymbol{\psi}, \nabla v\rangle_\Gamma = -\sum_{E \subset \Gamma_s} \big((\boldsymbol{\psi} \cdot \boldsymbol{n})\ \partial_{\boldsymbol{n}} v\big)\big|_{\partial E} - \sum_{E \subset \Gamma_f} \big(\boldsymbol{\psi} \cdot \nabla v - \partial_{\boldsymbol{t}}(\boldsymbol{\psi} \cdot \boldsymbol{t})\, v\big)\big|_{\partial E},$$

which vanishes for all $v \in C^\infty(\overline{\Omega}) \cap W$ for the following reasons:

- $\partial_{\boldsymbol{n}} v(x) = 0$ on all interior corner points of Γ_s,
  (Note, $\partial_{\boldsymbol{n}} v(x) = 0$ on all interior corner points of Γ_s, since $v = 0$ on Γ_s. This makes the tangential derivative vanish at this corner points for two linear independent tangential directions. Therefore, we obtain $\nabla v(x) = 0$ and $\partial_{\boldsymbol{n}} v(x) = 0$.)

- $\boldsymbol{\psi} \cdot \nabla v - \partial_{\boldsymbol{t}}(\boldsymbol{\psi} \cdot \boldsymbol{t}) v$ is continuous on Γ_f, hence the corresponding jump vanishes at interior corner points of Γ_f,

- $v(x) = 0$ and $\nabla v(x) = 0$ on corner points on the interface of Γ_s or Γ_f with Γ_c,
- and the compatibility condition on corner points on the interface of Γ_s and Γ_f, see (2.65).

For the last part, let a set of data $c_E \in \mathbb{R}$ for each edge $E \subset \Gamma_s$ and $r_C \in RT_0$ for each connected component $C \subset \Gamma_f$ be given, which satisfy the compatibility conditions. We first construct a continuous function $\hat{\boldsymbol{\psi}}_\Gamma$ on Γ with $\hat{\boldsymbol{\psi}}_\Gamma|_E \in P_1$ for each edge $E \in \mathcal{E}_\Gamma$ by prescribing its values on all corner points $x \in \mathcal{V}_\Gamma$ as follows:

$$\hat{\boldsymbol{\psi}}_\Gamma(x) = \begin{cases} r_C(x) & \text{if } x \text{ is a corner points of the connected component } C \subset \Gamma_f, \\ & \text{(including corner points on the interfaces } \Gamma_f,\, \Gamma_s \text{ and } \Gamma_f,\, \Gamma_c \text{)} \\ \boldsymbol{\psi}_s(x) & \text{if } x \text{ is an interior corner point of } \Gamma_s, \\ \boldsymbol{\psi}_{sc}(x) & \text{if } x \text{ is a corner point on the interface of } \Gamma_s \text{ and } \Gamma_c, \\ 0 & \text{if } x \text{ is an interior corner point of } \Gamma_c. \end{cases}$$

Here $\boldsymbol{\psi}_s(x)$ and $\boldsymbol{\psi}_{cs}(x)$ are defined as follows: For a common corner point x of $E, E' \subset \Gamma_s$, $\boldsymbol{\psi}_s(x)$ is uniquely given by

$$\boldsymbol{\psi}_s(x) \cdot \boldsymbol{n}_E = c_E \quad \text{and} \quad \boldsymbol{\psi}_s(x) \cdot \boldsymbol{n}_{E'} = c_{E'}.$$

For a common corner point x of $E \subset \Gamma_s$ and $E' \subset \Gamma_c$, $\boldsymbol{\psi}_{sc}(x)$ is uniquely given by

$$\boldsymbol{\psi}_{sc}(x) \cdot \boldsymbol{n}_E = c_E \quad \text{and} \quad \boldsymbol{\psi}_{sc}(x) \cdot \boldsymbol{t}_E = 0.$$

Let $\hat{\boldsymbol{\psi}} \in (H^1(\Omega))^2$ be the harmonic extension of $\hat{\boldsymbol{\psi}}_\Gamma$. It is easy to verify that the boundary conditions (2.61) are satisfied. □

In order to eliminate c_E and r_C in (2.61) we introduce the following Clément-type projection operator on Γ.

Definition 2.27. *Let $\boldsymbol{\psi}_\Gamma \in L^2(\Gamma)$. Then the projection $\Pi_\Gamma \colon L^2(\Gamma) \to L^2(\Gamma)$ is given by $\hat{\boldsymbol{\psi}}_\Gamma = \Pi_\Gamma \boldsymbol{\psi}_\Gamma$, where $\hat{\boldsymbol{\psi}}_\Gamma$ is constructed as in the last part of the proof of Lemma 2.26 from the data $c_E(\boldsymbol{\psi}_\Gamma)$ and $r_C(\boldsymbol{\psi}_\Gamma)$, which are the L^2-projection of $\boldsymbol{\psi}_\Gamma \cdot \boldsymbol{n}_E$ onto the set P_0 of constant functions and the L^2-projection of $\boldsymbol{\psi}_\Gamma|_C$ onto RT_0, respectively. Except if $E \subset \Gamma_s$ is an adjacent edge to a component C of Γ_f, then we set instead $c_E(\boldsymbol{\psi}_\Gamma) = r_C(\boldsymbol{\psi}_\Gamma)(x) \cdot \boldsymbol{n}_E$, where x is the enclosed corner point and $\boldsymbol{n}_E$ is the normal on E, in order to enforce the compatibility conditions.*

With the just introduced notation we have for $\boldsymbol{\psi} \in (H^1(\Omega))^2$ satisfying the boundary conditions (2.61), that

$$c_E(\boldsymbol{\psi}) = c_E, \quad \text{and} \quad r_C(\boldsymbol{\psi}) = r_C,$$

hence

$$\boldsymbol{\psi} \cdot \boldsymbol{n} = \Pi_\Gamma \boldsymbol{\psi} \cdot \boldsymbol{n} \quad \text{on } \Gamma_s, \qquad \boldsymbol{\psi} = \Pi_\Gamma \boldsymbol{\psi} \quad \text{on } \Gamma_f.$$

Therefore, the boundary conditions (2.61) can be rewritten as

$$(\mathrm{P}\,\boldsymbol{\psi}) \cdot \boldsymbol{n} = 0 \quad \text{on } \Gamma_s, \qquad \mathrm{P}\,\boldsymbol{\psi} = 0 \quad \text{on } \Gamma_f, \tag{2.66}$$

with $\mathrm{P}\,\boldsymbol{\psi} = (I - \Pi_\Gamma)\boldsymbol{\psi}$.

2.5.2 Decoupled formulation

The representation of $\mathcal{V}$ as a direct sum in (2.57) leads to the following equivalent formulation of (2.50): Find $p \in Q$, $\boldsymbol{\phi} \in \Psi_p = \boldsymbol{\psi}[p] + \Psi_0$ and $u \in Q$ such that

$$\begin{aligned}
&(p\boldsymbol{I} + \operatorname{symCurl}\boldsymbol{\phi}, q\boldsymbol{I} + \operatorname{symCurl}\boldsymbol{\psi}[q])_{\mathcal{C}^{-1}} - (\nabla u, \nabla q) &&= 0\\
&(p\boldsymbol{I} + \operatorname{symCurl}\boldsymbol{\phi}, \operatorname{symCurl}\boldsymbol{\psi}_0)_{\mathcal{C}^{-1}} &&= 0\\
&-(\nabla p, \nabla v) &&= -\langle F, v\rangle.
\end{aligned}$$

for all $q \in Q$, $\boldsymbol{\psi}_0 \in \Psi_0$ and $v \in Q$. Therefore, the mixed formulation of the Kirchhoff plate bending problem is equivalent to three (consecutively to solve) elliptic second-order problems:

1. The p-problem: Find $p \in Q$ such that
$$(\nabla p, \nabla v) = \langle F, v\rangle \quad \text{for all } v \in Q. \tag{2.67}$$

2. The $\boldsymbol{\phi}$-problem: For given $p \in Q$, find $\boldsymbol{\phi} \in \Psi_p = \boldsymbol{\psi}[p] + \Psi_0/RT_0$ such that
$$(\operatorname{symCurl}\boldsymbol{\phi}, \operatorname{symCurl}\boldsymbol{\psi}_0)_{\mathcal{C}^{-1}} = -(p\boldsymbol{I}, \operatorname{symCurl}\boldsymbol{\psi}_0)_{\mathcal{C}^{-1}} \quad \text{for all } \boldsymbol{\psi}_0 \in \Psi_0/RT_0. \tag{2.68}$$
In order to obtain uniqueness of the solution we have replaced Ψ_0 by Ψ_0/RT_0. Note that the right-hand side vanishes for $\boldsymbol{\psi}_0 \in RT_0$, which guarantees equivalence to the original problem.

3. The u-problem: For given $\boldsymbol{M} = p\boldsymbol{I} + \operatorname{symCurl}\boldsymbol{\phi}$, find $u \in Q$ such that
$$(\nabla u, \nabla q) = (\boldsymbol{M}, q\boldsymbol{I} + \operatorname{symCurl}\boldsymbol{\psi}[q])_{\mathcal{C}^{-1}} \quad \text{for all } q \in Q. \tag{2.69}$$

Remark 2.28. *For the rotated function $\boldsymbol{\phi}^\perp = (-\boldsymbol{\phi}_2, \boldsymbol{\phi}_1)^T$ the $\boldsymbol{\phi}$-problem becomes a linear elasticity problem*

$$(\varepsilon(\boldsymbol{\phi}^\perp), \varepsilon(\boldsymbol{\psi}_0^\perp))_{\tilde{\mathcal{C}}^{-1}} = -(p\boldsymbol{I}, \varepsilon(\boldsymbol{\psi}_0^\perp))_{\tilde{\mathcal{C}}^{-1}}$$

with an appropriately rotated material tensor $\tilde{\mathcal{C}}^{-1}$.

Remark 2.29. *If we only have clamped and simply supported boundary parts, we know from Section 2.5.1 that $\boldsymbol{\psi}[q] = 0$ on Ω. Therefore, the terms involving $\boldsymbol{\psi}[q]$ just vanish. Furthermore, in the purely clamped case Ψ_p and Ψ_0 become $(H^1(\Omega))^2$.*

Remark 2.30. *1. The p-problem and the u-problem are standard Poisson problems with mixed boundary conditions on $\Gamma_c \cup \Gamma_s$ and Γ_f.*

2. From (2.68) *we obtain using the symmetry of $\boldsymbol{M}$ that*

$$(p\boldsymbol{I} + \operatorname{symCurl}\boldsymbol{\phi}, \operatorname{symCurl}\boldsymbol{\psi})_{\mathcal{C}^{-1}} = (\boldsymbol{M}, \operatorname{symCurl}\boldsymbol{\psi})_{\mathcal{C}^{-1}} = (\boldsymbol{M}, \operatorname{Curl}\boldsymbol{\psi})_{\mathcal{C}^{-1}} = 0$$

for all $\boldsymbol{\psi} \in C_0^\infty(\Omega)$*. Here we used that* (2.68) *holds for all* $\boldsymbol{\psi}_0 \in \Psi_0$*, since* $\operatorname{symCurl} \boldsymbol{r} = 0$ *for* $\boldsymbol{r} \in RT_0$*. Hence (see Appendix A),*

$$\operatorname{Rot}(\mathcal{C}^{-1}\boldsymbol{M}) = 0 \quad \textit{in } L^2(\Omega), \tag{2.70}$$

therefore, $\mathcal{C}^{-1}\boldsymbol{M} \in \boldsymbol{H}(\operatorname{Rot}, \Omega) = \{\boldsymbol{L} \in \boldsymbol{L}^2(\Omega) : \operatorname{Rot} \boldsymbol{L} \in (L^2(\Omega))^2\}$*. With the representation* $\boldsymbol{M} = p\boldsymbol{I} + \operatorname{symCurl} \boldsymbol{\phi}$*, using* $p \in Q \subset H^1(\Omega)$*, we receive that the solution of the* $\boldsymbol{\phi}$*-problem satisfies the second-order differential equation*

$$\operatorname{Rot}\left(\mathcal{C}^{-1} \operatorname{symCurl} \boldsymbol{\phi}\right) = -\operatorname{Rot}\left(\mathcal{C}^{-1}(p\boldsymbol{I})\right) \quad \textit{in } L^2(\Omega) \tag{2.71}$$

and the (essential) boundary conditions (see (2.66)*)*

$$(\mathrm{P}\,\boldsymbol{\phi}) \cdot \boldsymbol{n} = (\mathrm{P}\,\boldsymbol{\psi}_\Gamma[p]) \cdot \boldsymbol{n} \quad \textit{on } \Gamma_s, \qquad \mathrm{P}\,\boldsymbol{\phi} = \mathrm{P}\,\boldsymbol{\psi}_\Gamma[p] \quad \textit{on } \Gamma_f. \tag{2.72}$$

In order to deduce the natural boundary conditions (which we need later), we multiply (2.71) *with a test function* $\boldsymbol{\psi}_0 \in \Psi_0$ *and integrate by parts*

$$\begin{aligned}(\operatorname{symCurl} \boldsymbol{\phi}, \operatorname{symCurl} \boldsymbol{\psi}_0)_{\mathcal{C}^{-1}} &+ \langle (\mathcal{C}^{-1} \operatorname{symCurl} \boldsymbol{\phi})\boldsymbol{t}, \boldsymbol{\psi}_0 \rangle_\Gamma \\ &= -(p\boldsymbol{I}, \operatorname{symCurl} \boldsymbol{\psi}_0)_{\mathcal{C}^{-1}} - ((\mathcal{C}^{-1} p\boldsymbol{I})\boldsymbol{t}, \boldsymbol{\psi}_0)_\Gamma.\end{aligned}$$

This leads comparing the last equation with (2.68) *to the following condition for the flux* $\boldsymbol{\chi} = (\mathcal{C}^{-1} \operatorname{symCurl} \boldsymbol{\phi})\, \boldsymbol{t}$*:*

$$\langle \boldsymbol{\chi}, \boldsymbol{\psi}_0 \rangle_\Gamma = -\big((\mathcal{C}^{-1} p\, \boldsymbol{I})\boldsymbol{t}, \boldsymbol{\psi}_0\big)_\Gamma \quad \textit{for all } \boldsymbol{\psi}_0 \in \Psi_0. \tag{2.73}$$

If $\boldsymbol{\chi} \in L^2(\Gamma)$*, then* (2.73) *can be rewritten using* $p = 0$ *on* $\Gamma_c \cup \Gamma_s$ *and incorporating the boundary conditions for* $\boldsymbol{\psi}_0 \in \Psi_0$

$$(\mathrm{P}\,\boldsymbol{\psi}_0) \cdot \boldsymbol{n} = 0 \quad \textit{on } \Gamma_s, \qquad \mathrm{P}\,\boldsymbol{\psi}_0 = 0 \quad \textit{on } \Gamma_f \quad \textit{(cf. (2.66))},$$

with $\boldsymbol{\psi}_0 = \mathrm{P}\,\boldsymbol{\psi}_0 + \Pi_\Gamma \boldsymbol{\psi}_0$ *as follows*

$$(\boldsymbol{\chi}, \boldsymbol{\psi}_0)_{\Gamma_c} + (\boldsymbol{\chi} \cdot \boldsymbol{t}, \boldsymbol{\psi}_0 \cdot \boldsymbol{t})_{\Gamma_s} + (\boldsymbol{\chi} \cdot \boldsymbol{n}, (\Pi_\Gamma \boldsymbol{\psi}_0) \cdot \boldsymbol{n})_{\Gamma_s} + (\boldsymbol{\chi} + (\mathcal{C}^{-1} p\boldsymbol{I})\boldsymbol{t}, \Pi_\Gamma \boldsymbol{\psi}_0)_{\Gamma_f} = 0.$$

Therefore, for $\boldsymbol{\chi} \in L^2(\Gamma)$ (2.73) *is equivalent to the following (natural) boundary conditions*

$$\boldsymbol{\chi} = 0 \quad \textit{on } \Gamma_c, \qquad \boldsymbol{\chi} \cdot \boldsymbol{t} = 0 \quad \textit{on } \Gamma_s, \tag{2.74}$$

and

$$(\boldsymbol{\chi} \cdot \boldsymbol{n}, (\Pi_\Gamma \boldsymbol{\psi}_0) \cdot \boldsymbol{n})_{\Gamma_s} + (\boldsymbol{\chi}, \Pi_\Gamma \boldsymbol{\psi}_0)_{\Gamma_f} = -((\mathcal{C}^{-1} p\boldsymbol{I})\boldsymbol{t}, \Pi_\Gamma \boldsymbol{\psi}_0)_{\Gamma_f} \quad \textit{for all } \boldsymbol{\psi}_0 \in \Psi_0.$$

Note, the projections Π_Γ *can be removed, as we have due to the boundary conditions* $\boldsymbol{\psi}_0 \cdot \boldsymbol{n} = (\Pi_\Gamma \boldsymbol{\psi}_0) \cdot \boldsymbol{n}$ *and* $\boldsymbol{\psi}_0 = \Pi_\Gamma \boldsymbol{\psi}_0$*, hence*

$$(\boldsymbol{\chi} \cdot \boldsymbol{n}, \boldsymbol{\psi}_0 \cdot \boldsymbol{n})_{\Gamma_s} + (\boldsymbol{\chi}, \boldsymbol{\psi}_0)_{\Gamma_f} = -((\mathcal{C}^{-1} p\boldsymbol{I})\boldsymbol{t}, \boldsymbol{\psi}_0)_{\Gamma_f} \quad \textit{for all } \boldsymbol{\psi}_0 \in \Psi_0. \tag{2.75}$$

3. *For the rotated function $\boldsymbol{\phi}^\perp$,* (2.71) *is a linear elasticity problem*

$$-\operatorname{Div}\left(\tilde{\mathcal{C}}^{-1}\varepsilon(\boldsymbol{\phi}^\perp)\right) = \operatorname{Div}\left(\tilde{\mathcal{C}}^{-1}(p\boldsymbol{I})\right) \quad \text{in } L^2(\Omega)$$

with the corresponding boundary conditions for $\boldsymbol{\phi}^\perp$ and $\left(\tilde{\mathcal{C}}^{-1}\varepsilon(\boldsymbol{\phi}^\perp)\right)\boldsymbol{n}$, which can be interpreted as displacement and traction.

The just obtained decomposition result of the Kirchhoff plate-bending problem on the continuous level allows for great flexibility in the construction of new discretization methods. Namely, any well-working discretization method and solution strategy for standard second-order problems can be used as modular building blocks of the new method. In the next subsections we propose a discretization method for the above introduced formulation using a Nitsche method to incorporate the boundary conditions in the $\boldsymbol{\phi}$-problem and present a numerical analysis of the method.
Keep in mind, the considerations of the current section heavily rely on a polygonal domain and it is not clear how to obtain standard boundary conditions in case of a curved boundary. This is our main motivation to investigate an alternative approach to incorporate the coupling condition (2.46) based on Lagrange multipliers, which we introduce in Section 2.6.

2.5.3 The discretization method

Let $(\mathcal{T}_h)_{h\in\mathcal{H}}$ be a shape-regular family of subdivisions of the domain Ω into polygonal elements. The diameter of an element $T \in \mathcal{T}_h$ is denoted by h_T and we define $h = \max\{h_T : T \in \mathcal{T}_h\}$. We denote the set of all edges e of elements $T \in \mathcal{T}_h$ with $e \subset \Gamma_s$ by $\mathcal{E}_{h,s}$ and with $e \subset \Gamma_f$ by $\mathcal{E}_{h,f}$. The length of an edge e is denoted by h_e. Moreover, we assume that the total number of edges in $\mathcal{E}_{h,s} \cup \mathcal{E}_{h,f}$ is bounded by $c\,h^{-1}$. Note, this assumption is weaker than requiring a quasi-uniform family of subdivisions. It can be viewed as a quasi-uniformity condition in a neighborhood of the boundary. Here and in the sequel c denotes a generic constant independent of h, possibly different at each occurrence.

Remark 2.31. *Observe that the symbol e is used to indicate element edges, while the symbol E used in the preceding sections is reserved for edges of the domain as introduced at the beginning of Section 2.1.3.*

Let $\mathcal{S}_h^p(\Omega)$ be a finite dimensional subspace of $H^1(\Omega)$ of piecewise polynomials with degree p associated with $\mathcal{T}_h$. For shorter notation and since no ambiguity is possible, we skip in the remainder of this section the explicit dependence on the polynomial degree p and the domain Ω and write simply $\mathcal{S}_h$. We set $\mathcal{S}_{h,0} = \mathcal{S}_h \cap H^1_{0,\Gamma_c\cup\Gamma_s}(\Omega)$.
For $v \in H^s(\Omega)$, with $1 \le s \le p+1$, we assume the standard approximation property

$$\inf_{v_h\in\mathcal{S}_h} \|v - v_h\|_l \le c\, h^{s-l}\|v\|_s, \tag{A1}$$

for all $l \in \{0,\dots,s\}$. Moreover, we require the discrete trace inequality

$$\|v_h\|_{0,e} \le c\, h_T^{-\frac{1}{2}}\|v_h\|_{0,T} \quad \text{for all } v_h \in \mathcal{S}_h,\ e \subset \partial T,\ T \in \mathcal{T}_h, \tag{A2}$$

and the continuous trace inequality

$$\|v\|_{0,e} \le c\, h_T^{-\frac{1}{2}} (\|v\|_{0,T} + h_T \|\nabla v\|_{0,T}) \quad \text{for all } v \in H^1(T),\ e \subset \partial T,\ T \in \mathcal{T}_h. \tag{A3}$$

Here and in the following the L^2-norm on an element T and an edge e are denoted by $\|.\|_{0,T}$ and $\|.\|_{0,e}$, respectively. The properties (A1), (A2), (A3) are satisfied for standard finite element spaces or isogeometric B-spline discretization spaces under standard assumptions.
The method we propose consists of three consecutive steps:

1. The discrete p-problem: Find $p_h \in \mathcal{S}_{h,0}$ such that

$$(\nabla p_h, \nabla v_h) = \langle F, v_h \rangle \quad \text{for all } v_h \in \mathcal{S}_{h,0}. \tag{2.76}$$

2. The discrete $\boldsymbol{\phi}$-problem: For given $p_h \in \mathcal{S}_{h,0}$, find $\boldsymbol{\phi}_h \in (\mathcal{S}_h)^2/RT_0$ such that

$$a_{\boldsymbol{\phi},h}(\boldsymbol{\phi}_h, \boldsymbol{\psi}_h) = \langle F_{\boldsymbol{\phi},h}, \boldsymbol{\psi}_h \rangle \quad \text{for all } \boldsymbol{\psi}_h \in (\mathcal{S}_h)^2/RT_0, \tag{2.77}$$

with

$$\begin{aligned} a_{\boldsymbol{\phi},h}(\boldsymbol{\phi}, \boldsymbol{\psi}) &= (\operatorname{symCurl} \boldsymbol{\phi}, \operatorname{symCurl} \boldsymbol{\psi})_{\mathcal{C}^{-1}} + s(\boldsymbol{\phi}, \boldsymbol{\psi}) + s(\boldsymbol{\psi}, \boldsymbol{\phi}) + r_h(\boldsymbol{\phi}, \boldsymbol{\psi}), \\ \langle F_{\boldsymbol{\phi},h}, \boldsymbol{\psi} \rangle &= -(p_h \boldsymbol{I}, \operatorname{symCurl} \boldsymbol{\psi})_{\mathcal{C}^{-1}} - c(p_h, \boldsymbol{\psi}) + s(\boldsymbol{\psi}, \boldsymbol{\psi}_\Gamma[p_h]) + r_h(\boldsymbol{\psi}_\Gamma[p_h], \boldsymbol{\psi}), \end{aligned}$$

where

$$\begin{aligned} s(\boldsymbol{\phi}, \boldsymbol{\psi}) &= (\boldsymbol{\chi} \cdot \boldsymbol{n}, \mathrm{P}\, \boldsymbol{\psi} \cdot \boldsymbol{n})_{\Gamma_s} + (\boldsymbol{\chi}, \mathrm{P}\, \boldsymbol{\psi})_{\Gamma_f} \quad \text{with} \quad \boldsymbol{\chi} = (\mathcal{C}^{-1} \operatorname{symCurl} \boldsymbol{\phi}) \boldsymbol{t}, \\ c(q, \boldsymbol{\psi}) &= \big((\mathcal{C}^{-1} q \boldsymbol{I}) \boldsymbol{t}, \mathrm{P}\, \boldsymbol{\psi} \big)_{\Gamma_f}, \\ r_h(\boldsymbol{\phi}, \boldsymbol{\psi}) &= \sum_{e \in \mathcal{E}_{h,s}} \frac{\eta}{h_e} \left(\mathrm{P}\, \boldsymbol{\phi} \cdot \boldsymbol{n}, \mathrm{P}\, \boldsymbol{\psi} \cdot \boldsymbol{n} \right)_e + \sum_{e \in \mathcal{E}_{h,f}} \frac{\eta}{h_e} \left(\mathrm{P}\, \boldsymbol{\phi}, \mathrm{P}\, \boldsymbol{\psi} \right)_e, \end{aligned}$$

for some penalty parameter $\eta > 0$.

3. The discrete u-problem: For given $\boldsymbol{M}_h = p_h \boldsymbol{I} + \operatorname{symCurl} \boldsymbol{\phi}_h$, find $u_h \in \mathcal{S}_{h,0}$ such that

$$(\nabla u_h, \nabla q_h) = \langle F_{u,h}, q_h \rangle \quad \text{for all } q_h \in \mathcal{S}_{h,0}, \tag{2.78}$$

with

$$\begin{aligned} \langle F_{u,h}, q \rangle = (\boldsymbol{M}_h, q\boldsymbol{I})_{\mathcal{C}^{-1}} &- s(\boldsymbol{\phi}_h, \boldsymbol{\psi}_\Gamma[q]) - c(p_h, \boldsymbol{\psi}_\Gamma[q]) \\ &- r_h(\boldsymbol{\phi}_h - \boldsymbol{\psi}_\Gamma[p_h], \boldsymbol{\psi}_\Gamma[q]). \end{aligned} \tag{2.79}$$

Remark 2.32. *Note, the above problems are derived as follows:*

1. *The discrete p-problem is the standard Galerkin method applied to* (2.67).

2. *The discrete $\boldsymbol{\phi}$-problem is a Nitsche method applied to* (2.68)*, which is derived as follows. We start from the identity*

$$(\boldsymbol{M}, \operatorname{symCurl} \boldsymbol{\psi})_{\mathcal{C}^{-1}} + \langle (\mathcal{C}^{-1}\boldsymbol{M})\boldsymbol{t}, \boldsymbol{\psi}\rangle_\Gamma = 0 \quad \text{for all } \boldsymbol{\psi} \in (H^1(\Omega))^2, \tag{2.80}$$

which follows from (2.70) by multiplying with a test function with no prescribed boundary conditions and using integration by parts (see Appendix A.3). For the second term of (2.80) *the representation of $\boldsymbol{M}$ leads to*

$$\langle (\mathcal{C}^{-1}\boldsymbol{M})\boldsymbol{t}, \boldsymbol{\psi}\rangle_\Gamma = \langle \boldsymbol{\chi}, \boldsymbol{\psi}\rangle_\Gamma + \big((\mathcal{C}^{-1}p\boldsymbol{I})\boldsymbol{t}, \boldsymbol{\psi}\big)_\Gamma \tag{2.81}$$

with $\boldsymbol{\chi} = (\mathcal{C}^{-1} \operatorname{symCurl} \boldsymbol{\phi})\boldsymbol{t}$. If $\boldsymbol{\chi} \in L^2(\Gamma)$, we can rewrite the expression using $p = 0$ on $\Gamma_c \cup \Gamma_s$ and $\boldsymbol{\psi} = \mathrm{P}\,\boldsymbol{\psi} + \Pi_\Gamma \boldsymbol{\psi}$ as follows

$$\begin{aligned}\langle (\mathcal{C}^{-1}\boldsymbol{M})\boldsymbol{t}, \boldsymbol{\psi}\rangle_\Gamma = {}& (\boldsymbol{\chi}, \boldsymbol{\psi})_{\Gamma_c} + (\boldsymbol{\chi}\cdot \boldsymbol{t}, \boldsymbol{\psi}\cdot\boldsymbol{t})_{\Gamma_s} + (\boldsymbol{\chi}\cdot\boldsymbol{n}, (\mathrm{P}\,\boldsymbol{\psi} + \Pi_\Gamma\boldsymbol{\psi})\cdot\boldsymbol{n})_{\Gamma_s} \\ &+ (\boldsymbol{\chi} + (\mathcal{C}^{-1}p\boldsymbol{I})\boldsymbol{t}, \mathrm{P}\,\boldsymbol{\psi} + \Pi_\Gamma\boldsymbol{\psi})_{\Gamma_f}.\end{aligned}$$

For the next step note that according to Lemma 2.26 there exists a $\boldsymbol{\psi}_0 \in \Psi_0$ such that $\boldsymbol{\psi}_0 = \Pi_\Gamma\boldsymbol{\psi}$ on Γ. Then the natural boundary conditions (2.74) *and* (2.75) *lead to*

$$\begin{aligned}\langle (\mathcal{C}^{-1}\boldsymbol{M})\boldsymbol{t}, \boldsymbol{\psi}\rangle_\Gamma &= (\boldsymbol{\chi}\cdot\boldsymbol{n}, \mathrm{P}\,\boldsymbol{\psi}\cdot\boldsymbol{n})_{\Gamma_s} + (\boldsymbol{\chi}, \mathrm{P}\,\boldsymbol{\psi})_{\Gamma_f} + \big((\mathcal{C}^{-1}p\boldsymbol{I})\boldsymbol{t}, \mathrm{P}\,\boldsymbol{\psi}\big)_{\Gamma_f} \\ &= s(\boldsymbol{\phi}, \boldsymbol{\psi}) + c(p, \boldsymbol{\psi}).\end{aligned} \tag{2.82}$$

Then the method is obtained by first extending (2.80) *by the terms $s(\boldsymbol{\psi}, \boldsymbol{\phi} - \boldsymbol{\psi}_\Gamma[p])$ and $r_h(\boldsymbol{\phi} - \boldsymbol{\psi}_\Gamma[p], \boldsymbol{\psi})$, which vanish for the exact solution (see* (2.72)*), and then by replacing p, $\boldsymbol{\phi}$, and $\boldsymbol{\psi}$ with p_h, $\boldsymbol{\phi}_h$, and $\boldsymbol{\psi}_h$. Following [31] we call $s(\boldsymbol{\phi}, \boldsymbol{\psi}) + c(p, \boldsymbol{\psi})$, $s(\boldsymbol{\psi}, \boldsymbol{\phi})$, and $r_h(\boldsymbol{\phi}, \boldsymbol{\psi})$ the consistency, symmetry and penalty terms, respectively.*

3. *The discrete u-problem is the standard Galerkin method applied to* (2.69)*, where the right-hand side is reformulated as follows. By using* (2.80) *and* (2.82) *with $\boldsymbol{\psi} = \boldsymbol{\psi}[q]$ we obtain for the right-hand side of (2.69):*

$$\begin{aligned}&(\boldsymbol{M}, q\boldsymbol{I} + \operatorname{symCurl} \boldsymbol{\psi}[q])_{\mathcal{C}^{-1}} \\ &\quad = (\boldsymbol{M}, q\boldsymbol{I})_{\mathcal{C}^{-1}} - s(\boldsymbol{\phi}, \boldsymbol{\psi}_\Gamma[q]) - c(p, \boldsymbol{\psi}_\Gamma[q]) - r_h(\boldsymbol{\phi} - \boldsymbol{\psi}_\Gamma[p], \boldsymbol{\psi}_\Gamma[q]),\end{aligned} \tag{2.83}$$

where we additionally added the term $r_h(\boldsymbol{\phi} - \boldsymbol{\psi}_\Gamma[p], \boldsymbol{\psi}_\Gamma[q])$, which vanishes for the exact solution. Then (2.79) *is obtained by replacing p, $\boldsymbol{\phi}$, and q with p_h, $\boldsymbol{\phi}_h$, and q_h.*

Remark 2.33. *Since only $\boldsymbol{\psi}_\Gamma[p_h]$ and $\boldsymbol{\psi}_\Gamma[q_h]$ appear in the numerical method, the extensions of $\boldsymbol{\psi}[p_h]$ and $\boldsymbol{\psi}[q_h]$ to the interior are not needed. Although the functions $\boldsymbol{\psi}_\Gamma[q_h]$ do not have local support, the linear systems can still be assembled with optimal complexity.*

Remark 2.34. *Our decomposition of the continuous problem also leads to a new interpretation of the well-known Hellan-Herrmann-Johnson (HHJ) method; see [44, 45, 49]. We can proceed similar as in Theorem 2.19 and derive a discrete regular decomposition for the approximation space of the auxiliary variable, leading to a reformulation of the HHJ method in form of three consecutively to solve discretized second-order problems, as it has already been worked out in details in the purely clamped situation in [54]. The HHJ method is mainly restricted to triangular meshes. In [70] a HHJ-type method on rectangular meshes is considered. The new method introduced above is more flexible in the sense that triangular and general quadrilateral meshes can be handled and also isogeometric B-spline discretization spaces can be used.*

2.5.4 Numerical analysis

The main result of this section is the following a-priori discretization error estimate for the proposed method.

Theorem 2.35. *Assume that* Ω *is convex. Let* $(u, \boldsymbol{M})$*, with* $\boldsymbol{M} = p\boldsymbol{I} + \operatorname{symCurl} \boldsymbol{\phi}$*, be the solution of the mixed formulation* (2.37) *and* $(u_h, \boldsymbol{M}_h)$*, with* $\boldsymbol{M}_h = p_h\boldsymbol{I} + \operatorname{symCurl} \boldsymbol{\phi}_h$*, be the approximate solution, given by* (2.76), (2.77), (2.78)*. For* $u \in H^{s_u}(\Omega)$*,* $p \in H^{s_p}(\Omega)$ *and* $\boldsymbol{\phi} \in (H^{s_{\boldsymbol{\phi}}}(\Omega))^2$*, with* $1 \leq s_u, s_p \leq p+1$ *and* $\frac{3}{2} < s_{\boldsymbol{\phi}} \leq p+1$*, where* p *appearing in the upper bounds of* s_u*,* s_p *and* $s_{\boldsymbol{\phi}}$ *refers to the polynomial degree, we have the estimate*

$$\|\boldsymbol{M} - \boldsymbol{M}_h\|_0 + \|u - u_h\|_1 \leq c \left(h^{s_u-1}\|u\|_{s_u} + h^{s_p-1}\|p\|_{s_p} + h^{s_{\boldsymbol{\phi}}-1}\|\boldsymbol{\phi}\|_{s_{\boldsymbol{\phi}}}\right).$$

Especially, for $s_p = p+1$*,* $s_{\boldsymbol{\phi}} = p+1$ *and* $s_u = p+1$ *we obtain*

$$\|\boldsymbol{M} - \boldsymbol{M}_h\|_0 + \|u - u_h\|_1 \leq c\ h^p \left(\|u\|_{p+1} + \|p\|_{p+1} + \|\boldsymbol{\phi}\|_{p+1}\right).$$

In the above theorem, the p appearing in the upper bounds and particular choices for s_p, $s_{\boldsymbol{\phi}}$ and s_u refers to the polynomial degree and is not to be confused with the function p from the regular decomposition of $\boldsymbol{M}$. Since most of the time it is clear from the context what is meant and otherwise stated explicitly, we use this slight abuse of notation in order to stick to the common notation p for the polynomial degree.
We derive these estimates by discussing the discretization errors of the p-problem, the $\boldsymbol{\phi}$-problem, and the u-problem consecutively.

Error estimates for the p-problem

We start with the p-problem (2.67) and its discretization (2.76). It is well-known that the following error estimates hold:

Lemma 2.36 (p-problem)**.** *Under the assumptions of Theorem 2.35 we have*

$$\|p - p_h\|_1 \leq c\ h^{s_p-1}\|p\|_{s_p}, \quad \|p - p_h\|_0 \leq c\ h^{s_p}\|p\|_{s_p}, \quad \|p - p_h\|_{0,\Gamma} \leq c\ h^{s_p-\frac{1}{2}}\|p\|_{s_p}.$$

Proof. For the first two estimates, we refer to standard literature for the proof. For the third estimate note that by means of the continuous trace inequality (A3) we have

$$
\begin{aligned}
\|p-p_h\|_{0,\Gamma}^2 &= \sum_{E\in\mathcal{E}_{h,f}} \|p-p_h\|_{0,E}^2 \\
&\le c \sum_{T\in\mathcal{T}_h} h_T^{-1}\Big(\|p-p_h\|_{0,T} + h_T\|\nabla(p-p_h)\|_{0,T}\Big)^2 \\
&\le c \sum_{T\in\mathcal{T}_h} h_T^{-1}\Big(h_T^{s_p}\|p\|_{s_p,T} + h_T h_T^{s_p-1}\|p\|_{s_p,T}\Big)^2 \\
&\le c \sum_{T\in\mathcal{T}_h} \Big(h_T^{s_p-\frac{1}{2}}\|p\|_{s_p,T}\Big)^2 \\
&\le c\,(h^{s_p-\frac{1}{2}}\|p\|_{s_p})^2.
\end{aligned}
$$

Here, $\|.\|_{m,T}$ denotes the H^m-norm on an element T for $m > 0$. □

Note that the convexity of Ω is used only for the L^2-estimates.

Error estimates for the $\boldsymbol{\phi}$-problem

We follow the standard approach as outlined, e.g., in [31, 36], where in the second reference the case of linear elasticity is considered which is similar to our problem. So we introduce two mesh-dependent semi-norms: the jump semi-norm $|\boldsymbol{\psi}|_{\frac{1}{2},h}$ and the average semi-norm $|\boldsymbol{\chi}|_{-\frac{1}{2},h}$, which are given by

$$
\begin{aligned}
|\boldsymbol{\psi}|_{\frac{1}{2},h}^2 &= \sum_{e\in\mathcal{E}_{h,s}} h_e^{-1}\,\|\boldsymbol{\psi}\cdot\boldsymbol{n}\|_{0,e}^2 + \sum_{e\in\mathcal{E}_{h,f}} h_e^{-1}\,\|\boldsymbol{\psi}\|_{0,e}^2, \\
|\boldsymbol{\chi}|_{-\frac{1}{2},h}^2 &= \sum_{e\in\mathcal{E}_{h,s}} h_e\,\|\boldsymbol{\chi}\cdot\boldsymbol{n}\|_{0,e}^2 + \sum_{e\in\mathcal{E}_{h,f}} h_e\,\|\boldsymbol{\chi}\|_{0,e}^2.
\end{aligned}
$$

The analysis relies on the discrete coercivity and the boundedness of the bilinear form $a_{\boldsymbol{\phi},h}$ in appropriate norms.

Lemma 2.37 (Coercivity)**.** *There is a constant $c > 0$ such that*

$$a_{\boldsymbol{\phi},h}(\boldsymbol{\psi}_h,\boldsymbol{\psi}_h) \ge c\,\|\boldsymbol{\psi}_h\|_h^2 \quad \text{for all } \boldsymbol{\psi}_h \in (\mathcal{S}_h)^2/RT_0,$$

provided η is sufficiently large, where the mesh-dependent norm $\|\boldsymbol{\phi}\|_h$ is given by

$$\|\boldsymbol{\phi}\|_h^2 = (\operatorname{symCurl}\boldsymbol{\phi}, \operatorname{symCurl}\boldsymbol{\phi})_{\mathcal{C}^{-1}} + |\operatorname{P}\boldsymbol{\phi}|_{\frac{1}{2},h}^2.$$

Lemma 2.38 (Boundedness)**.** *There is a constant $c > 0$ such that*

$$a_{\boldsymbol{\phi},h}(\boldsymbol{\phi},\boldsymbol{\psi}_h) \le c\,\|\boldsymbol{\phi}\|_{h,*}\|\boldsymbol{\psi}_h\|_h$$

for all $\boldsymbol{\phi} \in (H^s(\Omega))^2 + (\mathcal{S}_h)^2$, *with* $s > \frac{3}{2}$ *and* $\boldsymbol{\psi}_h \in (\mathcal{S}_h)^2$, *where the mesh-dependent norm* $\|\boldsymbol{\phi}\|_{h,*}$ *is given by*

$$\|\boldsymbol{\phi}\|_{h,*}^2 = \|\boldsymbol{\phi}\|_h^2 + |\boldsymbol{\chi}|_{-\frac{1}{2},h}^2 \quad \textit{with} \quad \boldsymbol{\chi} = \left(\mathcal{C}^{-1}\,\mathrm{symCurl}\,\boldsymbol{\phi}\right)\boldsymbol{t}.$$

The proofs of these two lemmas are analogous to the proofs of similar results in [31, 36], but for completeness we outline them in the sequel. They are based on the following fundamental estimates for the symmetry, consistency, and penalty terms which we explicitly formulate in an auxiliary lemma for later use.

Lemma 2.39. *For all* $\boldsymbol{\psi} \in (H^1(\Omega))^2$, $\boldsymbol{\xi} \in (L^2(\Gamma))^2$, $q \in L^2(\Gamma)$, *and* $\boldsymbol{\psi}_h \in (\mathcal{S}_h)^2$ *we have*

$$|s(\boldsymbol{\psi}, \boldsymbol{\xi})| \le \|\boldsymbol{\psi}\|_{h,*}\, |\,\mathrm{P}\,\boldsymbol{\xi}|_{\frac{1}{2},h}, \tag{2.84}$$

$$\begin{aligned} |s(\boldsymbol{\psi}_h, \boldsymbol{\xi})| &\le c\,\left((\mathrm{symCurl}\,\boldsymbol{\psi}_h, \mathrm{symCurl}\,\boldsymbol{\psi}_h)_{\mathcal{C}^{-1}}\right)^{\frac{1}{2}}\, |\,\mathrm{P}\,\boldsymbol{\xi}|_{\frac{1}{2},h} \qquad (2.85)\\ &\le c\,\|\boldsymbol{\psi}_h\|_h\, |\,\mathrm{P}\,\boldsymbol{\xi}|_{\frac{1}{2},h}, \end{aligned}$$

$$|s(\boldsymbol{\psi}, \boldsymbol{\xi})| \le c\,\big(\inf_{\boldsymbol{\psi}_h \in (\mathcal{S}_h)^2} \|\boldsymbol{\psi} - \boldsymbol{\psi}_h\|_{h,*} + \|\boldsymbol{\psi}\|_h\big)\, |\,\mathrm{P}\,\boldsymbol{\xi}|_{\frac{1}{2},h}. \tag{2.86}$$

Moreover,

$$|c(q, \boldsymbol{\psi})| \le |(\mathcal{C}^{-1} q \boldsymbol{I})\boldsymbol{t}|_{-\frac{1}{2},h}\, \|\boldsymbol{\psi}\|_h, \tag{2.87}$$

$$|r_h(\boldsymbol{\xi}, \boldsymbol{\psi})| \le \eta\, |\,\mathrm{P}\,\boldsymbol{\xi}|_{\frac{1}{2},h}\, \|\boldsymbol{\psi}\|_h. \tag{2.88}$$

Proof. First we proof (2.84) and (2.85). Using Cauchy-Schwarz inequality we obtain

$$\begin{aligned}
&|s(\boldsymbol{\psi}, \boldsymbol{\xi})| = |\,(\boldsymbol{\chi}\cdot\boldsymbol{n}, \mathrm{P}\,\boldsymbol{\xi}\cdot\boldsymbol{n})_{\Gamma_s} + (\boldsymbol{\chi}, \mathrm{P}\,\boldsymbol{\xi})_{\Gamma_f}\,| \\
&\le \sum_{e\in\mathcal{E}_{h,s}} \|\boldsymbol{\chi}\cdot\boldsymbol{n}\|_{0,e} \|\,\mathrm{P}\,\boldsymbol{\xi}\cdot\boldsymbol{n}\|_{0,e} + \sum_{e\in\mathcal{E}_{h,f}} \|\boldsymbol{\chi}\|_{0,e} \|\,\mathrm{P}\,\boldsymbol{\xi}\|_{0,e} \\
&= \sum_{e\in\mathcal{E}_{h,s}} {h_e}^{\frac{1}{2}} \|\boldsymbol{\chi}\cdot\boldsymbol{n}\|_{0,e}\, {h_e}^{-\frac{1}{2}} \|\,\mathrm{P}\,\boldsymbol{\xi}\cdot\boldsymbol{n}\|_{0,e} + \sum_{e\in\mathcal{E}_{h,f}} {h_e}^{\frac{1}{2}} \|\boldsymbol{\chi}\|_{0,e}\, {h_e}^{-\frac{1}{2}} \|\,\mathrm{P}\,\boldsymbol{\xi}\|_{0,e} \\
&\le \left[\sum_{e\in\mathcal{E}_{h,s}} h_e \|\boldsymbol{\chi}\cdot\boldsymbol{n}\|_{0,e}^2\right]^{\frac{1}{2}} \left[\sum_{e\in\mathcal{E}_{h,s}} {h_e}^{-1} \|\,\mathrm{P}\,\boldsymbol{\xi}\cdot\boldsymbol{n}\|_{0,e}^2\right]^{\frac{1}{2}} + \left[\sum_{e\in\mathcal{E}_{h,f}} h_e \|\boldsymbol{\chi}\|_{0,e}^2\right]^{\frac{1}{2}} \left[\sum_{e\in\mathcal{E}_{h,f}} {h_e}^{-1} \|\,\mathrm{P}\,\boldsymbol{\xi}\|_{0,e}^2\right]^{\frac{1}{2}} \\
&\le \left[\sum_{e\in\mathcal{E}_{h,s}} h_e \|\boldsymbol{\chi}\cdot\boldsymbol{n}\|_{0,e}^2 + \sum_{e\in\mathcal{E}_{h,f}} h_e \|\boldsymbol{\chi}\|_{0,e}^2\right]^{\frac{1}{2}} \left[\sum_{e\in\mathcal{E}_{h,s}} {h_e}^{-1} \|\,\mathrm{P}\,\boldsymbol{\xi}\cdot\boldsymbol{n}\|_{0,e}^2 + \sum_{e\in\mathcal{E}_{h,f}} {h_e}^{-1} \|\,\mathrm{P}\,\boldsymbol{\xi}\|_{0,e}^2\right]^{\frac{1}{2}} \\
&\le |\boldsymbol{\chi}|_{-\frac{1}{2},h} |\,\mathrm{P}\,\boldsymbol{\xi}|_{\frac{1}{2},h}.
\end{aligned}$$

This immediately shows (2.84). For discrete $\boldsymbol{\psi}_h$, the discrete trace inequality (A2) provides

$$\begin{aligned} h_e \|\boldsymbol{\chi}\|_{0,e}^2 &= h_e \|(\mathcal{C}^{-1}\,\mathrm{symCurl}\,\boldsymbol{\psi}_h)\boldsymbol{t}\|_{0,e}^2 \le h_T \|(\mathcal{C}^{-1}\,\mathrm{symCurl}\,\boldsymbol{\psi}_h)\|_{0,e}^2 \\ &\le c\,\|\mathcal{C}^{-1}\,\mathrm{symCurl}\,\boldsymbol{\psi}_h\|_{0,T}^2 \end{aligned}$$

and analogous

$$h_e\|\boldsymbol{\chi}\cdot\boldsymbol{n}\|_{0,e}^2 \le c\,\|\mathcal{C}^{-1}\,\mathrm{symCurl}\,\boldsymbol{\psi}_h\|_{0,T}^2.$$

In summary we obtain

$$|s(\boldsymbol{\psi}_h,\boldsymbol{\xi})|^2 \le c\,\|\mathcal{C}^{-1}\,\mathrm{symCurl}\,\boldsymbol{\psi}_h\|_{0,\Omega}^2|\,\mathrm{P}\,\boldsymbol{\xi}|_{\frac12,h}^2 \le c\,(\mathrm{symCurl}\,\boldsymbol{\psi}_h,\mathrm{symCurl}\,\boldsymbol{\psi}_h)_{\mathcal{C}^{-1}}|\,\mathrm{P}\,\boldsymbol{\xi}|_{\frac12,h}^2,$$

which implies (2.85). Using (2.84) and (2.85) we receive for arbitrary $\boldsymbol{\psi}_h \in (\mathcal{S}_h)^2$

$$\begin{aligned}
s(\boldsymbol{\psi},\boldsymbol{\xi}) &= s(\boldsymbol{\psi}-\boldsymbol{\psi}_h+\boldsymbol{\psi}_h,\boldsymbol{\xi}) = s(\boldsymbol{\psi}-\boldsymbol{\psi}_h,\boldsymbol{\xi}) + s(\boldsymbol{\psi}_h,\boldsymbol{\xi})\\
&\le c\,(\|\boldsymbol{\psi}-\boldsymbol{\psi}_h\|_{h,*}+\|\boldsymbol{\psi}_h\|_h)|\,\mathrm{P}\,\boldsymbol{\xi}|_{\frac12,h}\\
&\le c\,(\|\boldsymbol{\psi}-\boldsymbol{\psi}_h\|_{h,*}+\|\boldsymbol{\psi}_h-\boldsymbol{\psi}+\boldsymbol{\psi}\|_h)|\,\mathrm{P}\,\boldsymbol{\xi}|_{\frac12,h}\\
&\le c\,(\|\boldsymbol{\psi}-\boldsymbol{\psi}_h\|_{h,*}+\|\boldsymbol{\psi}\|_h)|\,\mathrm{P}\,\boldsymbol{\xi}|_{\frac12,h},
\end{aligned}$$

which shows (2.86). For the estimates (2.87) and (2.88) note that by Cauchy-Schwarz inequality we have

$$\begin{aligned}
|c(q,\boldsymbol{\psi})| &= \sum_{e\in\mathcal{E}_{h,f}}((\mathcal{C}^{-1}q\boldsymbol{I})\boldsymbol{t},\mathrm{P}\,\boldsymbol{\psi})_e \le c \sum_{e\in\mathcal{E}_{h,f}} h_e^{\frac12}\|(\mathcal{C}^{-1}q\boldsymbol{I})\boldsymbol{t}\|_{e,0}\; h_e^{-\frac12}\|\,\mathrm{P}\,\boldsymbol{\psi}\|_{e,0}\\
&\le \left[\sum_{e\in\mathcal{E}_{h,f}} h_e\|(\mathcal{C}^{-1}q\boldsymbol{I})\boldsymbol{t}\|_{e,0}^2\right]^{\frac12}\left[\sum_{e\in\mathcal{E}_{h,f}} h_e^{-1}\|\,\mathrm{P}\,\boldsymbol{\psi}\|_{e,0}^2\right]^{\frac12}\\
&\le |(\mathcal{C}^{-1}q\boldsymbol{I})\boldsymbol{t}|_{-\frac12,h}\,|\,\mathrm{P}\,\boldsymbol{\psi}|_{\frac12,h}
\end{aligned}$$

and

$$\begin{aligned}
|r(\boldsymbol{\xi},\boldsymbol{\psi})| &= \sum_{e\in\mathcal{E}_{h,s}}\frac{\eta}{h_e}(\mathrm{P}\,\boldsymbol{\xi}\cdot\boldsymbol{n},\mathrm{P}\,\boldsymbol{\psi}\cdot\boldsymbol{n})_e + \sum_{e\in\mathcal{E}_{h,f}}\frac{\eta}{h_e}(\mathrm{P}\,\boldsymbol{\xi},\mathrm{P}\,\boldsymbol{\psi})_e\\
&\le \eta\left(\sum_{e\in\mathcal{E}_{h,s}} h_e^{-\frac12}\|\,\mathrm{P}\,\boldsymbol{\xi}\cdot\boldsymbol{n}\|_{e,0}h_e^{-\frac12}\|\,\mathrm{P}\,\boldsymbol{\psi}\cdot\boldsymbol{n}\|_{e,0} + \sum_{e\in\mathcal{E}_{h,f}} h_e^{-\frac12}\|\,\mathrm{P}\,\boldsymbol{\xi}\|_{e,0}h_e^{-\frac12}\|\,\mathrm{P}\,\boldsymbol{\psi}\|_{e,0}\right)\\
&\le \eta\left(\left[\sum_{e\in\mathcal{E}_{h,s}} h_e^{-1}\|\,\mathrm{P}\,\boldsymbol{\xi}\cdot\boldsymbol{n}\|_{e,0}^2\right]^{\frac12}\left[\sum_{e\in\mathcal{E}_{h,s}} h_e^{-1}\|\,\mathrm{P}\,\boldsymbol{\psi}\cdot\boldsymbol{n}\|_{e,0}^2\right]^{\frac12}\right.\\
&\qquad\left. + \left[\sum_{e\in\mathcal{E}_{h,f}} h_e^{-1}\|\,\mathrm{P}\,\boldsymbol{\xi}\|_{e,0}^2\right]^{\frac12}\left[\sum_{e\in\mathcal{E}_{h,f}} h_e^{-1}\|\,\mathrm{P}\,\boldsymbol{\psi}\|_{e,0}^2\right]^{\frac12}\right)\\
&\le \eta\left[\sum_{e\in\mathcal{E}_{h,s}} h_e^{-1}\|\,\mathrm{P}\,\boldsymbol{\xi}\cdot\boldsymbol{n}\|_{e,0}^2 + \sum_{e\in\mathcal{E}_{h,f}} h_e^{-1}\|\,\mathrm{P}\,\boldsymbol{\xi}\|_{e,0}^2\right]^{\frac12}\\
&\qquad\left[\sum_{e\in\mathcal{E}_{h,s}} h_e^{-1}\|\,\mathrm{P}\,\boldsymbol{\psi}\cdot\boldsymbol{n}\|_{e,0}^2 + \sum_{e\in\mathcal{E}_{h,f}} h_e^{-1}\|\,\mathrm{P}\,\boldsymbol{\psi}\|_{e,0}^2\right]^{\frac12}
\end{aligned}$$

$$\leq \eta \, |\, \mathrm{P}\, \boldsymbol{\xi}|_{\frac{1}{2},h} \, |\, \mathrm{P}\, \boldsymbol{\psi}|_{\frac{1}{2},h}.$$

□

Proof of Lemma 2.37. For $\boldsymbol{\psi}_h \in (\mathcal{S}_h)^2/RT_0$ we have

$$\begin{aligned} a_{\boldsymbol{\phi},h}(\boldsymbol{\psi}_h, \boldsymbol{\psi}_h) &= (\operatorname{symCurl} \boldsymbol{\psi}_h, \operatorname{symCurl} \boldsymbol{\psi}_h)_{\mathcal{C}^{-1}} + 2s(\boldsymbol{\psi}_h, \boldsymbol{\psi}_h) + r_h(\boldsymbol{\psi}_h, \boldsymbol{\psi}_h) \\ &\geq (\operatorname{symCurl} \boldsymbol{\psi}_h, \operatorname{symCurl} \boldsymbol{\psi}_h)_{\mathcal{C}^{-1}} - 2|s(\boldsymbol{\psi}_h, \boldsymbol{\psi}_h)| + \eta |\, \mathrm{P}\, \boldsymbol{\psi}_h|^2_{\frac{1}{2},h}. \end{aligned}$$

From (2.85) we obtain

$$|s(\boldsymbol{\psi}_h, \boldsymbol{\psi}_h)| \leq c \left((\operatorname{symCurl} \boldsymbol{\psi}_h, \operatorname{symCurl} \boldsymbol{\psi}_h)_{\mathcal{C}^{-1}}\right)^{\frac{1}{2}} |\, \mathrm{P}\, \boldsymbol{\psi}_h|_{\frac{1}{2},h}. \tag{2.89}$$

For the next step we use Young's inequality, which reads: Let $x, y \in \mathbb{R}$ and $\varepsilon > 0$. Then we have

$$xy \leq \frac{1}{2} \left(\varepsilon^{-1} x^2 + \varepsilon y^2\right).$$

By applying Young's inequality with $\varepsilon = 2$ to the right-hand side of (2.89) we get

$$|s(\boldsymbol{\psi}_h, \boldsymbol{\psi}_h)| \leq \frac{1}{2} \left(\frac{1}{2} (\operatorname{symCurl} \boldsymbol{\psi}_h, \operatorname{symCurl} \boldsymbol{\psi}_h)_{\mathcal{C}^{-1}} + 2\, c^2 \, |\, \mathrm{P}\, \boldsymbol{\psi}_h|^2_{\frac{1}{2},h} \right).$$

In summary we end up with

$$\begin{aligned} a_{\boldsymbol{\phi},h}(\boldsymbol{\psi}_h, \boldsymbol{\psi}_h) \geq{}& (\operatorname{symCurl} \boldsymbol{\psi}_h, \operatorname{symCurl} \boldsymbol{\psi}_h)_{\mathcal{C}^{-1}} + \eta |\, \mathrm{P}\, \boldsymbol{\psi}_h|^2_{\frac{1}{2},h} \\ &- \frac{1}{2} (\operatorname{symCurl} \boldsymbol{\psi}_h, \operatorname{symCurl} \boldsymbol{\psi}_h)_{\mathcal{C}^{-1}} - 2\, c^2 |\, \mathrm{P}\, \boldsymbol{\psi}_h|^2_{\frac{1}{2},h}. \end{aligned}$$

With the assumption $\eta \geq 2c^2 + \frac{1}{2}$ we obtain

$$a_{\boldsymbol{\phi},h}(\boldsymbol{\psi}_h, \boldsymbol{\psi}_h) \geq \frac{1}{2} (\operatorname{symCurl} \boldsymbol{\psi}_h, \operatorname{symCurl} \boldsymbol{\psi}_h)_{\mathcal{C}^{-1}} + (\eta - 2c^2) |\, \mathrm{P}\, \boldsymbol{\psi}_h|^2_{\frac{1}{2},h} \geq \frac{1}{2} \|\boldsymbol{\psi}_h\|_h^2.$$

□

Proof of Lemma 2.38. For $\boldsymbol{\phi} \in (H^s(\Omega))^2 + (\mathcal{S}_h)^2$ with $s > \frac{3}{2}$ and $\boldsymbol{\psi}_h \in (\mathcal{S}_h)^2$ we have

$$a_{\boldsymbol{\phi},h}(\boldsymbol{\phi}, \boldsymbol{\psi}_h) = (\operatorname{symCurl} \boldsymbol{\phi}, \operatorname{symCurl} \boldsymbol{\psi}_h)_{\mathcal{C}^{-1}} + s(\boldsymbol{\phi}, \boldsymbol{\psi}_h) + s(\boldsymbol{\psi}_h, \boldsymbol{\phi}) + r(\boldsymbol{\phi}, \boldsymbol{\psi}_h).$$

For the first term we have using Cauchy inequality on Ω

$$\begin{aligned} (\operatorname{symCurl} \boldsymbol{\phi}, \operatorname{symCurl} \boldsymbol{\psi}_h)_{\mathcal{C}^{-1}} &\leq (\operatorname{symCurl} \boldsymbol{\phi}, \operatorname{symCurl} \boldsymbol{\phi})_{\mathcal{C}^{-1}} (\operatorname{symCurl} \boldsymbol{\psi}_h, \operatorname{symCurl} \boldsymbol{\psi}_h)_{\mathcal{C}^{-1}} \\ &\leq \|\boldsymbol{\phi}\|_{h,*} \|\boldsymbol{\psi}_h\|_h. \end{aligned}$$

The remaining terms can also be bounded from above by $c\, \|\boldsymbol{\phi}\|_{h,*} \|\boldsymbol{\psi}_h\|_h$ using (2.84) for $s(\boldsymbol{\phi}, \boldsymbol{\psi}_h)$, (2.85) for $s(\boldsymbol{\psi}_h, \boldsymbol{\phi})$, and (2.88) for $r(\boldsymbol{\phi}, \boldsymbol{\psi}_h)$. □

Additionally we need an estimate of the consistency error.

Lemma 2.40 (Consistency error)**.** *Under the assumptions of Theorem 2.35 we have*

$$\sup_{0\neq\boldsymbol{\psi}_h\in(\mathcal{S}_h)^2} \frac{|a_{\boldsymbol{\phi},h}(\boldsymbol{\phi},\boldsymbol{\psi}_h) - \langle F_{\boldsymbol{\phi},h},\boldsymbol{\psi}_h\rangle|}{\|\boldsymbol{\psi}_h\|_h} \leq c\, h^{s_p-1}\|p\|_{s_p}.$$

Proof. As outlined in Remark 2.32, by construction $\boldsymbol{\phi}$ satisfies the discrete $\boldsymbol{\phi}$-problem (2.77) with p instead of p_h in the right-hand side, i.e.,

$$a_{\boldsymbol{\phi},h}(\boldsymbol{\phi},\boldsymbol{\psi}_h) = -(p\boldsymbol{I}, \operatorname{symCurl}\boldsymbol{\psi}_h)_{\mathcal{C}^{-1}} - c(p,\boldsymbol{\psi}_h) + s(\boldsymbol{\psi}_h, \boldsymbol{\psi}_\Gamma[p]) + r_h(\boldsymbol{\psi}_\Gamma[p],\boldsymbol{\psi}_h)$$

for all $\boldsymbol{\psi}_h \in (\mathcal{S}_h)^2/RT_0$. Therefore, the difference to the discrete right-hand side $\langle F_{\boldsymbol{\phi},h},\boldsymbol{\psi}_h\rangle$ in (2.77) reads

$$\begin{aligned} a_{\boldsymbol{\phi},h}(\boldsymbol{\phi},\boldsymbol{\psi}_h) - \langle F_{\boldsymbol{\phi},h},\boldsymbol{\psi}_h\rangle = &-((p-p_h)\boldsymbol{I}, \operatorname{symCurl}\boldsymbol{\psi}_h)_{\mathcal{C}^{-1}} - c(p-p_h,\boldsymbol{\psi}_h) \\ &+ s(\boldsymbol{\psi}_h, \boldsymbol{\psi}_\Gamma[p-p_h]) + r_h(\boldsymbol{\psi}_\Gamma[p-p_h],\boldsymbol{\psi}_h). \end{aligned}$$

From the Cauchy inequality on Ω and (2.87), (2.85), (2.88) we obtain

$$\begin{aligned} &|a_{\boldsymbol{\phi},h}(\boldsymbol{\phi},\boldsymbol{\psi}_h) - \langle F_{\boldsymbol{\phi},h},\boldsymbol{\psi}_h\rangle| \\ &\quad\leq c\left(\|p-p_h\|_0 + |(\mathcal{C}^{-1}(p-p_h)\boldsymbol{I})\boldsymbol{t}|_{-\frac{1}{2},h} + |\operatorname{P}\boldsymbol{\psi}_\Gamma[p-p_h]|_{\frac{1}{2},h}\right)\|\boldsymbol{\psi}_h\|_h. \end{aligned} \tag{2.90}$$

The second term on the right-hand side is estimated as follows: From the continuous trace inequality (A3) it follows that

$$\begin{aligned} |(\mathcal{C}^{-1}(p-p_h)\boldsymbol{I})\boldsymbol{t}|^2_{-\frac{1}{2},h} &\leq c \sum_{e\in\mathcal{E}_{h,f}} h_e\,\|p-p_h\|^2_{0,e} \\ &\leq c \sum_{T\in\mathcal{T}_h} h_T\; h_T^{-1}\Big(\|p-p_h\|_{0,T} + h_T\|\nabla(p-p_h)\|_{0,T}\Big)^2 \\ &\leq c\,(\|p-p_h\|^2_0 + h^2\|\nabla(p-p_h)\|^2_0). \end{aligned} \tag{2.91}$$

It remains to estimate the third term on the right-hand side of (2.90). For this we use that

$$h_e^{-1}\|\Pi_\Gamma\boldsymbol{\xi}\|^2_{0,e} \leq c\,h\,|\boldsymbol{\xi}|^2_{\frac{1}{2},h} \quad\text{and}\quad h_e^{-1}\|\boldsymbol{\psi}_\Gamma[q]\|^2_{0,e} \leq c\,\|q\|^2_{0,\Gamma} \tag{2.92}$$

for all $e\in\mathcal{E}_{h,s}\cup\mathcal{E}_{h,f}$, $\boldsymbol{\xi}\in(L^2(\Gamma))^2$, and $q\in L^2(\Gamma)$. For proofing the first inequality note that

$$\|\Pi_\Gamma\boldsymbol{\xi}\|_{L^\infty(\Gamma)} \leq \max(|c_E(\boldsymbol{\xi})|, |a_C(\boldsymbol{\xi})|, |b_C(\boldsymbol{\xi})|),$$

where $c_E(\boldsymbol{\xi})\in\mathbb{R}$ and $r_C(\boldsymbol{\xi})(x) = a_C(\boldsymbol{\xi})\,x + b_C(\boldsymbol{\xi})$ with $a_C(\boldsymbol{\xi})\in\mathbb{R}$ and $b_C(\boldsymbol{\xi})\in\mathbb{R}^2$ are the data used in the construction of $\Pi_\Gamma\boldsymbol{\xi}$ in Definition 2.27. On an edge $E\subset\Gamma_s$ we have

$$|c_E(\boldsymbol{\xi})| = \left|\frac{1}{|E|}\int_E \boldsymbol{\xi}\cdot\boldsymbol{n}\;ds\right| \leq c\sum_{e\subset E}\int_e |\boldsymbol{\xi}\cdot\boldsymbol{n}|\;ds \leq c\sum_{e\subset E} h_e^{1/2}\;\|\boldsymbol{\xi}\cdot\boldsymbol{n}\|_{0,e}$$

$$= c\sum_{e\subset E} h_e\;(h_e^{-1/2}\|\boldsymbol{\xi}\cdot\boldsymbol{n}\|_{0,e}) \leq c\left[\sum_{e\subset E} h_e^2\right]^{\frac{1}{2}}\left[\sum_{e\subset E} h_e^{-1}\,\|\boldsymbol{\xi}\cdot\boldsymbol{n}\|^2_{0,e}\right]^{\frac{1}{2}} \leq c\,\Bigg[\underbrace{\sum_{e\subset E} h_e}_{\leq|E|}\Bigg]^{\frac{1}{2}} h^{1/2}\,|\boldsymbol{\xi}|_{\frac{1}{2},h}.$$

By similar arguments analogous results hold for $|a_C(\boldsymbol{\xi})|$ and $|b_C(\boldsymbol{\xi})|$. Combining these estimates with

$$\|\Pi_\Gamma\boldsymbol{\xi}\|_{0,e}^2 = \int_e |\Pi_\Gamma\boldsymbol{\xi}|^2 \, ds \le h_e \|\Pi_\Gamma\boldsymbol{\xi}\|_{L^\infty(\Gamma)}^2$$

provides the first inequality. The second inequality in (2.92) holds since

$$\|\boldsymbol{\psi}_\Gamma[q]\|_{0,e}^2 = \int_e \left|\int_0^\sigma q\boldsymbol{n} \, ds\right|^2 d\sigma \le \int_e \left(\int_\Gamma |q| \, ds\right)^2 d\sigma \le h_e|\Gamma|\|q\|_{0,\Gamma}^2 \le c\, h_e\|q\|_{0,\Gamma}^2.$$

Using that the total number of edges in $\mathcal{E}_{h,s} \cup \mathcal{E}_{h,f}$ is bounded by $c\,h^{-1}$ it follows that

$$|\Pi_\Gamma\boldsymbol{\xi}|_{\frac{1}{2},h} \le c\,|\boldsymbol{\xi}|_{\frac{1}{2},h} \quad \text{and} \quad |\boldsymbol{\psi}_\Gamma[q]|_{\frac{1}{2},h} \le c\,h^{-\frac{1}{2}}\,\|q\|_{0,\Gamma}, \tag{2.93}$$

since using (2.92) we have

$$|\Pi_\Gamma\boldsymbol{\xi}|_{\frac{1}{2},h}^2 = \sum_{e\in\mathcal{E}_{h,s}} h_e^{-1}\,\|\Pi_\Gamma\boldsymbol{\xi}\cdot\boldsymbol{n}\|_{0,e}^2 + \sum_{e\in\mathcal{E}_{h,f}} h_e^{-1}\,\|\Pi_\Gamma\boldsymbol{\xi}\|_{0,e}^2 \le c\;h\,|\boldsymbol{\xi}|_{\frac{1}{2},h}^2 \sum_{e\in\mathcal{E}_{h,s}\cup\mathcal{E}_{h,f}} 1 = c\;|\boldsymbol{\xi}|_{\frac{1}{2},h}^2$$

and similar

$$|\boldsymbol{\psi}_\Gamma[q]|_{\frac{1}{2},h}^2 = \sum_{e\in\mathcal{E}_{h,s}} h_e^{-1}\,\|\boldsymbol{\psi}_\Gamma[q]\cdot\boldsymbol{n}\|_{0,e}^2 + \sum_{e\in\mathcal{E}_{h,f}} h_e^{-1}\,\|\boldsymbol{\psi}_\Gamma[q]\|_{0,e}^2 \le c\,\|q\|_{0,\Gamma}^2 \sum_{e\in\mathcal{E}_{h,s}\cup\mathcal{E}_{h,f}} 1 = c\,h^{-1}\|q\|_{0,\Gamma}^2.$$

Therefore, using $\mathrm{P}\,\boldsymbol{\psi} = (I - \Pi_\Gamma)\boldsymbol{\psi}$ we can estimate the remaining third term on the right-hand side of (2.90) by

$$|\,\mathrm{P}\,\boldsymbol{\psi}_\Gamma[p-p_h]|_{\frac{1}{2},h}^2 = |\boldsymbol{\psi}_\Gamma[p-p_h] - \Pi_\Gamma(\boldsymbol{\psi}_\Gamma[p-p_h])|_{\frac{1}{2},h}^2 \le c\,h^{-1}\|p-p_h\|_{0,\Gamma}^2. \tag{2.94}$$

Then the estimate immediately follows from the error estimates in Lemma 2.36. □

From the last three lemmas we obtain the following error estimate for the $\boldsymbol{\phi}$-problem:

Lemma 2.41 ($\boldsymbol{\phi}$-problem)**.** *Under the assumptions of Theorem 2.35 we have*

$$\|\boldsymbol{\phi} - \boldsymbol{\phi}_h\|_h \le c\;(h^{s_\phi - 1}\|\boldsymbol{\phi}\|_{s_\phi} + h^{s_p-1}\|p\|_{s_p}).$$

Proof. In the following we rely on standard arguments as in the second lemma of Strang to estimate the discretization error in terms of approximation error and consistency error. We briefly recall them here for completeness. With the triangular inequality we split the error into two parts

$$\|\boldsymbol{\phi} - \boldsymbol{\phi}_h\|_h \le \|\boldsymbol{\phi} - \boldsymbol{\psi}_h\|_h + \|\boldsymbol{\psi}_h - \boldsymbol{\phi}_h\|_h \quad \text{for all } \boldsymbol{\psi}_h \in (\mathcal{S}_h)^2.$$

Using the coercivity estimate, see Lemma 2.37, we receive for arbitrary $\boldsymbol{\xi}_h = \boldsymbol{\psi}_h - \boldsymbol{\phi}_h$

$$\begin{aligned} c\|\boldsymbol{\psi}_h - \boldsymbol{\phi}_h\|_h^2 &\le a_{\boldsymbol{\phi},h}(\boldsymbol{\psi}_h - \boldsymbol{\phi}_h, \boldsymbol{\psi}_h - \boldsymbol{\phi}_h) = a_{\boldsymbol{\phi},h}(\boldsymbol{\psi}_h - \boldsymbol{\phi}_h + \boldsymbol{\phi} - \boldsymbol{\phi}, \boldsymbol{\psi}_h - \boldsymbol{\phi}_h) \\ &= a_{\boldsymbol{\phi},h}(\boldsymbol{\psi}_h - \boldsymbol{\phi}, \boldsymbol{\xi}_h) + a_{\boldsymbol{\phi},h}(\boldsymbol{\phi}, \boldsymbol{\xi}_h) - a_{\boldsymbol{\phi},h}(\boldsymbol{\phi}_h, \boldsymbol{\xi}_h). \end{aligned}$$

Using for the first term the boundedness estimate in Lemma 2.38 and for the last term that $\boldsymbol{\phi}_h$ is solution of the discrete $\boldsymbol{\phi}$-problem (2.77) we obtain

$$c\|\boldsymbol{\xi}_h\|_h^2 \le c\|\boldsymbol{\phi} - \boldsymbol{\psi}_h\|_{h,*}\|\boldsymbol{\xi}_h\|_h + \sup_{0\neq\boldsymbol{\xi}_h\in(\mathcal{S}_h)^2} \frac{|a_{\boldsymbol{\phi},h}(\boldsymbol{\phi},\boldsymbol{\xi}_h) - \langle F_{\boldsymbol{\phi},h},\boldsymbol{\xi}_h\rangle|}{\|\boldsymbol{\xi}_h\|_h}\|\boldsymbol{\xi}_h\|_h.$$

Next we divide by $\|\boldsymbol{\xi}_h\|_h$. Summing up, we have

$$\|\boldsymbol{\phi} - \boldsymbol{\phi}_h\|_h \le \underbrace{\|\boldsymbol{\phi} - \boldsymbol{\psi}_h\|_h}_{\le\|\boldsymbol{\phi}-\boldsymbol{\psi}_h\|_{h,*}} + c\left(\|\boldsymbol{\phi} - \boldsymbol{\psi}_h\|_{h,*} \sup_{0\neq\boldsymbol{\xi}_h\in(\mathcal{S}_h)^2} \frac{|a_{\boldsymbol{\phi},h}(\boldsymbol{\phi},\boldsymbol{\xi}_h) - \langle F_{\boldsymbol{\phi},h},\boldsymbol{\xi}_h\rangle|}{\|\boldsymbol{\xi}_h\|_h}\right).$$

Since $\boldsymbol{\psi}_h \in (\mathcal{S}_h)^2$ is arbitrary, we end up with

$$\|\boldsymbol{\phi} - \boldsymbol{\phi}_h\|_h \le c\left(\inf_{\boldsymbol{\psi}_h\in(\mathcal{S}_h)^2}\|\boldsymbol{\psi}_h - \boldsymbol{\phi}\|_{h,*} + \sup_{0\neq\boldsymbol{\xi}_h\in(\mathcal{S}_h)^2} \frac{|a_{\boldsymbol{\phi},h}(\boldsymbol{\phi},\boldsymbol{\xi}_h) - \langle F_{\boldsymbol{\phi},h},\boldsymbol{\xi}_h\rangle|}{\|\boldsymbol{\xi}_h\|_h}\right).$$

Since $\boldsymbol{\phi} \in (H^s(\Omega))^2$, with $\frac{3}{2} < s \le p+1$, it follows from assumptions (A1) and (A3) that

$$\inf_{\boldsymbol{\psi}_h\in(\mathcal{S}_h)^2}\|\boldsymbol{\phi} - \boldsymbol{\psi}_h\|_{h,*} \le c\, h^{s-1}\|\boldsymbol{\phi}\|_s. \tag{A3*}$$

This approximation property and Lemma 2.40 directly imply the error estimate. □

Error estimates for the u-problem

Lemma 2.42 (u-problem)**.** *Under the assumptions of Theorem 2.35 we have*

$$\|u - u_h\|_1 \le c\left(h^{s_u-1}\|u\|_{s_u} + h^{s_p-1}\|p\|_{s_p} + h^{s_{\boldsymbol{\phi}}-1}\|\boldsymbol{\phi}\|_{s_{\boldsymbol{\phi}}}\right).$$

Proof. The first lemma of Strang provides

$$\|u - u_h\|_1 \le c\left(\inf_{v_h\in\mathcal{S}_{h,0}}\|u - v_h\|_1 + \sup_{0\neq q_h\in\mathcal{S}_{h,0}} \frac{|(\nabla u,\nabla q_h) - \langle F_{u,h},q_h\rangle|}{\|q_h\|_1}\right).$$

The first term can be estimated by the approximation property (A1). It remains to estimate the consistency error. As outlined in Remark 2.32, by construction u satisfies the discrete u-problem (2.78) with p instead of p_h and $\boldsymbol{\phi}$ instead of $\boldsymbol{\phi}_h$ in the right-hand side, i.e.,

$$(\nabla u,\nabla q_h) = (\boldsymbol{M}, q_h\boldsymbol{I})_{\mathcal{C}^{-1}} - s(\boldsymbol{\phi},\boldsymbol{\psi}_\Gamma[q_h]) - c(p,\boldsymbol{\psi}_\Gamma[q_h]) - r_h(\underbrace{\boldsymbol{\phi} - \boldsymbol{\psi}_\Gamma[p]}_{=0},\boldsymbol{\psi}_\Gamma[q_h])$$

for all $q_h \in \mathcal{S}_h$. Hence, the difference to the discrete right-hand side $\langle F_{u,h}, q_h\rangle$ given by (2.79) reads

$$\begin{aligned}(\nabla u,\nabla q_h) - \langle F_{u,h},q_h\rangle = {}& (\boldsymbol{M} - \boldsymbol{M}_h, q_h\boldsymbol{I})_{\mathcal{C}^{-1}} - s(\boldsymbol{\phi} - \boldsymbol{\phi}_h,\boldsymbol{\psi}_\Gamma[q_h]) - c(p - p_h,\boldsymbol{\psi}_\Gamma[q_h]) \\ & + r_h\big(\boldsymbol{\phi}_h - \boldsymbol{\psi}_\Gamma[p_h],\boldsymbol{\psi}_\Gamma[q_h]\big).\end{aligned} \tag{2.95}$$

As outlined in Remark 2.32, by construction $\boldsymbol{\phi}$ satisfies the discrete $\boldsymbol{\phi}$-problem (2.77) with p instead of p_h in the right-hand side, i.e.,

$$(\operatorname{symCurl}\boldsymbol{\phi}, \operatorname{symCurl}\boldsymbol{\psi}_h)_{\mathcal{C}^{-1}} + s(\boldsymbol{\phi}, \boldsymbol{\psi}_h) = -(p\boldsymbol{I}, \operatorname{symCurl}\boldsymbol{\psi}_h)_{\mathcal{C}^{-1}} - c(p, \boldsymbol{\psi}_h)$$

for all $\boldsymbol{\psi}_h \in (\mathcal{S}_h)^2/RT_0$, where we used that $s(\boldsymbol{\psi}_h, \boldsymbol{\phi} - \boldsymbol{\psi}_\Gamma[p])$ and $r_h(\boldsymbol{\phi} - \boldsymbol{\psi}_\Gamma[p], \boldsymbol{\psi}_h)$ are zero. For the next step we bring all terms to the left-hand side

$$(p\boldsymbol{I} + \operatorname{symCurl}\boldsymbol{\phi}, \operatorname{symCurl}\boldsymbol{\psi}_h)_{\mathcal{C}^{-1}} + s(\boldsymbol{\phi}, \boldsymbol{\psi}_h) + c(p, \boldsymbol{\psi}_h) = 0.$$

Subtracting the discrete $\boldsymbol{\phi}$-problem (2.77) rewritten as

$$\begin{aligned}(p_h\boldsymbol{I} + \operatorname{symCurl}\boldsymbol{\phi}_h, \operatorname{symCurl}\boldsymbol{\psi}_h)_{\mathcal{C}^{-1}} + s(\boldsymbol{\phi}_h, \boldsymbol{\psi}_h) + c(p_h, \boldsymbol{\psi}_h)\\ + s(\boldsymbol{\psi}_h, \boldsymbol{\phi}_h - \boldsymbol{\psi}_\Gamma[p_h]) + r_h(\boldsymbol{\phi}_h - \boldsymbol{\psi}_\Gamma[p_h], \boldsymbol{\psi}_h) = 0\end{aligned}$$

from the continuous $\boldsymbol{\phi}$-problem given by the last equation leads to

$$\begin{aligned}&(\boldsymbol{M} - \boldsymbol{M}_h, \operatorname{symCurl}\boldsymbol{\psi}_h)_{\mathcal{C}^{-1}} + s(\boldsymbol{\phi} - \boldsymbol{\phi}_h, \boldsymbol{\psi}_h) + c(p - p_h, \boldsymbol{\psi}_h)\\ &\quad - s(\boldsymbol{\psi}_h, \boldsymbol{\phi}_h - \boldsymbol{\psi}_\Gamma[p_h]) - r_h(\boldsymbol{\phi}_h - \boldsymbol{\psi}_\Gamma[p_h], \boldsymbol{\psi}_h) = 0 \quad \text{for all } \boldsymbol{\psi}_h \in (\mathcal{S}_h)^2.\end{aligned}$$

By adding the last equation and (2.95) we obtain

$$\begin{aligned}(\nabla u, \nabla q_h) - \langle F_{u,h}, q_h\rangle &= (\boldsymbol{M} - \boldsymbol{M}_h, q_h\boldsymbol{I} + \operatorname{symCurl}\boldsymbol{\psi}_h)_{\mathcal{C}^{-1}}\\ &\quad - s(\boldsymbol{\phi} - \boldsymbol{\phi}_h, \boldsymbol{\psi}_\Gamma[q_h] - \boldsymbol{\psi}_h) - c(p - p_h, \boldsymbol{\psi}_\Gamma[q_h] - \boldsymbol{\psi}_h)\\ &\quad - s(\boldsymbol{\psi}_h, \boldsymbol{\phi}_h - \boldsymbol{\psi}_\Gamma[p_h]) + r_h\big(\boldsymbol{\phi}_h - \boldsymbol{\psi}_\Gamma[p_h], \boldsymbol{\psi}_\Gamma[q_h] - \boldsymbol{\psi}_h\big).\end{aligned}$$

The five terms on the right-hand side, denoted by T_1, T_2, $\dots T_5$ in consecutive order of their appearance, are estimated as follows: From the Cauchy inequality on Ω and (2.86), (2.87), (2.88), (2.85) we obtain

$$\begin{aligned}|T_1| &\le c\,(\|p - p_h\|_0 + \|\boldsymbol{\phi} - \boldsymbol{\phi}_h\|_h)\,(\|q_h\|_0 + \|\operatorname{symCurl}\boldsymbol{\psi}_h\|_0),\\ |T_2| &\le c\,(\inf_{\boldsymbol{\xi}_h \in (\mathcal{S}_h)^2} \|\boldsymbol{\phi} - \boldsymbol{\xi}_h\|_{h,*} + \|\boldsymbol{\phi} - \boldsymbol{\phi}_h\|_h)\,\|\boldsymbol{\psi}_\Gamma[q_h] - \boldsymbol{\psi}_h\|_h,\\ |T_3| &\le |(\mathcal{C}^{-1}(p - p_h)\boldsymbol{I})\boldsymbol{t}|_{-\frac{1}{2},h}\|\boldsymbol{\psi}[q_h] - \boldsymbol{\psi}_h\|_h,\\ |T_4| &\le c\,|\operatorname{P}(\boldsymbol{\phi}_h - \boldsymbol{\psi}_\Gamma[p_h])|_{\frac{1}{2},h}\,\|\operatorname{symCurl}\boldsymbol{\psi}_h\|_0,\\ |T_5| &\le c\,|\operatorname{P}(\boldsymbol{\phi}_h - \boldsymbol{\psi}_\Gamma[p_h])|_{\frac{1}{2},h}\,\|\boldsymbol{\psi}_\Gamma[q_h] - \boldsymbol{\psi}_h\|_h\end{aligned}$$

for all $\boldsymbol{\psi}_h \in (\mathcal{S}_h)^2$. In particular, we choose $\boldsymbol{\psi}_h = \Pi_h(\boldsymbol{\psi}[q_h])$, where Π_h denotes the L^2-orthogonal projection onto $(\mathcal{S}_h)^2$. Then we have using (2.93) and denoting the approximation error of the L^2-orthogonal projection by

$$e_{\Pi_h} = \boldsymbol{\psi}[q_h] - \Pi_h(\boldsymbol{\psi}[q_h])$$

the following estimate

$$\begin{aligned}\|\boldsymbol{\psi}[q_h] - \boldsymbol{\psi}_h\|_h^2 = \|e_{\Pi_h}\|_h^2 &= (\operatorname{symCurl} e_{\Pi_h}, \operatorname{symCurl} e_{\Pi_h})_{\mathcal{C}^{-1}} + |\operatorname{P} e_{\Pi_h}|^2_{\frac{1}{2},h}\\ &\le c\,(\|e_{\Pi_h}\|_1^2 + |e_{\Pi_h}|^2_{\frac{1}{2},h}),\end{aligned}$$

where using the continuous trace inequality (A3) the second term becomes

$$\begin{aligned}
|e_{\Pi_h}|^2_{\frac{1}{2},h} &= \sum_{e\in\mathcal{E}_{h,s}} h_e^{-1}\,\|(e_{\Pi_h})\cdot\boldsymbol{n}\|^2_{0,e} + \sum_{e\in\mathcal{E}_{h,f}} h_e^{-1}\,\|e_{\Pi_h}\|^2_{0,e} \\
&\le c \sum_{T\in\mathcal{T}_h} h_e^{-1}h_T^{-1}\Big(\|e_{\Pi_h}\|_{0,T} + h_T\|\nabla(e_{\Pi_h})\|_{0,T}\Big)^2 \\
&\le c \sum_{T\in\mathcal{T}_h} \Big(h_T^{-1}\|e_{\Pi_h}\|_{0,T} + \|\nabla(e_{\Pi_h})\|_{0,T}\Big)^2 .
\end{aligned}$$

Using the optimal approximation order of the L^2-orthogonal projection, see, e.g., [31], we end up with

$$\|\boldsymbol{\psi}[q_h] - \boldsymbol{\psi}_h\|_h \le c\,h^{s-1}\|\boldsymbol{\psi}[q_h]\|_s$$

and, hence, for $s = 1$

$$\|\boldsymbol{\psi}[q_h] - \boldsymbol{\psi}_h\|_h \le c\,\|\boldsymbol{\psi}[q_h]\|_1 .$$

Therefore,

$$\begin{aligned}
\|\operatorname{symCurl}\boldsymbol{\psi}_h\|_0 &\le \|\operatorname{symCurl}(\boldsymbol{\psi}_h - \boldsymbol{\psi}[q_h])\|_0 + \|\operatorname{symCurl}\boldsymbol{\psi}[q_h]\|_0 \\
&\le c\,(\|\boldsymbol{\psi}[q_h] - \boldsymbol{\psi}_h\|_h + \|\operatorname{symCurl}\boldsymbol{\psi}[q_h]\|_0) \le c\,\|\boldsymbol{\psi}[q_h]\|_1 .
\end{aligned}$$

With the stability estimate (2.58) it follows that

$$\|\boldsymbol{\psi}[q_h] - \boldsymbol{\psi}_h\|_h \le c\ \|q_h\|_1 \quad \text{and} \quad \|\operatorname{symCurl}\boldsymbol{\psi}_h\|_0 \le c\ \|q_h\|_1 .$$

Hence, the second factor in all five terms T_1, T_2, $\ldots T_5$ can be bounded by $\|q_h\|_1$. It remains to show error bounds for the first factors.
According to the estimates for the jump and average semi-norm in (2.91) and (2.94) we have:

$$|(\mathcal{C}^{-1}(p-p_h)\boldsymbol{I})\boldsymbol{t}|_{-\frac{1}{2},h} \le c(\|p-p_h\|_0 + h\|\nabla(p-p_h)\|_0)$$

and

$$\begin{aligned}
|\operatorname{P}(\boldsymbol{\phi}_h - \boldsymbol{\psi}_\Gamma[p_h])|_{\frac{1}{2},h} &= |\operatorname{P}(\boldsymbol{\phi}_h - \boldsymbol{\phi}) + \operatorname{P}(\boldsymbol{\psi}_\Gamma[p-p_h])|_{\frac{1}{2},h} \\
&\le \|\boldsymbol{\phi}_h - \boldsymbol{\phi}\|_h + |\operatorname{P}\boldsymbol{\psi}_\Gamma[p-p_h]|_{\frac{1}{2},h} \le \|\boldsymbol{\phi}_h - \boldsymbol{\phi}\|_h + ch^{-\frac{1}{2}}\|p-p_h\|_{0,\Gamma} .
\end{aligned}$$

Then the result follows directly from the estimates in Lemma 2.36, Lemma 2.41, and (A3*). □

Finally, we obtain the proof of the main result:

Proof of Theorem 2.35. By combining the results of Lemma 2.36 and Lemma 2.41 we obtain

$$\|\boldsymbol{M} - \boldsymbol{M}_h\|_0 \le \|p-p_h\|_0 + \|\operatorname{symCurl}(\boldsymbol{\phi} - \boldsymbol{\phi}_h)\|_0 \le c\ (h^{s_p-1}\|p\|_{s_p} + h^{s_\phi-1}\|\boldsymbol{\phi}\|_{s_\phi}).$$

Together with Lemma 2.42 this completes the proof. □

2.6 Coupling condition via Lagrange multipliers

In this section we make the following assumption on the domain Ω:

Assumption 4. We consider a domain Ω, whose boundary is a curvilinear polygon of class C^∞. This means that

$$\Gamma = \bigcup_{k=1}^{K} \overline{E}_k,$$

where the edges E_k are C^∞ curves for $k = 1, 2, \ldots, K$ and $\overline{E}_k$ denotes the closure of E_k.
Recall, as before, the edges are numbered consecutively in counterclockwise direction and we denote the vertex at the start point of $\overline{E}_k$ by x_k. Note, since we consider a closed boundary curve, the index $k = 0$ is in the following always identified with $k = K$. Additionally we need for the following considerations the interior angle at x_k, which we refer to as ω_k.

Recall from Section 2.4.1, in order to guarantee that the solution $(p, \boldsymbol{\phi})$ is in $\mathcal{V}$, p and $\boldsymbol{\phi}$ have to satisfy the coupling condition (2.46), which reads in terms of p and $\boldsymbol{\phi}$

$$\langle \partial_{\boldsymbol{t}} \boldsymbol{\phi}, \nabla v \rangle_\Gamma = - \int_\Gamma p \; \partial_{\boldsymbol{n}} v \; ds \quad \text{for all } v \in W.$$

As in Section 2.4.2, by using the representation $\nabla v = (\partial_{\boldsymbol{n}} v)\, \boldsymbol{n} + (\partial_{\boldsymbol{t}} v)\, \boldsymbol{t}$ and incorporating the boundary conditions for $v \in W$, the coupling condition becomes

$$(\partial_{\boldsymbol{t}} \boldsymbol{\phi} \cdot \boldsymbol{n} + p, \partial_{\boldsymbol{n}} v)_{\Gamma_s \cup \Gamma_f} + (\partial_{\boldsymbol{t}} \boldsymbol{\phi} \cdot \boldsymbol{t}, \partial_{\boldsymbol{t}} v)_{\Gamma_f} = 0 \quad \text{for all } v \in W,$$

provided $\partial_{\boldsymbol{t}} \boldsymbol{\phi} \in (L^2(\Gamma))^2$. We can rewrite the condition as follows

$$\sum_{E_k \subset \Gamma_s \cup \Gamma_f} (\partial_{\boldsymbol{t}} \boldsymbol{\phi} \cdot \boldsymbol{n} \; + \; p, \mu_n^k)_{E_k} + \sum_{E_k \subset \Gamma_f} (\partial_{\boldsymbol{t}} \boldsymbol{\phi} \cdot \boldsymbol{t}, \mu_t^k)_{E_k} = 0, \tag{2.96}$$

for all $\boldsymbol{\mu} = ((\mu_t^1, \mu_t^2, \ldots, \mu_t^K), (\mu_n^1, \mu_n^2, \ldots, \mu_n^K)) \in \boldsymbol{\Lambda}$, where

$$\boldsymbol{\Lambda} = \{(\partial_{\boldsymbol{t}} v, \partial_{\boldsymbol{n}} v) \colon \text{for } v \in W\}$$

with

$$\partial_{\boldsymbol{t}} v = (\partial_{\boldsymbol{t}} v|_{E_1}, \partial_{\boldsymbol{t}} v|_{E_2}, \ldots, \partial_{\boldsymbol{t}} v|_{E_K}), \quad \partial_{\boldsymbol{n}} v = (\partial_{\boldsymbol{n}} v|_{E_1}, \partial_{\boldsymbol{n}} v|_{E_2}, \ldots, \partial_{\boldsymbol{n}} v|_{E_K}).$$

2.6.1 Decoupled formulation

We view the original formulation (2.50) as optimality system of the optimality problem: For $F \in Q^*$, find $(p, \boldsymbol{\phi}) \in \mathcal{V}$ such that

$$J(p, \boldsymbol{\phi}) = \inf_{(q, \boldsymbol{\psi}) \in \mathcal{V}} J(q, \boldsymbol{\psi}) = \frac{1}{2} \int_\Omega \mathcal{C}^{-1}(q\boldsymbol{I} + \operatorname{symCurl} \boldsymbol{\psi}) : (q\boldsymbol{I} + \operatorname{symCurl} \boldsymbol{\psi}) \; dx$$

subject to the constraint

$$\int_\Omega \nabla p \cdot \nabla v \; dx = \langle F, v \rangle \quad \text{for all } v \in Q.$$

Next, we replace the space $\mathcal{V}$ by $Q \times (H^1(\Omega))^2$ and add (2.96) as additional constraint and obtain an equivalent optimization problem. The corresponding optimality system leads to the following equivalent formulation of (2.50), provided $\partial_{\boldsymbol{t}} \boldsymbol{\phi} \in (L^2(\Gamma))^2$: Find $p \in Q$, $\boldsymbol{\phi} \in (H^1(\Omega))^2$, $u \in Q$ and $\boldsymbol{\lambda} \in \boldsymbol{\Lambda}$ such that

$$\begin{aligned}
&(p\boldsymbol{I} + \operatorname{symCurl} \boldsymbol{\phi}, q\boldsymbol{I})_{\mathcal{C}^{-1}} && - (\nabla u, \nabla q) + l_p(q, \boldsymbol{\lambda}) &&= 0 \\
&(p\boldsymbol{I} + \operatorname{symCurl} \boldsymbol{\phi}, \operatorname{symCurl} \boldsymbol{\psi})_{\mathcal{C}^{-1}} && + l_{\boldsymbol{\phi}}(\boldsymbol{\psi}, \boldsymbol{\lambda}) &&= 0 \\
&- (\nabla p, \nabla v) && &&= -\langle F, v \rangle \\
&l_p(p, \boldsymbol{\mu}) + l_{\boldsymbol{\phi}}(\boldsymbol{\phi}, \boldsymbol{\mu}) && &&= 0
\end{aligned}$$

for all $q \in Q$, $\boldsymbol{\psi} \in (H^1(\Omega))^2$, $v \in Q$ and $\boldsymbol{\mu} \in \boldsymbol{\Lambda}$, where

$$\begin{aligned}
l_{\boldsymbol{\phi}}(\boldsymbol{\phi}, \boldsymbol{\mu}) &= \sum_{E_k \subset \Gamma_s \cup \Gamma_f} (\partial_{\boldsymbol{t}} \boldsymbol{\phi} \cdot \boldsymbol{n}, \mu_n^k)_{E_k} + \sum_{E_k \subset \Gamma_f} (\partial_{\boldsymbol{t}} \boldsymbol{\phi} \cdot \boldsymbol{t}, \mu_t^k)_{E_k}, \\
l_p(p, \boldsymbol{\mu}) &= \sum_{E_k \subset \Gamma_f} (p, \mu_n^k)_{E_k},
\end{aligned}$$

for $\boldsymbol{\mu} = ((\mu_t^1, \mu_t^2, \ldots, \mu_t^K), (\mu_n^1, \mu_n^2, \ldots, \mu_n^K))$.
Therefore, provided $\partial_{\boldsymbol{t}} \boldsymbol{\phi} \in (L^2(\Gamma))^2$, the mixed formulation of the Kirchhoff plate bending problem is equivalent to three (consecutively to solve) second-order problems:

1. The p-problem: Find $p \in Q$ such that
$$(\nabla p, \nabla v) = \langle F, v \rangle \quad \text{for all } v \in Q. \tag{2.97}$$

2. The $(\boldsymbol{\phi}, \boldsymbol{\lambda})$-problem: For given $p \in Q$, find $\boldsymbol{\phi} \in (H^1(\Omega))^2 / RT_0$ and $\boldsymbol{\lambda} \in \boldsymbol{\Lambda}$ such that
$$\begin{aligned}
&(\operatorname{symCurl} \boldsymbol{\phi}, \operatorname{symCurl} \boldsymbol{\psi})_{\mathcal{C}^{-1}} + l_{\boldsymbol{\phi}}(\boldsymbol{\psi}, \boldsymbol{\lambda}) = -(p\boldsymbol{I}, \operatorname{symCurl} \boldsymbol{\psi})_{\mathcal{C}^{-1}} \\
&l_{\boldsymbol{\phi}}(\boldsymbol{\phi}, \boldsymbol{\mu}) \qquad\qquad\qquad\qquad\qquad\quad = -l_p(p, \boldsymbol{\mu})
\end{aligned} \tag{2.98}$$
for all $\boldsymbol{\psi} \in (H^1(\Omega))^2 / RT_0$ and $\boldsymbol{\mu} \in \boldsymbol{\Lambda}$.

3. The u-problem: For given $\boldsymbol{M} = p\boldsymbol{I} + \operatorname{symCurl} \boldsymbol{\phi}$ and $\boldsymbol{\lambda} \in \boldsymbol{\Lambda}$, find $u \in Q$ such that
$$(\nabla u, \nabla q) = (\boldsymbol{M}, q\boldsymbol{I})_{\mathcal{C}^{-1}} + l_p(q, \boldsymbol{\lambda}) \quad \text{for all } q \in Q. \tag{2.99}$$

In comparison with the decoupled formulation in Section 2.5.2, here the second problem, the $(\boldsymbol{\phi}, \boldsymbol{\lambda})$-problem, is a saddle point problem.
The just stated decoupled formulation is the starting point for the discretization method we introduce in Section 2.6.3. First, we discuss the characterization of the space for the Lagrange multiplier $\boldsymbol{\Lambda}$.

2.6.2 Characterization of $\boldsymbol{\Lambda}$

In this subsection we provide an explicit characterization of $\boldsymbol{\Lambda}$.

Lemma 2.43. *Let $\boldsymbol{\mu}$ be of the form $\boldsymbol{\mu} = ((\mu_t^1, \mu_t^2, \dots, \mu_t^K), (\mu_n^1, \mu_n^2, \dots, \mu_n^K))$, where μ_t^k and μ_n^k for $k = 1, 2, \dots, K$ are Lipschitz continuous functions on $\overline{E}_k$. Then $\boldsymbol{\mu} \in \boldsymbol{\Lambda}$ if and only if the following three conditions are satisfied:*

1. *The conditions $\mu_t^k = 0$ on edges $E_k \subset \Gamma_s \cup \Gamma_c$ and $\mu_n^k = 0$ on edges $E_k \subset \Gamma_c$ have to hold.*
2. *On each connected component C of Γ_f the compatibility condition*

$$\sum_{E_k \subset C} \int_{E_k} \mu_t^k \, ds = 0$$

has to be satisfied.

3. *For the case $\omega_k \neq \pi$, the four corner values $\mu_t^{k-1}(x_k), \mu_t^k(x_k), \mu_n^{k-1}(x_k), \mu_n^k(x_k)$ satisfy the following conditions:*

$$\begin{aligned} \mu_t^{k-1}(x_k) + \cos\omega_k \; \mu_t^k(x_k) - \sin\omega_k \; \mu_n^k(x_k) &= 0, \\ \cos\omega_k \; \mu_t^{k-1}(x_k) + \sin\omega_k \; \mu_n^{k-1}(x_k) + \mu_t^k(x_k) &= 0, \end{aligned} \tag{2.100}$$

for all $k = 1, 2, \dots, K$.

Sketch of proof. Condition 1 follows directly from the boundary conditions for $v \in W$. In order to deduce condition 2, keep in mind, for $v \in W$ the boundary condition $v = 0$ on $\Gamma_c \cup \Gamma_s$ is prescribed. Then continuity of v at corner points on the interface of Γ_f with Γ_c or Γ_s is equivalent to the stated compatibility condition. Concerning condition 3, note that the four corner values $\mu_t^{k-1}(x_k), \mu_t^k(x_k), \mu_n^{k-1}(x_k), \mu_n^k(x_k)$ have to be coupled appropriately. For $\omega_k = \frac{\pi}{2}$ (and analogous $\omega_k = \frac{3\pi}{2}$) these conditions are directly implied by [39, Theorem 1.4.3]. Using a linear change of variables, the case of an arbitrary angle ω_k can be reduced to one of the two already known situations with $\omega_k = \frac{\pi}{2}$ and $\omega_k = \frac{3\pi}{2}$. The result of this considerations in a quite general framework (higher-order derivatives, no edgewise continuity) is stated in [39, Theorem 1.4.6], from which our conditions follow as special case. □

In the following we derive in detail the implications of condition 3. We fix a corner x_k and work out the relation implied by the corresponding boundary conditions and the conditions (2.100) for the four involved quantities $\mu_t^{k-1}(x_k)$, $\mu_t^k(x_k)$, $\mu_n^{k-1}(x_k)$, $\mu_n^k(x_k)$, where we skip in the following the argument x_k for better readability. We distinguish three situations:

1. Let x_k be an interior corner point of Γ_f. Then the conditions (2.100) lead to

$$\mu_n^{k-1} = -\frac{1}{\sin w_k}(\cos w_k \; \mu_t^{k-1} + \mu_t^k), \quad \mu_n^k = \frac{1}{\sin w_k}(\mu_t^{k-1} + \cos w_k \; \mu_t^k),$$

for arbitrary μ_t^{k-1} and μ_t^k.

2. Let x_k be a corner point on the interface of $E_{k-1} \subset \Gamma_s$ and $E_k \subset \Gamma_f$. Then the conditions (2.100) and the boundary condition $\mu_t^{k-1} = 0$ on E_{k-1} provide

$$\mu_t^{k-1} = 0, \quad \mu_n^{k-1} = -\frac{1}{\sin w_k}\mu_t^k, \quad \mu_n^k = \frac{1}{\sin w_k}\cos w_k\ \mu_t^k,$$

where μ_t^k can be freely chosen. For the reverse case $E_{k-1} \subset \Gamma_f$ and $E_k \subset \Gamma_s$ an analogous result holds.

3. In all other cases, we obtain by incorporating the boundary conditions $\mu_t^k = 0$ on edges $E_k \subset \Gamma_s \cup \Gamma_c$ and $\mu_n^k = 0$ on edges $E_k \subset \Gamma_c$ that $\mu_t^{k-1} = \mu_t^k = \mu_n^{k-1} = \mu_n^k = 0$.

Remark 2.44. *In order to describe a change of boundary condition we may also consider an interior angle ω_k of π. A corresponding adaption of the conditions* (2.100) *can be found in [40].*

2.6.3 The discretization method

Recall the notations introduced at the beginning of Section 2.5.3. As before, let $\mathcal{S}_h^p(\Omega)$ be a finite dimensional subspace of $H^1(\Omega)$ of piecewise polynomials with degree p associated with $\mathcal{T}_h$ and we set $\mathcal{S}_{h,0}^p(\Omega) = \mathcal{S}_h^p(\Omega) \cap H^1_{0,\Gamma_c\cup\Gamma_s}(\Omega)$. The restriction of functions from $\mathcal{S}_h^p(\Omega)$ to E_k is defined as

$$\mathcal{S}_h^p(E_k) = \{v|_{E_k} : v \in \mathcal{S}_h^p(\Omega)\}.$$

For the discrete space $\mathbf{\Lambda}_h$ we choose all

$$\boldsymbol{\mu}_h = ((\mu_t^1, \mu_t^2, \dots, \mu_t^K), (\mu_n^1, \mu_n^2, \dots, \mu_n^K))$$

where

$$\mu_t^k \in \mathcal{S}_h^{p-1}(E_k) \quad \text{and} \quad \mu_n^k \in \mathcal{S}_h^{p-1}(E_k) \quad \text{for } k = 1, 2, \dots, K$$

subject to the constraints derived in Lemma 2.43.
In the discrete setting the original formulation (2.50) is as in the continuous case equivalent to three (consecutively to solve) second-order problems:

1. The discrete p-problem: Find $p_h \in \mathcal{S}_{h,0}^p(\Omega)$ such that

$$(\nabla p_h, \nabla v_h) = \langle F, v_h\rangle \quad \text{for all } v_h \in \mathcal{S}_{h,0}^p(\Omega).$$

2. The discrete $(\boldsymbol{\phi}, \boldsymbol{\lambda})$-problem:
For given $p_h \in \mathcal{S}_{h,0}^p(\Omega)$, find $\boldsymbol{\phi}_h \in (\mathcal{S}_h^p(\Omega))^2/RT_0$ and $\boldsymbol{\lambda}_h \in \mathbf{\Lambda}_h$ such that

$$\begin{aligned}(\operatorname{symCurl}\boldsymbol{\phi}_h, \operatorname{symCurl}\boldsymbol{\psi}_h)_{\mathcal{C}^{-1}} + l_{\boldsymbol{\phi}}(\boldsymbol{\psi}_h, \boldsymbol{\lambda}_h) &= -(p_h\boldsymbol{I}, \operatorname{symCurl}\boldsymbol{\psi}_h)_{\mathcal{C}^{-1}}\\ l_{\boldsymbol{\phi}}(\boldsymbol{\phi}_h, \boldsymbol{\mu}_h) &= -l_p(p_h, \boldsymbol{\mu}_h),\end{aligned}$$

for all $\boldsymbol{\psi}_h \in (\mathcal{S}_h^p(\Omega))^2/RT_0$ and $\boldsymbol{\mu}_h \in \mathbf{\Lambda}_h$.

3. The discrete u-problem: For given $\boldsymbol{M}_h = p_h\boldsymbol{I} + \operatorname{symCurl}\boldsymbol{\phi}_h$ and $\boldsymbol{\lambda}_h \in \boldsymbol{\Lambda}_h$, find $u_h \in \mathcal{S}^p_{h,0}(\Omega)$ such that
$$(\nabla u_h, \nabla q_h) = (\boldsymbol{M}_h, q_h\boldsymbol{I})_{\mathcal{C}^{-1}} + l_p(q_h, \boldsymbol{\lambda}_h) \quad \text{for all } q_h \in \mathcal{S}^p_{h,0}(\Omega).$$

Remark 2.45. *The motivation for choosing the discrete space $\boldsymbol{\Lambda}_h$ as a space with polynomial degree $p-1$ is that $\boldsymbol{\mu}_h \in \boldsymbol{\Lambda}_h$ are discrete representations of $(\partial_{\boldsymbol{t}}v, \partial_{\boldsymbol{n}}v)$ for $v \in W$. However, in order to give a satisfactory answer to the question of the appropriate choice for the discrete space $\boldsymbol{\Lambda}_h$, a rigorous numerical analysis is required.*

2.7 Treatment of inhomogeneous boundary conditions

So far we have only considered homogeneous boundary conditions, cf. Assumption 1. In this section we outline how to adapt the presented approach to inhomogeneous boundary conditions of the following form (cf. (2.18) and (2.19)):
$$\begin{aligned}
&u = \hat{u}_3, && \partial_{\boldsymbol{n}}u = \hat{\theta}_n && \text{on } \Gamma_c,\\
&u = \hat{u}_3, && M_{nn} = \hat{M}_n && \text{on } \Gamma_s,\\
&M_{nn} = \hat{M}_n, && \partial_{\boldsymbol{t}}M_{nt} + \operatorname{Div}\boldsymbol{M}\cdot\boldsymbol{n} = \hat{V}_n && \text{on } \Gamma_f,
\end{aligned}$$
and the corner conditions
$$[\![M_{nt}]\!]_x = \hat{R}_x \quad \text{for all } x \in \mathcal{V}_{\Gamma,f}.$$
For this, recall the primal variational formulation with inhomogeneous boundary conditions (2.20), given by: find $u \in W_g$ such that
$$\int_\Omega \mathcal{C}\nabla^2 u : \nabla^2 v \; dx = \langle \hat{F}, v\rangle \quad \text{for all } v \in W_0 \tag{2.101}$$
with the right-hand side
$$\langle \hat{F}, v\rangle = \langle F^1, v\rangle - \int_{\Gamma_s\cup\Gamma f} \hat{M}_n\, \partial_{\boldsymbol{n}}v \; ds - \sum_{x\in\mathcal{V}_{\Gamma,f}} \hat{R}_x\, v(x),$$
where
$$\langle F^1, v\rangle = \langle F, v\rangle + \int_{\Gamma_f} \hat{V}_n\, v \; ds.$$
Here, for later use we split the right-hand side $\hat{F}$ into a part F^1 and the remaining terms. In F^1 we include beside F also the term resulting from the inhomogeneous boundary conditions that is well defined even for test functions $v \in H^1(\Omega)$ (not only $H^2(\Omega)$). The function spaces are given by
$$\begin{aligned}
W_0 &= \{v \in H^2(\Omega) :\; v = 0,\; \partial_{\boldsymbol{n}}v = 0 \text{ on } \Gamma_c, \quad v = 0 \text{ on } \Gamma_s\},\\
W_g &= \{v \in H^2(\Omega) :\; v = \hat{u}_3,\; \partial_{\boldsymbol{n}}v = \hat{\theta}_n \text{ on } \Gamma_c, \quad v = \hat{u}_3 \text{ on } \Gamma_s\}.
\end{aligned}$$

2.7.1 Mixed variational formulation

As in Section 2.3, we introduce as a new unknown the bending moment tensor $\boldsymbol{M}$, which is given by

$$\boldsymbol{M} = -\mathcal{C}\nabla^2 u.$$

This leads to a first mixed variational formulation, which reads: find $\boldsymbol{M} \in \boldsymbol{L}^2(\Omega)_{\mathrm{sym}}$ and $u \in W_g$ such that

$$\begin{aligned}
&\int_\Omega \mathcal{C}^{-1}\boldsymbol{M} : \boldsymbol{L}\, dx + \int_\Omega \nabla^2 u : \boldsymbol{L}\, dx = 0 && \text{for all } \boldsymbol{L} \in \boldsymbol{L}^2(\Omega)_{\mathrm{sym}},\\
&\int_\Omega \boldsymbol{M} : \nabla^2 v\, dx \qquad\qquad\qquad = -\langle \hat{F}, v\rangle && \text{for all } v \in W_0.
\end{aligned}$$

From (2.24) we obtain the representation

$$u = \overline{u} + u_0.$$

Hence, the above formulation can be rewritten as follows: Find $\boldsymbol{M} \in \boldsymbol{L}^2(\Omega)_{\mathrm{sym}}$ and $u_0 \in W_0$, such that

$$\begin{aligned}
&\int_\Omega \mathcal{C}^{-1}\boldsymbol{M} : \boldsymbol{L}\, dx + \int_\Omega \nabla^2 u_0 : \boldsymbol{L}\, dx = -\int_\Omega \nabla^2 \overline{u} : \boldsymbol{L}\, dx,\\
&\int_\Omega \boldsymbol{M} : \nabla^2 v\, dx \qquad\qquad\qquad = -\langle F^1, v\rangle + \int_{\Gamma_s \cup \Gamma_f} \hat{M}_n \partial_{\boldsymbol{n}} v\, ds + \sum_{x \in \mathcal{V}_{\Gamma,f}} \hat{R}_x\, v(x),
\end{aligned} \tag{2.102}$$

for all $\boldsymbol{L} \in \boldsymbol{L}^2(\Omega)_{\mathrm{sym}}$ and $v \in W_0$.
After homogenization, we search for u_0 in the homogeneous space W_0. Therefore, we use the same choice for Q_0 as in Section 2.3 given by the following $H^1(\Omega)$ subspace

$$Q_0 = Q = H^1_{0,\Gamma_c \cup \Gamma_s}(\Omega) = \{v \in H^1(\Omega) : v = 0 \text{ on } \Gamma_c \cup \Gamma_s\} \quad \text{(cf. (2.31))}.$$

For the test functions $\boldsymbol{L}$ we use the space

$$\boldsymbol{V}_0 = \boldsymbol{H}(\operatorname{div}\operatorname{Div}, \Omega; Q_0^*)_{\mathrm{sym}},$$

which is defined as in Section 2.3. According to Theorem 2.15 sufficiently smooth functions $\boldsymbol{L} \in \boldsymbol{V}_0$ satisfy homogeneous essential boundary conditions of the form

$$L_{nn} = 0 \quad \text{on } \Gamma_s \cup \Gamma_f \quad \text{and} \quad [\![L_{nt}]\!]_x = 0 \quad \text{for all } x \in \mathcal{V}_{\Gamma,f}. \tag{2.103}$$

Therefore, we call $\boldsymbol{V}_0$ in the following the homogeneous space.
Let us recall from Section 2.3 the definition of the homogeneous space $\boldsymbol{V}_0$ using the notation of the current section. Elements of the space $\boldsymbol{V}_0 = \boldsymbol{H}(\operatorname{div}\operatorname{Div}, \Omega; Q_0^*)_{\mathrm{sym}}$ are characterized in the following way (cf.(2.33)): $\boldsymbol{L} \in \boldsymbol{H}(\operatorname{div}\operatorname{Div}, \Omega; Q_0^*)_{\mathrm{sym}}$ if and only if $\boldsymbol{L} \in \boldsymbol{L}^2(\Omega)_{\mathrm{sym}}$ and there is a linear functional $G \in Q_0^*$ such that

$$\langle \nabla^2 v, \boldsymbol{L}\rangle = \int_\Omega \nabla^2 v : \boldsymbol{L}\, dx = \langle G, v\rangle \quad \text{for all } v \in W_0.$$

For $\boldsymbol{L} \in \boldsymbol{H}(\operatorname{div}\operatorname{Div}, \Omega; Q_0^*)_{\mathrm{sym}}$ we set $\operatorname{div}\operatorname{Div}\boldsymbol{L} = G$.
It remains to define an appropriate affine space $\boldsymbol{V}_g$ with inhomogeneous boundary conditions for the auxiliary variable $\boldsymbol{M}$. Motivated by the second line of (2.102) we characterize in analogy to the definition of the homogeneous space $\boldsymbol{V}_0$ elements of the affine space $\boldsymbol{V}_g = \boldsymbol{H}(\operatorname{div}\operatorname{Div}_g, \Omega; Q_0^*)_{\mathrm{sym}}$ in the following way: $\boldsymbol{M} \in \boldsymbol{H}(\operatorname{div}\operatorname{Div}_g, \Omega; Q_0^*)_{\mathrm{sym}}$ if and only if $\boldsymbol{M} \in \boldsymbol{L}^2(\Omega)_{\mathrm{sym}}$ and there is a linear functional $G \in Q_0^*$ such that

$$\int_\Omega \nabla^2 v : \boldsymbol{M}\, dx - \int_{\Gamma_s \cup \Gamma_f} \hat{M}_n \partial_{\boldsymbol{n}} v\, ds - \sum_{x \in \mathcal{V}_{\Gamma,f}} \hat{R}_x\, v(x) = \langle G, v\rangle \quad \text{for all } v \in W_0. \tag{2.104}$$

For $\boldsymbol{M} \in \boldsymbol{H}(\operatorname{div}\operatorname{Div}_g, \Omega; Q_0^*)_{\mathrm{sym}}$ we set $\operatorname{div}\operatorname{Div}_g \boldsymbol{M} = G$. Note that in contrast to the linear space $\boldsymbol{V}_0$, $\boldsymbol{V}_g$ is an affine space, i.e., for $\boldsymbol{M}, \boldsymbol{L} \in \boldsymbol{V}_g$ we have $((1-\alpha)\boldsymbol{M} + \alpha\boldsymbol{L}) \in \boldsymbol{V}_g$ for all α. More explicitly the affine space $\boldsymbol{H}(\operatorname{div}\operatorname{Div}_g, \Omega; Q_0^*)_{\mathrm{sym}}$ is given by

$$\begin{aligned}\boldsymbol{H}(\operatorname{div}\operatorname{Div}_g, \Omega; Q_0^*)_{\mathrm{sym}} = \{&\boldsymbol{L} \in \boldsymbol{L}^2(\Omega)_{\mathrm{sym}} \colon \text{ the functional} \\ &G \colon v \mapsto \int_\Omega \nabla^2 v : \boldsymbol{L}\, dx - \int_{\Gamma_s \cup \Gamma_f} \hat{M}_n \partial_{\boldsymbol{n}} v\, ds - \sum_{x \in \mathcal{V}_{\Gamma,f}} \hat{R}_x\, v(x),\ v \in W_0, \\ &\text{is bounded w.r.t. the } Q\text{-norm}\}.\end{aligned} \tag{2.105}$$

Analogous to Theorem 2.15 one can show that sufficiently smooth functions $\boldsymbol{M} \in \boldsymbol{V}_g$ satisfy essential boundary conditions of the form

$$M_{nn} = \hat{M}_n \quad \text{on } \Gamma_s \cup \Gamma_f \quad \text{and} \quad [\![M_{nt}]\!]_x = R_x \quad \text{for all } x \in \mathcal{V}_{\Gamma,f}. \tag{2.106}$$

Therefore, we call $\boldsymbol{V}_g$ in the following the inhomogeneous space.
This motivates the new mixed formulation as follows: For $F^1 \in Q_0^*$, find $\boldsymbol{M} \in \boldsymbol{V}_g$ and $u = \overline{u} + u_0 \in Q_g$ such that

$$\begin{aligned} \int_\Omega \mathcal{C}^{-1}\boldsymbol{M} : \boldsymbol{L}\, dx + \langle \operatorname{div}\operatorname{Div}\boldsymbol{L}, u_0\rangle &= -\int_\Omega \nabla^2 \overline{u} : \boldsymbol{L}\, dx && \text{for all } \boldsymbol{L} \in \boldsymbol{V}_0, \\ \langle \operatorname{div}\operatorname{Div}_g \boldsymbol{M}, v\rangle &= -\langle F^1, v\rangle && \text{for all } v \in Q_0 \end{aligned} \tag{2.107}$$

with the function spaces

$$\begin{aligned} \boldsymbol{V}_0 &= \boldsymbol{H}(\operatorname{div}\operatorname{Div}, \Omega; Q_0^*)_{\mathrm{sym}}, & Q_0 &= H^1_{0,\Gamma_c \cup \Gamma_s}(\Omega), \\ \boldsymbol{V}_g &= \boldsymbol{H}(\operatorname{div}\operatorname{Div}_g, \Omega; Q_0^*)_{\mathrm{sym}}, & Q_g &= \overline{u} + Q_0. \end{aligned} \tag{2.108}$$

2.7.2 Mathematical analysis

Theorem 2.46. *Let $u = \overline{u} + u_0 \in W_g$ be the solution of the primal problem* (2.101) *for $F^1 \in Q_0^*$. Then we have $\boldsymbol{M} = -\mathcal{C}\nabla^2 u \in \boldsymbol{V}_g$ and $(\boldsymbol{M}, u)$ solves the mixed problem* (2.107).

Proof. Since $u = \overline{u} + u_0 \in W_g$ solves (2.101), it follows that $\boldsymbol{M} \in \boldsymbol{L}^2(\Omega)_{\text{sym}}$ and

$$\int_\Omega \boldsymbol{M} : \nabla^2 v \, dx - \int_{\Gamma_s \cup \Gamma_f} \hat{M}_n \partial_{\boldsymbol{n}} v \, ds - \sum_{x \in \mathcal{V}_{\Gamma,f}} \hat{R}_x \, v(x) = -\langle F^1, v \rangle \quad \text{for all } v \in W_0.$$

From the definition of $\boldsymbol{V}_g$ in (2.104) we obtain that $\boldsymbol{M} \in \boldsymbol{V}_g$ with $\operatorname{div} \operatorname{Div}_g \boldsymbol{M} = -F^1 \in Q_0^*$, which shows that the second equation of (2.107) is satisfied. Since $u_0 \in W_0$, we receive from (2.36) that

$$\begin{aligned} \langle \operatorname{div} \operatorname{Div} \boldsymbol{L}, u_0 \rangle = \langle \nabla^2 u_0, \boldsymbol{L} \rangle = \int_\Omega \boldsymbol{L} : \nabla^2 u_0 \, dx = \int_\Omega \boldsymbol{L} : \nabla^2 (u - \overline{u}) \, dx \\ = -\int_\Omega \boldsymbol{L} : \mathcal{C}^{-1} \boldsymbol{M} \, dx - \int_\Omega \boldsymbol{L} : \nabla^2 \overline{u} \, dx \end{aligned}$$

for all $\boldsymbol{L} \in \boldsymbol{V}_0$, which proves the first equation of (2.107). □

Let $\overline{\boldsymbol{M}}$ be an arbitrary fixed element in $\boldsymbol{V}_g$. Then we have

$$\boldsymbol{V}_g = \overline{\boldsymbol{M}} + \boldsymbol{V}_0,$$

and we can rewrite the mixed formulation (2.107) as follows: For $F^1 \in Q_0^*$, find $\boldsymbol{M} = \overline{\boldsymbol{M}} + \boldsymbol{M}_0 \in \boldsymbol{V}_g$ and $u = \overline{u} + u_0 \in Q_g$ such that

$$\begin{aligned} \int_\Omega \mathcal{C}^{-1} \boldsymbol{M}_0 : \boldsymbol{L} \, dx + \langle \operatorname{div} \operatorname{Div} \boldsymbol{L}, u_0 \rangle &= -\int_\Omega \nabla^2 \overline{u} : \boldsymbol{L} \, dx - \int_\Omega \mathcal{C}^{-1} \overline{\boldsymbol{M}} : \boldsymbol{L} \, dx, \\ \langle \operatorname{div} \operatorname{Div} \boldsymbol{M}_0, v \rangle &= -\langle F^1, v \rangle - \langle \operatorname{div} \operatorname{Div}_g \overline{\boldsymbol{M}}, v \rangle, \end{aligned} \tag{2.109}$$

for all $\boldsymbol{L} \in \boldsymbol{V}_0$ and $v \in Q_0$.
This problem has the typical structure of a saddle point problem, see beginning of Section 2.3.4. From Theorem 2.10 we obtain that the associated operator $\mathcal{A}$ is an isomorphism from $\boldsymbol{V}_0 \times Q_0$ onto $(\boldsymbol{V}_0 \times Q_0)^*$. Since, we have $(F_{\boldsymbol{L}}, F_v) \in (\boldsymbol{V}_0 \times Q_0)^*$ for our right-hand sides given by

$$\begin{aligned} \langle F_{\boldsymbol{L}}, \boldsymbol{L} \rangle &= -\int_\Omega \nabla^2 \overline{u} : \boldsymbol{L} \, dx - \int_\Omega \mathcal{C}^{-1} \overline{\boldsymbol{M}} : \boldsymbol{L} \, dx, \\ \langle F_v, v \rangle &= -\langle F^1, v \rangle - \langle \operatorname{div} \operatorname{Div}_g \overline{\boldsymbol{M}}, v \rangle \end{aligned}$$

this provides well-posedness of our new mixed formulation (2.107).

Corollary 2.47. *For $F^1 \in Q_0^*$, the primal problem* (2.101) *and the mixed problem* (2.107) *are equivalent in the following sense: If $u \in W_g$ solves* (2.101)*, then $\boldsymbol{M} = -\mathcal{C} \nabla^2 u \in \boldsymbol{V}_g$ and $(\boldsymbol{M}, u)$ solves* (2.107)*. Vice versa, if $(\boldsymbol{M}, u) \in \boldsymbol{V}_g \times Q_g$ solves* (2.107)*, then $u \in W_g$ and solves* (2.101)*.*

Proof. The first part has already been shown in Theorem 2.46. Since, both problems are uniquely solvable the reverse direction is true as well. □

The mixed formulation in (2.109) provides a first, quite obvious way to incorporate inhomogeneous boundary conditions. However, for the numerical realization appropriate extensions $\overline{u} \in H^2(\Omega)$ and $\overline{\boldsymbol{M}} \in \boldsymbol{V}_g$ of the corresponding boundary data have to be constructed. In the following we outline how the explicit construction of $\overline{u}$ and $\overline{\boldsymbol{M}}$ can be avoided by means of the regular decomposition in Theorem 2.19 and the corresponding adaption for the inhomogeneous space $\boldsymbol{V}_g$.

2.7.3 Regular decomposition

A straight-forward adaption of Theorem 2.19 provides the following regular decomposition of the inhomogeneous space $\boldsymbol{V}_g = \boldsymbol{H}(\operatorname{div}\operatorname{Div}_g, \Omega; Q_0^*)_{\mathrm{sym}}$, which makes this space computationally accessible.

Theorem 2.48. *Let Ω be simply connected. For each $\boldsymbol{L} \in \boldsymbol{H}(\operatorname{div}\operatorname{Div}_g, \Omega; Q_0^*)_{\mathrm{sym}}$ there exists a decomposition*

$$\boldsymbol{L} = q\boldsymbol{I} + \operatorname{symCurl}\boldsymbol{\psi} \tag{2.110}$$

with $q \in Q_0 = H^1_{0,\Gamma_c\cup\Gamma_s}(\Omega)$ and $\boldsymbol{\psi} \in (H^1(\Omega))^2$ satisfying the inhomogeneous coupling condition

$$\langle \partial_{\boldsymbol{t}}\boldsymbol{\psi}, \nabla v\rangle_\Gamma = -\int_\Gamma q\, \partial_{\boldsymbol{n}} v\, ds + \int_{\Gamma_s\cup\Gamma_f} \hat{M}_n \partial_{\boldsymbol{n}} v\, ds + \sum_{x\in\mathcal{V}_{\Gamma,f}} \hat{R}_x\, v(x) \quad \text{for all } v \in W_0. \tag{2.111}$$

The function $q \in Q_0$ is the unique solution of the Poisson problem

$$\int_\Omega \nabla q\cdot\nabla v\, dx \quad = -\langle \operatorname{div}\operatorname{Div}_g \boldsymbol{L}, v\rangle \quad \text{for all } v \in Q_0, \tag{2.112}$$

and $\boldsymbol{\psi} \in (H^1(\Omega))^2$ is unique up to an element from RT_0. Vice versa, for each $\boldsymbol{L}$ given by (2.110) *with $q \in Q_0$ and $\boldsymbol{\psi} \in (H^1(\Omega))^2$ satisfying* (2.111)*, it follows that $\boldsymbol{L} \in \boldsymbol{H}(\operatorname{div}\operatorname{Div}_g, \Omega; Q_0^*)_{\mathrm{sym}}$ with the same representation of* $\operatorname{div}\operatorname{Div}_g \boldsymbol{L}$ *as in* (2.112).

Proof. The proof is completely analogous to the proof of Theorem 2.19. The only difference are the additional terms in the definition of $\operatorname{div}\operatorname{Div}_g \boldsymbol{L}$ (cf. (2.104)), which now appear unmodified in the inhomogeneous coupling condition. □

The resulting mixed formulation

Recall the representation of elements $\boldsymbol{L}$ in the homogeneous space $\boldsymbol{V}_0$ in terms of a pair of functions $(q, \boldsymbol{\psi}) \in \mathcal{V}$ stated at the beginning of Section 2.4.1. Analogously we obtain from Theorem 2.48 that elements of the inhomogeneous space $\boldsymbol{V}_g = \boldsymbol{H}(\operatorname{div}\operatorname{Div}_g, \Omega; Q_0^*)_{\mathrm{sym}}$ can be characterized in the following way: $\boldsymbol{M} \in \boldsymbol{V}_g$ if and only if it can be represented by (2.110) with a pair of functions $(p, \boldsymbol{\phi}) \in \mathcal{V}_g$, where

$$\mathcal{V}_g = \{(p, \boldsymbol{\phi}) \in Q \times (H^1(\Omega))^2 \colon p, \boldsymbol{\phi} \text{ satisfy the inhom. coupling condition (2.111)}\}.$$

Then proceeding as in Section 2.4.1 leads to an equivalent formulation of (2.107): For $F^1 \in Q_0^*$, find $(p, \boldsymbol{\phi}) \in \mathcal{V}_g$ and $u_0 \in Q_0$, where $u = \overline{u} + u_0$, such that

$$\begin{aligned}\int_\Omega \mathcal{C}^{-1}(p\boldsymbol{I} + \operatorname{symCurl}\boldsymbol{\phi}) : (q\boldsymbol{I} + \operatorname{symCurl}\boldsymbol{\psi})\, dx - \int_\Omega \nabla u_0 \cdot \nabla q\, dx \\ = -\int_\Omega \nabla^2 \overline{u} : (q\boldsymbol{I} + \operatorname{symCurl}\boldsymbol{\psi})\, dx, \\ -\int_\Omega \nabla p \cdot \nabla v\, dx = -\langle F, v\rangle - \int_{\Gamma_f} \hat{V}_n\, v\, ds\end{aligned} \tag{2.113}$$

for all $(q, \boldsymbol{\psi}) \in \mathcal{V}$ and $v \in Q_0$. In comparison with (2.50), we search now for $(p, \boldsymbol{\phi}) \in \mathcal{V}_g$ and in both equations we have additional contributions on the right-hand side.
Note, the contribution to the right-hand side in the first equation involves the H^2 extension $\overline{u}$, which is not in accordance with our goal to derive a mixed formulation only based on standard components for second-order problems. However, integration by parts (as in the proof of Theorem 2.19) provides

$$\int_\Omega (q\boldsymbol{I} + \operatorname{symCurl}\boldsymbol{\psi}) : \nabla^2 \overline{u}\, dx = -\int_\Omega \nabla\overline{u} \cdot \nabla q\, dx + \langle \partial_{\boldsymbol{t}}\boldsymbol{\psi}, \nabla\overline{u}\rangle_\Gamma + \int_\Gamma q\, \partial_{\boldsymbol{n}}\overline{u}\, ds.$$

Provided $\partial_{\boldsymbol{t}}\boldsymbol{\psi} \in (L^2(\Omega))^2$, we can rewrite the boundary term using $\nabla\overline{u} = (\partial_{\boldsymbol{n}}\overline{u})\,\boldsymbol{n} + (\partial_{\boldsymbol{t}}\overline{u})\,\boldsymbol{t}$ and $q = 0$ on $\Gamma_c \cup \Gamma_s$ as follows

$$\begin{aligned}\langle \partial_{\boldsymbol{t}}\boldsymbol{\psi}, \nabla\overline{u}\rangle_\Gamma + \int_\Gamma q\, \partial_{\boldsymbol{n}}\overline{u}\, ds &= \int_\Gamma \big((\partial_{\boldsymbol{t}}\boldsymbol{\psi} \cdot \boldsymbol{n} + q)\, \partial_{\boldsymbol{n}}\overline{u} + \partial_{\boldsymbol{t}}\boldsymbol{\psi} \cdot \boldsymbol{t}\, \partial_{\boldsymbol{t}}\overline{u}\big)\, ds \\ = \int_{\Gamma_c} \big(\partial_{\boldsymbol{t}}\boldsymbol{\psi} \cdot \boldsymbol{n}\, \underbrace{\partial_{\boldsymbol{n}}\overline{u}}_{=\hat{\theta}_n} + \partial_{\boldsymbol{t}}\boldsymbol{\psi} \cdot \boldsymbol{t}\, \underbrace{\partial_{\boldsymbol{t}}\overline{u}}_{=\partial_t \hat{u}_3}\big)\, ds &+ \int_{\Gamma_s} \big(\underbrace{\partial_{\boldsymbol{t}}\boldsymbol{\psi} \cdot \boldsymbol{n}}_{=0}\, \partial_{\boldsymbol{n}}\overline{u} + \partial_{\boldsymbol{t}}\boldsymbol{\psi} \cdot \boldsymbol{t}\, \underbrace{\partial_{\boldsymbol{t}}\overline{u}}_{=\partial_t \hat{u}_3}\big)\, ds \\ + \int_{\Gamma_f} \big(\underbrace{(\partial_{\boldsymbol{t}}\boldsymbol{\psi} \cdot \boldsymbol{n} + q)}_{=0}\, \partial_{\boldsymbol{n}}\overline{u} + \partial_{\boldsymbol{t}}\boldsymbol{\psi} \cdot \boldsymbol{t}\, \partial_{\boldsymbol{t}}\overline{u}\big)\, ds.\end{aligned}$$

Here the terms $\partial_{\boldsymbol{t}}\boldsymbol{\psi} \cdot \boldsymbol{n}$ on Γ_s and $\partial_{\boldsymbol{t}}\boldsymbol{\psi} \cdot \boldsymbol{n} + q$ on Γ_f vanish due to the boundary conditions (2.52), which are a consequence of the coupling condition (2.46). For the remaining term $\partial_{\boldsymbol{t}}\boldsymbol{\psi} \cdot \boldsymbol{t}\, \partial_{\boldsymbol{t}}\overline{u}$, the boundary conditions (2.52) and the corner conditions (2.53) imply

$$\begin{aligned}\int_{\Gamma_f} \partial_{\boldsymbol{t}}\boldsymbol{\psi} \cdot \boldsymbol{t}\, \partial_{\boldsymbol{t}}\overline{u}\, ds &= -\int_{\Gamma_f} \underbrace{\partial_{\boldsymbol{t}}(\partial_{\boldsymbol{t}}\boldsymbol{\psi} \cdot \boldsymbol{t})}_{=0}\, \overline{u}\, ds + \sum_{E \subset \Gamma_f} \big(\partial_{\boldsymbol{t}}\boldsymbol{\psi} \cdot \boldsymbol{t}\, \overline{u}\big)\big|_{\partial E} \\ &= \sum_{C \subset \Gamma_f} \big(\partial_{\boldsymbol{t}}\boldsymbol{\psi} \cdot \boldsymbol{t}\, \underbrace{\overline{u}}_{=\hat{u}_3}\big)\big|_{\partial C} + \sum_{x \in \mathcal{V}_{\Gamma,f}} \underbrace{[\![\partial_{\boldsymbol{t}}\boldsymbol{\psi} \cdot \boldsymbol{t}]\!]_x}_{=0}\, \overline{u}(x).\end{aligned}$$

Here, $C \subset \Gamma_f$ denotes a connected component C of Γ_f and the symbol $|_{\partial C}$ denotes the restriction of a function to the boundary of the connected component C, i.e.,

$$f|_{\partial C} = f(x_e) - f(x_s),$$

where x_s and x_e denote the start and end point of the connected component C (in counterclockwise direction). Summing up, we obtain

$$\int_\Omega (q\boldsymbol{I} + \operatorname{symCurl} \boldsymbol{\psi}) : \nabla^2 \overline{u}\ dx = -\int_\Omega \nabla\overline{u} \cdot \nabla q\ dx - \langle F[\hat{u}_3, \hat{\theta}_n], \boldsymbol{\psi}\rangle_\Gamma$$

where

$$\langle F[\hat{u}_3, \hat{\theta}_n], \boldsymbol{\psi}\rangle_\Gamma = -\int_{\Gamma_c} \partial_{\boldsymbol{t}}\boldsymbol{\psi} \cdot \boldsymbol{n}\, \hat{\theta}_n + \partial_{\boldsymbol{t}}\boldsymbol{\psi} \cdot \boldsymbol{t}\, \partial_{\boldsymbol{t}}\hat{u}_3\ ds - \int_{\Gamma_s} \partial_{\boldsymbol{t}}\boldsymbol{\psi} \cdot \boldsymbol{t}\, \partial_{\boldsymbol{t}}\hat{u}_3\ ds - \sum_{C\subset\Gamma_f} \left(\partial_{\boldsymbol{t}}\boldsymbol{\psi} \cdot \boldsymbol{t}\ \hat{u}_3\right)\big|_{\partial C}.$$

Using this representation we can undo the homogenization in u and the formulation (2.113) can be rewritten using

$$\int_\Omega \nabla u_0 \cdot \nabla q\ dx + \int_\Omega \nabla\overline{u} \cdot \nabla q\ dx = \int_\Omega \nabla \underbrace{(u_0 + \overline{u})}_{=u} \cdot \nabla q\ dx$$

as follows: For $F^1 \in Q_0^*$, find $(p, \boldsymbol{\phi}) \in \mathcal{V}_g$ and $u \in Q_g$ such that

$$\begin{aligned}
\int_\Omega \mathcal{C}^{-1}(p\boldsymbol{I} + \operatorname{symCurl} \boldsymbol{\phi}) : (q\boldsymbol{I} + \operatorname{symCurl} \boldsymbol{\psi})\ dx - \int_\Omega \nabla u \cdot \nabla q\ dx &= \langle F[\hat{u}_3, \hat{\theta}_n], \boldsymbol{\psi}\rangle_\Gamma, \\
-\int_\Omega \nabla p \cdot \nabla v\ dx \qquad &= -\langle F, v\rangle - \int_{\Gamma_f} \hat{V}_n\, v\ ds,
\end{aligned}$$

for all $(q, \boldsymbol{\psi}) \in \mathcal{V}$ and $v \in Q_0$. Recall, Q_g is introduced as the affine space $Q_g = \overline{u} + Q_0$ with a particular extension $\overline{u} \in H^2(\Omega)$ satisfying the boundary conditions $\overline{u} = \hat{u}_3$ on $\Gamma_c \cup \Gamma_s$ and $\partial_{\boldsymbol{n}}\overline{u} = \hat{\theta}_n$ on Γ_c. Nevertheless, every other function $\overline{u}^1 \in H^1(\Omega)$ satisfying the boundary condition $\overline{u}^1 = \hat{u}_3$ on $\Gamma_c \cup \Gamma_s$ leads to the same affine space Q_g. Hence, we can represent Q_g as

$$Q_g = \overline{u}^1 + Q_0,$$

where $\overline{u}^1 \in H^1(\Omega)$ is an extension of the boundary data $\hat{u}_3$ such that $\overline{u}^1 = \hat{u}_3$ on $\Gamma_c \cup \Gamma_s$. So only an H^1 extension is needed.
It remains to consider the incorporation of the inhomogeneous coupling condition (2.111).

2.7.4 Inhomogeneous coupling condition as standard boundary conditions

In this section we extend the approach presented in Section 2.5. For this we construct a specific function $\boldsymbol{\psi}_\Gamma[p, \hat{M}_n, \hat{R}_x]$ that satisfies the inhomogeneous coupling condition (2.111). With the same notations as used in (2.56) we set

$$\boldsymbol{\psi}_\Gamma[p, \hat{M}_n, \hat{R}_x](x) = \int_0^\sigma (-p\,\boldsymbol{n} + \hat{M}_n\,\boldsymbol{n} + c\,\boldsymbol{t})\ ds \quad \text{with} \quad x = \gamma(\sigma) \quad \text{on } \Gamma \setminus E,$$

where c is edgewise constant, given by

$$c|_{E_k} = -\sum_{i=1}^{k} \hat{R}_{x_i}, \quad \text{for all } k = 1, \dots, K.$$

As before, the edges E_k for $k = 1, \dots, K$ are numbered consecutively in counterclockwise direction with E_1 starting from $x_B = \gamma(0)$. Furthermore, we denote the vertex at the start point of $\overline{E}_k$ by x_k and use the convention $\hat{R}_{x_k} = 0$ for corner points $x_k \notin \mathcal{V}_{\Gamma,f}$. For all $v \in W_0$ we have using integration by parts

$$\begin{aligned}
\langle \partial_{\boldsymbol{t}} \boldsymbol{\psi}_\Gamma[p, \hat{M}_n, \hat{R}_x], \nabla v \rangle_\Gamma &= \int_\Gamma (-p + \hat{M}_n)\ \boldsymbol{n} \cdot \nabla v + c\ \boldsymbol{t} \cdot \nabla v\ ds \\
&= \int_\Gamma (-p + \hat{M}_n)\ \partial_{\boldsymbol{n}} v + c\ \partial_{\boldsymbol{t}} v\ ds \\
&= \int_\Gamma (-p + \hat{M}_n)\ \partial_{\boldsymbol{n}} v\ ds - \int_\Gamma \partial_{\boldsymbol{t}} c\ v\ ds + \sum_{x \in \mathcal{V}_{\Gamma,f}} [\![c]\!]_x v(v) \\
&= \int_\Gamma (-p + \hat{M}_n)\ \partial_{\boldsymbol{n}} v\ ds + \sum_{x \in \mathcal{V}_{\Gamma,f}} \hat{R}_x\, v(x),
\end{aligned}$$

which shows that $\boldsymbol{\psi}_\Gamma[p, \hat{M}_n, \hat{R}_x]$ satisfies the inhomogeneous coupling condition (2.111). The remaining considerations can be directly carried over from Section 2.5, leading to the following decoupled formulation:

Decoupled formulation

1. In the p-problem (cf. (2.67)) we have an addition contribution on the right-hand side: Find $p \in Q_0$ such that

$$(\nabla p, \nabla v) = \langle F, v \rangle + \int_{\Gamma_f} \hat{V}_n v\ ds \quad \text{for all } v \in Q_0.$$

2. In the $\boldsymbol{\phi}$-problem (cf. (2.68)) we have a different space Ψ_p and an addition contribution on the right-hand side: For given $p \in Q_0$, find $\boldsymbol{\phi} \in \Psi_p = \boldsymbol{\psi}[p, \hat{M}_n, \hat{R}_x] + \Psi_0 / RT_0$ such that

$$(\operatorname{symCurl} \boldsymbol{\phi}, \operatorname{symCurl} \boldsymbol{\psi}_0)_{\mathcal{C}^{-1}} = -(p\boldsymbol{I}, \operatorname{symCurl} \boldsymbol{\psi}_0)_{\mathcal{C}^{-1}} + \langle F[\hat{u}_3, \hat{\theta}_n], \boldsymbol{\psi}_0 \rangle_\Gamma,$$

for all $\boldsymbol{\psi}_0 \in \Psi_0 / RT_0$.

3. In the u-problem (cf. (2.69)) we also get an additional contribution on the right-hand side: For given $\boldsymbol{M} = p\boldsymbol{I} + \operatorname{symCurl} \boldsymbol{\phi}$, find $u \in Q_g = \overline{u}^1 + Q_0$ such that

$$(\nabla u, \nabla q) = (\boldsymbol{M}, q\boldsymbol{I} + \operatorname{symCurl} \boldsymbol{\psi}[q])_{\mathcal{C}^{-1}} + \langle F[\hat{u}_3, \hat{\theta}_n], \boldsymbol{\psi}[q] \rangle_\Gamma \quad \text{for all } q \in Q_0.$$

Starting from this decoupled formulation we proceed analogously to Section 2.5.3 to obtain a discretization method based on Nitsche's method.

Remark 2.49. *In Section 2.6 we proposed an alternative approach to incorporate the coupling condition via Lagrange multipliers. The extension of this approach to the inhomogeneous coupling condition* (2.111) *is completely straight-forward for inhomogeneous data $\hat{M}_n$. However, in order to incorporate inhomogeneous data $\hat{R}_x$ and $\hat{V}_n$ an extension of the characterization of the involved traces, see Section 2.6.2, is necessary.*

2.8 Numerical experiments

For the proposed discretization methods in Section 2.5.3 and Section 2.6.3 we only need a conforming discretization space of $H^1(\Omega)$, i.e.,

$$\mathcal{S}_h^p(\Omega) \subset H^1(\Omega).$$

Note, only C^0 basis functions are required, so the continuity requirements are easily satisfied with standard basis functions. In order to illustrate the flexibility of our discretization methods we consider isogeometric B-spline discretization spaces with degree $p \geq 1$. For $p = 1$, the discretization space coincides with the standard isoparametric finite element space of continuous and piecewise bilinear elements.
We denote by $S^{p_1,p_2}_{\alpha_1,\alpha_2}$ the tensor product B-spline space defined as

$$S^{p_1,p_2}_{\alpha_1,\alpha_2} = S^{p_1}_{\alpha_1} \otimes S^{p_2}_{\alpha_2}, \tag{2.114}$$

where S^p_α is the one-dimensional B-spline space with degree p and α continuous derivatives across interior knots; see, e.g, [29, 6] for further information.
In all numerical experiments we use tensor product B-spline spaces of degree p with maximum smoothness at interior knots, i.e., $\alpha = p - 1$. Hence, $\mathcal{S}_h^p(\Omega)$ is defined by the space $S^{p,p}_{\alpha,\alpha}$, which is in case of a non-rectangular domain Ω appropriately mapped to the physical domain via a geometry mapping.
Our approach can easily be extended to domains given as union of non-overlapping subdomains (patches), because continuity between patches does not require extra attention, since standard C^0-coupling is sufficient.
In all experiments a sparse direct solver is used for each of the three sub-problems. The implementation is done in the framework of the object-oriented C++ library G+Smo ("Geometry + Simulation Modules"), see [57]. Recall, in both discretization methods, see Section 2.5.3 and Section 2.6.3, the bilinear form of the $\boldsymbol{\phi}$-problem has a non-trivial kernel, namely RT_0. In the numerical experiments we eliminate this kernel by imposing additional constraints.
In the first part of this section (Section 2.8.1 and Section 2.8.2) we consider homogeneous boundary conditions and illustrate the theoretical results obtained for the discretization approach based on Nitsche's method and show promising numerical results for the approach based on Lagrange multipliers (also for a non-polygonal domain). In the second part of this section (Section 2.8.3) we show first numerical experiments of the extended discretization method for inhomogeneous boundary conditions.

2.8.1 Square plate with clamped, simply supported and free boundary (homogeneous)

We consider a square plate $\Omega = (-1,1)^2$ with simply supported north and south boundaries, clamped west boundary and free east boundary. The material tensor $\mathcal{C}$ is given as in (2.21) with $D = 1$, $\nu = 0$, which implies that $\mathcal{C}$ is the identity, and the load is

$$f(x,y) = 4\pi^4 \sin(\pi x)\sin(\pi y).$$

The exact solution is of the form

$$u(x,y) = \big((a+bx)\cosh(\pi x) + (c+dx)\sinh(\pi x) + \sin(\pi x)\big)\sin(\pi y),$$

which automatically satisfies the boundary conditions on the simply supported boundary parts, see [69]. The constants a, b, c and d are chosen such that the four remaining boundary conditions (on the clamped and free boundary parts) are satisfied, which leads to

$$\begin{aligned}
a &= -\frac{2(\sinh(\pi) - 3\sinh(3\pi) + \pi(4\pi\sinh(\pi) + 7\cosh(\pi) - 3\cosh(3\pi)))}{5 + 8\pi^2 + 3\cosh(4\pi)},\\
b &= -\frac{8\pi(2\pi\sinh(\pi) + \cosh(\pi))}{5 + 8\pi^2 + 3\cosh(4\pi)},\\
c &= \frac{10\cosh(\pi) + 6\cosh(3\pi) + 16\pi(\sinh(\pi) + \pi\cosh(\pi))}{5 + 8\pi^2 + 3\cosh(4\pi)},\\
d &= \frac{2\pi(5\sinh(\pi) - 3\sinh(3\pi) + 4\pi\cosh(\pi))}{5 + 8\pi^2 + 3\cosh(4\pi)}.
\end{aligned}$$

Discretization approach based on Nitsche's method (see Section 2.5.3)

The following results are obtained with the discretization method of Section 2.5.3. In Table 2.1 the discretization errors for $p = 1, 2, 3$ are shown. The first column shows the refinement level L, i.e. the number of uniform h-refinements of Ω. The next five pairs of columns show the respective discretization error and the error reduction relative to the previous level.

The experiments show optimal convergence rates for u and $\boldsymbol{M}$ as predicted by the analysis. For the columns containing the errors for p and $\boldsymbol{\phi}$, the (analytically not available) exact solutions p and $\boldsymbol{\phi}$ are replaced by their numerical solutions on level $L = 9$. Note that also for p and $\boldsymbol{\phi}$ optimal convergence rates are observed.

Discretization approach based on Lagrange multipliers (see Section 2.6.3)

The following results are obtained with the discretization method of Section 2.6.3.

In Table 2.2 the discretization errors for $p = 1, 2, 3$ are presented. The results show optimal convergence rates for u, $\boldsymbol{M}$, p and $\boldsymbol{\phi}$.

	L	$\|u-u_h\|_0$	order	$\|u-u_h\|_1$	order	$\|\boldsymbol{M}-\boldsymbol{M}_h\|_0$	order	$\|p-p_h\|_0$	order	$\|\boldsymbol{\phi}-\boldsymbol{\phi}_h\|_1$	order
p=1	4	$2.83\cdot10^{-2}$	1.886	$6.83\cdot10^{-1}$	0.992	$2.83\cdot10^{0}$	0.974	$3.44\cdot10^{-1}$	1.985	$3.24\cdot10^{0}$	1.050
	5	$7.20\cdot10^{-3}$	1.978	$3.42\cdot10^{-1}$	0.998	$1.42\cdot10^{0}$	0.993	$8.61\cdot10^{-2}$	1.999	$1.60\cdot10^{0}$	1.017
	6	$1.80\cdot10^{-3}$	1.996	$1.71\cdot10^{-1}$	0.999	$7.13\cdot10^{-1}$	0.998	$2.13\cdot10^{-2}$	2.011	$7.94\cdot10^{-1}$	1.012
	7	$4.51\cdot10^{-4}$	1.999	$8.55\cdot10^{-2}$	0.999	$3.56\cdot10^{-1}$	0.999	$5.15\cdot10^{-3}$	2.052	$3.87\cdot10^{-1}$	1.036
p=2	4	$8.57\cdot10^{-4}$	3.167	$4.33\cdot10^{-2}$	2.071	$1.75\cdot10^{-1}$	2.104	$1.22\cdot10^{-2}$	3.160	$2.04\cdot10^{-1}$	2.055
	5	$1.03\cdot10^{-4}$	3.043	$1.06\cdot10^{-2}$	2.018	$4.29\cdot10^{-2}$	2.030	$1.49\cdot10^{-3}$	3.042	$5.05\cdot10^{-2}$	2.017
	6	$1.28\cdot10^{-5}$	3.010	$2.66\cdot10^{-3}$	2.004	$1.06\cdot10^{-2}$	2.008	$1.85\cdot10^{-4}$	3.010	$1.25\cdot10^{-2}$	2.005
	7	$1.60\cdot10^{-6}$	3.002	$6.65\cdot10^{-4}$	2.001	$2.66\cdot10^{-3}$	2.002	$2.30\cdot10^{-5}$	3.002	$3.13\cdot10^{-3}$	2.003
p=3	4	$5.49\cdot10^{-5}$	4.179	$2.75\cdot10^{-3}$	3.084	$1.10\cdot10^{-2}$	3.105	$7.69\cdot10^{-4}$	4.219	$1.27\cdot10^{-2}$	3.019
	5	$3.42\cdot10^{-6}$	4.003	$3.46\cdot10^{-4}$	2.994	$1.38\cdot10^{-3}$	2.989	$4.63\cdot10^{-5}$	4.054	$1.62\cdot10^{-3}$	2.966
	6	$2.16\cdot10^{-7}$	3.984	$4.37\cdot10^{-5}$	2.985	$1.75\cdot10^{-4}$	2.978	$2.86\cdot10^{-6}$	4.012	$2.07\cdot10^{-4}$	2.972
	7	$1.36\cdot10^{-8}$	3.989	$5.50\cdot10^{-6}$	2.989	$2.22\cdot10^{-5}$	2.985	$1.78\cdot10^{-7}$	4.005	$2.60\cdot10^{-5}$	2.995

Table 2.1: Square plate, Nitsche's method, discretization errors.

	L	$\|u-u_h\|_0$	order	$\|u-u_h\|_1$	order	$\|\boldsymbol{M}-\boldsymbol{M}_h\|_0$	order	$\|p-p_h\|_0$	order	$\|\boldsymbol{\phi}-\boldsymbol{\phi}_h\|_1$	order
p=1	4	$2.84\cdot10^{-2}$	1.951	$6.83\cdot10^{-1}$	0.989	$2.87\cdot10^{0}$	1.014	$3.44\cdot10^{-1}$	1.985	$3.68\cdot10^{0}$	1.235
	5	$7.21\cdot10^{-3}$	1.982	$3.42\cdot10^{-1}$	0.998	$1.43\cdot10^{0}$	1.005	$8.61\cdot10^{-2}$	1.999	$1.67\cdot10^{0}$	1.138
	6	$1.81\cdot10^{-3}$	1.993	$1.71\cdot10^{-1}$	0.999	$7.14\cdot10^{-1}$	1.001	$2.13\cdot10^{-2}$	2.011	$8.05\cdot10^{-1}$	1.057
	7	$4.53\cdot10^{-4}$	1.997	$8.55\cdot10^{-2}$	0.999	$3.56\cdot10^{-1}$	1.000	$5.15\cdot10^{-3}$	2.052	$3.88\cdot10^{-1}$	1.049
p=2	4	$8.78\cdot10^{-4}$	3.204	$4.33\cdot10^{-2}$	2.068	$2.50\cdot10^{-1}$	1.996	$1.22\cdot10^{-2}$	3.160	$4.96\cdot10^{-1}$	1.862
	5	$1.04\cdot10^{-4}$	3.067	$1.06\cdot10^{-2}$	2.019	$6.34\cdot10^{-2}$	1.980	$1.49\cdot10^{-3}$	3.042	$1.30\cdot10^{-1}$	1.932
	6	$1.29\cdot10^{-5}$	3.020	$2.66\cdot10^{-3}$	2.005	$1.60\cdot10^{-2}$	1.985	$1.85\cdot10^{-4}$	3.010	$3.30\cdot10^{-2}$	1.975
	7	$1.60\cdot10^{-6}$	3.005	$6.65\cdot10^{-4}$	2.001	$4.02\cdot10^{-3}$	1.991	$2.30\cdot10^{-5}$	3.002	$8.10\cdot10^{-3}$	2.029
p=3	4	$5.47\cdot10^{-5}$	4.149	$2.75\cdot10^{-3}$	3.071	$1.54\cdot10^{-2}$	3.026	$7.69\cdot10^{-4}$	4.219	$2.63\cdot10^{-2}$	2.970
	5	$3.41\cdot10^{-6}$	4.002	$3.46\cdot10^{-4}$	2.993	$1.95\cdot10^{-3}$	2.985	$4.63\cdot10^{-5}$	4.054	$3.33\cdot10^{-3}$	2.980
	6	$2.15\cdot10^{-7}$	3.984	$4.37\cdot10^{-5}$	2.985	$2.46\cdot10^{-4}$	2.987	$2.86\cdot10^{-6}$	4.012	$4.19\cdot10^{-4}$	2.992
	7	$1.35\cdot10^{-8}$	3.989	$5.50\cdot10^{-6}$	2.99	$3.09\cdot10^{-5}$	2.992	$1.79\cdot10^{-7}$	4.002	$5.21\cdot10^{-5}$	3.006

Table 2.2: Square plate, Lagrange multipliers, discretization errors

2.8.2 Circular plate with simply supported boundary (homogeneous)

As a second example, we consider the simply supported circular plate with radius $r = 1$ and uniform loading $f = 1$. The material tensor $\mathcal{C}$ is given as in (2.21) with $D = 1$ and $\nu = 0.3$. The exact solution is given by

$$u(x) = c_1 + c_2 r^2 + c_3 r^4,$$

where $r^2 = x_1^2 + x_2^2$, $c_3 = 1/64$ and c_1, c_2 are determined from the boundary conditions as

$$c_1 = \frac{1}{64}\frac{5+\nu}{1+\nu}, \; c_2 = -\frac{1}{32}\frac{3+\nu}{1+\nu}.$$

For this test reproducing the exact geometry is essential, see the discussion of the so-called Babuška paradox in [4]. Therefore, we use an exact geometry representation by means of non-uniform rational B-splines (NURBS). As discretization space $\mathcal{S}_h^p(\Omega)$ we still use the tensor product B-spline space as defined in the beginning of this section. For this non-polygonal domain only the discretization approach based on Lagrange multipliers introduced in Section 2.6.3 can be applied. The discretization errors for

$p = 1, 2, 3$ are presented in Table 2.3. The results show optimal convergence rates for u, $\boldsymbol{M}$, p and $\boldsymbol{\phi}$.

	L	$\|u-u_h\|_0$	order	$\|u-u_h\|_1$	order	$\|\boldsymbol{M}-\boldsymbol{M}_h\|_0$	order	$\|p-p_h\|_0$	order	$\|\boldsymbol{\phi}-\boldsymbol{\phi}_h\|_1$	order
p=1	4	$3.58 \cdot 10^{-4}$	1.984	$8.37 \cdot 10^{-3}$	1.002	$8.79 \cdot 10^{-3}$	1.020	$8.87 \cdot 10^{-4}$	1.997	$8.99 \cdot 10^{-3}$	1.004
	5	$8.98 \cdot 10^{-5}$	1.996	$4.18 \cdot 10^{-3}$	1.000	$4.38 \cdot 10^{-3}$	1.005	$2.21 \cdot 10^{-4}$	2.001	$4.48 \cdot 10^{-3}$	1.003
	6	$2.24 \cdot 10^{-5}$	1.999	$2.09 \cdot 10^{-3}$	1.000	$2.18 \cdot 10^{-3}$	1.001	$5.50 \cdot 10^{-5}$	2.010	$2.23 \cdot 10^{-3}$	1.008
	7	$5.62 \cdot 10^{-6}$	1.999	$1.04 \cdot 10^{-3}$	1.000	$1.09 \cdot 10^{-3}$	1.000	$1.33 \cdot 10^{-5}$	2.044	$1.08 \cdot 10^{-3}$	1.035
p=2	4	$6.90 \cdot 10^{-6}$	3.162	$3.63 \cdot 10^{-4}$	2.062	$3.42 \cdot 10^{-4}$	2.121	$1.77 \cdot 10^{-5}$	3.092	$3.47 \cdot 10^{-4}$	2.075
	5	$8.38 \cdot 10^{-7}$	3.041	$8.97 \cdot 10^{-5}$	2.016	$8.38 \cdot 10^{-5}$	2.031	$2.18 \cdot 10^{-6}$	3.023	$8.57 \cdot 10^{-5}$	2.019
	6	$1.04 \cdot 10^{-7}$	3.010	$2.23 \cdot 10^{-5}$	2.004	$2.08 \cdot 10^{-5}$	2.008	$2.71 \cdot 10^{-7}$	3.005	$2.13 \cdot 10^{-5}$	2.005
	7	$1.29 \cdot 10^{-8}$	3.002	$5.59 \cdot 10^{-6}$	2.001	$5.20 \cdot 10^{-6}$	2.002	$3.39 \cdot 10^{-8}$	3.001	$5.32 \cdot 10^{-6}$	2.003
p=3	4	$4.05 \cdot 10^{-7}$	4.319	$1.93 \cdot 10^{-5}$	3.163	$2.01 \cdot 10^{-5}$	3.206	$7.83 \cdot 10^{-7}$	4.182	$2.04 \cdot 10^{-5}$	3.160
	5	$2.38 \cdot 10^{-8}$	4.083	$2.35 \cdot 10^{-6}$	3.034	$2.42 \cdot 10^{-6}$	3.054	$4.74 \cdot 10^{-8}$	4.045	$2.48 \cdot 10^{-6}$	3.040
	6	$1.47 \cdot 10^{-9}$	4.019	$2.93 \cdot 10^{-7}$	3.004	$3.00 \cdot 10^{-7}$	3.013	$2.94 \cdot 10^{-9}$	4.010	$3.08 \cdot 10^{-7}$	3.009
	7	$9.17 \cdot 10^{-11}$	4.004	$3.67 \cdot 10^{-8}$	2.999	$3.74 \cdot 10^{-8}$	3.003	$1.83 \cdot 10^{-10}$	4.001	$3.85 \cdot 10^{-8}$	3.002

Table 2.3: Circular plate, Lagrange multipliers, discretization errors.

2.8.3 Inhomogeneous boundary conditions

In this section we show first numerical experiments of the extended discretization method for inhomogeneous boundary conditions, as outlined in Section 2.7.
In all tests we consider the following setup: The material tensor $\mathcal{C}$ is given as in (2.21) with $D = 1$, $\nu = 0$, which implies that $\mathcal{C}$ is the identity. The load

$$f(x, y) = 48\left(x^4 + 12x^2y^2 + y^4\right) + 25\pi^4 \cos(\pi x)\cos(2\pi y).$$

is chosen such that the exact solution is given by

$$u(x, y) = 2x^4y^4 + \cos(\pi x)\cos(2\pi y).$$

In following we consider inhomogeneous values of the boundary data $\hat{u}_3, \hat{\theta}_n, \hat{M}_n, \hat{V}_n$, cf. Section 2.7, resulting from the stated manufactured solution. With a slight abuse of notation we refer to the type of boundary condition we impose on Γ_c and Γ_s also in case of inhomogeneous data as clamped and simply supported.

Inhomogeneous boundary data $\hat{u}_3$ and $\hat{\theta}_n$ (clamped)

In order to test this type of inhomogeneous boundary data, we consider the simple situation of a fully clamped boundary (with inhomogeneous data), where no coupling condition has to be incorporated. The incorporation of inhomogeneous boundary conditions enables us to easily compare results for the same right-hand side (and exact solution) on different domains, namely:

- the square plate $\Omega_S = (-1, 1)^2$ (used in Section 2.8.1), representing a domain with corners,

- and the circular plate Ω_C with radius $r = 1$ (used in Section 2.8.2), representing a domain with smooth boundary.

In Table 2.4 the discretization errors for $p = 1, 2, 3, 4$ are shown for $\Omega = \Omega_S$. For u we obtain optimal convergence rates. However, for $\boldsymbol{M}$ the convergence rate seems to stagnate at the order 3, and, hence, for $p = 4$ the optimal convergence order of 4 is not reached. This behavior is in accordance with the results for p and $\boldsymbol{\phi}$, where for both the order stagnates at 3, resulting in non-optimal convergence orders for p and $\boldsymbol{\phi}$ for polynomial degrees $p = 3, 4$ and $p = 4$, respectively.

	L	$\Vert u - u_h\Vert_0$	order	$\Vert u - u_h\Vert_1$	order	$\Vert\boldsymbol{M} - \boldsymbol{M}_h\Vert_0$	order	$\Vert p - p_h\Vert_0$	order	$\Vert\boldsymbol{\phi} - \boldsymbol{\phi}_h\Vert_1$	order
p=1	4	$6.58 \cdot 10^{-2}$	1.922	$1.54 \cdot 10^{0}$	0.990	$1.21 \cdot 10^{1}$	1.01	$2.50 \cdot 10^{0}$	1.955	$1.51 \cdot 10^{1}$	1.020
	5	$1.66 \cdot 10^{-2}$	1.979	$7.76 \cdot 10^{-1}$	0.997	$6.04 \cdot 10^{0}$	1.003	$6.26 \cdot 10^{-1}$	2.001	$7.48 \cdot 10^{0}$	1.014
	6	$4.18 \cdot 10^{-3}$	1.995	$3.88 \cdot 10^{-1}$	0.999	$3.02 \cdot 10^{0}$	1.000	$1.51 \cdot 10^{-1}$	2.051	$3.64 \cdot 10^{0}$	1.037
	7	$1.04 \cdot 10^{-3}$	1.998	$1.94 \cdot 10^{-1}$	0.999	$1.51 \cdot 10^{0}$	1.000	$3.18 \cdot 10^{-2}$	2.245	$1.62 \cdot 10^{0}$	1.161
p=2	4	$3.37 \cdot 10^{-3}$	3.506	$1.59 \cdot 10^{-1}$	2.244	$1.28 \cdot 10^{0}$	2.346	$1.52 \cdot 10^{-1}$	3.374	$1.44 \cdot 10^{0}$	2.211
	5	$3.75 \cdot 10^{-4}$	3.166	$3.80 \cdot 10^{-2}$	2.071	$3.00 \cdot 10^{-1}$	2.096	$1.79 \cdot 10^{-2}$	3.089	$3.49 \cdot 10^{-1}$	2.052
	6	$4.55 \cdot 10^{-5}$	3.045	$9.38 \cdot 10^{-3}$	2.018	$7.37 \cdot 10^{-2}$	2.025	$2.23 \cdot 10^{-3}$	3.005	$8.62 \cdot 10^{-2}$	2.017
	7	$5.64 \cdot 10^{-6}$	3.011	$2.33 \cdot 10^{-3}$	2.004	$1.83 \cdot 10^{-2}$	2.006	$2.81 \cdot 10^{-4}$	2.993	$2.08 \cdot 10^{-2}$	2.047
p=3	4	$4.84 \cdot 10^{-4}$	4.690	$1.96 \cdot 10^{-2}$	3.530	$1.49 \cdot 10^{-1}$	3.609	$1.95 \cdot 10^{-2}$	4.513	$1.62 \cdot 10^{-1}$	3.414
	5	$2.64 \cdot 10^{-5}$	4.196	$2.23 \cdot 10^{-3}$	3.136	$1.71 \cdot 10^{-2}$	3.132	$1.51 \cdot 10^{-3}$	3.686	$1.94 \cdot 10^{-2}$	3.061
	6	$1.59 \cdot 10^{-6}$	4.046	$2.73 \cdot 10^{-4}$	3.027	$2.14 \cdot 10^{-3}$	2.992	$1.60 \cdot 10^{-4}$	3.239	$2.47 \cdot 10^{-3}$	2.973
	7	$9.93 \cdot 10^{-8}$	4.008	$3.41 \cdot 10^{-5}$	3.002	$2.75 \cdot 10^{-4}$	2.963	$1.91 \cdot 10^{-5}$	3.068	$3.17 \cdot 10^{-4}$	2.959
p=4	4	$6.35 \cdot 10^{-5}$	6.081	$2.78 \cdot 10^{-3}$	4.790	$2.03 \cdot 10^{-2}$	4.803	$3.99 \cdot 10^{-3}$	5.334	$2.26 \cdot 10^{-2}$	4.538
	5	$1.51 \cdot 10^{-6}$	5.393	$1.45 \cdot 10^{-4}$	4.257	$1.44 \cdot 10^{-3}$	3.815	$3.95 \cdot 10^{-4}$	3.334	$1.77 \cdot 10^{-3}$	3.669
	6	$4.34 \cdot 10^{-8}$	5.118	$8.64 \cdot 10^{-6}$	4.072	$1.47 \cdot 10^{-4}$	3.293	$4.88 \cdot 10^{-5}$	3.017	$1.87 \cdot 10^{-4}$	3.243
	7	$1.32 \cdot 10^{-9}$	5.031	$5.33 \cdot 10^{-7}$	4.018	$1.73 \cdot 10^{-5}$	3.091	$6.24 \cdot 10^{-6}$	2.967	$2.20 \cdot 10^{-5}$	3.093

Table 2.4: Square plate, discretization errors.

For $\Omega = \Omega_C$, the results for $p = 1, 2$ are identical, and are therefore not shown. In Table 2.5 the discretization errors for $p = 3, 4$ are presented. In contrast to the results for $\Omega = \Omega_S$, now optimal convergence rates are obtained beside u also for $\boldsymbol{M}$, p and $\boldsymbol{\phi}$.

	L	$\Vert u - u_h\Vert_0$	order	$\Vert u - u_h\Vert_1$	order	$\Vert\boldsymbol{M} - \boldsymbol{M}_h\Vert_0$	order	$\Vert p - p_h\Vert_0$	order	$\Vert\boldsymbol{\phi} - \boldsymbol{\phi}_h\Vert_1$	order
p=3	4	$6.97 \cdot 10^{-4}$	5.347	$3.00 \cdot 10^{-2}$	3.901	$2.18 \cdot 10^{-1}$	4.154	$3.29 \cdot 10^{-2}$	5.315	$2.09 \cdot 10^{-1}$	3.897
	5	$3.27 \cdot 10^{-5}$	4.413	$3.15 \cdot 10^{-3}$	3.254	$2.11 \cdot 10^{-2}$	3.373	$1.52 \cdot 10^{-3}$	4.430	$2.20 \cdot 10^{-2}$	3.244
	6	$1.88 \cdot 10^{-6}$	4.116	$3.76 \cdot 10^{-4}$	3.066	$2.46 \cdot 10^{-3}$	3.098	$8.78 \cdot 10^{-5}$	4.122	$2.64 \cdot 10^{-3}$	3.061
	7	$1.15 \cdot 10^{-7}$	4.030	$4.65 \cdot 10^{-5}$	3.016	$3.03 \cdot 10^{-4}$	3.022	$5.36 \cdot 10^{-6}$	4.031	$3.27 \cdot 10^{-4}$	3.013
p=4	4	$1.39 \cdot 10^{-4}$	6.890	$5.64 \cdot 10^{-3}$	5.434	$4.28 \cdot 10^{-2}$	5.641	$6.73 \cdot 10^{-3}$	6.841	$3.95 \cdot 10^{-2}$	5.424
	5	$2.79 \cdot 10^{-6}$	5.639	$2.57 \cdot 10^{-4}$	4.453	$1.76 \cdot 10^{-3}$	4.602	$1.34 \cdot 10^{-4}$	5.648	$1.80 \cdot 10^{-3}$	4.449
	6	$7.61 \cdot 10^{-8}$	5.195	$1.47 \cdot 10^{-5}$	4.128	$9.74 \cdot 10^{-5}$	4.179	$3.65 \cdot 10^{-6}$	5.199	$1.03 \cdot 10^{-4}$	4.125
	7	$2.29 \cdot 10^{-9}$	5.052	$8.99 \cdot 10^{-7}$	4.033	$5.89 \cdot 10^{-6}$	4.046	$1.09 \cdot 10^{-7}$	5.054	$6.32 \cdot 10^{-6}$	4.034

Table 2.5: Circular plate, discretization errors.

From this observations we can conclude that the incorporation of inhomogeneous boundary data $\hat{u}_3, \hat{\theta}_n$ works.
The reduced convergence order of p and $\boldsymbol{\phi}$ in case of the square domain is probably due to reduced regularity of the exact solutions p and $\boldsymbol{\phi}$. Although the exact solution

u is in $C^\infty(\Omega)$, and, hence also $\boldsymbol{M} = -\mathcal{C}\nabla^2 u$, we cannot directly deduce regularity of the exact p and $\boldsymbol{\phi}$. The reason is the following: according to the regular decomposition in Theorem 2.19 we have

$$\boldsymbol{M} = p\boldsymbol{I} + \operatorname{symCurl}\boldsymbol{\phi}$$

and the function $p \in Q_0$ is the unique solution of the Poisson problem

$$\int_\Omega \nabla p \cdot \nabla v \; dx \quad = -\langle \operatorname{div}\operatorname{Div}\boldsymbol{M}, v\rangle \quad \text{for all } v \in Q_0.$$

Therefore, in order to deduce smoothness of p from smoothness of $\boldsymbol{M}$ we need results from regularity theory. In case of a smooth boundary smoothness of the right-hand side can directly be transferred to smoothness of the solution. However, this is in the presence of corners not true anymore. For more details on the fairly complicated topic of regularity theory we refer to [40, 39].
This experiment indicates, that we can only expect optimal convergence orders for $\boldsymbol{M}$ in case of sufficiently smooth exact solutions p and $\boldsymbol{\phi}$. This observation is not inconsistent with the a-priori discretization error estimate of Theorem 2.35, since we additionally to u also assumed p and $\boldsymbol{\phi}$ to be smooth enough. However, in the numerical experiments with homogeneous data we have never encountered such a behavior.

The observed difficulties due to reduced regularity for p and $\boldsymbol{\phi}$ seem to be even more pronounced for inhomogeneous data $\hat{M}_n$ and $\hat{V}_n$. For the incorporation of the coupling condition we use the approach based on Nitsche's method, as outlined in Section 2.7.4. Therefore, we only consider polygonal domains.

Inhomogeneous boundary data $\hat{u}_3$ and $\hat{M}_n$ (simply supported)

In order to test this type of inhomogeneous data, we consider a square plate $\Omega_S = (-1,1)^2$ with simply supported east boundary with applied force $\hat{M}_n$ and clamped remaining boundaries. In Table 2.6 the discretization errors for $p = 1$ are shown. For higher polynomial degrees stagnated convergence orders both in u and $\boldsymbol{M}$ have been observed. As indicated above, the reason for this is probably missing regularity in p and $\boldsymbol{\phi}$, which in this case also effects the convergence order of u not only $\boldsymbol{M}$, as before.

	L	$\|u-u_h\|_0$	order	$\|u-u_h\|_1$	order	$\|\boldsymbol{M}-\boldsymbol{M}_h\|_0$	order	$\|p-p_h\|_0$	order	$\|\boldsymbol{\phi}-\boldsymbol{\phi}_h\|_1$	order
p=1	4	$6.55\cdot10^{-2}$	1.921	$1.54\cdot10^{0}$	0.990	$1.21\cdot10^{1}$	1.011	$2.50\cdot10^{0}$	1.955	$1.51\cdot10^{1}$	1.018
	5	$1.66\cdot10^{-2}$	1.978	$7.76\cdot10^{-1}$	0.997	$6.04\cdot10^{0}$	1.003	$6.26\cdot10^{-1}$	2.001	$7.47\cdot10^{0}$	1.013
	6	$4.17\cdot10^{-3}$	1.994	$3.88\cdot10^{-1}$	0.999	$3.02\cdot10^{0}$	1.000	$1.51\cdot10^{-1}$	2.051	$3.64\cdot10^{0}$	1.037
	7	$1.04\cdot10^{-3}$	1.998	$1.94\cdot10^{-1}$	0.999	$1.51\cdot10^{0}$	1.000	$3.18\cdot10^{-2}$	2.245	$1.62\cdot10^{0}$	1.161

Table 2.6: Square plate, Nitsche's method, discretization errors.

Inhomogeneous boundary data $\hat{M}_n$ and $\hat{V}_n$ (free without corner forces)

In order to test this type of inhomogeneous data, we consider a square plate $\Omega_S = (-1,1)^2$ with free east boundary with applied forces $\hat{M}_n$ and $\hat{V}_n$ and clamped remaining boundaries. In Table 2.7 the discretization errors for $p = 1$ are shown. For higher polynomial degrees stagnated convergence orders both in u and $\boldsymbol{M}$ have been observed.

	L	$\|u - u_h\|_0$	order	$\|u - u_h\|_1$	order	$\|\boldsymbol{M} - \boldsymbol{M}_h\|_0$	order	$\|p - p_h\|_0$	order	$\|\boldsymbol{\phi} - \boldsymbol{\phi}_h\|_1$	order
p=1	4	$7.19 \cdot 10^{-2}$	1.901	$1.55 \cdot 10^{0}$	1.022	$1.18 \cdot 10^{1}$	1.007	$2.63 \cdot 10^{0}$	1.962	$1.45 \cdot 10^{1}$	1.005
	5	$1.88 \cdot 10^{-2}$	1.933	$7.81 \cdot 10^{-1}$	0.994	$5.99 \cdot 10^{0}$	0.979	$6.59 \cdot 10^{-1}$	2.000	$7.51 \cdot 10^{0}$	0.955
	6	$4.88 \cdot 10^{-3}$	1.945	$3.90 \cdot 10^{-1}$	0.998	$3.19 \cdot 10^{0}$	0.910	$1.59 \cdot 10^{-1}$	2.048	$4.15 \cdot 10^{0}$	0.853
	7	$1.28 \cdot 10^{-3}$	1.922	$1.95 \cdot 10^{-1}$	0.999	$1.93 \cdot 10^{0}$	0.723	$3.37 \cdot 10^{-2}$	2.239	$2.48 \cdot 10^{0}$	0.741

Table 2.7: Square plate, Nitsche's method, discretization errors.

In order to better understand the observed behavior, more extensive testing is needed. However, this is out of the scope of this thesis, since we only outline a first idea how to adapt the presented approach to inhomogeneous boundary conditions and the full understanding and numerical analysis of the proposed extension is a topic for future work.

Chapter 3

Kirchhoff-Love shells

In this chapter we show the extensibility of our approach to the Kirchhoff-Love shell problem, which is a more general fourth-order problem. In contrast to plates, the Kirchhoff-Love shell problem does not separate into a membrane and bending problem, since membrane and bending strain depend on both the tangential and transverse part of the displacement. Therefore, all three components of the displacement have to be considered at once, leading to a vector-valued space of displacements.
This chapter is organized similar as Chapter 2. In Section 3.1 the Kirchhoff-Love shell model is introduced. The (primal) variational formulation of the Kirchhoff-Love shell problem is stated in Section 3.2. Section 3.3 contains a new mixed formulation and in Section 3.4 an extended mixed formulation to circumvent membrane locking is presented. For both mixed formulations well-posedness and equivalence to the original problem is shown. In Section 3.5 we reformulate both mixed formulations as formulations in standard H^1 spaces using the regular decomposition of Section 2.4. The new mixed formulation leads in a natural way to the construction of a new discretization method, which is introduced in Section 3.6. The paper closes with numerical experiments in Section 3.7.

3.1 Kirchhoff-Love shell model

The description of the Kirchhoff-Love shell model is based on the presentations in [28, 24, 13].
A shell is a thin-walled structure, where one dimension is considerably smaller that the other two. A plate is the special case of a shell with no curvature. The 3D shell in the undeformed configuration is described by a mapping $\vec{\Phi}$ from a 3D parameter domain

$$\Omega^{3\mathrm{D}} = \left\{ (\xi^1, \xi^2, \xi^3) \in \mathbb{R}^3 \colon (\xi^1, \xi^2) \in \Omega,\ \xi^3 \in \left(-\frac{t}{2}, \frac{t}{2} \right) \right\},$$

where $\Omega \subset \mathbb{R}^2$ and t denotes the thickness, into the 3D physical space, $\vec{\Phi} : \Omega^{3\mathrm{D}} \to \mathbb{R}^3$.

For simplicity we assume the thickness t to be constant. The map $\vec{\Phi}$ is of the form

$$\vec{\Phi}(\xi^1, \xi^2, \xi^3) = \vec{r}(\xi^1, \xi^2) + \xi^3 \, \vec{a}_3(\xi^1, \xi^2), \tag{3.1}$$

where the mapping $\vec{r} : \Omega \to \mathbb{R}^3$ defines the midsurface of the shell, and $\vec{a}_3$ is the unit director, which is normal to the midsurface. The map $\vec{r}$ is assumed to be sufficiently smooth, e.g., $\mathcal{C}^3(\overline{\Omega})$, cf. [28]. The 3D map $\vec{\Phi}$ with the parameter domain Ω^{3D} describes the shell with a natural curvilinear coordinate system (based on the midsurface). In order to work with tensors in this specific coordinate system, we introduce in the sequel basic concepts of surface differential geometry.
Based on the map $\vec{\Phi}$ the 3D covariant base vectors $\vec{g}_i$ are defined by

$$\vec{g}_i = \partial_i \vec{\Phi},$$

where we use here and in the following the notation $\partial_i = \partial_{\xi^i}$. For the specific form of the map $\vec{\Phi}$ we obtain

$$\vec{g}_\alpha = \partial_\alpha \vec{\Phi} = \partial_\alpha \vec{r} + \xi^3 \, \partial_\alpha \vec{a}_3 = \vec{a}_\alpha + \xi^3 \, \partial_\alpha \vec{a}_3,$$

with the covariant base vectors $\vec{a}_\alpha$ of the midsurface given by

$$\vec{a}_\alpha = \partial_\alpha \vec{r}.$$

We assume that the map $\vec{r}$ is such that at each point of the midsurface the vectors $\vec{a}_\alpha$ are linearly independent, and, hence, form a basis of the tangent plane to the midsurface. Moreover, we have

$$\vec{g}_3 = \partial_3 \vec{\Phi} = \vec{a}_3,$$

with

$$\vec{a}_3 = \frac{\vec{a}_1 \times \vec{a}_2}{|\vec{a}_1 \times \vec{a}_2|}.$$

Contravariant base vectors are defined via the orthogonality conditions

$$\vec{g}_i \cdot \vec{g}^j = \delta_i^j \quad \text{and} \quad \vec{a}_i \cdot \vec{a}^j = \delta_i^j \quad \text{for all } i, j = 1, 2, 3.$$

Note that $\vec{g}^3 = \vec{g}_3$ and $\vec{a}^3 = \vec{a}_3$.
Before we continue with further basic concepts of surface differential geometry we introduce some notations.

3.1.1 Notations

We adapt the notations from Section 2.1.1 to the needs of shell modeling. In this chapter, scalars are denoted by simple letters and for tensors we place over the corresponding letter a number of arrows equal to their order. As in Chapter 2, Greek indices take values $\{1, 2\}$ and Latin indices $\{1, 2, 3\}$.

A first-order tensor $\vec{u}$ is of the form

$$\vec{u} = u^i \vec{a}_i = u_i \vec{a}^{\,i},$$

where the components (u^i) of the tensor in the covariant basis $(\vec{a}_1, \vec{a}_2, \vec{a}_3)$ are called the contravariant components and the components (u_i) in the contravariant basis $(\vec{a}^{\,1}, \vec{a}^{\,2}, \vec{a}^{\,3})$ are called the covariant components. Note, in the special case of an orthonormal covariant basis the contravariant basis coincides with the covariant one, and, hence the covariant and contravariant components of tensors are equal. In the following, superscripts always indicate contravariant components and subscripts mark covariant components. Furthermore, Einstein's summation convention is applied to indices appearing once as a subscript and once as a superscript in an expression. For a second-order tensor $\vec{\vec{T}}$ we have

$$\vec{\vec{T}} = T^{ij} \vec{a}_i \otimes \vec{a}_j = T_{ij} \vec{a}^{\,i} \otimes \vec{a}^{\,j} = T_i^{\cdot j} \vec{a}^{\,i} \otimes \vec{a}_j = T^i_{\cdot j} \vec{a}_i \otimes \vec{a}^{\,j},$$

where we have in addition to contravariant and covariant components also mixed components, namely, covariant-contravariant components $T_i^{\cdot j}$ and contravariant-covariant components $T^i_{\cdot j}$ (in general $T_i^{\cdot j} \neq T^i_{\cdot j}$). The transformation of one set of components into another is done by means of the standard transformation formulas, see, e.g., [24]. In the following we use both surface tensors on the midsurface of the shell and 3D tensors. For components of surface tensors we use Greek indices, in order to distinguish them from components of 3D tensors denoted by Latin indices.
In order to allow for a (relatively straight forward) application of the results obtained in Chapter 2, we choose in the following for each tensor $\vec{u}$, $\vec{\vec{T}}$ a particular basis. We view the components with respect to the chosen basis as entries of a vector or matrix, denoted by $\boldsymbol{u}$ and $\boldsymbol{T}$.
For two first-order tensors $\vec{u} = u^i \vec{a}_i$ and $\vec{v} = v_j \vec{a}^{\,j}$ we have

$$\vec{u} \cdot \vec{v} = (u^i \vec{a}_i) \cdot (v_j \vec{a}^{\,j}) = u^i v_j (\vec{a}_i \cdot \vec{a}^{\,j}) = u^i v_j \delta_i^j = u^i v_i = \boldsymbol{u} \cdot \boldsymbol{v},$$

where $\boldsymbol{u} = (u^i)$ and $\boldsymbol{v} = (v_i)$ are vectors that store the corresponding components. For two second-order tensors $\vec{\vec{M}} = M^{ij} \vec{a}_i \otimes \vec{a}_j$ and $\vec{\vec{N}} = N_{km} \vec{a}^{\,k} \otimes \vec{a}^{\,m}$ we have

$$\vec{\vec{M}} : \vec{\vec{N}} = M^{ij} N_{km} (\vec{a}_j \cdot \vec{a}^{\,k})(\vec{a}_i \cdot \vec{a}^{\,m}) = M^{ij} N_{km} \delta_j^k \delta_i^m = M^{ij} N_{ji} = \boldsymbol{M} : \boldsymbol{N},$$

where $\boldsymbol{M} = (M^{ij})$ and $\boldsymbol{N} = (N_{ji})$ are matrices that store the corresponding components. We use the same notations in 2D. Usually the dimension is clear from the context and otherwise stated explicitly.
Let the parameter domain Ω of the shell midsurface be a domain with a Lipschitz boundary Γ. In what follows let $\boldsymbol{n} = (n_\alpha)$ and $\boldsymbol{t} = (t_\alpha)$, with $t_1 = -n_2$ and $t_2 = -n_1$, represent the unit outer normal vector and the unit counterclockwise tangent vector to Γ, respectively. For a first-order tensor $\vec{w} = w_\alpha \vec{a}^{\,\alpha}$ with covariant components $\boldsymbol{w} = (w_\alpha)$ we introduce the following notations

$$w_n = \boldsymbol{w} \cdot \boldsymbol{n}, \qquad w_t = \boldsymbol{w} \cdot \boldsymbol{t}$$

for the normal component and the tangential component.

3.1.2 Differential geometry

The 3D metric tensor $\vec{\vec{g}}$ is defined by its covariant components or alternatively contravariant components

$$\begin{aligned} g_{\alpha\beta} &= \vec{g}_\alpha \cdot \vec{g}_\beta, & g_{\alpha 3} &= \vec{g}_\alpha \cdot \vec{g}_3 = 0, & g_{33} &= \vec{g}_3 \cdot \vec{g}_3 = 1, \\ g^{\alpha\beta} &= \vec{g}^{\,\alpha} \cdot \vec{g}^{\,\beta}, & g^{\alpha 3} &= \vec{g}^{\,\alpha} \cdot \vec{g}^{\,3} = 0, & g^{33} &= \vec{g}^{\,3} \cdot \vec{g}^{\,3} = 1. \end{aligned} \tag{3.2}$$

For later use we define $\sqrt{g} = \sqrt{\det(g_{ij})}$, where we assume in the following that $\det(g_{ij}) > 0$. The first fundamental form $\vec{\vec{a}}$ of the shell midsurface is given by its covariant components or alternatively contravariant components

$$a_{\alpha\beta} = \vec{a}_\alpha \cdot \vec{a}_\beta, \qquad a^{\alpha\beta} = \vec{a}^{\,\alpha} \cdot \vec{a}^{\,\beta}.$$

The first fundamental form is the restriction of the 3D metric tensor to the tangent plane. For later use we define $\sqrt{a} = \sqrt{\det(a_{\alpha\beta})}$, where we assume in the following that $\det(a_{\alpha\beta}) > 0$. The second fundamental form $\vec{\vec{b}}$ of the shell midsurface is defined by its covariant components or alternatively mixed components

$$b_{\alpha\beta} = \vec{a}_3 \cdot \partial_\beta \vec{a}_\alpha, \qquad b^\alpha_\beta = a^{\alpha\lambda} b_{\lambda\beta}.$$

The second fundamental form is called curvature tensor because it contains all the information on the curvature of the surface. We define the third fundamental form $\vec{\vec{c}}$ by

$$c_{\alpha\beta} = b^\lambda_\alpha b_{\lambda\beta}.$$

Next we briefly recall the covariant derivatives of a first-order surface tensors $\vec{u} = u_i \vec{a}^{\,i}$, for details see [24, 28]. The (surface) covariant derivative of u_β, denoted by $u_{\beta|\alpha}$, is given by

$$u_{\beta|\alpha} = \partial_\alpha u_\beta - \Gamma^\lambda_{\beta\alpha} u_\lambda, \tag{3.3}$$

where the (surface) Christoffel symbols are given by

$$\Gamma^\lambda_{\beta\alpha} = \partial_\alpha \vec{a}_\beta \cdot \vec{a}^{\lambda}.$$

Furthermore, we define the second-order covariant derivative of u_3 by

$$u_{3|\alpha\beta} = \partial_{\alpha\beta} u_3 - \Gamma^\lambda_{\alpha\beta} \partial_\lambda u_3,$$

and the covariant derivative of the second fundamental form

$$b^\mu_{\alpha|\beta} = \partial_\beta b^\mu_\alpha + \Gamma^\mu_{\beta\lambda} b^\lambda_\alpha - \Gamma^\lambda_{\beta\alpha} b^\mu_\lambda.$$

3.1.3 Shell kinematics and constitutive equation

In this section we proceed analogously to Section 2.1.3, where the special case of plates was considered.
The classical plate theory for thin plates was extended to the classical shell theory in 1888 by August E. H. Love (1863-1940). It is called the Kirchhoff-Love shell model since it is based on the Kirchhoff kinematical assumptions stated in Section 2.1. As in case of plates, there exists for thick shells a Reissner-Mindlin shell model based on the Reissner-Mindlin kinematical assumptions.
The Reissner-Mindlin kinematical assumptions imply the following form of the 3D displacement:

$$\vec{U}(\xi^1,\xi^2,\xi^3) = \vec{u}(\xi^1,\xi^2) + \xi^3\vec{\theta}(\xi^1,\xi^2), \tag{3.4}$$

where $\vec{u}$ denotes the displacement of the midsurface of the shell and $\vec{\theta}$ the rotation of a line in the direction of $\vec{a}_3$. Note, $\vec{u}$, $\vec{U}$ and $\vec{\theta}$ are tensors given by

$$\vec{U} = U_i\vec{g}^i = U_\alpha\vec{g}^\alpha + U_3\vec{g}^3 \tag{3.5}$$

and

$$\vec{u} = u_i\vec{a}^i = u_\alpha\vec{a}^\alpha + u_3\vec{a}^3 \quad \text{and} \quad \vec{\theta} = \theta_\alpha\vec{a}^\alpha. \tag{3.6}$$

The unknowns we search for are the covariant components of the displacement denoted by

$$\boldsymbol{u} = (u_i), \tag{3.7}$$

and for the 3D displacement by

$$\boldsymbol{U} = (U_i).$$

In the following we refer to $\underline{\boldsymbol{u}} = (u_\alpha)$ and u_3 as tangential and transverse part of the displacement, respectively.
As in Chapter 2, we consider small displacements, i.e., linear analysis. Then the 3D strain tensor is given by its covariant components $\boldsymbol{E} = (E_{ij})$ with

$$E_{ij} = \frac{1}{2}(\vec{g}_i\cdot\partial_j\vec{U} + \vec{g}_j\cdot\partial_i\vec{U}). \tag{3.8}$$

For the specific displacement in (3.4) we obtain

$$\begin{aligned} E_{\alpha\beta}(\boldsymbol{U}) &= \varepsilon_{\alpha\beta}(\boldsymbol{u}) + \xi^3\kappa_{\alpha\beta}(\boldsymbol{u},\boldsymbol{\theta}) - (\xi^3)^2\,\chi_{\alpha\beta}(\boldsymbol{\theta}),\\ E_{\alpha3}(\boldsymbol{U}) &= \gamma_\alpha(\boldsymbol{u},\boldsymbol{\theta}), \end{aligned} \tag{3.9}$$

where

$$\varepsilon_{\alpha\beta}(\boldsymbol{u}) = \frac{1}{2}(u_{\alpha|\beta} + u_{\beta|\alpha}) - b_{\alpha\beta}\,u_3 \tag{3.10}$$

$$\kappa_{\alpha\beta}(\boldsymbol{u},\boldsymbol{\theta}) = \frac{1}{2}(\theta_{\alpha|\beta} + \theta_{\beta|\alpha} - b^\lambda_\beta u_{\lambda|\alpha} - b^\lambda_\alpha u_{\lambda|\beta}) + c_{\alpha\beta}u_3 \tag{3.11}$$

$$\gamma_\alpha(\boldsymbol{u},\boldsymbol{\theta}) = \frac{1}{2}(\theta_\alpha + \partial_\alpha u_3 + b^\lambda_\alpha u_\lambda) \tag{3.12}$$

$$\chi_{\alpha\beta}(\boldsymbol{\theta}) = \frac{1}{2}(b^\lambda_\beta\theta_{\lambda|\alpha} + b^\lambda_\alpha\theta_{\lambda|\beta}). \tag{3.13}$$

The tensors $\vec{\vec{\epsilon}}$, $\vec{\vec{\kappa}}$ and $\vec{\vec{\gamma}}$ with covariant components $\boldsymbol{\varepsilon}(\boldsymbol{u}) = (\varepsilon_{\alpha\beta})$, $\boldsymbol{\kappa}(\boldsymbol{\theta}) = (\kappa_{\alpha\beta})$ and $\boldsymbol{\gamma}(u_3, \boldsymbol{\theta}) = (\gamma_\alpha)$ are called membrane strain, bending strain, and shear strain tensor, respectively.
The third part of the Kirchhoff kinematical assumptions implies

$$\theta_\lambda = -\partial_\lambda u_3 - b^\mu_\lambda u_\mu.$$

Substituting this expression for the rotation in (3.9) leads to

$$\begin{aligned} E_{\alpha\beta}(\boldsymbol{U}) &= \varepsilon_{\alpha\beta}(\boldsymbol{u}) + \xi^3 \kappa_{\alpha\beta}(\boldsymbol{u}) - (\xi^3)^2 \chi_{\alpha\beta}(\boldsymbol{u}), \\ E_{\alpha 3}(\boldsymbol{U}) &= 0 \end{aligned} \tag{3.14}$$

with

$$\kappa_{\alpha\beta}(\boldsymbol{u}) = -\bar{\kappa}_{\alpha\beta}(\boldsymbol{u}), \tag{3.15}$$

and

$$\bar{\kappa}_{\alpha\beta}(\boldsymbol{u}) = u_{3|\alpha\beta} + b^\mu_{\alpha|\beta} u_\mu + b^\mu_\alpha u_{\mu|\beta} + b^\mu_\beta u_{\mu|\alpha} - c_{\alpha\beta} u_3. \tag{3.16}$$

Note, the tensor $\vec{\vec{\bar{\kappa}}}$ with covariant components $\bar{\boldsymbol{\kappa}}(\boldsymbol{u}) = (\bar{\kappa}_{\alpha\beta})$ can be interpreted as the linearized change of curvature of the midsurface provided the displacement is smooth, see [24]. Since the quadratic term in ξ^3 is dropped in the next step, we skip rewriting $\chi_{\alpha\beta}(\boldsymbol{\theta})$ in terms of $\boldsymbol{u}$. For a thorough derivation of these expressions see, e.g., [24].
For an isotropic linear elastic material Hooke's law in a general curvilinear coordinate system provides for the 3D stress tensor in terms of its contravariant components $\boldsymbol{\sigma} = (\sigma^{ij})$

$$\sigma^{ij} = H^{ijkl} E_{kl},$$

where

$$H^{ijkl} = \lambda g^{ij} g^{kl} + \mu (g^{ik} g^{jl} + g^{il} g^{jk}),$$

with the Lamé constants λ and μ given by

$$\lambda = \frac{\nu E}{(1+\nu)(1-2\nu)} \quad \text{and} \quad \mu = \frac{E}{2(1+\nu)}$$

if we denote Young's modulus by E and Poisson's ratio by ν, as usual. By specializing the constitutive equation to our considered curvilinear coordinate system attached to the midsurface, where we have (3.2), and applying the plane stress assumption $\sigma^{33} = 0$ leads to the 2D constitutive equation

$$\sigma^{\alpha\beta} = C^{\alpha\beta\lambda\mu} E_{\lambda\mu} \tag{3.17}$$

with

$$C^{\alpha\beta\lambda\mu} = \frac{E}{2(1+\nu)} \left(g^{\alpha\lambda} g^{\beta\mu} + g^{\alpha\mu} g^{\beta\lambda} + \frac{2\nu}{1-\nu} g^{\alpha\beta} g^{\lambda\mu} \right). \tag{3.18}$$

The fourth order material tensor is defined by its contravariant components $C^{\alpha\beta\lambda\mu}$. In short we write for $\boldsymbol{\sigma}_{2D} = (\sigma^{\alpha\beta})$ and $\boldsymbol{E}_{2D} = (E_{\alpha\beta})$

$$\boldsymbol{\sigma}_{2D} = \mathcal{C} \boldsymbol{E}_{2D}$$

with the application of the material tensor given by (3.17). The theory presented in the following is independent of the particular structure of the material tensor. We only assume that the material tensor is symmetric and positive definite on symmetric matrices, i.e.,

$$\begin{aligned} \mathcal{C}\boldsymbol{A} : \boldsymbol{B} &= \boldsymbol{A} : \mathcal{C}\boldsymbol{B} && \text{for all } \boldsymbol{A}, \boldsymbol{B} \in \mathbb{R}^{2\times 2}_{sym} \\ \mathcal{C}\boldsymbol{A} : \boldsymbol{A} &> 0 && \text{for all } 0 \neq \boldsymbol{A} \in \mathbb{R}^{2\times 2}_{sym}, \end{aligned}$$

where as before $\mathbb{R}^{2\times 2}_{sym}$ denotes the space of symmetric matrices in $\mathbb{R}^{2\times 2}$. Furthermore, $\lambda_{min}(\mathcal{C})$ and $\lambda_{max}(\mathcal{C})$ denote the minimal and maximal eigenvalue of $\mathcal{C}$, respectively, then

$$\lambda_{min}(\mathcal{C})\, \boldsymbol{A} : \boldsymbol{A} \;\leq\; \mathcal{C}\boldsymbol{A} : \boldsymbol{A} \;\leq\; \lambda_{max}(\mathcal{C})\, \boldsymbol{A} : \boldsymbol{A} \quad \text{for all } \boldsymbol{A} \in \mathbb{R}^{2\times 2}_{sym}.$$

3.1.4 Derivation of the shell model

In this section we proceed analogously to Section 2.1.3, where the special case of plates was considered. Throughout this section all functions are assumed to be sufficiently smooth.

We consider the standard linear elasticity problem in the domain $\vec{\Phi}(\Omega^{3D}) \subset \mathbb{R}^3$ with boundary $\vec{\Phi}(\Sigma)$. The parameter domain $\Omega \subset \mathbb{R}^2$ of the undeformed midsurface is assumed to have a Lipschitz boundary Γ. Let the boundary Γ be written in the form

$$\Gamma = \mathcal{V}_\Gamma \cup \mathcal{E}_\Gamma \quad \text{with } \mathcal{E}_\Gamma = \bigcup_{k=1}^{K} E_k,$$

where E_k, $k = 1, 2, \ldots, K$, are the edges of Γ, considered as open line segments and $\mathcal{V}_\Gamma$ denotes the set of corner points in Γ.

The 3D shell is considered to be clamped on a part $\vec{\Phi}(\Sigma_c) = \vec{\Phi}(\Gamma_c \times (-t/2, t/2))$, simply supported on a part $\vec{\Phi}(\Sigma_s) = \vec{\Phi}(\Gamma_s \times (-t/2, t/2))$, and free on a part $\vec{\Phi}(\Sigma_f) = \vec{\Phi}(\Gamma_f \times (-t/2, t/2))$, with $\Gamma = \Gamma_c \cup \Gamma_s \cup \Gamma_f$ and pairwise disjoint subsets Γ_c, Γ_s and Γ_f. We assume that each edge $E \in \mathcal{E}_\Gamma$ is contained in exactly one of the sets Γ_c, Γ_s, Γ_f, and the edges are maximal in the sense that two edges with the same type of boundary condition do not meet at an angle of π. We suppose that no kinematical boundary conditions for the displacement are prescribed on the top and bottom boundaries of Ω^{3D}. For simplicity, no tractions are prescribed. Body forces are given in terms of contravariant components $\boldsymbol{f} = (f^i)$ in Ω^{3D}.

Using the same arguments as in Section 2.1.3, we obtain the variational formulation: find $\boldsymbol{U} = (\underline{\boldsymbol{U}}, U)$ satisfying the kinematical boundary conditions

$$\begin{aligned} \underline{\boldsymbol{U}} = 0 \quad U_3 = 0 \quad &\text{on } \Sigma_c, \\ \underline{U}_t = 0,\, U_3 = 0 \quad &\text{on } \Sigma_s, \end{aligned} \tag{3.19}$$

such that

$$\int_{\Omega^{3D}} \boldsymbol{\sigma}(\boldsymbol{U}) : \boldsymbol{E}(\boldsymbol{V})\, \sqrt{g}\, dV = \int_{\Omega^{3D}} \boldsymbol{f} \cdot \boldsymbol{V}\, \sqrt{g}\, dV \tag{3.20}$$

for all $\boldsymbol{V}$ satisfying the kinematical boundary conditions.
The Kirchhoff-Love shell model is obtained from (3.20) by substituting the strain representation (3.14) and the constitutive equation for the plane stress state (3.17) and truncating the expressions to the lowest order terms of their expansion with respect to the transverse coordinate ξ^3. Namely, we substitute

$$\begin{aligned} &\varepsilon_{\alpha\beta}(\boldsymbol{u}) - \xi^3\bar{\kappa}_{\alpha\beta}(\boldsymbol{u}) \quad \text{for } E_{\alpha\beta}(\boldsymbol{U}) \quad (\text{see } (3.14)) \\ &\sqrt{a} \quad \text{for } \sqrt{g} \end{aligned}$$

and

$$\frac{E}{2(1+\nu)}\left(a^{\alpha\lambda}a^{\beta\mu} + a^{\alpha\mu}a^{\beta\lambda} + \frac{2\nu}{1-\nu}a^{\alpha\beta}a^{\lambda\mu}\right) \quad \text{for } C^{\alpha\beta\lambda\mu} \quad (\text{cf. } (3.18)), \tag{3.21}$$

where we refer to the truncated components of the material tensor again as $C^{\alpha\beta\lambda\mu}$.
Using these simplifications and performing explicit integration with respect to ξ^3 the following variational formulation is obtained: Find $\boldsymbol{u} = (\underline{\boldsymbol{u}}, u_3)$ satisfying the essential boundary conditions

$$\begin{aligned} \underline{\boldsymbol{u}} &= 0, \quad u_3 = 0,\ \partial_{\boldsymbol{n}} u_3 = 0 \quad && \text{on } \Gamma_c, \\ \underline{u}_t &= 0, \quad u_3 = 0 && \text{on } \Gamma_s \end{aligned} \tag{3.22}$$

such that

$$\int_\Omega \left(t\,\mathcal{C}\boldsymbol{\varepsilon}(\boldsymbol{u}) : \boldsymbol{\varepsilon}(\boldsymbol{v}) + \frac{t^3}{12}\mathcal{C}\bar{\boldsymbol{\kappa}}(\boldsymbol{u}) : \bar{\boldsymbol{\kappa}}(\boldsymbol{v}) \right) \sqrt{a}\; d\xi = \int_\Omega \hat{\boldsymbol{f}} \cdot \boldsymbol{v}\, \sqrt{a}\; d\xi \tag{3.23}$$

for all $\boldsymbol{v}$ satisfying the essential boundary conditions. Here the applied distributed forces are given in terms of their contravariant components $\hat{\boldsymbol{f}} = (\hat{f}^i)$ by

$$\hat{f}^i = \int_{-\frac{t}{2}}^{\frac{t}{2}} f^i\; d\xi^3.$$

Note that for simplicity, we do not include terms involving couples in the just stated formulation (in contrast Section 2.1.3 for plates).
In contrast to plates, the variational problem (3.23) does not separate into a membrane and bending problem, since membrane and bending strain depend on both the tangential and transverse part of the displacement.

Plates as special case of shells

For the special case of a planar midsurface the just introduced Kirchhoff-Love shell model reduces to the plate model of Section 2.1.3. Since there is no curvature in this case, we have a constant surface normal $\vec{a}_3$ and the second and third fundamental form are zero tensors. As a consequence we have

$$g_{\alpha\beta} = a_{\alpha\beta} \quad \text{and} \quad \sqrt{g} = \sqrt{a}.$$

Furthermore, the strain measures reduce to

$$\begin{aligned}\varepsilon_{\alpha\beta}(\boldsymbol{u}) &= \frac{1}{2}(u_{\alpha|\beta} + u_{\beta|\alpha}) \quad \text{(cf. (3.11))}\\ \kappa_{\alpha\beta}(\boldsymbol{u}) &= -u_{3|\alpha\beta} \quad \text{(cf. (3.15))}.\end{aligned}$$

Unlike for a curved midsurface in case of shells, for plates it is always possible to choose an orthonormal (Cartesian) coordinate system to describe the midsurface. The quantities in the shell model become much simpler as then the Christoffel symbols vanish, and, therefore, the covariant derivatives become usual derivatives and $a^{\alpha\beta} = a_{\alpha\beta} = \delta^{\alpha}_{\beta}$. Hence,

$$\begin{aligned}\varepsilon_{\alpha\beta}(\boldsymbol{u}) &= \frac{1}{2}(\partial_\beta u_\alpha + \partial_\beta u_\alpha)\\ \kappa_{\alpha\beta}(\boldsymbol{u}) &= -\partial_{\alpha\beta} u_3\end{aligned}$$

which coincide with the strain components obtained in case of plates in (2.3) and (2.5).

3.2 Variational formulation

So far we have assumed all functions to be sufficiently smooth. In this section we state mathematically precisely what is meant by the term smoothness. To do this, we equip the variational formulation of the shell with appropriate function spaces. For simplicity of notation, we drop in the following the bar in the notation of the bending strain and write $\boldsymbol{\kappa}(\boldsymbol{v})$, where the covariant components are still given by (3.16).
Then the (primal) variational formulation of the shell derived in (3.23) becomes: find $\boldsymbol{u} \in \boldsymbol{W}$ such that

$$\int_\Omega \left(t\, \mathcal{C}\boldsymbol{\varepsilon}(\boldsymbol{u}) : \boldsymbol{\varepsilon}(\boldsymbol{v}) + \frac{t^3}{12}\, \mathcal{C}\boldsymbol{\kappa}(\boldsymbol{u}) : \boldsymbol{\kappa}(\boldsymbol{v}) \right) \sqrt{a}\, d\xi = \langle F, \boldsymbol{v} \rangle \quad \text{for all } \boldsymbol{v} \in \boldsymbol{W}, \tag{3.24}$$

with the right-hand side $\langle F, \boldsymbol{v}\rangle = \int_\Omega \hat{\boldsymbol{f}} \cdot \boldsymbol{v}\, \sqrt{a}\, d\xi$. Here the function space is given by

$$\begin{aligned}\boldsymbol{W} &= W_1 \times W_2 \times W_3\\ &= \{\boldsymbol{v} = (v_i) \in H^1(\Omega) \times H^1(\Omega) \times H^2(\Omega), \quad v_i = 0,\ \partial_{\boldsymbol{n}} v_3 = 0 \text{ on } \Gamma_c,\\ &\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \underline{v}_t = v_3 = 0 \text{ on } \Gamma_s\}\end{aligned} \tag{3.25}$$

with associated norm

$$\|\boldsymbol{v}\|^2_{\boldsymbol{W}} = \|v_1\|^2_1 + \|v_2\|^2_1 + \|v_3\|^2_2.$$

Keep in mind, for the tangential displacement $\underline{\boldsymbol{u}} = (u_1, u_2)$ the space $(H^1(\Omega))^2$ is sufficient, only for the transverse displacement u_3 we need higher smoothness, namely, $H^2(\Omega)$. Furthermore, the space W_3 coincides with the space W in (2.22) of the plate bending problem.
In order to avoid technicalities, we assume that the essential boundary conditions in $\boldsymbol{W}$ are such that no rigid body motion is possible, i.e., the only element $\boldsymbol{v} \in \boldsymbol{W}$ that

satisfies $\boldsymbol{\varepsilon}(\boldsymbol{v}) = \boldsymbol{\kappa}(\boldsymbol{v}) = 0$ is $\boldsymbol{v} = 0$. Then existence and uniqueness of a solution $\boldsymbol{u}$ to (3.24) are guaranteed by the theorem of Lax-Milgram (see, e.g., [24, 28]) for even more general right-hand sides $\langle F, \boldsymbol{v}\rangle$, where $F \in \boldsymbol{W}^*$. Moreover, the solution $\boldsymbol{u}$ depends continuously on F

$$\|\boldsymbol{u}\|_{\boldsymbol{W}} \leq c_{st} \|F\|_{\boldsymbol{W}^*}. \tag{3.26}$$

3.3 The $\boldsymbol{M}$-mixed variational formulation

We introduce as a new unknown the bending moment tensor $\vec{\vec{M}}$, whose contravariant components $\boldsymbol{M} = (M^{\alpha\beta})$ are related to the bending strain through the constitutive equation

$$M^{\alpha\beta} = \sqrt{a}\frac{t^3}{12}\, C^{\alpha\beta\lambda\mu}\kappa_{\lambda\mu}(\boldsymbol{u}),$$

or in short

$$\boldsymbol{M} = \sqrt{a}\frac{t^3}{12}\, \mathcal{C}\boldsymbol{\kappa}(\boldsymbol{u}) = \hat{\mathcal{C}}_{\boldsymbol{M}}\boldsymbol{\kappa}(\boldsymbol{u}), \text{ with } \hat{\mathcal{C}}_{\boldsymbol{M}} = \sqrt{a}\frac{t^3}{12}\, \mathcal{C}.$$

Note, in contrast to standard notation we additionally include in $\boldsymbol{M}$ the geometry measure $\sqrt{a}$. This leads, analogously as in Section 2.3, to a first $\boldsymbol{M}$-mixed formulation: find $\boldsymbol{M} \in \boldsymbol{L}^2(\Omega)_{\text{sym}}$ and $\boldsymbol{u} \in \boldsymbol{W}$ such that

$$\begin{aligned} \int_\Omega \hat{\mathcal{C}}_{\boldsymbol{M}}^{-1}\boldsymbol{M} : \boldsymbol{L}\, d\xi \quad - \int_\Omega \boldsymbol{\kappa}(\boldsymbol{u}) : \boldsymbol{L}\, d\xi &= 0 && \text{for all } \boldsymbol{L} \in \boldsymbol{L}^2(\Omega)_{\text{sym}}, \\ -\int_\Omega \boldsymbol{M} : \boldsymbol{\kappa}(\boldsymbol{v})\, d\xi - c(\boldsymbol{u}, \boldsymbol{v}) &= -\langle F, \boldsymbol{v}\rangle && \text{for all } \boldsymbol{v} \in \boldsymbol{W} \end{aligned} \tag{3.27}$$

with $c(\boldsymbol{u}, \boldsymbol{v})$ given by the membrane part

$$c(\boldsymbol{u}, \boldsymbol{v}) = \int_\Omega t\, \mathcal{C}\boldsymbol{\varepsilon}(\boldsymbol{u}) : \boldsymbol{\varepsilon}(\boldsymbol{v})\, \sqrt{a}\, d\xi,$$

where the first line results from the relation $\hat{\mathcal{C}}_{\boldsymbol{M}}^{-1}\boldsymbol{M} - \boldsymbol{\kappa}(\boldsymbol{u}) = 0$ and the second line originates from the primal variational formulation (3.24). This first mixed formulation is trivially equivalent to the original problem, but brings no advantage.

The goal of this section is, as in Section 2.3, to derive a reformulation of (3.27) that allows us to use a different space for the displacement $\boldsymbol{u}$ (and test functions $\boldsymbol{v}$) with less smoothness. The obtained new mixed formulation will be based on a displacement space that uses $H^1(\Omega)$ for all three components of the displacement. In Chapter 2 a new mixed formulation for Kirchhoff plates using $H^1(\Omega)$ for the transverse displacement u_3 was introduced. An extension of this approach to shells is possible, since the only term involving second-order derivatives is the Hessian of the transverse displacement u_3.

We divide the bending strain $\boldsymbol{\kappa}(\boldsymbol{v})$ into the Hessian of the transverse displacement v_3 and the remaining terms that only involve first-order derivatives of $\boldsymbol{v}$ denoted by $\boldsymbol{\kappa}^1(\boldsymbol{v})$, i.e.,

$$\boldsymbol{\kappa}(\boldsymbol{v}) = \nabla^2 v_3 + \boldsymbol{\kappa}^1(\boldsymbol{v}).$$

With this notation we can rewrite the bilinear form

$$b(\boldsymbol{L}, \boldsymbol{v}) = -\int_\Omega \boldsymbol{\kappa}(\boldsymbol{v}) : \boldsymbol{L}\ d\xi$$

which appears in (3.27) separating the integral involving $\nabla^2 v_3$ and the remaining terms

$$b(\boldsymbol{L}, \boldsymbol{v}) = b_0(\boldsymbol{L}, v_3) + b_1(\boldsymbol{L}, \boldsymbol{v})$$

with

$$b_0(\boldsymbol{L}, v_3) = -\int_\Omega \nabla^2 v_3 : \boldsymbol{L}\ d\xi \quad \text{and} \quad b_1(\boldsymbol{L}, \boldsymbol{v}) = -\int_\Omega \boldsymbol{\kappa}^1(\boldsymbol{v}) : \boldsymbol{L}\ d\xi.$$

For the further considerations observe the following:

- For the first part b_0 the associated operator B_0 given by

$$\langle B_0 v_3, \boldsymbol{L}\rangle = b_0(\boldsymbol{L}, v_3) = \int_\Omega \nabla^2 v_3 : \boldsymbol{L}\ d\xi$$

  coincides with the operator B considered in Section 2.3. Therefore, we will apply for this term the techniques developed there.

- For the second part b_1 we do not need further considerations since it is well defined for $\boldsymbol{v} \in (H^1(\Omega))^3$ and $\boldsymbol{L} \in \boldsymbol{L}^2(\Omega)_{\text{sym}}$ and bounded: There is a constant $\|b_1\|$ such that

$$|b_1(\boldsymbol{L}, \boldsymbol{v})| = \left| \int_\Omega \boldsymbol{\kappa}^1(\boldsymbol{v}) : \boldsymbol{L}\ d\xi \right| \leq \|b_1\| \|\boldsymbol{v}\|_1 \|\boldsymbol{L}\|_0 \tag{3.28}$$

  for all $\boldsymbol{v} \in (H^1(\Omega))^3$ and $\boldsymbol{L} \in \boldsymbol{L}^2(\Omega)_{\text{sym}}$, where the constant $\|b_1\|$ depends on geometry quantities. Note, an estimate of this forms holds since $\boldsymbol{\kappa}^1(\boldsymbol{v})$ is a linear expression in terms of $\boldsymbol{v}$ and first-order derivatives of $\boldsymbol{v}$ with $L^\infty(\Omega)$ coefficients.

- The c bilinear form is well defined for $\boldsymbol{u} \in (H^1(\Omega))^3$ and $\boldsymbol{v} \in (H^1(\Omega))^3$ and bounded: There is a constant $\|c\|$ such that

$$|c(\boldsymbol{u}, \boldsymbol{v})| = \left| \int_\Omega t\ \mathcal{C}\boldsymbol{\varepsilon}(\boldsymbol{u}) : \boldsymbol{\varepsilon}(\boldsymbol{v})\ \sqrt{a}\ d\xi \right| \leq \|c\|\ \|\boldsymbol{u}\|_1 \|\boldsymbol{v}\|_1 \tag{3.29}$$

  for all $\boldsymbol{u} \in (H^1(\Omega))^3$ and $\boldsymbol{v} \in (H^1(\Omega))^3$, where the constant $\|c\|$ depends on geometry quantities and material parameters. As above, an estimate of this forms holds since $\boldsymbol{\varepsilon}(\boldsymbol{v})$ is a linear expression in terms of $\boldsymbol{v}$ and first-order derivatives of $\boldsymbol{v}$ with $L^\infty(\Omega)$ coefficients.

3.3.1 Derivation of the new mixed formulation

The approach to derive the new mixed formulation is analogous to Section 2.3.3 based on the abstract framework of densely defined (possibly unbounded) linear operators introduced in Section 2.3.2. Therefore, we just state the needed definitions using the notation of the current chapter and refer for motivation and discussion of the setting to Section 2.3.3.
We choose $D(B_0) = W_3$ and define $B_0 = \nabla^2$ as an operator mapping from W_3 to $Y^* = (\boldsymbol{L}^2(\Omega)_{\text{sym}})^*$ given by

$$\langle \nabla^2 v, \boldsymbol{L} \rangle = \int_\Omega \nabla^2 v : \boldsymbol{L} \, d\xi \quad \text{for } v \in D(B_0) = W_3, \; \boldsymbol{L} \in Y = \boldsymbol{L}^2(\Omega)_{\text{sym}}. \tag{3.30}$$

Furthermore, we set the base space X equal to Q_3 given by

$$Q_3 = H^1_{0,\Gamma_c \cup \Gamma_s}(\Omega) = \{v \in H^1(\Omega) : v = 0 \text{ on } \Gamma_c \cup \Gamma_s\},$$

equipped with the norm $\|v\|_{Q_3} = \|v\|_1$. Note, the space Q_3 coincides with the space Q in (2.31) for the plate bending problem. From now on we use the notations

$$\operatorname{div}\operatorname{Div} \quad \text{and} \quad \boldsymbol{H}(\operatorname{div}\operatorname{Div}, \Omega; Q_3^*)_{\text{sym}} \tag{3.31}$$

instead of B_0^* and $D(B_0^*)$, respectively. In consistence with the abstract framework elements of the Hilbert space $\boldsymbol{H}(\operatorname{div}\operatorname{Div}, \Omega; Q_3^*)_{\text{sym}}$ are characterized in the following way (cf.(2.27)): $\boldsymbol{L} \in \boldsymbol{H}(\operatorname{div}\operatorname{Div}, \Omega; Q_3^*)_{\text{sym}}$ if and only if $\boldsymbol{L} \in \boldsymbol{L}^2(\Omega)_{\text{sym}}$ and there is a linear functional $G \in Q_3^*$ such that

$$\langle \nabla^2 v, \boldsymbol{L} \rangle = \int_\Omega \nabla^2 v : \boldsymbol{L} \, dx = \langle G, v \rangle \quad \text{for all } v \in W_3. \tag{3.32}$$

For $\boldsymbol{L} \in \boldsymbol{H}(\operatorname{div}\operatorname{Div}, \Omega; Q_3^*)_{\text{sym}}$ we set $\operatorname{div}\operatorname{Div}\boldsymbol{L} = G$. More explicitly the space $\boldsymbol{H}(\operatorname{div}\operatorname{Div}, \Omega; Q_3^*)_{\text{sym}}$ is given by (cf. (2.28))

$$\begin{aligned} \boldsymbol{H}(\operatorname{div}\operatorname{Div}, \Omega; Q_3^*)_{\text{sym}} = \{&\boldsymbol{L} \in \boldsymbol{L}^2(\Omega)_{\text{sym}} \colon \text{ the functional} \\ &G\colon v \mapsto \int_\Omega \nabla^2 v : \boldsymbol{L} \, d\xi, \; v \in W_3, \text{ is bounded w.r.t. the } Q_3\text{-norm}\}, \end{aligned} \tag{3.33}$$

equipped with the norm

$$\|\boldsymbol{L}\|_{\operatorname{div}\operatorname{Div};Q_3^*} = (\|\boldsymbol{L}\|_0^2 + \|\operatorname{div}\operatorname{Div}\boldsymbol{L}\|_{Q_3^*}^2)^{1/2}.$$

Furthermore, we have the following well-known relation for an operator and its adjoint (cf. (2.29)):

$$\langle \operatorname{div}\operatorname{Div}\boldsymbol{L}, v \rangle = \langle \nabla^2 v, \boldsymbol{L} \rangle \quad \text{for all } v \in W_3, \; \boldsymbol{L} \in \boldsymbol{H}(\operatorname{div}\operatorname{Div}, \Omega; Q_3^*)_{\text{sym}}. \tag{3.34}$$

This motivates the $\boldsymbol{M}$-mixed formulation as follows: For $F \in \boldsymbol{Q}^*$, find $\boldsymbol{M} \in \boldsymbol{V}$ and $\boldsymbol{u} \in \boldsymbol{Q}$ such that

$$\begin{aligned} &\int_\Omega \hat{\mathcal{C}}_{\boldsymbol{M}}^{-1} \boldsymbol{M} : \boldsymbol{L} \, d\xi && - \langle \operatorname{div} \operatorname{Div} \boldsymbol{L}, u_3 \rangle + b_1(\boldsymbol{L}, \boldsymbol{u}) = 0 \\ &- \langle \operatorname{div} \operatorname{Div} \boldsymbol{M}, v_3 \rangle + b_1(\boldsymbol{M}, \boldsymbol{v}) - c(\boldsymbol{u}, \boldsymbol{v}) && = -\langle F, \boldsymbol{v} \rangle \end{aligned} \tag{3.35}$$

for all $\boldsymbol{L} \in \boldsymbol{V}$ and $\boldsymbol{v} \in \boldsymbol{Q}$ with the function spaces

$$\begin{aligned} \boldsymbol{V} &= \boldsymbol{H}(\operatorname{div} \operatorname{Div}, \Omega; Q_3^*)_{\text{sym}}, \\ \boldsymbol{Q} &= W_1 \times W_2 \times Q_3 \subset H^1(\Omega) \times H^1(\Omega) \times H^1(\Omega), \end{aligned} \tag{3.36}$$

equipped with the norms

$$\|\boldsymbol{L}\|_{\boldsymbol{V}} = \|\boldsymbol{L}\|_{\operatorname{div} \operatorname{Div}; Q_3^*} \quad \text{and} \quad \|\boldsymbol{v}\|_{\boldsymbol{Q}} = \|\boldsymbol{v}\|_1. \tag{3.37}$$

Note, in comparison with the first mixed formulation (3.27) the bilinear forms c and b_1 remain unchanged in the new mixed formulation (3.35).

Remark 3.1. *Note, in $\boldsymbol{Q}$ compared with $\boldsymbol{W}$ only those boundary conditions for the transverse displacement u_3 which are also available in $H^1(\Omega)$ are prescribed. In the formulation* (3.36) *the originally essential boundary condition $\partial_{\boldsymbol{n}} u_3 = 0$ becomes a natural condition. In the primal formulation* (3.24) *only natural boundary conditions are imposed for the bending moment tensor $\boldsymbol{M}$. Those natural boundary conditions corresponding to the normal-normal component of $\boldsymbol{M}$ and the corner conditions for the normal-tangential component of $\boldsymbol{M}$ become essential conditions and the remaining condition stays natural. For further information we refer the reader to the detailed discussion of the nonstandard Sobolev space $\boldsymbol{H}(\operatorname{div} \operatorname{Div}, \Omega; Q_3^*)_{\text{sym}}$ in Section 2.3.5.*

3.3.2 Mathematical analysis

In this section we show well-posedness of our new mixed formulation. Problem (3.35) has the typical structure of a saddle point problem

$$\begin{aligned} a(\boldsymbol{M}, \boldsymbol{L}) + b(\boldsymbol{L}, \boldsymbol{u}) &= \langle F_{\boldsymbol{L}}, \boldsymbol{L} \rangle \quad \text{for all } \boldsymbol{L} \in \boldsymbol{V}, \\ b(\boldsymbol{M}, \boldsymbol{v}) \ - c(\boldsymbol{u}, \boldsymbol{v}) &= \langle F_{\boldsymbol{v}}, \boldsymbol{v} \rangle \quad \text{for all } \boldsymbol{v} \in \boldsymbol{Q}, \end{aligned}$$

whose associated linear operator $\mathcal{A}\colon \boldsymbol{V} \times \boldsymbol{Q} \longrightarrow (\boldsymbol{V} \times \boldsymbol{Q})^*$ is given by

$$\langle \mathcal{A}(\boldsymbol{M}, \boldsymbol{u}), (\boldsymbol{L}, \boldsymbol{v}) \rangle = a(\boldsymbol{M}, \boldsymbol{L}) + b(\boldsymbol{L}, \boldsymbol{u}) + b(\boldsymbol{M}, \boldsymbol{v}) - c(\boldsymbol{u}, \boldsymbol{v}).$$

Theorem 3.2. *If the bilinear forms a and c are symmetric, i.e., $a(\boldsymbol{M}, \boldsymbol{L}) = a(\boldsymbol{L}, \boldsymbol{M})$ and $c(\boldsymbol{u}, \boldsymbol{v}) = c(\boldsymbol{v}, \boldsymbol{u})$, and non-negative, i.e., $a(\boldsymbol{L}, \boldsymbol{L}) \geq 0$ and $c(\boldsymbol{v}, \boldsymbol{v}) \geq 0$, then $\mathcal{A}$ is an isomorphism from $\boldsymbol{V} \times \boldsymbol{Q}$ onto $(\boldsymbol{V} \times \boldsymbol{Q})^*$, if and only if the following conditions are satisfied:*

1. *a is bounded: There is a constant $\|a\| > 0$ such that*

$$|a(\boldsymbol{M}, \boldsymbol{L})| \leq \|a\| \, \|\boldsymbol{M}\|_{\boldsymbol{V}} \, \|\boldsymbol{L}\|_{\boldsymbol{V}} \quad \textit{for all } \boldsymbol{M}, \ \boldsymbol{L} \in \boldsymbol{V}.$$

2. *b is bounded: There is a constant $\|b\| > 0$ such that*

$$|b(\boldsymbol{L}, \boldsymbol{v})| \leq \|b\| \, \|\boldsymbol{L}\|_{\boldsymbol{V}} \|\boldsymbol{v}\|_{\boldsymbol{Q}} \quad \textit{for all } \boldsymbol{L} \in \boldsymbol{V}, \ \boldsymbol{v} \in \boldsymbol{Q}.$$

3. *c is bounded: There is a constant $\|c\| > 0$ such that*

$$|c(\boldsymbol{u}, \boldsymbol{v})| \leq \|c\| \, \|\boldsymbol{u}\|_{\boldsymbol{Q}} \|\boldsymbol{v}\|_{\boldsymbol{Q}} \quad \textit{for all } \boldsymbol{u} \in \boldsymbol{Q}, \ \boldsymbol{v} \in \boldsymbol{Q}.$$

4. *There is a constant $\alpha > 0$ such that*

$$a(\boldsymbol{L}, \boldsymbol{L}) + \left(\sup_{0 \neq \boldsymbol{v} \in \boldsymbol{Q}} \frac{b(\boldsymbol{L}, \boldsymbol{v})}{\|\boldsymbol{v}\|_{\boldsymbol{Q}}} \right)^2 \geq \alpha \, \|\boldsymbol{L}\|^2_{\boldsymbol{V}} \quad \textit{for all } \boldsymbol{L} \in \boldsymbol{V}. \tag{3.38}$$

5. *There is a constant $\beta > 0$ such that*

$$c(\boldsymbol{v}, \boldsymbol{v}) + \left(\sup_{0 \neq \boldsymbol{L} \in \boldsymbol{V}} \frac{b(\boldsymbol{L}, \boldsymbol{v})}{\|\boldsymbol{L}\|_{\boldsymbol{V}}} \right)^2 \geq \beta \, \|\boldsymbol{v}\|^2_{\boldsymbol{Q}} \quad \textit{for all } \boldsymbol{v} \in \boldsymbol{Q}.$$

Proof. For the proof see [74] and also [15, Theorem 4.3.1]. □

(As in Chapter 2, we tacitly assume that $\|a\|$, $\|b\|$, and $\|c\|$ are the smallest constants for estimating the bilinear forms a, b, and c. Then $\|a\|$, $\|b\|$, and $\|c\|$ match the standard notation for the norms of the bilinear forms a, b, and c.)
Note that the assumption in Theorem 3.2 concerning symmetry and non-negativity of the bilinear forms a and c are satisfied for (3.35). In order to verify the just stated well-posedness conditions for (3.35), we need the following result on the relation between the primal problem (3.24) and the new mixed problem (3.35).

Theorem 3.3. *Let $\boldsymbol{u} \in \boldsymbol{W}$ be the solution of the primal problem* (3.24) *for $F \in \boldsymbol{Q}^*$. Then we have $\boldsymbol{M} = \hat{\mathcal{C}}_{\boldsymbol{M}} \boldsymbol{\kappa}(\boldsymbol{u}) \ \in \boldsymbol{V}$ and $(\boldsymbol{M}, \boldsymbol{u})$ solves the mixed problem* (3.35).

Proof. Since $\boldsymbol{u} \in \boldsymbol{W}$ solves (3.24), it follows that $\boldsymbol{M} \in \boldsymbol{L}^2(\Omega)_{\text{sym}}$ and using the splitting $\boldsymbol{\kappa}(\boldsymbol{v}) = \nabla^2 v_3 + \boldsymbol{\kappa}^1(\boldsymbol{v})$ we obtain

$$\langle \nabla^2 v_3, \boldsymbol{M} \rangle = \int_\Omega \boldsymbol{M} : \nabla^2 v_3 \, d\xi = \langle F, \boldsymbol{v} \rangle - c(\boldsymbol{u}, \boldsymbol{v}) - \int_\Omega \boldsymbol{M} : \boldsymbol{\kappa}^1(\boldsymbol{v}) \, d\xi \quad \text{for all } \boldsymbol{v} \in \boldsymbol{W}. \tag{3.39}$$

In the first step, we show that the second equation of (3.35) is satisfied for all $\boldsymbol{v} \in \boldsymbol{Q}$. Note, $\boldsymbol{Q}$ is given by $W_1 \times W_2 \times Q_3$, hence, for $\boldsymbol{v} \in \boldsymbol{Q}$ we have

$$\boldsymbol{v} = (v_1, v_2, v_3) = (v_1, v_2, 0) + (0, 0, v_3),$$

with $v_1 \in W_1$, $v_2 \in W_2$ and $v_3 \in Q_3$. The equation will be obtained by combining results for $\boldsymbol{v} = (v_1, v_2, 0) \in \boldsymbol{W}$ and $\boldsymbol{v} = (0, 0, v_3)$ with $v_3 \in Q_3$.
First, we consider (3.39) for test functions $\boldsymbol{v}$ of the form $\boldsymbol{v} = (0, 0, v_3) \in \boldsymbol{W}$. Note, the functional on the right hand side of (3.39) given by

$$\langle G, v_3 \rangle = \langle F, \boldsymbol{v} \rangle - c(\boldsymbol{u}, \boldsymbol{v}) - \int_\Omega \boldsymbol{M} : \boldsymbol{\kappa}^1(\boldsymbol{v}) \, d\xi \quad \text{for } \boldsymbol{v} = (0, 0, v_3)$$

is bounded with respect to the H^1-norm, i.e., $G \in Q_3^*$. Hence, we obtain from the definition of $\boldsymbol{V}$ in (3.32) that $\boldsymbol{M} \in \boldsymbol{V}$ with

$$\langle \operatorname{div} \operatorname{Div} \boldsymbol{M}, v_3 \rangle = \langle G, v_3 \rangle = \langle F, \boldsymbol{v} \rangle - c(\boldsymbol{u}, \boldsymbol{v}) - \int_\Omega \boldsymbol{M} : \boldsymbol{\kappa}^1(\boldsymbol{v}) \, d\xi$$

for all $\boldsymbol{v} = (0, 0, v_3)$ with $v_3 \in Q_3$. Next, we consider (3.39) with test functions of the form $\boldsymbol{v} = (v_1, v_2, 0)$ and obtain

$$0 = \langle F, \boldsymbol{v} \rangle - c(\boldsymbol{u}, \boldsymbol{v}) - \int_\Omega \boldsymbol{M} : \boldsymbol{\kappa}^1(\boldsymbol{v}) \, d\xi \quad \text{for all } \boldsymbol{v} = (v_1, v_2, 0) \in \boldsymbol{W}.$$

By combining the last two equations we obtain

$$\langle \operatorname{div} \operatorname{Div} \boldsymbol{M}, v_3 \rangle = \langle F, \boldsymbol{v} \rangle - c(\boldsymbol{u}, \boldsymbol{v}) - \int_\Omega \boldsymbol{M} : \boldsymbol{\kappa}^1(\boldsymbol{v}) \, d\xi \quad \text{for all } \boldsymbol{v} \in \boldsymbol{Q}.$$

This shows that the second equation of (3.35) is satisfied.
Since $u_3 \in W_3$, we receive from (3.34) and $\boldsymbol{M} = \hat{\mathcal{C}}_{\boldsymbol{M}}(\nabla^2 u_3 + \boldsymbol{\kappa}^1(\boldsymbol{u}))$ that

$$\langle \operatorname{div} \operatorname{Div} \boldsymbol{L}, u_3 \rangle = \langle \nabla^2 u_3, \boldsymbol{L} \rangle = \int_\Omega \boldsymbol{L} : \nabla^2 u_3 \, d\xi = \int_\Omega \boldsymbol{L} : \hat{\mathcal{C}}_{\boldsymbol{M}}^{-1} \boldsymbol{M} \, d\xi - \int_\Omega \boldsymbol{L} : \boldsymbol{\kappa}^1(\boldsymbol{u}) \, d\xi$$

for all $\boldsymbol{L} \in \boldsymbol{V}$, which proves the first equation of (3.35). □

In the following we use a second norm for $\boldsymbol{L} \in \boldsymbol{V} = \boldsymbol{H}(\operatorname{div} \operatorname{Div}, \Omega; Q_3^*)_{\text{sym}}$ beside the graph norm

$$\|\boldsymbol{L}\|_{\boldsymbol{V}}^2 = \|\boldsymbol{L}\|_{\operatorname{div} \operatorname{Div}; Q_3^*}^2 = \|\boldsymbol{L}\|_0^2 + \sup_{0 \neq v_3 \in Q_3} \left(\frac{\langle \operatorname{div} \operatorname{Div} \boldsymbol{L}, v_3 \rangle}{\|v_3\|_1} \right)^2 \quad \text{(cf. (3.37))},$$

namely:

$$\|\boldsymbol{L}\|_{\boldsymbol{V},b}^2 = \|\boldsymbol{L}\|_0^2 + \sup_{0 \neq \boldsymbol{v} \in \boldsymbol{Q}} \left(\frac{b(\boldsymbol{L}, \boldsymbol{v})}{\|\boldsymbol{v}\|_1} \right)^2 = \|\boldsymbol{L}\|_0^2 + \sup_{0 \neq \boldsymbol{v} \in \boldsymbol{Q}} \left(\frac{\langle \operatorname{div} \operatorname{Div} \boldsymbol{L}, v_3 \rangle + b_1(\boldsymbol{L}, \boldsymbol{v})}{\|\boldsymbol{v}\|_1} \right)^2, \tag{3.40}$$

whose definition is motivated by (3.38). The next lemma provides the equivalence of the two norms for $\boldsymbol{L} \in \boldsymbol{V} = \boldsymbol{H}(\operatorname{div} \operatorname{Div}, \Omega; Q_3^*)_{\text{sym}}$.

Lemma 3.4. *For* $\boldsymbol{L} \in \boldsymbol{V} = \boldsymbol{H}(\operatorname{div}\operatorname{Div}, \Omega; Q_3^*)_{\mathrm{sym}}$ *the norm* $\|\boldsymbol{L}\|_{\boldsymbol{V},b}$ *and the norm* $\|\boldsymbol{L}\|_{\boldsymbol{V}}$ *are equivalent. More precisely*

$$\underline{c}_{\boldsymbol{V},b}\, \|\boldsymbol{L}\|_{\operatorname{div}\operatorname{Div};Q_3^*} \le \|\boldsymbol{L}\|_{\boldsymbol{V},b} \le \overline{c}_{\boldsymbol{V},b}\, \|\boldsymbol{L}\|_{\operatorname{div}\operatorname{Div};Q_3^*} \quad \textit{for all}\ \boldsymbol{L} \in \boldsymbol{V} = \boldsymbol{H}(\operatorname{div}\operatorname{Div}, \Omega; Q_3^*)_{\mathrm{sym}}$$

with the constants

$$\overline{c}^2_{\boldsymbol{V},b} = \max\{1 + 2\,\|b_1\|^2, 2\} \quad \textit{and} \quad \underline{c}^2_{\boldsymbol{V},b} = \frac{1}{2(2 + \|b_1\|^2)}.$$

Proof. Let $\boldsymbol{L} \in \boldsymbol{V} = \boldsymbol{H}(\operatorname{div}\operatorname{Div}, \Omega; Q_3^*)_{\mathrm{sym}}$. Then we obtain using (3.28) and $\|\boldsymbol{v}\|_1 \ge \|v_3\|_1$ the following estimate from above

$$\begin{aligned}
\|\boldsymbol{L}\|^2_{\boldsymbol{V},b} &= \|\boldsymbol{L}\|_0^2 + \sup_{0 \ne \boldsymbol{v} \in \boldsymbol{Q}} \left(\frac{\langle \operatorname{div}\operatorname{Div} \boldsymbol{L}, v_3 \rangle + b_1(\boldsymbol{L}, \boldsymbol{v})}{\|\boldsymbol{v}\|_1} \right)^2 \\
&\le \|\boldsymbol{L}\|_0^2 + \sup_{0 \ne \boldsymbol{v} \in \boldsymbol{Q}} \left(\frac{\langle \operatorname{div}\operatorname{Div} \boldsymbol{L}, v_3 \rangle}{\|\boldsymbol{v}\|_1} + \|b_1\| \frac{\|\boldsymbol{L}\|_0 \|\boldsymbol{v}\|_1}{\|\boldsymbol{v}\|_1} \right)^2 \\
&\le \|\boldsymbol{L}\|_0^2 + 2 \left(\sup_{0 \ne \boldsymbol{v} \in \boldsymbol{Q}} \left(\frac{\langle \operatorname{div}\operatorname{Div} \boldsymbol{L}, v_3 \rangle}{\|\boldsymbol{v}\|_1} \right)^2 + \|b_1\|^2 \|\boldsymbol{L}\|_0^2 \right) \\
&\le (1 + 2\,\|b_1\|^2) \|\boldsymbol{L}\|_0^2 + 2 \sup_{0 \ne v_3 \in Q_3} \left(\frac{\langle \operatorname{div}\operatorname{Div} \boldsymbol{L}, v_3 \rangle}{\|v_3\|_1} \right)^2 \\
&\le \max\{1 + 2\,\|b_1\|^2, 2\} \left(\|\boldsymbol{L}\|_0^2 + \sup_{0 \ne v_3 \in Q_3} \left(\frac{\langle \operatorname{div}\operatorname{Div} \boldsymbol{L}, v_3 \rangle}{\|v_3\|_1} \right)^2 \right).
\end{aligned}$$

For the estimate from below note that $\|\boldsymbol{L}\|_{\boldsymbol{V},b}$ can be bounded from below by its first part

$$\|\boldsymbol{L}\|^2_{\boldsymbol{V},b} \ge \|\boldsymbol{L}\|_0^2 \tag{3.41}$$

and its second part

$$\|\boldsymbol{L}\|^2_{\boldsymbol{V},b} \ge \sup_{0 \ne \boldsymbol{v} \in \boldsymbol{Q}} \frac{1}{\|\boldsymbol{v}\|_1^2} \left(\langle \operatorname{div}\operatorname{Div} \boldsymbol{L}, v_3 \rangle + b_1(\boldsymbol{L}, \boldsymbol{v}) \right)^2.$$

For further estimating the second part we use that for $x, y \in \mathbb{R}$ there holds

$$(x + y)^2 \ge \frac{1}{2} x^2 - y^2 \tag{3.42}$$

since

$$x^2 = (x + y - y)^2 \le 2 \left((x + y)^2 + y^2 \right).$$

Then we can estimate the second part from below using (3.28) as follows

$$\begin{aligned}
\|\boldsymbol{L}\|^2_{\boldsymbol{V},b} &\ge \sup_{0 \ne \boldsymbol{v} \in \boldsymbol{Q}} \frac{1}{\|\boldsymbol{v}\|_1^2} \left(\frac{1}{2} \langle \operatorname{div}\operatorname{Div} \boldsymbol{L}, v_3 \rangle^2 - b_1(\boldsymbol{L}, \boldsymbol{v})^2 \right) \\
&\ge \sup_{0 \ne \boldsymbol{v} \in \boldsymbol{Q}} \left(\frac{1}{2} \left(\frac{\langle \operatorname{div}\operatorname{Div} \boldsymbol{L}, v_3 \rangle}{\|\boldsymbol{v}\|_1} \right)^2 - \|b_1\|^2 \|\boldsymbol{L}\|_0^2 \right).
\end{aligned} \tag{3.43}$$

By multiplying the estimate (3.41) with $(1+\|b_1\|^2)$ and adding it to (3.43) we receive

$$(2+\|b_1\|^2)\,\|\boldsymbol{L}\|^2_{\boldsymbol{V},b} \geq \|\boldsymbol{L}\|_0^2 + \sup_{0\neq \boldsymbol{v}\in \boldsymbol{Q}} \frac{1}{2}\left(\frac{\langle \operatorname{div}\operatorname{Div}\boldsymbol{L}, v_3\rangle}{\|\boldsymbol{v}\|_1}\right)^2.$$

With the choice $\boldsymbol{v}=(0,0,v_3)$ we end up with

$$\|\boldsymbol{L}\|^2_{\boldsymbol{V},b} \geq \frac{1}{2(2+\|b_1\|^2)}\left(\|\boldsymbol{L}\|_0^2 + \sup_{0\neq v_3\in Q_3}\left(\frac{\langle \operatorname{div}\operatorname{Div}\boldsymbol{L}, v_3\rangle}{\|v_3\|_1}\right)^2\right).$$

□

Theorem 3.5. *The bilinear forms*

$$\begin{aligned}
a(\boldsymbol{M},\boldsymbol{L}) &= \int_\Omega \hat{\mathcal{C}}_{\boldsymbol{M}}^{-1}\boldsymbol{M} : \boldsymbol{L}\; d\xi,\\
b(\boldsymbol{L},\boldsymbol{v}) &= -\langle \operatorname{div}\operatorname{Div}\boldsymbol{L}, v_3\rangle - \int_\Omega \boldsymbol{L} : \boldsymbol{\kappa}^1(\boldsymbol{v})\; d\xi,\\
c(\boldsymbol{u},\boldsymbol{v}) &= \int_\Omega t\; \mathcal{C}\boldsymbol{\varepsilon}(\boldsymbol{u}) : \boldsymbol{\varepsilon}(\boldsymbol{v})\; \sqrt{a}\; d\xi
\end{aligned}$$

satisfy the conditions for well-posedness stated in Theorem 3.2 on the spaces $\boldsymbol{V} = \boldsymbol{H}(\operatorname{div}\operatorname{Div},\Omega;Q_3^*)_{\text{sym}}$ *and* $\boldsymbol{Q}$*, equipped with the norms* $\|\boldsymbol{L}\|_{\boldsymbol{V}} = \|\boldsymbol{L}\|_{\operatorname{div}\operatorname{Div};Q_3^*}$ *and* $\|\boldsymbol{v}\|_{\boldsymbol{Q}} = \|\boldsymbol{v}\|_1$*, respectively, with the constants* $\|a\| = 1/\lambda_{min}(\hat{\mathcal{C}}_{\boldsymbol{M}})$*,* $\|b\| = \overline{c}_{\boldsymbol{V},b}$*,* $\|c\|$ *as in* (3.29)*,*

$$\alpha = \underline{c}^2_{\boldsymbol{V},b}\min\{1, 1/\lambda_{max}(\hat{\mathcal{C}}_{\boldsymbol{M}})\} \quad \textit{and} \quad \beta = \frac{c'\,\underline{c}^2_{\boldsymbol{V},b}}{2(1+c'\,\underline{c}^2_{\boldsymbol{V},b}\,c_{st}^2\|c\|)},$$

where $c' = \Big((1+\|c\|\,c_{st})(\lambda_{max}(\hat{\mathcal{C}}_{\boldsymbol{M}})c_{st}+1+\|c\|c_{st})\Big)^{-1}$ *with* c_{st} *as in the stability estimate of the primal problem* (3.26)*.*

Proof. 1. Let $\boldsymbol{M},\boldsymbol{L}\in \boldsymbol{H}(\operatorname{div}\operatorname{Div},\Omega;Q_3^*)_{\text{sym}}$. Then

$$\begin{aligned}
|(\hat{\mathcal{C}}_{\boldsymbol{M}}^{-1}\boldsymbol{M},\boldsymbol{L})| &\leq (1/\lambda_{min}(\hat{\mathcal{C}}_{\boldsymbol{M}}))\|\boldsymbol{M}\|_0\|\boldsymbol{L}\|_0\\
&\leq (1/\lambda_{min}(\hat{\mathcal{C}}_{\boldsymbol{M}}))\|\boldsymbol{M}\|_{\operatorname{div}\operatorname{Div};Q_3^*}\|\boldsymbol{L}\|_{\operatorname{div}\operatorname{Div};Q_3^*}.
\end{aligned}$$

2. Let $\boldsymbol{L}\in \boldsymbol{H}(\operatorname{div}\operatorname{Div},\Omega;Q_3^*)_{\text{sym}}$ and $\boldsymbol{v}\in\boldsymbol{Q}$. Then using Lemma 3.4 we have

$$|b(\boldsymbol{L},\boldsymbol{v})| \leq \|\boldsymbol{L}\|_{\boldsymbol{V},b}\|\boldsymbol{v}\|_1 \leq \overline{c}_{\boldsymbol{V},b}\|\boldsymbol{L}\|_{\operatorname{div}\operatorname{Div};Q_3^*}\|\boldsymbol{v}\|_1.$$

3. Let $\boldsymbol{u},\boldsymbol{v}\in\boldsymbol{Q}$. Then according to (3.29)

$$|c(\boldsymbol{u},\boldsymbol{v})| \leq \|c\|\;\|\boldsymbol{u}\|_1\|\boldsymbol{v}\|_1.$$

4. Let $\boldsymbol{L} \in \boldsymbol{H}(\operatorname{div}\operatorname{Div},\Omega;Q_3^*)_{\mathrm{sym}}$. Then using Lemma 3.4 we have

$$a(\boldsymbol{L},\boldsymbol{L}) + \left(\sup_{0\neq \boldsymbol{v}\in\boldsymbol{Q}} \frac{b(\boldsymbol{L},\boldsymbol{v})}{\|\boldsymbol{v}\|_1}\right)^2 \geq (1/\lambda_{max}(\hat{\mathcal{C}}_{\boldsymbol{M}}))\|\boldsymbol{L}\|_0^2 + \left(\sup_{0\neq \boldsymbol{v}\in\boldsymbol{Q}} \frac{b(\boldsymbol{L},\boldsymbol{v})}{\|\boldsymbol{v}\|_1}\right)^2$$
$$\geq \min\{1,(1/\lambda_{max}(\hat{\mathcal{C}}_{\boldsymbol{M}}))\}\ \|\boldsymbol{L}\|_{\boldsymbol{V},b}^2 \geq \underline{c}_{\boldsymbol{V},b}^2 \min\{1,1/\lambda_{max}(\hat{\mathcal{C}}_{\boldsymbol{M}})\}\ \|\boldsymbol{L}\|_{\operatorname{div}\operatorname{Div};Q_3^*}^2.$$

5. Let $\boldsymbol{v} \in \boldsymbol{Q}$. Then we have to show

$$c(\boldsymbol{v},\boldsymbol{v}) + \left(\sup_{0\neq\boldsymbol{L}\in\boldsymbol{H}(\operatorname{div}\operatorname{Div},\Omega;Q_3^*)_{\mathrm{sym}}} \frac{b(\boldsymbol{L},\boldsymbol{v})}{\|\boldsymbol{L}\|_{\operatorname{div}\operatorname{Div};Q_3^*}}\right)^2 \geq \beta\ \|\boldsymbol{v}\|_1^2 \quad \text{for all } \boldsymbol{v}\in\boldsymbol{Q}.$$

For a fixed but arbitrary $\boldsymbol{v} \in \boldsymbol{Q}$, let $\boldsymbol{u}^{\boldsymbol{v}}$ be the solution of the primal problem (3.24) with the right-hand side $\langle F^{\boldsymbol{v}}, .\rangle = -(\boldsymbol{v}, .)_{\boldsymbol{Q}} \in \boldsymbol{Q}^*$. From Theorem 3.3 it follows that $\boldsymbol{M}^{\boldsymbol{v}} = \hat{\mathcal{C}}_{\boldsymbol{M}}\boldsymbol{\kappa}(\boldsymbol{u}^{\boldsymbol{v}}) \in \boldsymbol{V}$, and $(\boldsymbol{M}^{\boldsymbol{v}}, \boldsymbol{u}^{\boldsymbol{v}})$ is solution of the corresponding mixed problem (3.35). The second line of the mixed formulation (3.35) provides

$$b(\boldsymbol{M}^{\boldsymbol{v}},\boldsymbol{w}) - c(\boldsymbol{u}^{\boldsymbol{v}},\boldsymbol{w}) = (\boldsymbol{v},\boldsymbol{w})_{\boldsymbol{Q}} \quad \text{for all } \boldsymbol{w}\in\boldsymbol{Q}.$$

Therefore,

$$b(\boldsymbol{M}^{\boldsymbol{v}},\boldsymbol{v}) - c(\boldsymbol{u}^{\boldsymbol{v}},\boldsymbol{v}) = (\boldsymbol{v},\boldsymbol{v})_{\boldsymbol{Q}} = \|\boldsymbol{v}\|_1^2$$

and using (3.29)

$$\sup_{\boldsymbol{w}\in\boldsymbol{Q}} \frac{b(\boldsymbol{M}^{\boldsymbol{v}},\boldsymbol{w})}{\|\boldsymbol{w}\|_1} = \sup_{\boldsymbol{w}\in\boldsymbol{Q}} \frac{(\boldsymbol{v},\boldsymbol{w})_{\boldsymbol{Q}} + c(\boldsymbol{u}^{\boldsymbol{v}},\boldsymbol{w})}{\|\boldsymbol{w}\|_1} \leq \|\boldsymbol{v}\|_1 + \|c\|\ \|\boldsymbol{u}^{\boldsymbol{v}}\|_1.$$

Using that $\boldsymbol{u}^{\boldsymbol{v}}$ is the solution of the primal problem (3.24) we obtain

$$\begin{aligned}\|\boldsymbol{M}^{\boldsymbol{v}}\|_0^2 = \|\hat{\mathcal{C}}_{\boldsymbol{M}}\boldsymbol{\kappa}(\boldsymbol{u}^{\boldsymbol{v}})\|_0^2 &\leq \lambda_{max}(\hat{\mathcal{C}}_{\boldsymbol{M}})(\hat{\mathcal{C}}_{\boldsymbol{M}}\boldsymbol{\kappa}(\boldsymbol{u}^{\boldsymbol{v}}),\boldsymbol{\kappa}(\boldsymbol{u}^{\boldsymbol{v}}))\\ &= \lambda_{max}(\hat{\mathcal{C}}_{\boldsymbol{M}})(-(\boldsymbol{v},\boldsymbol{u}^{\boldsymbol{v}})_{\boldsymbol{Q}} - c(\boldsymbol{u}^{\boldsymbol{v}},\boldsymbol{u}^{\boldsymbol{v}}))\\ &\leq \lambda_{max}(\hat{\mathcal{C}}_{\boldsymbol{M}})(\|\boldsymbol{v}\|_1\|\boldsymbol{u}^{\boldsymbol{v}}\|_1 + \|c\|\|\boldsymbol{u}^{\boldsymbol{v}}\|_1^2).\end{aligned}$$

Furthermore, the stability estimate (3.26) provides

$$\|\boldsymbol{u}^{\boldsymbol{v}}\|_1 \leq \|\boldsymbol{u}^{\boldsymbol{v}}\|_{\boldsymbol{W}} \leq c_{st}\|F^{\boldsymbol{v}}\|_{\boldsymbol{W}^*} \leq c_{st}\|F^{\boldsymbol{v}}\|_{\boldsymbol{Q}^*} = c_{st}\|\boldsymbol{v}\|_1. \tag{3.44}$$

Hence, using Lemma 3.4 we obtain from the last three inequalities

$$\begin{aligned}&\underline{c}_{\boldsymbol{V},b}^2\|\boldsymbol{M}^{\boldsymbol{v}}\|_{\operatorname{div}\operatorname{Div};Q_3^*}^2 \leq \|\boldsymbol{M}^{\boldsymbol{v}}\|_{\boldsymbol{V},b}^2 = \|\boldsymbol{M}^{\boldsymbol{v}}\|_0^2 + \sup_{\boldsymbol{w}\in\boldsymbol{Q}}\left(\frac{b(\boldsymbol{M}^{\boldsymbol{v}},\boldsymbol{w})}{\|\boldsymbol{w}\|_1}\right)^2\\ &\leq \lambda_{max}(\hat{\mathcal{C}}_{\boldsymbol{M}})c_{st}(1+\|c\|\,c_{st})\|\boldsymbol{v}\|_1^2 + (1+\|c\|\,c_{st})^2\|\boldsymbol{v}\|_1^2\\ &\leq (1+\|c\|\,c_{st})(\lambda_{max}(\hat{\mathcal{C}}_{\boldsymbol{M}})c_{st} + 1 + \|c\|\,c_{st})\|\boldsymbol{v}\|_1^2 = (c')^{-1}\ \|\boldsymbol{v}\|_1^2,\end{aligned}$$

where $c' = \Big((1+\|c\|\,c_{st})(\lambda_{max}(\hat{\mathcal{C}}_{\boldsymbol{M}})c_{st}+1+\|c\|\,c_{st})\Big)^{-1}$. Therefore,

$$c(\boldsymbol{v},\boldsymbol{v}) + \sup_{\boldsymbol{L}\in\boldsymbol{V}} \left(\frac{b(\boldsymbol{L},\boldsymbol{v})}{\|\boldsymbol{L}\|_{\operatorname{div}\operatorname{Div};Q_3^*}}\right)^2 \geq c(\boldsymbol{v},\boldsymbol{v}) + \frac{b(\boldsymbol{M}^{\boldsymbol{v}},\boldsymbol{v})^2}{\|\boldsymbol{M}^{\boldsymbol{v}}\|^2_{\operatorname{div}\operatorname{Div};Q_3^*}}$$
$$\geq c(\boldsymbol{v},\boldsymbol{v}) + c'\,\underline{c}^2_{\boldsymbol{V},b}\,\frac{\big(\|\boldsymbol{v}\|_1^2 + c(\boldsymbol{u}^{\boldsymbol{v}},\boldsymbol{v})\big)^2}{\|\boldsymbol{v}\|_1^2}.$$

The expression on the left-hand side, denoted by I, can be bounded from below by its first part

$$I \geq c(\boldsymbol{v},\boldsymbol{v}) = |\boldsymbol{v}|_c^2, \tag{3.45}$$

where for the positive semi-definite bilinear form c we define the associated semi-norm by $|\boldsymbol{v}|_c^2 = c(\boldsymbol{v},\boldsymbol{v})$, and second part

$$I \geq c'\,\underline{c}^2_{\boldsymbol{V},b}\,\frac{1}{\|\boldsymbol{v}\|_1^2}\big(\|\boldsymbol{v}\|_1^2 + c(\boldsymbol{u}^{\boldsymbol{v}},\boldsymbol{v})\big)^2.$$

Using (3.42) we can estimate the second part from below as follows

$$\begin{aligned} I &\geq c'\,\underline{c}^2_{\boldsymbol{V},b}\,\frac{1}{\|\boldsymbol{v}\|_1^2}\left(\frac{1}{2}\|\boldsymbol{v}\|_1^4 - c(\boldsymbol{u}^{\boldsymbol{v}},\boldsymbol{v})^2\right) \\ &\geq c'\,\underline{c}^2_{\boldsymbol{V},b}\,\frac{1}{\|\boldsymbol{v}\|_1^2}\left(\frac{1}{2}\|\boldsymbol{v}\|_1^4 - |\boldsymbol{u}^{\boldsymbol{v}}|_c^2|\boldsymbol{v}|_c^2\right) \\ &\geq c'\,\underline{c}^2_{\boldsymbol{V},b}\left(\frac{1}{2}\|\boldsymbol{v}\|_1^2 - c_{st}^2\|c\||\boldsymbol{v}|_c^2\right). \end{aligned} \tag{3.46}$$

Here we used

$$\begin{aligned} &|c(\boldsymbol{u}^{\boldsymbol{v}},\boldsymbol{v})| \leq |\boldsymbol{u}^{\boldsymbol{v}}|_c|\boldsymbol{v}|_c, \\ &|\boldsymbol{u}^{\boldsymbol{v}}|_c^2 = c(\boldsymbol{u}^{\boldsymbol{v}},\boldsymbol{u}^{\boldsymbol{v}}) \leq \|c\|\|\boldsymbol{u}^{\boldsymbol{v}}\|_1^2 \leq c_{st}^2\|c\|\|\boldsymbol{v}\|_1^2, \end{aligned}$$

where the first estimate holds since the bilinear form c is positive semi-definite and the second estimate follows from (3.29) and (3.44). By multiplying the estimate (3.45) with $c'\,\underline{c}^2_{\boldsymbol{V},b}c_{st}^2\|c\|$ and adding it to (3.46) we receive

$$(1 + c'\,\underline{c}^2_{\boldsymbol{V},b}c_{st}^2\|c\|)\left(c(\boldsymbol{v},\boldsymbol{v}) + \left(\sup_{\boldsymbol{L}\in\boldsymbol{V}}\frac{b(\boldsymbol{L},\boldsymbol{v})}{\|\boldsymbol{L}\|_{\operatorname{div}\operatorname{Div};Q_3^*}}\right)^2\right) \geq \frac{1}{2}c'\,\underline{c}^2_{\boldsymbol{V},b}\|\boldsymbol{v}\|_1^2.$$

and, hence,

$$\left(c(\boldsymbol{v},\boldsymbol{v}) + \left(\sup_{\boldsymbol{L}\in\boldsymbol{V}}\frac{b(\boldsymbol{L},\boldsymbol{v})}{\|\boldsymbol{L}\|_{\operatorname{div}\operatorname{Div};Q_3^*}}\right)^2\right) \geq \frac{c'\,\underline{c}^2_{\boldsymbol{V},b}}{2(1 + c'\,\underline{c}^2_{\boldsymbol{V},b}c_{st}^2\|c\|)}\|\boldsymbol{v}\|_1^2.$$

$\square$

Corollary 3.6. *For $F \in \boldsymbol{Q}^*$, the primal problem* (3.24) *and the mixed problem* (3.35) *are equivalent in the following sense: If $\boldsymbol{u} \in \boldsymbol{W}$ solves* (3.24)*, then $\boldsymbol{M} = \hat{\mathcal{C}}_{\boldsymbol{M}}\boldsymbol{\kappa}(\boldsymbol{u}) \in \boldsymbol{V}$ and $(\boldsymbol{M}, \boldsymbol{u})$ solves* (3.35)*. Vice versa, if $(\boldsymbol{M}, \boldsymbol{u}) \in \boldsymbol{V} \times \boldsymbol{Q}$ solves* (3.35)*, then $\boldsymbol{u} \in \boldsymbol{W}$ and solves* (3.24)*.*

Proof. The first part has already been shown in Theorem 3.3. Since, both problems are uniquely solvable the reverse direction is true as well. □

3.4 Membrane locking and the $\boldsymbol{M}$-$\boldsymbol{N}$-mixed formulation

Membrane locking is a well known difficulty in the construction of shell elements. As it is out of the scope of this thesis to give a thorough overview of membrane locking and corresponding discretization technologies, we only provide a brief introduction and refer for further information to [13, 32] and references therein.
Mechanically, membrane locking refers to the inability of a discrete formulation to represent pure bending without unwanted, parasitic membrane strains. As the parasitic membrane strains are large compared to the bending strains for thin shells, the occurring deformations are severely underestimated.
Mathematically, the reason for membrane locking is an ill-conditioning. The ratio of membrane stiffness to bending stiffness tends to infinity in the limit $t \to 0$, see (3.24). For plates this phenomenon does not occur, since the membrane part and the bending part decouple into two independent problems, as outlined in Section 2.1.3.
In order to alleviate membrane locking one popular concept (among many others) is to consider a mixed formulation with the membrane force tensor $\vec{\vec{N}}$ as new unknown, see, e.g., [33, 25]. An application of this technique leads to a well matching extension of the $\boldsymbol{M}$-mixed formulation introduced in Section 3.3. We refer to [32] for alternative approaches.
So, we additionally introduce as a new unknown the membrane force tensor $\vec{\vec{N}}$, whose contravariant components $\boldsymbol{N} = (N^{\alpha\beta})$ are connected to the membrane strain through the constitutive equation

$$N^{\alpha\beta} = \sqrt{a}\; t\; C^{\alpha\beta\lambda\mu}\varepsilon_{\lambda\mu}(\boldsymbol{u}),$$

or in short

$$\boldsymbol{N} = \sqrt{a}\; t\; \mathcal{C}\boldsymbol{\varepsilon}(\boldsymbol{u}) = \hat{\mathcal{C}}_{\boldsymbol{N}}\boldsymbol{\varepsilon}(\boldsymbol{u}), \text{ with } \hat{\mathcal{C}}_{\boldsymbol{N}} = \sqrt{a}\; t\; \mathcal{C}.$$

Note, again we include in $\boldsymbol{N}$ the geometry measure $\sqrt{a}$.
This motivates the extension of the $\boldsymbol{M}$-mixed formulation in (3.35) to the $\boldsymbol{M}$-$\boldsymbol{N}$-mixed formulation as follows: For $F \in \boldsymbol{Q}^*$, find $\boldsymbol{M} \in \boldsymbol{V}$, $\boldsymbol{N} \in \boldsymbol{L}^2(\Omega)_{\mathrm{sym}}$ and $\boldsymbol{u} \in \boldsymbol{Q}$

such that

$$
\begin{aligned}
&\int_\Omega (\hat{\mathcal{C}}_{\boldsymbol{M}}^{-1} : \boldsymbol{M}) : \boldsymbol{L} \, d\xi && && - \langle \operatorname{div} \operatorname{Div} \boldsymbol{L}, u_3 \rangle + b_1(\boldsymbol{L}, \boldsymbol{u}) = 0 \\
& && \int_\Omega (\hat{\mathcal{C}}_{\boldsymbol{N}}^{-1} : \boldsymbol{N}) : \boldsymbol{K} \, d\xi - \int_\Omega \boldsymbol{\varepsilon}(\boldsymbol{u}) : \boldsymbol{K} \, d\xi && = 0 \\
&- \langle \operatorname{div} \operatorname{Div} \boldsymbol{M}, v_3 \rangle + b_1(\boldsymbol{M}, \boldsymbol{v}) - \int_\Omega \boldsymbol{N} : \boldsymbol{\varepsilon}(\boldsymbol{v}) \, d\xi && && = -\langle F, \boldsymbol{v} \rangle
\end{aligned}
\tag{3.47}
$$

for all $\boldsymbol{L} \in \boldsymbol{V}$, $\boldsymbol{K} \in \boldsymbol{L}^2(\Omega)_{\mathrm{sym}}$, and $\boldsymbol{v} \in \boldsymbol{Q}$ with the function spaces as in (3.36)

$$
\begin{aligned}
\boldsymbol{V} &= \boldsymbol{H}(\operatorname{div} \operatorname{Div}, \Omega; Q_3^*)_{\mathrm{sym}}, \\
\boldsymbol{Q} &= W_1 \times W_2 \times Q_3 \subset H^1(\Omega) \times H^1(\Omega) \times H^1(\Omega).
\end{aligned}
$$

Note, the terms involving the membrane strain $\boldsymbol{\varepsilon}$ are well defined for functions $\boldsymbol{u}$ (or $\boldsymbol{v}$) in $\boldsymbol{Q}$, as already noted in (3.29). Hence, there is no need to modify them and we stick to the space $\boldsymbol{L}^2(\Omega)_{\mathrm{sym}}$ for the new unknown $\boldsymbol{N}$.
Well-posedness of $\boldsymbol{M}$-$\boldsymbol{N}$-mixed formulation formulation follows directly from the well-posedness of the $\boldsymbol{M}$-mixed formulation. Since the space for the test functions $\boldsymbol{K}$ corresponding to the auxiliary variable $\boldsymbol{N}$ is $\boldsymbol{L}^2(\Omega)_{\mathrm{sym}}$, the second line of this mixed formulation can be used to eliminate $\boldsymbol{N}$ leading again to the $\boldsymbol{M}$-mixed formulation in (3.35).

3.5 Regular decomposition and resulting mixed formulations

In this section we proceed analogously to Section 2.4.1. Based on the regular decomposition in Theorem 2.19 we equivalently reformulate the mixed formulations (3.35) and (3.47) involving the nonstandard Sobolev space $\boldsymbol{H}(\operatorname{div} \operatorname{Div}, \Omega; Q_3^*)_{\mathrm{sym}}$ as mixed formulations solely based on standard $H^1(\Omega)$ spaces.
According to Theorem 2.19 elements of the space $\boldsymbol{V} = \boldsymbol{H}(\operatorname{div} \operatorname{Div}, \Omega; Q_3^*)_{\mathrm{sym}}$ can be characterized in the following way: $\boldsymbol{L} \in \boldsymbol{V} = \boldsymbol{H}(\operatorname{div} \operatorname{Div}, \Omega; Q_3^*)_{\mathrm{sym}}$ if and only if it can be represented by

$$
\boldsymbol{L} = q\boldsymbol{I} + \operatorname{symCurl} \boldsymbol{\psi} \quad \text{(cf. (2.45))} \tag{3.48}
$$

with a pair of functions $(q, \boldsymbol{\psi}) \in \mathcal{V}$, where

$$
\begin{aligned}
\mathcal{V} = \{(q, \boldsymbol{\psi}) \in Q_3 \times (H^1(\Omega))^2 \colon & \, q, \boldsymbol{\psi} \text{ satisfy the coupling condition} \\
& \langle \partial_{\boldsymbol{t}} \boldsymbol{\psi}, \nabla v \rangle_\Gamma = - \int_\Gamma q \, \partial_{\boldsymbol{n}} v \, ds \quad \text{for all } v \in W_3 \quad \text{(cf. (2.46))}\}.
\end{aligned}
$$

Moreover, for $\boldsymbol{L} \in \boldsymbol{V} = \boldsymbol{H}(\operatorname{div} \operatorname{Div}, \Omega; Q_3^*)_{\mathrm{sym}}$ we have the following description of $\operatorname{div} \operatorname{Div} \boldsymbol{L}$

$$
\langle \operatorname{div} \operatorname{Div} \boldsymbol{L}, v \rangle = - \int_\Omega \nabla q \cdot \nabla v \, d\xi \quad \text{for all } v \in Q_3.
$$

3.5.1 The $\boldsymbol{M}$-mixed formulation

Using the following representations of the solution $\boldsymbol{M}$ and test functions $\boldsymbol{L}$

$$\boldsymbol{M} = p\boldsymbol{I} + \operatorname{symCurl}\boldsymbol{\phi}, \quad \boldsymbol{L} = q\boldsymbol{I} + \operatorname{symCurl}\boldsymbol{\psi},$$

with $(p,\boldsymbol{\phi}) \in \mathcal{V}$ and $(q,\boldsymbol{\psi}) \in \mathcal{V}$, the problem in (3.35) becomes a variational problem in $(p,\boldsymbol{\phi},\boldsymbol{u})$ given as follows: Find $(p,\boldsymbol{\phi}) \in \mathcal{V}$ and $\boldsymbol{u} \in \boldsymbol{Q}$ such that

$$\begin{aligned}
&(p\boldsymbol{I}, q\boldsymbol{I})_{\hat{\mathcal{C}}_{\boldsymbol{M}}^{-1}} &&+ (\operatorname{symCurl}\boldsymbol{\phi}, q\boldsymbol{I})_{\hat{\mathcal{C}}_{\boldsymbol{M}}^{-1}} &&+ (\nabla u_3, \nabla q) + b_1(q\boldsymbol{I}, \boldsymbol{u}) &&= 0\\
&(p\boldsymbol{I}, \operatorname{symCurl}\boldsymbol{\psi})_{\hat{\mathcal{C}}_{\boldsymbol{M}}^{-1}} &&+ (\operatorname{symCurl}\boldsymbol{\phi}, \operatorname{symCurl}\boldsymbol{\psi})_{\hat{\mathcal{C}}_{\boldsymbol{M}}^{-1}} &&+ b_1(\operatorname{symCurl}\boldsymbol{\psi}, \boldsymbol{u}) &&= 0\\
&(\nabla p, \nabla v_3) + b_1(p\boldsymbol{I}, \boldsymbol{v}) + b_1(\operatorname{symCurl}\boldsymbol{\phi}, \boldsymbol{v}) && &&- c(\boldsymbol{u}, \boldsymbol{v}) &&= -\langle F, \boldsymbol{v}\rangle.
\end{aligned} \tag{3.49}$$

for all $(q,\boldsymbol{\psi}) \in \mathcal{V}$ and $\boldsymbol{v} \in \boldsymbol{Q}$.

Remark 3.7. *The mixed formulation* (3.49) *is different from the formulation obtained for plates in Chapter 2. In case of plates membrane and bending parts decouple and for the bending part a decomposition into three consecutively to solve second-order problems is obtained. This is no longer possible for shells.*

Problem (3.49) has the typical structure of a saddle point problem: find $\boldsymbol{x} = (p,\boldsymbol{\phi}) \in \mathcal{V}$ and $\boldsymbol{u} \in \boldsymbol{Q}$ such that

$$\begin{aligned}
a(\boldsymbol{x},\boldsymbol{y}) + b(\boldsymbol{y},\boldsymbol{u}) &= 0 && \text{for all } \boldsymbol{y} = (q,\boldsymbol{\psi}) \in \mathcal{V},\\
b(\boldsymbol{x},\boldsymbol{v}) - c(\boldsymbol{u},\boldsymbol{v}) &= -\langle F, \boldsymbol{v}\rangle && \text{for all } \boldsymbol{v} \in \boldsymbol{Q}.
\end{aligned}$$

3.5.2 The $\boldsymbol{M}$-$\boldsymbol{N}$-mixed formulation

Analogously as above, we can reformulate the problem (3.47) in terms of (p, $\boldsymbol{\phi}$, $\boldsymbol{N}$, $\boldsymbol{u}$): find $(p,\boldsymbol{\phi}) \in \mathcal{V}$, $\boldsymbol{N} \in \boldsymbol{L}^2(\Omega)_{\text{sym}}$ and $\boldsymbol{u} \in \boldsymbol{Q}$ such that

$$\begin{aligned}
&(p\boldsymbol{I}, q\boldsymbol{I})_{\hat{\mathcal{C}}_{\boldsymbol{M}}^{-1}} &&+ (\operatorname{symCurl}\boldsymbol{\phi}, q\boldsymbol{I})_{\hat{\mathcal{C}}_{\boldsymbol{M}}^{-1}} &&+ (\nabla u_3, \nabla q) + b_1(q\boldsymbol{I}, \boldsymbol{u}) &&= 0\\
&(p\boldsymbol{I}, \operatorname{symCurl}\boldsymbol{\psi})_{\hat{\mathcal{C}}_{\boldsymbol{M}}^{-1}} &&+ (\operatorname{symCurl}\boldsymbol{\phi}, \operatorname{symCurl}\boldsymbol{\psi})_{\hat{\mathcal{C}}_{\boldsymbol{M}}^{-1}} &&+ b_1(\operatorname{symCurl}\boldsymbol{\psi}, \boldsymbol{u}) &&= 0\\
& && (\boldsymbol{N}, \boldsymbol{K})_{\hat{\mathcal{C}}_{\boldsymbol{N}}^{-1}} &&- (\boldsymbol{\varepsilon}(\boldsymbol{u}), \boldsymbol{K}) &&= 0\\
&(\nabla p, \nabla v_3) + b_1(p\boldsymbol{I}, \boldsymbol{v}) + b_1(\operatorname{symCurl}\boldsymbol{\phi}, \boldsymbol{v}) &&- (\boldsymbol{N}, \boldsymbol{\varepsilon}(\boldsymbol{v})) && &&= -\langle F, \boldsymbol{v}\rangle
\end{aligned}$$

for all $(q,\boldsymbol{\psi}) \in \mathcal{V}$, $\boldsymbol{K} \in \boldsymbol{L}^2(\Omega)_{\text{sym}}$ and $\boldsymbol{v} \in \boldsymbol{Q}$.

3.6 The discretization method

In this section we first construct a conforming discretization space for the displacement $\boldsymbol{u}$, i.e.,

$$\boldsymbol{Q}_h \subset \boldsymbol{Q} \subset (H^1(\Omega))^3.$$

Note, only C^0 basis functions are required, so the continuity requirements are easily satisfied with standard basis functions. In the following we consider isogeometric B-spline discretization spaces with degree $p \geq 1$. We define discretization spaces on patch level and for our formulation continuity between patches does not require extra attention, since standard C^0-coupling is sufficient.
We use equal order discretization spaces for the three components of the displacement $\boldsymbol{u}$ and incorporate the essential boundary conditions, which brings us to the definition:

$$\boldsymbol{Q}_h = (S^{p,p}_{\alpha,\alpha})^3 \cap \boldsymbol{Q}$$

with $\alpha = p-1$, i.e., maximum smoothness at interior knots, where $S^{p,p}_{\alpha,\alpha}$ is defined as in (2.114).
For $\mathcal{V}$, the space of the auxiliary variables $(p, \boldsymbol{\phi})$, the construction of a conforming discretization space is more involved, since the coupling condition (2.46) has to be taken into account. For this we adapt the approach based on Lagrange multipliers introduced in Section 2.6. Therefore, we first disregard the coupling condition and construct a conforming discretization space of $\hat{\mathcal{V}} = Q_3 \times (H^1(\Omega))^2$. Using equal order discretization spaces for p and $\boldsymbol{\phi}$ we receive

$$\hat{\mathcal{V}}_h = (S^{p,p}_{\alpha,\alpha} \times (S^{p,p}_{\alpha,\alpha})^2) \cap \hat{\mathcal{V}}.$$

Using the notation introduced at the beginning of Section 2.6, the space $\mathcal{V}_h$ is defined as the subset of $\boldsymbol{y}_h = (q_h, \boldsymbol{\psi}_h) \in \hat{\mathcal{V}}_h$ satisfying a discrete version of the coupling condition (2.96) (with $(p, \boldsymbol{\phi})$ replaces by $(q, \boldsymbol{\psi})$)

$$\mathcal{V}_h = \{\boldsymbol{y}_h = (q_h, \boldsymbol{\psi}_h) \in \hat{\mathcal{V}}_h : d(\boldsymbol{y}_h, \boldsymbol{\mu}_h) = 0 \quad \text{for all } \boldsymbol{\mu}_h \in \boldsymbol{\Lambda}_h\},$$

with $\boldsymbol{\mu}_h = ((\mu_t^1, \mu_t^2, \ldots, \mu_t^K), (\mu_n^1, \mu_n^2, \ldots, \mu_n^K))$ and

$$d(\boldsymbol{y}_h, \boldsymbol{\mu}_h) = \sum_{E_k \subset \Gamma_s \cup \Gamma_f} (\partial_{\boldsymbol{t}} \boldsymbol{\psi}_h \cdot \boldsymbol{n} \ + \ q_h, \mu_n^k)_{E_k} + \sum_{E_k \subset \Gamma_f} (\partial_{\boldsymbol{t}} \boldsymbol{\psi}_h \cdot \boldsymbol{t}, \mu_t^k)_{E_k}. \tag{3.50}$$

Recall from Section 2.6 that the test functions $\boldsymbol{\mu}_h = ((\mu_t^1, \mu_t^2, \ldots, \mu_t^K), (\mu_n^1, \mu_n^2, \ldots, \mu_n^K))$ are discrete representations of $(\partial_{\boldsymbol{t}} v, \partial_{\boldsymbol{n}} v)$ for $v \in W_3$ with

$$\partial_{\boldsymbol{t}} v = (\partial_{\boldsymbol{t}} v|_{E_1}, \partial_{\boldsymbol{t}} v|_{E_2}, \ldots, \partial_{\boldsymbol{t}} v|_{E_K}), \quad \partial_{\boldsymbol{n}} v = (\partial_{\boldsymbol{n}} v|_{E_1}, \partial_{\boldsymbol{n}} v|_{E_2}, \ldots, \partial_{\boldsymbol{n}} v|_{E_K}).$$

Hence, according to Section 2.6.3, the discrete space $\boldsymbol{\Lambda}_h$ consists of all

$$\boldsymbol{\mu}_h = ((\mu_t^1, \mu_t^2, \ldots, \mu_t^K), (\mu_n^1, \mu_n^2, \ldots, \mu_n^K)),$$

where

$$\mu_t^k \in \mathcal{S}_h^{p-1}(E_k) \quad \text{and} \quad \mu_n^k \in \mathcal{S}_h^{p-1}(E_k) \quad \text{for } k = 1, 2, \ldots, K,$$

subject to the constraints derived in Lemma 2.43. Here, $\mathcal{S}_h^{p-1}(E_k)$ denotes the restriction of functions from $S^{p-1,p-1}_{\alpha-1,\alpha-1}$ to E_k, defined by

$$\mathcal{S}_h^{p-1}(E_k) = \{v|_{E_k} : v \in S^{p-1,p-1}_{\alpha-1,\alpha-1}\}.$$

Then the discrete version of (3.49) reads: find $\boldsymbol{x}_h = (p_h, \boldsymbol{\phi}_h) \in \mathcal{V}_h$ and $\boldsymbol{u}_h \in \boldsymbol{Q}_h$ such that

$$\begin{aligned} a(\boldsymbol{x}_h, \boldsymbol{y}_h) + b(\boldsymbol{y}_h, \boldsymbol{u}_h) &= 0 && \text{for all } \boldsymbol{y}_h = (q_h, \boldsymbol{\psi}_h) \in \mathcal{V}_h, \\ b(\boldsymbol{x}_h, \boldsymbol{v}_h) - c(\boldsymbol{u}_h, \boldsymbol{v}_h) &= -\langle F, \boldsymbol{v}_h \rangle && \text{for all } \boldsymbol{v}_h \in \boldsymbol{Q}_h. \end{aligned}$$

We do not explicitly build in the coupling condition by constructing a basis of the space $\mathcal{V}_h$, but incorporate it implicitly, by replacing $\boldsymbol{V}_h$ by $\hat{\boldsymbol{V}}_h$ and adding (3.50) as additional constraint. Since $\hat{\boldsymbol{V}}_h$, $\boldsymbol{Q}_h$ and $\boldsymbol{\Lambda}_h$ are finite dimensional, the bilinear forms a, b, c and d can be represented as matrices $\boldsymbol{A}_h$, $\boldsymbol{B}_h$, $\boldsymbol{C}_h$ and $\boldsymbol{D}_h$ acting on vectors of real numbers $\underline{\boldsymbol{x}}_h$, $\underline{\boldsymbol{u}}_h$ and $\underline{\boldsymbol{\lambda}}_h$ representing the elements in $\hat{\boldsymbol{V}}_h$, $\boldsymbol{Q}_h$ and $\boldsymbol{\Lambda}_h$, respectively, with respect to the chosen basis. In this matrix-vector notation the resulting system reads

$$\begin{aligned} \boldsymbol{A}_h \underline{\boldsymbol{x}}_h + \boldsymbol{B}_h^T \underline{\boldsymbol{u}}_h + \boldsymbol{D}_h^T \underline{\boldsymbol{\lambda}}_h &= 0, \\ \boldsymbol{B}_h \underline{\boldsymbol{x}}_h - \boldsymbol{C}_h \underline{\boldsymbol{u}}_h &= \boldsymbol{f}_h, \\ \boldsymbol{D}_h \underline{\boldsymbol{x}}_h &= 0, \end{aligned}$$

where $\boldsymbol{B}_h^T$ and $\boldsymbol{D}_h^T$ denote the transposed matrices.
For the second, the $\boldsymbol{M}$-$\boldsymbol{N}$-mixed formulation we consider for $\boldsymbol{u}$ and $(p, \boldsymbol{\phi})$ the discretization introduced above. For the additional unknown, the membrane force $\boldsymbol{N}$, we use the discretization space proposed in the Hybrid Stress (HS) method presented in [33]. The basis for the contravariant components of $\boldsymbol{N}$ is given by

$$N_h^{11} \in S_{\alpha-1,\alpha}^{p-1,p}, \quad N_h^{22} \in S_{\alpha,\alpha-1}^{p,p-1}, \quad N_h^{12} \in S_{\alpha-1,\alpha-1}^{p-1,p-1}.$$

3.7 Numerical experiments

Throughout the section, when speaking about the $\boldsymbol{M}$-mixed and $\boldsymbol{M}$-$\boldsymbol{N}$-mixed formulation we always refer to their reformulated versions based on standard $H^1(\Omega)$ spaces introduced in Section 3.5.1 and Section 3.5.2, respectively, with the $H^1(\Omega)$ conforming discretizations proposed in the previous section.
In the first part of this section we demonstrate that the $\boldsymbol{M}$-mixed formulation works by testing it with the three benchmark problems of the well-known shell obstacle course [11], consisting of two cylindrical shells and one spherical shell. In the second part we consider the $\boldsymbol{M}$-$\boldsymbol{N}$-mixed formulation.
For all problems in this section the undeformed midsurface is modeled exactly by non-uniform rational B-splines (NURBS). For the discretization of the unknowns standard B-spline spaces are used, see Section 3.6. Mesh density is characterized by the number of control points per edge. In case several congruent patches are used to describe the surface the corresponding number for one patch is used.
In all experiments a sparse direct solver is used. The implementation is done in the framework of the object-oriented C++ library G+Smo (“Geometry + Simulation Modules”), see [57].
In the description of the numerical experiments the term displacement always refers to the displacement given by the components of $\vec{u}$ with respect to the standard Cartesian

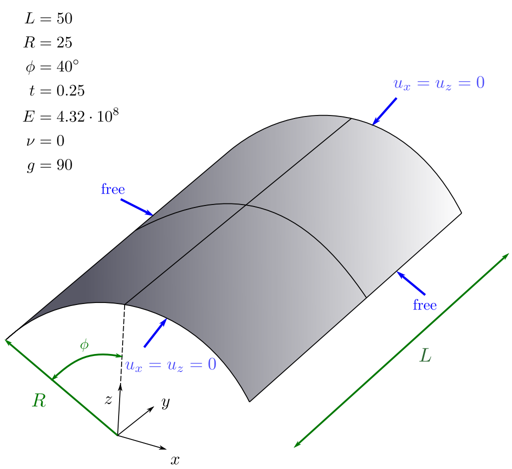
basis $e^x = (1,0,0)^T, e^y = (0,1,0)^T, e^z = (0,0,1)^T$, i.e.,

$$\vec{u} = u_x e^x + u_y e^y + u_z e^z.$$

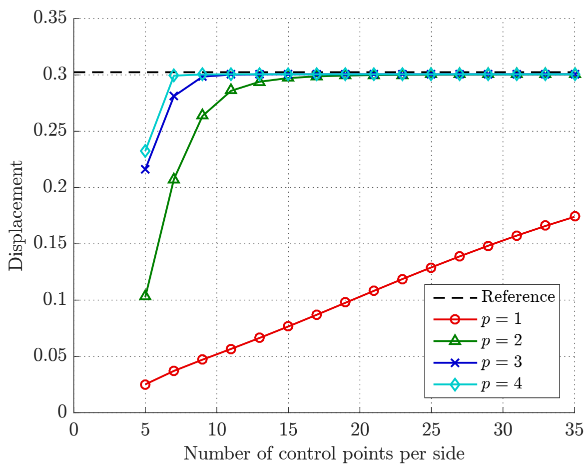
3.7.1 The $\boldsymbol{M}$-mixed formulation

Scordelis-Lo roof

The problem setup of the Scordelis-Lo roof benchmark is shown in Figure 3.1a. The structure is supported with rigid diaphragms at both ends and the side edges are free. This setup is realized by imposing homogeneous boundary conditions for the displacement of the form $u_x = u_z = 0$, leading to the conditions $u_1 = u_3 = 0$ for the covariant components of the displacement. The roof has moderate slenderness $\frac{R}{t} = 100$, with radius R and thickness t as defined in Figure 3.1a, and is subject to a uniform vertical load of $g = 90$ per unit area. This configuration yields a membrane dominated load-carrying behavior.
Due to the rectangular topology of the roof domain a single patch representation is quite natural. The surveyed quantity is the vertical displacement u_z at the midpoint of the free edges, with the value of the reference solution reported as 0.3024 in [11].
In Figure 3.1b the displacement convergence of the $\boldsymbol{M}$-mixed formulation for $p = 1,2,3,4$ is shown. As expected, it turns out that the convergence becomes considerably faster with higher polynomial degree. For $p = 3$ and $p = 4$ already the fourth refinement step with 11 control points per side yields a relative error of less than 1%, whereas the discretizations with $p = 2$ and especially $p = 1$ require a much finer mesh (19 and 400 control points per side, respectively) to provide the same accuracy.



(a) Problem setup.



(b) Displacement convergence of $\boldsymbol{M}$-mixed formulation.

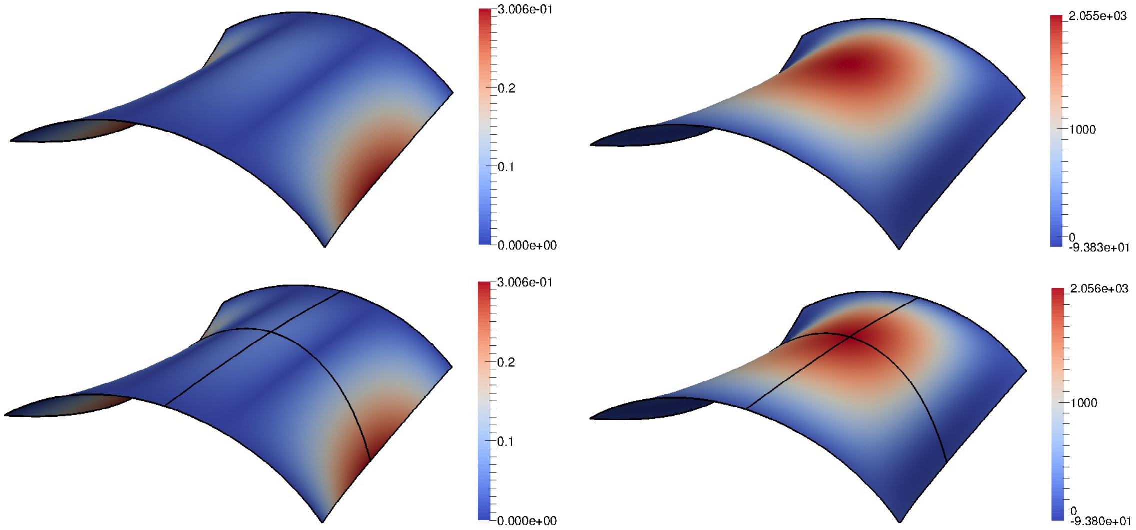
Figure 3.1: Scordelis-Lo roof.

According to Table 3.1, for $p = 2$ the results obtained in [32] for the standard purely displacement-based 3-parameter Kirchhoff-Love shell formulation (3p) conform well

Control points per edge	5	9	13	20	25	30
$p = 1$						
$\boldsymbol{M}$-mixed	0.0252	0.0471	0.0665	0.1027	0.1291	0.1528
$p = 2$						
$\boldsymbol{M}$-mixed	0.1028	0.2636	0.2940	0.2997	0.3003	0.3004
3p (Echter [32])	0.0440	0.2077	0.2801	0.2975	0.2994	0.3004

Table 3.1: Scordelis-Lo roof, displacements ($\boldsymbol{M}$-mixed, 3p).

with the $\boldsymbol{M}$-mixed shell elements developed in this thesis for fine discretizations but provide worse results for coarse meshes.
For comparison we consider a second representation of the midsurface using four patches, as illustrated in Figure 3.1a. In Figure 3.2 vertical displacements and bending moments for the one patch and four patch geometry representation are compared. Visually no difference like discontinuity across patch interfaces can be seen. As in [33], in order to obtain $M^{\overline{x}\overline{x}}$ the components of the bending moment tensor defined in the curvilinear coordinate system are first transformed into a local Cartesian basis with $\overline{x}$ and $\overline{z}$-axis aligned with ξ^1- and ξ^3-directions, for details see [13]. The obtained



(a) Displacement u_z.

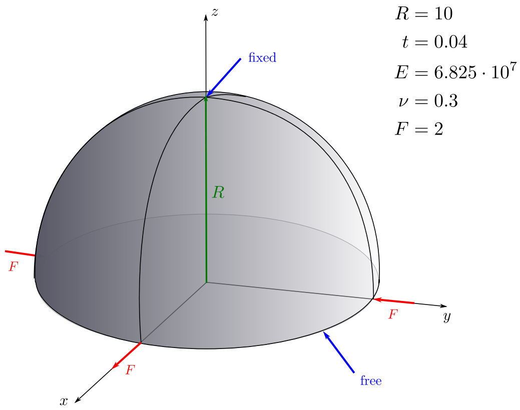
(b) Bending moment component $M^{\overline{x}\overline{x}}$.

Figure 3.2: Scordelis-Lo roof, analysis results for one patch (upper) and four patch (lower) geometry representations.

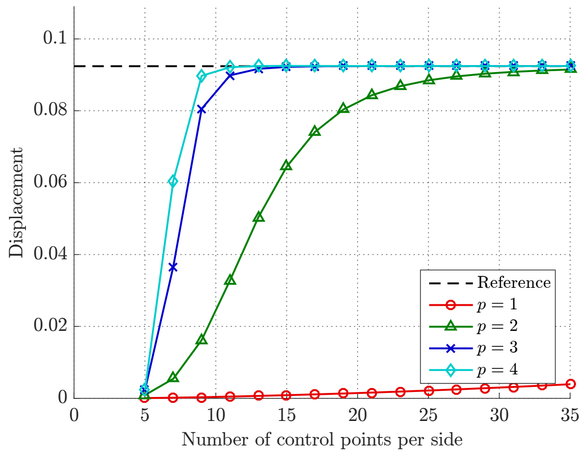
convergence results are quite similar to the ones received for the one patch geometry representation in Figure 3.1b, and are therefore not shown.

Pinched hemisphere

The problem setup of the pinched hemisphere benchmark is shown in Figure 3.3a. The structure is fixed at the top and free along the bottom circumferential edge. The shell has a slenderness of $\frac{R}{t} = 250$, with radius R and thickness t as defined in Figure 3.3a, and is subject to four radial point loads $F = \pm 2$ at its bottom. This configuration yields a bending dominated behavior with almost no membrane strains.
The undeformed midsurface is modeled using four patches, as illustrated in Figure 3.3a. The investigated quantity is the radial displacement at the points where the loads are applied, with the value of the reference solution reported as 0.0924 in [11].
In Figure 3.3b the displacement convergence of the $\boldsymbol{M}$-mixed formulation for $p = 1, 2, 3, 4$ is shown. Most observations carry over from the Scordelis-Lo roof benchmark to the pinched hemisphere. The slowed convergence of low order discretizations with $p = 1$ and $p = 2$ becomes even more amplified. The discretization with $p = 2$ requires 35 control points per edge (for each of the four patches) to reach a relative error of less than 1%. The discretization with $p = 1$ shows a considerably slower convergence. With a reasonable number of control points no acceptable accuracy could be achieved. For $p = 3$ and $p = 4$ only 13 control points per edge are needed to obtain the same accuracy.
The reason for the slow convergence in case of $p = 1$ and $p = 2$ for the Scordelis-Lo roof and (even more severe) for the pinched hemisphere is membrane locking, cf. Section 3.4. Similar results showing evidence of membrane locking have already been observed in [52, 32]. We come back to the important issue of membrane locking in Section 3.7.2.



(a) Problem setup.

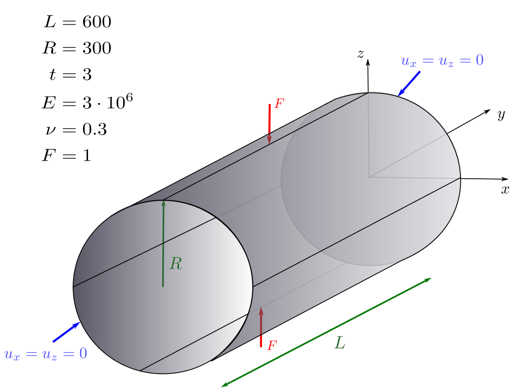


(b) Displacement convergence of $\boldsymbol{M}$-mixed formulation.

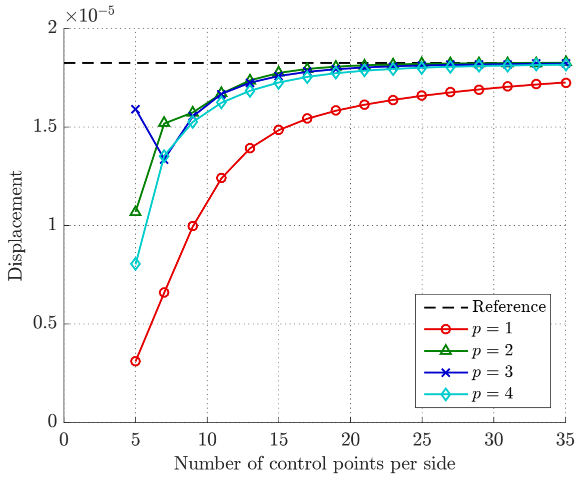
Figure 3.3: Pinched hemisphere.

Pinched cylinder

The last benchmark of the shell obstacle course is the pinched cylinder. The problem setup is shown in Figure 3.4a. The structure is supported with rigid diaphragms at both ends. The cylinder has moderate slenderness $\frac{R}{t} = 100$, with radius R and thickness t as defined in Figure 3.4a, and is subject to two opposite point loads $F = \pm 1$ in the middle. This configuration yields a severe test for both inextensional bending and complex membrane states.
The undeformed midsurface is modeled using four patches, as illustrated in Figure 3.4a. The analyzed quantity is the radial displacement at the points where the loads are applied, with the value of the reference solution reported as $1.8248 \cdot 10^{-5}$ in [11].
In Figure 3.4b the displacement convergence of the $\boldsymbol{M}$-mixed formulation for $p = 1, 2, 3, 4$ is shown. The convergence behavior differs from the ones obtained for the first two benchmarks. The differences in the results due to the usage of different polynomial orders are significantly smaller, especially discretizations with $p = 2, 3, 4$ provide comparable results. Note, already the first-order discretization shows an acceptable performance.



(a) Problem setup.



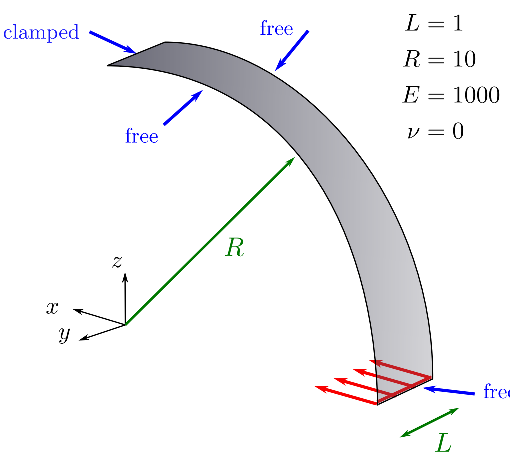
(b) Displacement convergence of $\boldsymbol{M}$-mixed formulation.

Figure 3.4: Pinched cylinder.

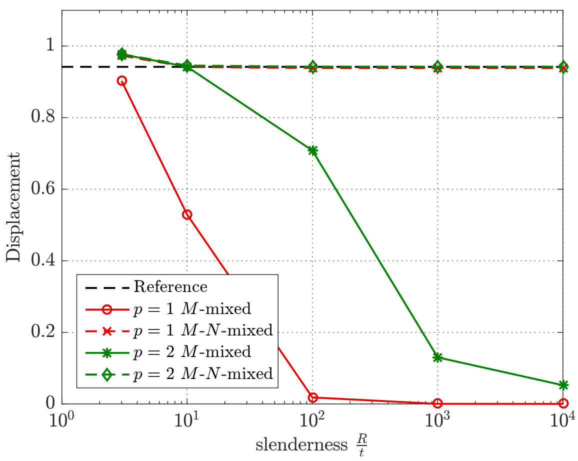
3.7.2 Membrane locking and the $\boldsymbol{M}$-$\boldsymbol{N}$-mixed formulation

In order to investigate membrane locking we consider first a simple model problem consisting of a cylindrical shell strip, see, e.g.,[13, 33]. The problem setup of this example is shown in Figure 3.5a. The structure is clamped along the edge $x = 0$ and subject to a constant line load in radial direction with magnitude $q_x = 0.1 \cdot t^3$ at the opposite free edge. This configuration yields a bending dominated behavior.

The quantity of interest is the radial displacement at the midpoint of the free edge, with exact value 0.942 independent of the slenderness $\frac{R}{t}$, according to an analytical solution based on the Bernoulli beam theory (cf. [33]). In order to ensure comparability with [33] the domain is discretized with a mesh of 10 elements in longitudinal direction and one element in the other direction.
In Figure 3.5b the influence of varying slenderness $\frac{R}{t}$ on the obtained displacement is investigated for the original $\boldsymbol{M}$-mixed formulation and the $\boldsymbol{M}$-$\boldsymbol{N}$-mixed formulation for $p = 1$ and $p = 2$. The results obtained with the $\boldsymbol{M}$-mixed formulation show severe membrane locking. The radial displacement tends to zero as the slenderness $\frac{R}{t}$ is increased. In case of $p = 2$ already for a moderate slenderness of $\frac{R}{t} = 100$ unphysical membrane strains lead to a considerable underestimation of the tip displacement of approximately 25% and for $p = 1$ the behavior is even worse. However, for both polynomial orders the results obtained with the $\boldsymbol{M}$-$\boldsymbol{N}$-mixed formulation indicate that this formulation completely removes the undesired membrane locking effects.



(a) Problem setup.



(b) Displacement convergence.

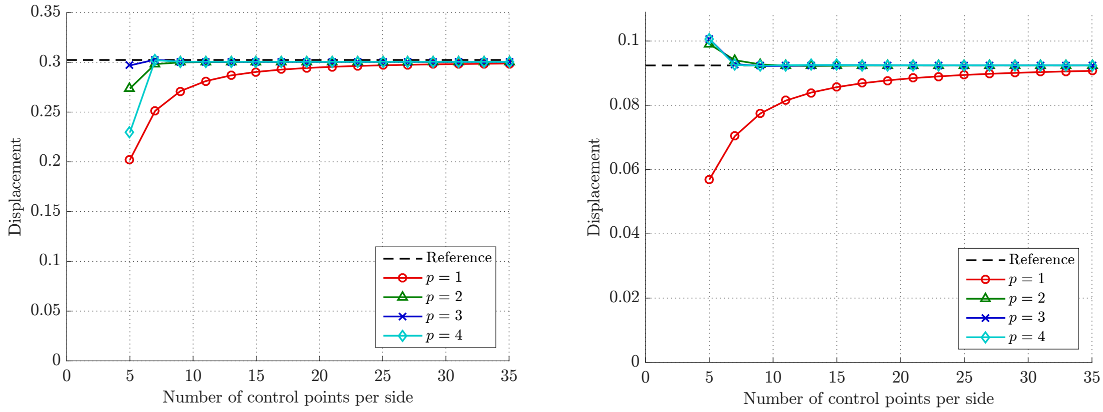
Figure 3.5: Cylindrical shell strip.

According to Table 3.2, for $p = 2$ the results in [33] for the standard purely displacement-based 3-parameter formulation (3p) and the corresponding formulation with a mixed Hybrid Stress modification of the membrane part (3p-HS) conform well with the $\boldsymbol{M}$-mixed and $\boldsymbol{M}$-$\boldsymbol{N}$-mixed shell elements proposed in this thesis.
The improved convergence behavior of the $\boldsymbol{M}$-$\boldsymbol{N}$-mixed formulation observed in the basic cylindrical shell strip problem carries over to the Scordelis-Lo roof and pinched hemisphere benchmark (cf. Section 3.7.1 and Section 3.7.1). In Figure 3.6a and Figure 3.6b the displacement convergence of the $\boldsymbol{M}$-$\boldsymbol{N}$-mixed formulation for $p = 1, 2, 3, 4$ is shown for the Scordelis-Lo roof and pinched hemisphere benchmark, respectively. For both problems the locking free $\boldsymbol{M}$-$\boldsymbol{N}$-mixed shell element converges considerably faster to the reference value for the same number of control points compared to the $\boldsymbol{M}$-mixed element in Figure 3.1b and Figure 3.3b. Furthermore, the differences in the

Slenderness $\frac{R}{t}$	10	100	1000	10000
$p = 2$				
$\boldsymbol{M}$-mixed	0.9420	0.7075	0.1156	0.0112
$\boldsymbol{M}$-$\boldsymbol{N}$-mixed	0.9454	0.9423	0.9422	0.9422
3p (Echter et al. [33])	0.9326	0.6635	0.0225	0.0002
3p-HS (Echter et al. [33])	0.9386	0.9425	0.9425	0.9425

Table 3.2: Cylindrical shell strip, displacements ($\boldsymbol{M}$-mixed, $\boldsymbol{M}$-$\boldsymbol{N}$-mixed, 3p, 3p-HS).

results for different polynomial orders are significantly reduced.



(a) Scordelis-Lo roof

(b) Pinched hemisphere

Figure 3.6: Displacement convergence of $\boldsymbol{M}$-$\boldsymbol{N}$-mixed formulation.

In Table 3.3 the results of the $\boldsymbol{M}$-$\boldsymbol{N}$-mixed formulation are compared to the results for the 3p-HS formulation of Echter [32]. Recall, our formulation does not require an H^2-conforming discretization in contrast to the one in [32]. A quantification of the improvements in numbers is obtained by comparing Table 3.3 with Table 3.1.

Control points per edge	5	9	13	20	25	30
$p = 1$						
$\boldsymbol{M}$-$\boldsymbol{N}$-mixed	0.2020	0.2711	0.2869	0.2941	0.2970	0.2978
$p = 2$						
$\boldsymbol{M}$-$\boldsymbol{N}$-mixed	0.2737	0.2999	0.3005	0.3006	0.3006	0.3006
3p-HS (Echter [32])	0.2517	0.3000	0.3005	0.3006	0.3006	0.3006

Table 3.3: Scordelis-Lo roof, displacements ($\boldsymbol{M}$-$\boldsymbol{N}$-mixed, 3p-HS).

Chapter 4

Conclusion and outlook

In this thesis, the starting point for the further considerations was in every studied situation a well-understood (continuous) new mixed variational formulation with the displacements in H^1 (with appropriate boundary conditions) and the auxiliary variable, the bending moment tensor, in the nonstandard Sobolev space $\boldsymbol{H}(\operatorname{div}\operatorname{Div}, \Omega; Q^*)_{\mathrm{sym}}$. More precisely, we performed a rigorous mathematical analysis of this formulation verifying Brezzi's conditions and equivalence to the original problem. For the Kirchhoff plate bending problem this fundamental continuous formulation was derived in Section 2.3 and extended to inhomogeneous boundary conditions in Section 2.7.1. For the Kirchoff-Love shell problem we showed this result in Section 3.3.

These important properties on the continuous level come at the cost of using the nonstandard Sobolev space $\boldsymbol{H}(\operatorname{div}\operatorname{Div}, \Omega; Q^*)_{\mathrm{sym}}$. In Section 2.4 we derived a regular decomposition of this nonstandard space, which makes it computationally accessible. The regular decomposition involves a coupling condition for the two components in the decomposition.

In case of a polygonal domain, the coupling condition was rewritten as essential boundary conditions that can be incorporated by a standard Nitsche method for second-order problems. For plates this allowed for an equivalent reformulation as a system of three (consecutively to solve) second-order elliptic problems. The decomposed formulation led in a natural way to the construction of a new discretization method based on Nitsche's method, for which a rigorous numerical analysis showing a-priori error estimates was carried out.

In order to extend our approach to curvilinear polygonal domains we proposed in Section 2.6 an alternative technique to incorporate the coupling condition based on Lagrange multipliers, where we added the coupling condition as additional constraint. In case of plates, we obtained a similar system that can be solved consecutively with the difference that the second problem is now a saddle point problem. For the resulting discretization method we showed promising numerical tests. An extension of the numerical analysis to curvilinear polygonal domains as well as to the incorporation of inhomogeneous boundary conditions is to be done.

For Kirchhoff-Love shells we obtained based on the regular decomposition of Section 2.4 again a mixed formulation only based on standard H^1 spaces. The new formulation led in a natural way to the construction of a new discretization method, where we incorporated the coupling condition via the approach based on Lagrange multipliers. Note that the system can no longer be solved consecutively due to additional lower-order derivative terms. Recall that the numerical analysis for plates, although only done for the discretization approach based on Nitsche's method (cf. Section 2.5.4), heavily relies on this decoupled structure. An extension of the numerical analysis to shells would be important, but requires new ideas.

Another important issue for the numerical analysis of our proposed methods (and in general for methods relying on a regular decomposition) is that smoothness assumptions on the solution of the original problem do not directly imply smoothness of the components of the regular decomposition, as discussed in more detail in Section 2.8.3. Hence, it would be desirable to derive a-priori error estimates that, in contrast to the result in Section 2.5.4, depend only on regularity of the original solution u and the auxiliary variable $\boldsymbol{M}$. However, the first experiments with inhomogeneous boundary data indicate that such an result is not possible. Note that with homogeneous boundary conditions we have never observed a behavior that contradicts such a result.

The extension of our approach to Kirchhoff-Love shells presented in Chapter 3 is a promising step towards the application of our technique to general fourth-order problems. Note that the procedure outlined in Chapter 3 is not independent of the chosen coordinate system, but relies on the representation of the Hessian on the parameter domain. In the context of shells, a more natural approach would be to work with the appropriate surface differential operators instead of classical differential operators on the parameter domain. The required extension of the regular decomposition in Section 2.4 to surfaces is another challenging area for further research.

In contrast to Kirchhoff-Love type thin shells, for conforming Reissner-Mindlin shell elements standard C^0-continuous shape functions are commonly used. In order to avoid transverse shear locking, in [56, 33] an H^2 based Mindlin-Reissner formulation obtained by a change of variables from rotations to transverse shear strains is proposed. This approach, which was later mathematically analyzed in the context of plates in [7], has the advantage that transverse shear locking is eliminated on the continuous formulation level, independent of a particular discretization. An application of our technique to obtain a mixed formulation in H^1 of this H^2 based Mindlin-Reissner formulation (representing another fourth-order problem in space dimension $d = 2$) seems possible and would be worth to investigate.

In this work, we have restricted our considerations to space dimension $d = 2$. The application of our approach to other fourth-order problems like, e.g., Cahn-Hilliard equation leads to the need of an extension to space dimension $d = 3$. The results of Section 2.3 carry over completely from the two-dimensional case to the three-dimensional setting. The main idea for the regular decomposition of the involved nonstandard Sobolev space is also the same, but the characterization of the kernel of the distributional $\operatorname{div}\operatorname{Div}$ used in this work is only valid for $d = 2$. In [61] a suitable representation of kernel elements for $d = 3$ has been presented.

Appendix A

Function spaces

In the following, we recall the definition of differential operators, corresponding functions spaces, and integration by parts formulas widely used throughout this thesis. For further information see, e.g., [1, 58].

A.1 Sobolev spaces

The differential operators in Section 2.1.1 are defined for smooth functions. These operators can also be defined in the weak sense, as exemplarily shown for ∇ and div: For $v \in L^2(\Omega)$, $\nabla v \in (L^2(\Omega))^2$ is called weak gradient of v if and only if

$$\int_\Omega \nabla v \cdot \boldsymbol{\psi} \; dx = -\int_\Omega v \operatorname{div} \boldsymbol{\psi} \; dx \quad \text{for all } \boldsymbol{\psi} \in (C_0^\infty(\Omega))^2.$$

For $\boldsymbol{\psi} \in (L^2(\Omega))^2$, $\operatorname{div} \boldsymbol{\psi} \in L^2(\Omega)$ is called weak divergence of $\boldsymbol{\psi}$ if and only if

$$\int_\Omega \operatorname{div} \boldsymbol{\psi} \; v \; dx = -\int_\Omega \boldsymbol{\psi} \cdot \nabla v \; dx \quad \text{for all } v \in C_0^\infty(\Omega).$$

Analogous definitions hold for their rotated versions curl and rot, and also for the corresponding row-wise versions Div, Curl and Rot.
The Sobolev spaces $H^1(\Omega), H(\operatorname{curl}, \Omega), H(\operatorname{Curl}, \Omega), \ldots$ are defined as the spaces, where the corresponding weak differential operator exists:

$$\begin{aligned}
H^1(\Omega) &= \{v \in L^2(\Omega) : \nabla v \in (L^2(\Omega))^2\}, \\
H(\operatorname{curl}, \Omega) &= \{v \in L^2(\Omega) : \operatorname{curl} v \in (L^2(\Omega))^2\}, \\
H(\operatorname{Curl}, \Omega) &= \{\boldsymbol{\psi} \in (L^2(\Omega))^2 : \operatorname{Curl} \boldsymbol{\psi} \in (L^2(\Omega))^{2\times 2}\}, \\
H(\operatorname{div}, \Omega) &= \{\boldsymbol{\psi} \in (L^2(\Omega))^2 : \operatorname{div} \boldsymbol{\psi} \in L^2(\Omega)\}, \\
H(\operatorname{rot}, \Omega) &= \{\boldsymbol{\psi} \in (L^2(\Omega))^2 : \operatorname{rot} \boldsymbol{\psi} \in L^2(\Omega)\}, \\
\boldsymbol{H}(\operatorname{Div}, \Omega) &= \{\boldsymbol{L} \in (L^2(\Omega))^{2\times 2} : \operatorname{Div} \boldsymbol{L} \in (L^2(\Omega))^2\}, \\
\boldsymbol{H}(\operatorname{Rot}, \Omega) &= \{\boldsymbol{L} \in (L^2(\Omega))^{2\times 2} : \operatorname{Rot} \boldsymbol{L} \in (L^2(\Omega))^2\}.
\end{aligned}$$

These spaces are Hilbert spaces, when equipped with the corresponding graph norms.

A.2 Trace theorems

Let Ω be a Lipschitz domain. Then the classical trace operator

$$\gamma_0 : C^\infty(\overline{\Omega}) \to C^\infty(\Gamma), \quad \gamma_0 v \mapsto v|_\Gamma$$

has a unique extension as a bounded linear operator from $H^1(\Omega)$ to $H^{\frac{1}{2}}(\Gamma)$, i.e., there exists a constant c_{tr} such that

$$\|\gamma_0 v\|_{\frac{1}{2},\Gamma} \leq c_{tr}\|v\|_1 \quad \text{for all } v \in H^1(\Omega).$$

Vice versa, for $g \in H^{\frac{1}{2}}(\Gamma)$ there exists an extension $v \in H^1(\Omega)$ such that

$$\gamma_0 v = g \quad \text{and} \quad \|v\|_1 \leq c_{ext}\|g\|_{\frac{1}{2},\Gamma}.$$

The second statement is ofter called inverse trace theorem or extension theorem.
Recall that on the boundary of a Lipschitz domain the outward unit normal vector $\boldsymbol{n}$ is defined almost everywhere. The classical normal trace operator

$$\gamma_n : (C^\infty(\overline{\Omega}))^2 \to C^\infty(\Gamma), \quad \gamma_n \boldsymbol{\psi} \mapsto \boldsymbol{\psi}|_\Gamma \cdot \boldsymbol{n}$$

has a unique extension as a bounded linear operator from $H(\text{div},\Omega)$ to $H^{-\frac{1}{2}}(\Gamma) = (H^{\frac{1}{2}}(\Gamma))^*$. In the following we just write $v|_\Gamma$ (or simply v) and $\boldsymbol{\psi}\cdot\boldsymbol{n}$ for the traces $\gamma_0 v$ and $\gamma_n\boldsymbol{\psi}$, respectively.
Analogously, we have well defined traces for the spaces with the rotated versions of the differential operators, i.e., for $v \in H(\text{curl},\Omega)$ and $\boldsymbol{\psi} \in H(\text{rot},\Omega)$

$$v|_\Gamma \in H^{\frac{1}{2}}(\Gamma) \quad \text{and} \quad \boldsymbol{\psi}\cdot\boldsymbol{t} \in H^{-\frac{1}{2}}(\Gamma).$$

Furthermore, for the spaces with corresponding row-wise versions of the differential operators we obtain the following traces: for $\boldsymbol{\psi} \in (H^1(\Omega))^2$ and $\boldsymbol{L} \in \boldsymbol{H}(\text{Div},\Omega)$

$$\boldsymbol{\psi}|_\Gamma \in (H^{\frac{1}{2}}(\Gamma))^2 \quad \text{and} \quad \boldsymbol{L}\boldsymbol{n} \in (H^{-\frac{1}{2}}(\Gamma))^2$$

and for $\boldsymbol{\psi} \in H(\text{Curl},\Omega)$ and $\boldsymbol{L} \in \boldsymbol{H}(\text{Rot},\Omega)$

$$\boldsymbol{\psi}|_\Gamma \in (H^{\frac{1}{2}}(\Gamma))^2 \quad \text{and} \quad \boldsymbol{L}\boldsymbol{t} \in (H^{-\frac{1}{2}}(\Gamma))^2.$$

A.3 Integration by parts formulas

We recall the standard integration by parts formulas for the differential operators introduced above. These identities are well-known in the classical sense. They also hold in the weak sense if all involved integrals and duality products for traces in Sobolev spaces of negative indices are well-defined. This is the case for the formulas stated below if the duality product on the boundary $\langle .,. \rangle_\Gamma$ is understood in the respective setting, as indicated above.

For $v \in H^1(\Omega)$ and $\boldsymbol{\psi} \in H(\operatorname{div}, \Omega)$ we have

$$(\nabla v, \boldsymbol{\psi}) = -(v, \operatorname{div} \boldsymbol{\psi}) + \langle \boldsymbol{\psi} \cdot \boldsymbol{n}, v \rangle_\Gamma \tag{A.1}$$

and analogous for $v \in H(\operatorname{curl}, \Omega)$ and $\boldsymbol{\psi} \in H(\operatorname{rot}, \Omega)$

$$(\operatorname{curl} v, \boldsymbol{\psi}) = (v, \operatorname{rot} \boldsymbol{\psi}) - \langle \boldsymbol{\psi} \cdot \boldsymbol{t}, v \rangle_\Gamma .$$

Furthermore, for $\boldsymbol{\psi} \in (H^1(\Omega))^2$ and $\boldsymbol{L} \in \boldsymbol{H}(\operatorname{Div}, \Omega)$ we have

$$(\nabla \boldsymbol{\psi}, \boldsymbol{L}) = -(\boldsymbol{\psi}, \operatorname{Div} \boldsymbol{L}) + \langle \boldsymbol{L}\boldsymbol{n}, \boldsymbol{\psi} \rangle_\Gamma$$

and analogous for $\boldsymbol{\psi} \in H(\operatorname{Curl}, \Omega)$ and $\boldsymbol{L} \in \boldsymbol{H}(\operatorname{Rot}, \Omega)$

$$(\operatorname{Curl} \boldsymbol{\psi}, \boldsymbol{L}) = (\boldsymbol{\psi}, \operatorname{Rot} \boldsymbol{L}) - \langle \boldsymbol{L}\boldsymbol{t}, \boldsymbol{\psi} \rangle_\Gamma .$$

A.4 Inequalities

We list two important inequalities in Sobolev spaces, Friedrichs' and Korn's inequality.

Lemma A.1 (Friedrichs' inequality)**.** *Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain. Let Γ_D be as subset of $\Gamma = \partial\Omega$ with positive measure. Then there exists a constant $c_F > 0$ depending only on Ω and Γ_D such that*

$$\|v\|_0 \leq c_F |v|_1 \quad \textit{for all } v \in H^1_{0,\Gamma_D}(\Omega),$$

where $|v|_1$ denotes the H^1-semi-norm and

$$H^1_{0,\Gamma_D} = \{v \in H^1(\Omega) \colon v = 0 \textit{ on } \Gamma_D\}.$$

Lemma A.2 (Korn's inequality)**.** *Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain. Then there exists a constant $c_K > 0$ depending only on Ω such that*

$$\|\varepsilon(\boldsymbol{\psi})\|_0 \leq c_K |\boldsymbol{\psi}|_1 \quad \textit{for all } \boldsymbol{\psi} \in (H^1(\Omega))^2/RM,$$

where $|\boldsymbol{\psi}|_1$ denotes the H^1-semi-norm and $(H^1(\Omega))^2/RM$ the L^2-orthogonal complement of RM in $(H^1(\Omega))^2$ with

$$RM = \{a(-x_2, x_1)^T + b \colon a \in \mathbb{R},\ b \in \mathbb{R}^2\}.$$

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Eidesstattliche Erklärung

Ich erkläre an Eides statt, dass ich die vorliegende Dissertation selbstständig und ohne fremde Hilfe verfasst, andere als die angegebenen Quellen und Hilfsmittel nicht benutzt bzw. die wörtlich oder sinngemäß entnommenen Stellen als solche kenntlich gemacht habe. Die vorliegende Dissertation ist mit dem elektronisch übermittelten Textdokument identisch.

Linz, Mai 2018

Katharina Rafetseder